

Analog Integrated Circuit Design

David A Johns & Ken Martin

Solution Manual

Chapter 1

1.9 PROBLEMS

Unless otherwise stated, assume the following hold throughout the problems section:

- Room temperature = 300 °K
- npn bipolar transistors:
 - $\beta = 100$
 - $V_A = 80 \text{ V}$
 - $\tau_b = 13 \text{ ps}$
 - $\tau_s = 4 \text{ ns}$
 - $r_b = 330 \Omega$
- n-channel MOS transistors:
 - $\mu_n C_{ox} = 92 \mu\text{A}/\text{V}^2$
 - $V_{tn} = 0.8 \text{ V}$
 - $\gamma = 0.5 \text{ V}^{1/2}$
 - $r_{ds} (\Omega) = 8000 L (\mu\text{m}) / I_D (\text{mA})$ in active region
 - $C_j = 2.4 \times 10^{-4} \text{ pF}/(\mu\text{m})^2$
 - $C_{j-sw} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$
 - $C_{ox} = 1.9 \times 10^{-3} \text{ pF}/(\mu\text{m})^2$
 - $C_{gs(overlap)} = C_{gd(overlap)} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$
- p-channel MOS transistors:
 - $\mu_p C_{ox} = 30 \mu\text{A}/\text{V}^2$
 - $V_{tp} = -0.9 \text{ V}$
 - $\gamma = 0.8 \text{ V}^{1/2}$
 - $r_{ds} (\Omega) = 12,000 L (\mu\text{m}) / I_D (\text{mA})$ in active region

$$\begin{aligned}
C_j &= 4.5 \times 10^{-4} \text{ pF}/(\mu\text{m})^2 \\
C_{j\text{-sw}} &= 2.5 \times 10^{-4} \text{ pF}/\mu\text{m} \\
C_{\text{ox}} &= 1.9 \times 10^{-3} \text{ pF}/(\mu\text{m})^2 \\
C_{\text{gs(overlap)}} &= C_{\text{gd(overlap)}} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}
\end{aligned}$$

- 1.1 Estimate the hole and electron concentrations in silicon doped with arsenic at a concentration of 10^{25} atoms/m³ at a temperature 22 °C above room temperature. Is the resulting material n type or p type?
- 1.2 For the pn junction of Example 1.2, does the built-in potential, Φ_0 , increase or decrease when the temperature is increased 11 °C above room temperature?
- 1.3 Calculate the amount of charge per $(\mu\text{m})^2$ in each of the n and p regions of the pn junction of Example 1.2 for a 5-V reverse-bias voltage. How much charge on each side would be present in a $10 \mu\text{m} \times 10 \mu\text{m}$ diode?
- 1.4 A silicon diode has $\tau_t = 12$ ps and $C_{j0} = 15$ fF. It is biased by a 43-k Ω resistor connected between the cathode of the diode and the input signal, as shown in Fig. P1.4. Initially, the input is 5 V, and then at time 0 it changes to 0 V. Estimate the time it takes for the output voltage to change from 5 V to 1.5 V (i.e., the $\Delta t_{-70\%}$ time). Repeat for an input voltage change from 0 V to 5 V and an output voltage change from 0 V to 3.5 V.

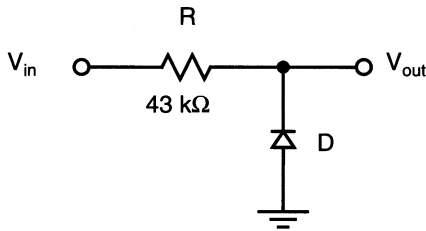


Fig. P1.4

- 1.5 Compare your answers for Problem 1.4 to those obtained using a SPICE simulation.
- 1.6 Verify that when $V_{\text{DS}} = V_{\text{eff}}$ is used in the triode equation for a MOS transistor, the current equals that of the active region equation given in (1.67).
- 1.7 Find I_D for an n-channel transistor having doping concentrations of $N_D = 10^{25}$ and $N_A = 10^{22}$ with $W = 50 \mu\text{m}$, $L = 1.5 \mu\text{m}$, $V_{\text{GS}} = 1.1$ V, and $V_{\text{DS}} = V_{\text{eff}}$. Assuming λ remains constant, estimate the new value of I_D if V_{DS} is increased by 0.3 V.
- 1.8 A MOS transistor in the active region is measured to have a drain current of 20 μA when $V_{\text{DS}} = V_{\text{eff}}$. When V_{DS} is increased by 0.5 V, I_D increases to 23 μA . Estimate the output impedance, r_{ds} , and the output impedance constant, λ .
- 1.9 Derive the low-frequency model parameters for an n-channel transistor having doping concentrations of $N_D = 10^{25}$ and $N_A = 10^{22}$ with $W = 10 \mu\text{m}$, $L = 1.2 \mu\text{m}$, $V_{\text{GS}} = 1.1$ V, and $V_{\text{DS}} = V_{\text{eff}}$. Assume that $V_{\text{SB}} = 1.0$ V.

- 1.10** Find the capacitances C_{gs} , C_{gd} , C_{db} , and C_{sb} for a transistor having $W = 50 \mu\text{m}$ and $L = 1.2 \mu\text{m}$. Assume that the source and drain junctions extend $4 \mu\text{m}$ beyond the gate, resulting in source and drain areas being $A_s = A_d = 200 (\mu\text{m})^2$ and the perimeter of each being $P_s = P_d = 58 \mu\text{m}$.
- 1.11** Consider the circuit shown in Fig. P1.11, where V_{in} is a dc signal of 1 V. Taking into account only the channel charge storage, determine the final value of V_{out} when the transistor is turned off, assuming half the channel charge goes to C_L .

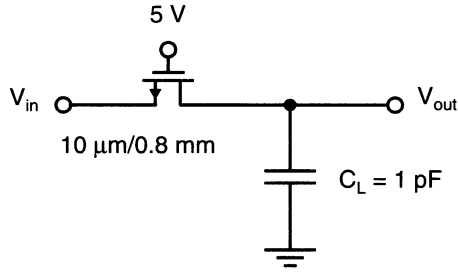


Fig. P1.11

- 1.12** For the same circuit as in Problem 1.11, the input voltage has a step voltage change at time 0 from 1 V to 1.2 V (the gate voltage remains at 5 V). Find its 99 percent settling time (the time it takes to settle to within 1 percent of the total voltage change). You may ignore the body effect and all capacitances except C_L . Also assume that $V_{tn} = V_{tn0}$. Repeat the question for V_{in} changing from 3 V to 3.1 V.
- 1.13** Repeat Problem 1.11, but now take into account the body effect on V_{tn} .
- 1.14** For an npn transistor having $I_C = 0.1 \text{ mA}$, calculate g_m , r_π , r_e , r_o , and $g_m r_o$.
- 1.15** A bipolar junction transistor has the following SPICE parameters (the SPICE name for the parameter is included in parentheses):
- I_S (IS) = 2.0×10^{-18}
 - B_F (BF) = 100
 - B_R (BR) = 1
 - V_A (VA) = 50 V
 - τ_F (TF) = $12 \times 10^{-12} \text{ s}$
 - τ_R (TR) = $4 \times 10^{-9} \text{ s}$
 - C_{je0} (CJE) = $15 \times 10^{-15} \text{ F}$
 - Φ_e (VJE) = 0.9 V
 - m_e (MJE) = 0.27
 - C_{jc0} (CJC) = $18 \times 10^{-15} \text{ F}$
 - Φ_c (VJC) = 0.7 V
 - m_c (MJC) = 0.37
 - C_{js0} (CJS) = $40 \times 10^{-15} \text{ F}$
 - Φ_s (VJS) = 0.64 V
 - m_s (MJS) = 0.29
 - R_e (RE) = 30

$$R_b (R_B) = 500$$

$$R_c (R_C) = 90$$

Initially, the circuit shown in Fig. P1.15 has a 0-V input. At time 0 its input changes to 5 V. Estimate the time it takes its output voltage to saturate, using the concepts of average capacitance and first-order transient solutions for each node. The time constants of the individual nodes can be added to arrive at an overall time constant for the approximate first-order transient response of the circuit. Next, assume that the input changes from 5 V to 0 V at time 0. How long does it take the output voltage to change to 3.5 V?

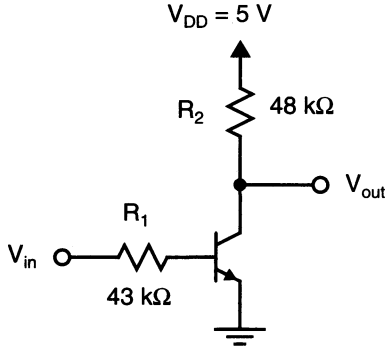


Fig. P1.15

- 1.16** Compare your answers to Problem 1.15 to those obtained using SPICE.
- 1.17** Verify that for $I_{BR} \gg I_B$ and $I_{BR} \gg I_C / \beta$, Eq. (1.140) simplifies to Eq. (1.139).

Chapter 1 - Problems

1.1) As is a pentavalent impurity

∴ n-type

$$n_n = N_D = \underline{10^{25} \text{ carriers/m}^3}$$

$$n_p = \frac{n_i^2}{N_D} \text{ for } n_i \Big|_{T=322K} = 4 \times n_i \Big|_{T=300K} = 6 \times 10^{16} \frac{\text{carriers}}{\text{m}^3}$$
$$= \underline{3.6 \times 10^8 \text{ carriers/m}^3}$$

1.2) From Example 1.2,

$$\Phi_0 = 0.88 \text{ V at } 300 \text{ K}$$

For $T = 311 \text{ K}$,

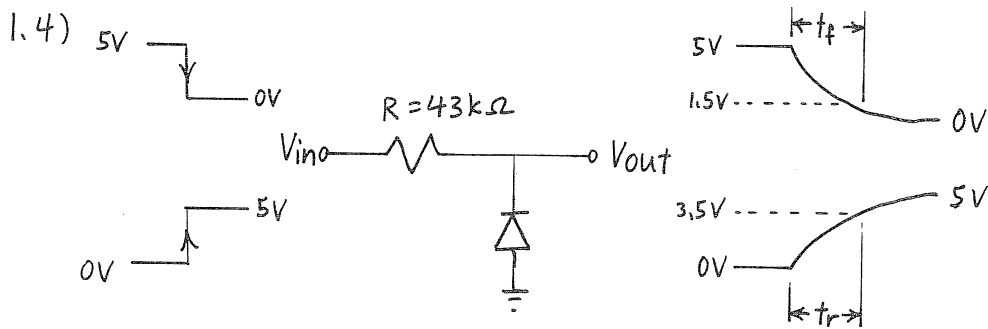
$$\Phi_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) = 26.8 \text{ mV} \times \ln\left(\frac{10^{25} \times 10^{22}}{(2 \times 1.5 \times 10^{16})^2}\right)$$
$$= \underline{0.87 \text{ V}}$$

∴ The built-in potential decreases when the temperature is increased.

$$1.3) \quad Q^+ = Q^- \cong [2q K_s \epsilon_0 (\Phi_0 + V_R) N_D]^{1/2}$$
$$= [2 \times 1.602 \times 10^{-19} \times 11.8 \times 8.85 \times 10^{-12} \times (0.88 + 5.0) \times 10^{22}]^{1/2}$$
$$= \underline{1.4 \text{ fC}/\mu\text{m}^2}$$

For a $10 \mu\text{m} \times 10 \mu\text{m} = 100 \mu\text{m}^2$ area,

140 fC of charge would be present.



Fall time, t_f :

$$\begin{aligned}
 C_{j-av} &= 2C_{j0} \Phi_0 \frac{\left(\sqrt{1 + \frac{V_2}{\Phi_0}} - \sqrt{1 + \frac{V_1}{\Phi_0}} \right)}{V_2 - V_1} \\
 &= 2 \times 15 \times 10^{-15} \times 0.9 \times \frac{\left(\sqrt{1 + \frac{5}{0.9}} - \sqrt{1 + \frac{1.5}{0.9}} \right)}{5 - 1.5} \\
 &= \underline{7.2 \text{ fF}}
 \end{aligned}$$

$$\begin{aligned}
 \therefore t_f &\approx 1.2\tau = 1.2 \times RC_{j-av} \\
 &= 1.2 \times 43 \times 10^3 \times 7.2 \times 10^{-15} \\
 &= \underline{0.37 \text{ nsec}}
 \end{aligned}$$

Rise time, t_r :

$$\begin{aligned}
 C_{j-av} &= 2 \times 15 \times 10^{-15} \times 0.9 \times \frac{\sqrt{1} - \sqrt{1 + \frac{3.5}{0.9}}}{0 - 3.5} \\
 &= \underline{9.3 \text{ fF}}
 \end{aligned}$$

$$\begin{aligned}
 \therefore t_r &\approx 1.2\tau = 1.2 RC_{j-av} \\
 &= 1.2 \times 43 \times 10^3 \times 9.3 \times 10^{-15} \\
 &= \underline{0.48 \text{ nsec}}
 \end{aligned}$$

1.5) Simulation results:

$$t_f = \underline{0.37 \text{ nsec}}$$

$$t_r = \underline{0.44 \text{ nsec}}$$

Results are consistent with our calculations in Problem 1.4.

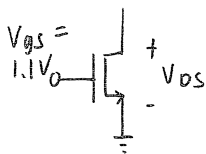
1.6) Active: $I_D = \frac{\mu_n C_{ox}}{2} \times \frac{W}{L} (V_{gs} - V_{tn})^2$ ①

Triode: $I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{gs} - V_{tn}) V_{DS} - \frac{V_{DS}^2}{2} \right]$

But $V_{os} = V_{eff} = V_{as} - V_{tn}$

$$\begin{aligned} \therefore I_D &= \mu_n C_{ox} \frac{W}{L} \left[(V_{gs} - V_{tn})^2 - \frac{1}{2} (V_{gs} - V_{tn})^2 \right] \\ &= \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{gs} - V_{tn})^2 \\ &= \text{①} \quad \text{Q.E.D.} \end{aligned}$$

1.7)



Estimate I_D when $V_{DS} = V_{eff} + 0.3V$.

$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{gs} - V_{tn})^2 (1 + \lambda (V_{DS} - V_{eff}))$$

Find $\lambda \triangleq \frac{k}{2L\sqrt{V_{DS} - V_{eff} + \Phi_0}}$

$$\begin{aligned} \text{where } k &= \sqrt{\frac{2K_s \epsilon_0}{qNA}} = \sqrt{\frac{2 \times 11.8 \times 8.85 \times 10^{-12}}{1.602 \times 10^{-19} \times 10^{22}}} \\ &= 3.612 \times 10^{-7} \text{ m}/\sqrt{V} \end{aligned}$$

For $V_{DS} = V_{eff}$,

$$\lambda = \frac{3.612 \times 10^{-7}}{2 \times 1.5 \times 10^{-6} \sqrt{0.9}} = 0.127 \text{ V}^{-1}$$

Now

$$\begin{aligned} I_D \big|_{V_{DS} = V_{eff}} &= \frac{q2}{2} \frac{\mu A}{V^2} \times \frac{50}{1.5} \times (1.1 - 0.8)^2 (1 + \cancel{\lambda \times 0}) \\ &= \underline{138 \mu A} \end{aligned}$$

$$\begin{aligned} I_D \big|_{V_{DS} = V_{eff} + 0.3V} &= \frac{q2}{2} \frac{\mu A}{V^2} \times \frac{50}{1.5} \times (1.1 - 0.8)^2 (1 + 0.127 \times 0.3) \\ &= \underline{143 \mu A} \end{aligned}$$

which translates into a 4% increase in current.

1.8) Find r_{DS} and λ .

Given: $I_D|_{V_{DS}=V_{eff}} = 20 \mu A$

$I_D|_{V_{DS}=V_{eff}+0.5V} = 23 \mu A$

Now $I_D|_{V_{DS}=V_{eff}+0.5V} = I_D|_{V_{DS}=V_{eff}} \times (1 + \lambda \times 0.5V)$

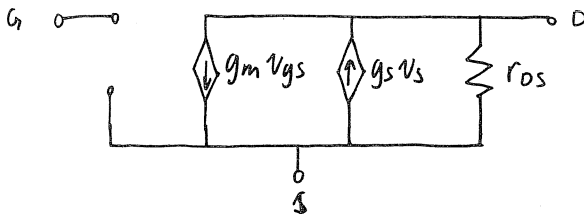
$\lambda = 2 \left(\frac{23 \mu A}{20 \mu A} - 1 \right)$

$\lambda = 0.3 V^{-1}$

$r_{DS} = \frac{\Delta V}{\Delta I}$

$= \frac{0.5V}{3 \mu A} = \underline{167 k\Omega}$

1.9) Derive model parameters g_m , g_s and r_{DS} .



$\lambda|_{V_{DS}=V_{eff}} = \frac{k}{2L\sqrt{\Phi_0}}$

from Problem 1.7, $k = 3.61 \times 10^{-7} m/V$

$\therefore \lambda|_{V_{DS}=V_{eff}} = \frac{3.61 \times 10^{-7}}{2 \times 1.2 \times 10^{-6} \sqrt{0.9}} = \underline{0.159 V^{-1}}$

Now $I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{DS} - V_{th})^2$

$= \frac{92 \mu A}{2} \times \frac{10}{1.2} (1.1 - 0.8)^2$

$= 34.5 \mu A$

$\therefore r_{DS} = \frac{1}{\lambda I_D} = \frac{1}{0.159 \times 34.5 \times 10^{-6}} = \underline{182 k\Omega}$

(cont.)

1.9) (cont.)

$$g_m = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_D} = \sqrt{2 \times 92 \frac{\mu A}{V^2} \times \frac{10}{1.2} \times 34.5 \times 10^{-6}} = \underline{230 \mu A/V}$$

$$g_s = \frac{\gamma g_m}{2 \sqrt{V_{SB} + 2|\phi_F|}}$$

$$= \frac{0.5 \times 230 \times 10^{-6}}{2 \sqrt{1 + 0.7}}$$

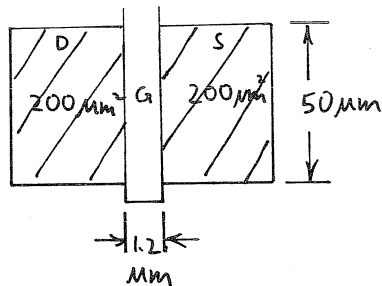
$$= \underline{44 \mu A/V}$$

Assume $\gamma = 0.5$

$$\phi_F = \frac{kT}{q} \ln\left(\frac{N_A}{n_i}\right) = \frac{k 300}{q} \ln\left(\frac{10^{22}}{1.1 \times 10^{10}}\right)$$

$$= 0.35 V$$

1.10) Find C_{gs} , C_{gd} , C_{db} , C_{sb}



$$C_{gs} = \frac{2}{3} W L C_{ox} + C_{ov} W$$

$$= \frac{2}{3} \times 50 \times 1.2 \times 1.9 \times 10^{-3} + 2 \times 10^{-4} \times 50$$

$$= \underline{86 fF}$$

$$C_{gd} = C_{gd-ov} \cdot W = 2 \times 10^{-4} \times 50$$

$$= \underline{10 fF}$$

$$C_{db} = A_d C_j + P_d C_{jsw}$$

$$= 200 \mu m^2 \cdot 2.4 \times 10^{-4} \frac{pF}{\mu m^2} + 58 \times 2 \times 10^{-4}$$

$$= \underline{60 fF}$$

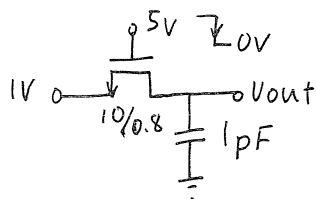
$$C_{sb} = C_j (A_s + W L) + C_{jsw} P_s$$

$$= 2.4 \times 10^{-4} (200 + 50 \times 1.2)$$

$$+ 2 \times 10^{-4} \times 58$$

$$= \underline{74 fF}$$

1.11)



Find V_{out}

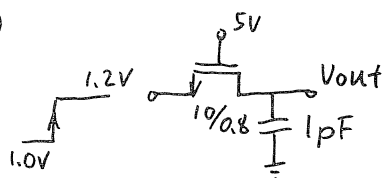
$$\begin{aligned} \text{channel charge } Q_{ch} &= WL C_{ox} (V_{gs} - V_{tn}) \\ &= 10 \times 0.8 \times 0.0019 ((5-1) - 0.8) \\ &= \underline{48.6 \text{ fC}} \end{aligned}$$

Assuming $\frac{1}{2} \times Q_{ch}$ is injected into the storage capacitor,

$$\Delta V = -\frac{\frac{1}{2} Q_{ch}}{C_L} = -\frac{\frac{1}{2} \times 48.6 \text{ fC}}{1 \text{ pF}} = -24 \text{ mV}$$

$$\therefore V_{out} = 1 \text{ V} - \Delta V = \underline{0.976 \text{ V}}$$

1.12)



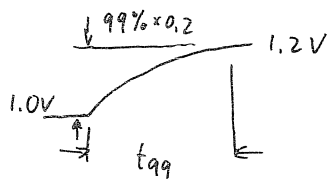
Find settling time.

$$\tau = r_{DS} C_L$$

$$\begin{aligned} \text{where } r_{DS} &= \frac{1}{\mu_n C_{ox} \frac{W}{L} V_{eff}} = \frac{1}{92 \times 10^{-6} \times \frac{10}{0.8} \times 3.2 \text{ V}} \\ &= 272 \Omega \end{aligned}$$

$$\begin{aligned} \therefore \tau &= 272 \Omega \times 1 \text{ pF} \\ &= \underline{0.27 \text{ nsec}} \end{aligned}$$

For a 99% settling time,



$$V_{out}(t) = 1 \text{ V} + 0.2 (1 - e^{-t/\tau})$$

$$\therefore \text{find } t_{99} \text{ where } 1 - e^{-t/\tau} = 0.99$$

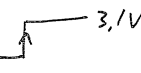
$$e^{-t/\tau} = 0.01$$

$$t_{99} = 4.6 \tau$$

$$= \underline{1.25 \text{ nsec}}$$

$\therefore$ the settling time is 1.25 nsec.

1.12) (cont.)

Repeat for $V_{in} = 3.0V$ 

$$r_{DS} = \frac{1}{92 \times 10^{-6} \times \frac{10}{0.8} (5 - 3 - 0.8)} = 725 \Omega$$

$$\therefore \tau = r_{DS} C_L = 725 \Omega \times 1 pF = 0.725 nsec$$

$$\text{and } t_{99\%} = 4.6 \tau = \underline{3.33 nsec}$$

1.13) Repeat Problem 1.12, this time accounting for the body effect.

 changes the threshold voltage

$$V_{tn} = V_{tn0} + \gamma (\sqrt{V_{SB} + 2|\phi_F|} - \sqrt{2|\phi_F|})$$

For $V_{in} = 1V$,

$$\begin{aligned} V_{tn} &= 0.8V + 0.5 (\sqrt{1 + 0.7} - \sqrt{0.7}) \\ &= 1.03V \end{aligned}$$

$$\begin{aligned} \therefore r_{DS} &= \frac{1}{\mu_n C_{ox} \frac{W}{L} V_{eff}} = \frac{1}{92 \times 10^{-6} \times \frac{10}{0.8} \times (4 - 1.03)} \\ &= 293 \Omega \end{aligned}$$

$$\therefore \tau = 293 \times 1 pF = 293 ps$$

$$\text{leading to } t_{99\%} = \underline{1.35 ns}$$

For $V_{in} = 3V \Rightarrow V_{tn} = 1.34V$

$$r_{DS} = 1.32 k\Omega \quad \tau = 1.32 ns$$

$$t_{99\%} = \underline{6.1 ns}$$

1.14)

$$g_m = \frac{I_C}{V_T} = \frac{0.1 \text{ mA}}{26 \text{ mV}} = 3.8 \text{ mA/V}$$

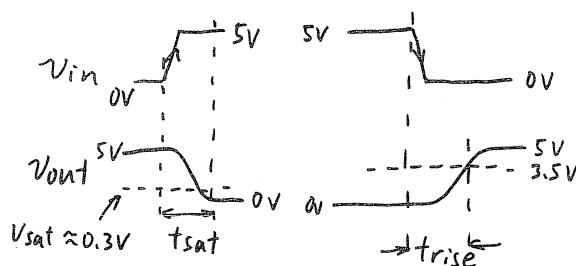
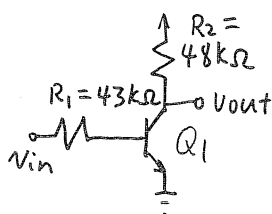
$$r_\pi = \frac{\beta}{g_m} = \frac{100}{3.8 \text{ mA/V}} = 26 \text{ k}\Omega$$

$$r_e = \alpha / g_m = \frac{100}{101} \times \frac{1}{3.8 \text{ mA/V}} = 260 \Omega$$

$$r_o = \frac{V_A}{I_C} = \frac{80 \text{ V}}{0.1 \text{ mA}} = 800 \text{ k}\Omega$$

$$g_m r_o = 3.8 \text{ mA/V} \times 800 \text{ k}\Omega = 3 \times 10^3$$

1.15)



Find t_{sat} and t_{rise} .

Part 1 : Estimating t_{sat}

The process of saturating the output is comprised of two phases :

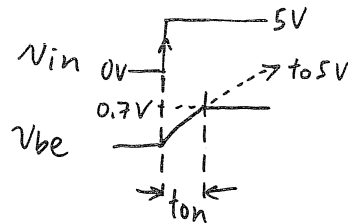
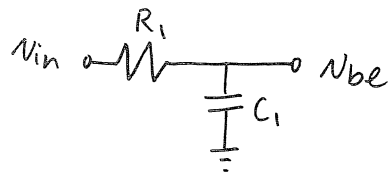
- ① The transistor must first require time, t_{on} , to turn on.
- ② The transistor then requires time, t_{fall} , to drive the output low.

(cont.)

1.15) (cont.) [Part 1, phase 1]

Phase ① - turning transistor on

The transistor is initially off and will remain so until $v_{be} \approx 0.7V$. Before then, the following circuit model can be used:



for a simple RC network

$$v_{be} = 5V \times (1 - e^{-t/\tau_{in}})$$

$$\text{where } \tau_{in} = R_1 C_1$$

$$R_1 = 43k\Omega$$

$$C_1 = C_{be}|_{v_{in}=0V} + C_{cb}|_{v_{cb}=5V}$$

$$= C_{je0} + \frac{C_{jco}}{(1 + \frac{v_{cb}}{V_{bc0}})^{1/3}} = 15 + \frac{18}{(1 + \frac{5V}{0.7})^{1/3}} \text{ fF}$$

$$= 24 \text{ fF}$$

$$\therefore \tau_{in} = 43k\Omega \times 24 \text{ fF} = 1.0 \text{ nsec}$$

To determine the time for v_{be} to reach $0.7V$, t_{on} ,

$$v_{be}(t) = 5(1 - e^{-t_{on}/\tau_{in}})V \equiv 0.7V$$

$$e^{-t_{on}/\tau_{in}} = 0.86$$

$$t_{on} = 0.15 \tau_{in}$$

$$\underline{t_{on} = 150 \text{ psec}}$$

(cont.)

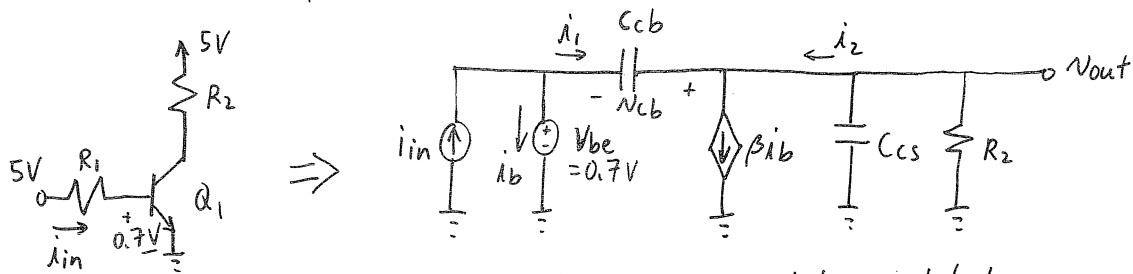
1.15 (cont.) [Part 1, phase 2]

Phase ② - Pulling the output, V_{out} , Low

During this phase, the transistor is on and in the active mode. To analyze the large signal behaviour, we take note of two properties of BJT devices in order to create the model below:

1) V_{be} is approximately constant and equal to $0.7V$.

2) $i_c = \beta i_b$



Schematic used to model large signal transients

Note that C_{be} is no longer significant as V_{be} is now constant.

$$C_{cb} = \frac{C_{jco}}{(1 + V_{cb}/\phi_{co})^{1/3}}$$

$$= \frac{18 \text{ fF}}{(1 + 2.5/0.7)^{1/3}}$$

Let $V_{cb} = 2.5V$ to represent the average value during t_{fall} .

$C_{cb} = 11 \text{ fF}$

$C_{cs} = \frac{C_{js}}{(1 + V_{cs}/\phi_{s0})^{1/2}} = \frac{40 \text{ fF}}{(1 + 2.5/0.64)^{1/2}} = 18 \text{ fF}$

Now $i_b = i_{in} - i_1$ [1]

$i_1 = -C_{cb} \frac{dV_{cb}}{dt} = -C_{cb} \frac{dV_{out}}{dt}$ [2]

$i_2 = -C_{cs} \frac{dV_{out}}{dt}$ [3]

and $\beta i_b = i_1 + i_2$ [4]

(assume effectively all of i_2 passes through C_{cs})

1.15 (cont.) [Part 1, phase 2]

① → ④

$$\beta(i_{in} - i_1) = i_1 + i_2$$

$$\beta i_{in} = (\beta + 1)i_1 + i_2$$

sub ②, ③

$$= -(\beta + 1)(C_{cb} - C_{cs}) \frac{dV_{out}}{dt}$$

$$\therefore \frac{dV_{out}}{dt} = \frac{-\beta i_{in}}{(\beta + 1)C_{cb} + C_{cs}}$$

$$= \frac{-100 \times 10^{-4}}{(101 \times 11 + 18) \times 10^{-15}}$$

$$\frac{dV_{out}}{dt} = -8.86 \text{ V/nsec}$$

$\therefore$ For the output to drop $(5 - 0.2)\text{V} = 4.8\text{V}$,

$$t_{fall} = \Delta V / \frac{dV_{out}}{dt} = -4.8\text{V} / -8.86 \text{ V/nsec}$$

$$t_{fall} = 540 \text{ psec}$$

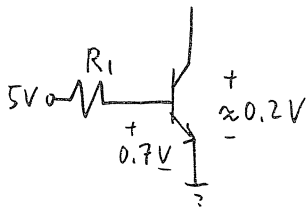
$\therefore$ The total time required to saturate Q_1 is

$$t_{sat} = t_{on} + t_{fall} = 150 + 540 \text{ psec}$$

$$t_{sat} = 690 \text{ psec}$$

Part 2: Estimating trise

For the case where v_{in} drops from 5V to 0V, Q_1 starts out in saturation. In order to pull the output up, the saturation charge, Q_s , must first be removed. After this, the base minority charge, Q_B , is removed. The times representing these two phases are t_s and t_b .



$$\therefore \underline{t_{rise} = t_s + t_b}$$

(cont.)

1.15 (cont.) [Part 2]

$$\text{Now } t_s = \tau_R \ln \left[\frac{I_{BR} + I_B}{I_{BR} + I_{C/2}} \right] \quad (1.140)$$

$$\text{where } I_{BR} = \frac{V_{fwd-diode}}{R_1} = \frac{0.7V}{43k\Omega} = 16\mu A$$

$$I_B = \frac{V_{in0} - V_{fwd-diode}}{R_1} = \frac{5 - 0.7V}{43k\Omega} = 0.1mA$$

$$I_C = \frac{V_{DD} - V_{sat}}{R_2} = \frac{5 - 0.2V}{48k\Omega} = 0.1mA$$

$$\tau_R = 4nsec$$

$$\therefore t_s = 4 \times 10^{-9} \ln \left[\frac{16 \times 10^{-6} + 10^{-4}}{16 \times 10^{-6} + 10^{-4}/100} \right] = \underline{7.7nsec}$$

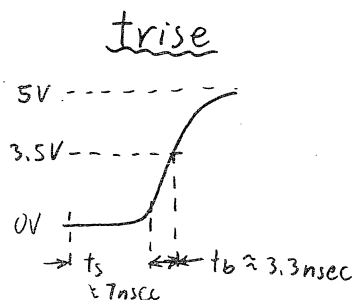
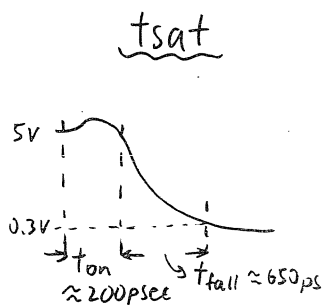
$$t_b = \frac{Q_b}{I_{BR}} = \frac{\tau_b I_C}{I_{BR}} = \frac{12 \times 10^{-12} \times 100 \times 10^{-6}}{16 \times 10^{-6}} = \underline{0.08nsec}$$

$$\left. \begin{array}{l} \therefore \underline{t_{rise} = 7.7 + 0.08 = 7.8nsec} \\ \text{and } \underline{t_{sat} = 690psec} \end{array} \right\}$$

1.16) SPICE results:

$$\left\{ \begin{array}{l} t_{rise} = 10.3nsec \\ t_{sat} = 850psec \end{array} \right\}$$

Thus our estimates are accurate to within about 25 percent. It is interesting to note that the different phases of each of the rise and fall times are clearly visible in the simulation results.



1.17) Verify Eq. (1.140) $\rightarrow$ Eq. (1.139) for $I_{BR} \gg (I_B \text{ \& } I_C/\beta)$

$$\begin{aligned}\text{Eq. (1.140): } t_s &= \tau_s \ln \left[\frac{I_{BR} + I_B}{I_{BR} + I_C/\beta} \right] \\ &= \tau_s \ln \left[1 + \frac{I_B - I_C/\beta}{I_{BR} + I_C/\beta} \right] \\ &\approx \tau_s \ln \left[1 + \underbrace{\frac{I_B - I_C/\beta}{I_{BR}}}_{\approx 0} \right] \\ &\quad \approx 0 \quad \because I_{BR} \gg I_B \text{ or } I_C/\beta\end{aligned}$$

$\therefore$ We can approximate this function with a first order Taylor Series Expansion

$$f(x) \Big|_{x \approx 0} \approx f(0) + f'(0) x$$

$$\text{For } f(x) = \tau_s \ln[1+x]$$

$$\begin{aligned}f(x) \Big|_{x \approx 0} &\approx \tau_s \left[\ln[1+0] + \frac{1}{1+0} x \right] \\ &= \tau_s x\end{aligned}$$

$$\therefore t_s = f\left(\frac{I_B - I_C/\beta}{I_{BR}}\right)$$

$$\therefore \underline{t_s \cong \tau_s \times \frac{I_B - I_C/\beta}{I_{BR}}} \quad \text{which is Eq. (1.139)}$$

Q.E.D.

Chapter 2

2.7 PROBLEMS

Unless otherwise stated, assume the following hold throughout the problems section:

- npn bipolar transistors:
 $\beta = 100$
 $V_A = 80 \text{ V}$
 $\tau_b = 13 \text{ ps}$
 $\tau_s = 4 \text{ ns}$
 $r_b = 330 \Omega$
- n-channel MOS transistors:
 $\mu_n C_{ox} = 92 \mu\text{A}/\text{V}^2$
 $V_{tn} = 0.8 \text{ V}$
 $\gamma = 0.5 \text{ V}^{1/2}$
 $r_{ds} (\Omega) = 8,000L (\mu\text{m})/I_D (\text{mA})$ in active region
 $C_j = 2.4 \times 10^{-4} \text{ pF}/(\mu\text{m})^2$
 $C_{j-sw} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$
 $C_{ox} = 1.9 \times 10^{-3} \text{ pF}/(\mu\text{m})^2$
 $C_{gs(overlap)} = C_{gd(overlap)} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$
- p-channel MOS transistors:
 $\mu_p C_{ox} = 30 \mu\text{A}/\text{V}^2$
 $V_{tp} = -0.9 \text{ V}$
 $\gamma = 0.8 \text{ V}^{1/2}$
 $r_{ds} (\Omega) = 12,000L (\mu\text{m})/I_D (\text{mA})$ in active region

$$C_j = 4.5 \times 10^{-4} \text{ pF}/(\mu\text{m})^2$$

$$C_{j\text{-sw}} = 2.5 \times 10^{-4} \text{ pF}/\mu\text{m}$$

$$C_{\text{ox}} = 1.9 \times 10^{-3} \text{ pF}/(\mu\text{m})^2$$

$$C_{\text{gs(overlap)}} = C_{\text{gd(overlap)}} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$$

- 2.1 Discuss briefly the relationships between an ion beam's acceleration potential, the beam current, and the time of implantation on the resulting doping profile.
- 2.2 Place the following processing steps in their correct order: metal deposition and patterning, field implantation, junction implantation, well implantation, polysilicon deposition and patterning, field-oxide growth.
- 2.3 What are the major problems associated with a single threshold-voltage-adjust implant?
- 2.4 What is the reason for using a field implant and why is it often not needed in the well regions?
- 2.5 What are the major trade-offs in using a wet process or a dry process when growing thermal SiO_2 ?
- 2.6 Why is polysilicon rather than metal used to realize gates of MOS transistors?
- 2.7 Why can't a microcircuit be annealed after metal has been deposited?
- 2.8 What minimum distance, in terms of λ , would you expect that metal should be separated from polysilicon? Why?
- 2.9 Find the circuit that the layout shown in Fig. P2.9 realizes. Simplify the circuit, if possible, and give the sizes of all transistors. Assume $L = 2\lambda$, where $\lambda = 1 \mu\text{m}$.

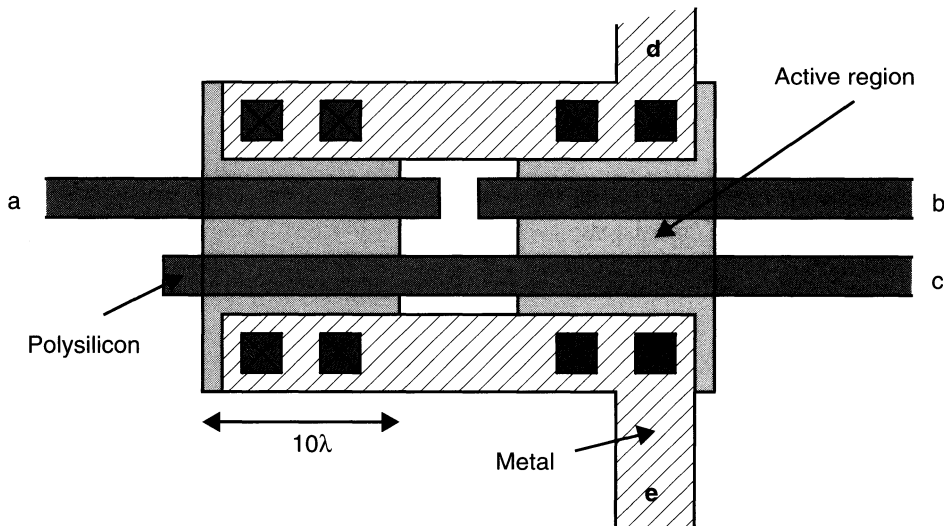


Fig. P2.9

- 2.10 Find the transistor schematic for the CMOS logic circuit realized by the layout shown in Fig. P2.10. Give the widths of all transistors. Assume $L = 2\lambda$,

where $\lambda = 0.4 \mu\text{m}$. In tabular form, give the area and perimeter of each junction that is not connected to V_{DD} or to ground.

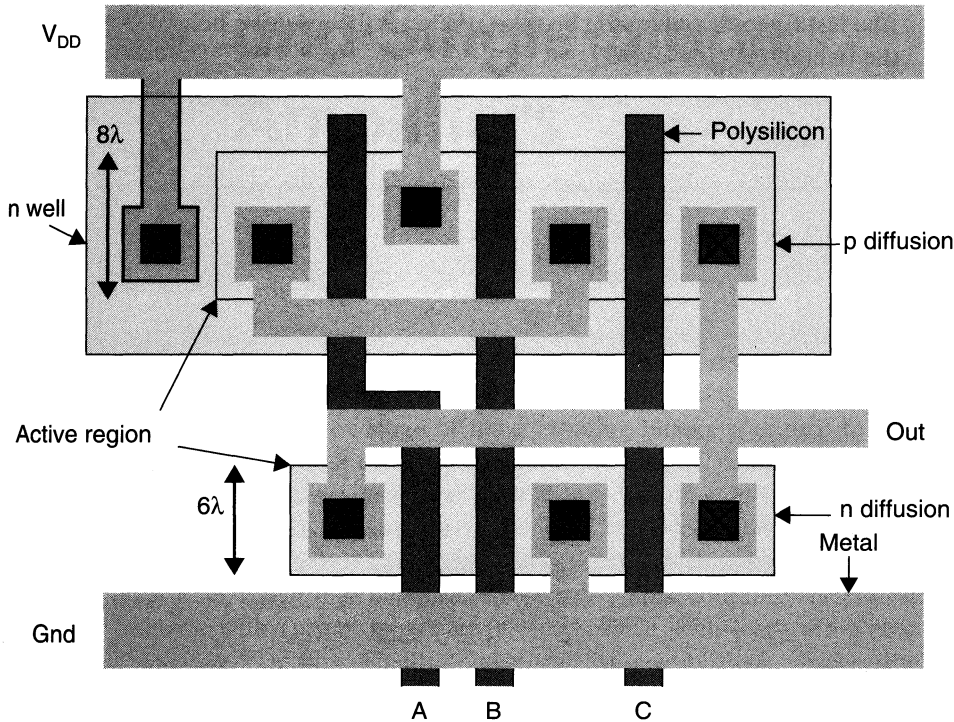


Fig. P2.10

- 2.11 Repeat Example 2.1 for the case in which the two transistors do not physically share any junction, but each junction is realized in a way similar to junction J_2 .
- 2.12 Repeat Example 2.2 where an overall transistor width of 80λ is still desired, but assume 8 parallel transistors, each of width 10λ , are used.
- 2.13 Repeat Example 2.2 where an overall transistor width of 80λ is still desired, but assume 2 parallel transistors, each of width 40λ , are used.
- 2.14 We desire to match two capacitors of relative sizes 9 and 4.523. Sketch a layout for the two capacitors such that their ratio will be maintained during overetching.
- 2.15 Given that a unit-sized capacitor is $5 \mu\text{m} \times 5 \mu\text{m}$ and that the e-beam lithography rounds all sizes to $0.1 \mu\text{m}$, what is the matching accuracy due to rounding of the capacitors found in Problem 2.14? What is the new matching accuracy if the capacitor sizes are doubled to 18 and 9.046?
- 2.16 Given that a polysilicon layer has $7 \Omega/\square$, what is the resistance of a long line that is $2 \mu\text{m}$ wide and $1,000 \mu\text{m}$ long? (Ignore any contact resistance.)

- 2.17** Repeat Problem 2.16 for the case in which the resistive line is laid out in a serpentine manner, as shown in Fig. 2.23, where enough bends are used such that a line drawn along the middle of the serpentine resistor has length $1,000\text{ }\mu\text{m}$ (the last finger length might be short). What is the resulting height and width of the overall resistor layout?

chapter 2 - Problems

2.1) The higher the beam's acceleration potential, the deeper the maximum ion concentration into the silicon wafer. The higher the beam current or the time of exposure, the larger the amount of dosage.

2.2) Well implantation, Field implantation, Field oxide growth, Polysilicon deposition & patterning, Junction implantation, Metal deposition & patterning.

2.3) Single threshold-voltage-adjust normally leads to an n-well doping level which is higher than optimum. This will increase the junction capacitance and the body effect of the transistors in the well.

2.4) It prohibits the silicon under the field oxide from inversion when a conductor running over the field oxide has a large voltage. The field implants are not critical for the wells, as the heavier doping of the well guarantees that the silicon will never invert under (cont.)

2.4) (cont.) the field oxide in this region.

2.5) Dry process is slower than the wet process. However, it results in denser, higher quality SiO_2 that is less porous.

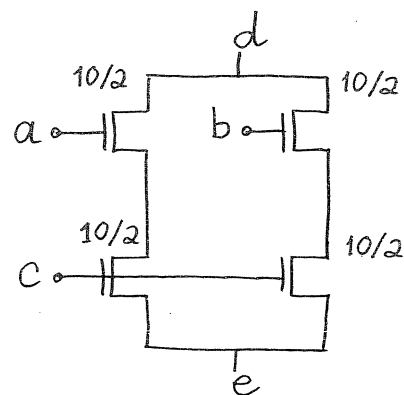
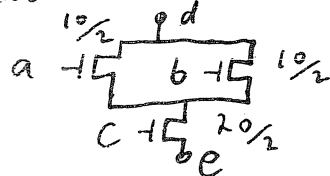
2.6) Polysilicon has a much higher melting point compared to metals, and that will allow annealing at higher temperature before metal deposition.

2.7) Annealing temperatures are normally higher than the metal melting point.

2.8) OK since there is no restriction on overlaying metal and polysilicon. However, many processes do not allow parallel metal and polysilicon lines to overlap each other due to metal breakage because of "hill" effects.

2.9) The circuit cannot be simplified.
(sizes are in μm).

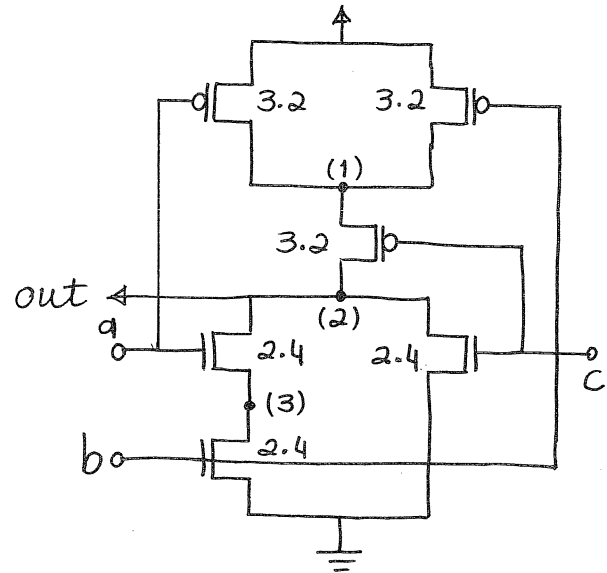
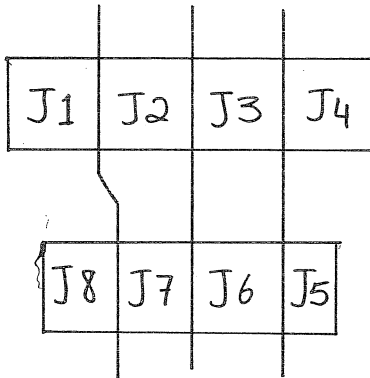
unless it is a digital circuit in which case



2.10) All the transistors have the same length equal to $0.8\mu\text{m}$.

The width are shown on the schematic.

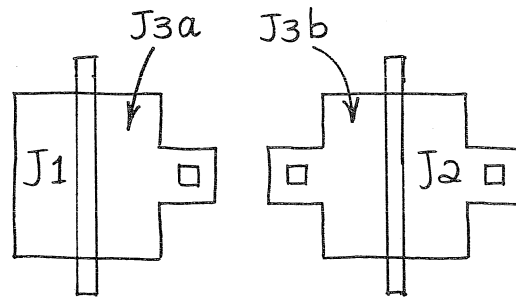
(all sizes are in μm .)



Junction	Node	Area (μm^2)	Peri. (μm)
J1	1	7.68	8
J2	VDD	—	—
J3	1	7.68	4.8
J4	2	7.68	8
J5	2	5.76	7.2
J6	GND	—	—
J7	3	1.92	1.6
J8	2	5.76	7.2

2.11) J_1 & J_2 remain unchanged.

J_3 is now splitted to J_{3a} & J_{3b} .

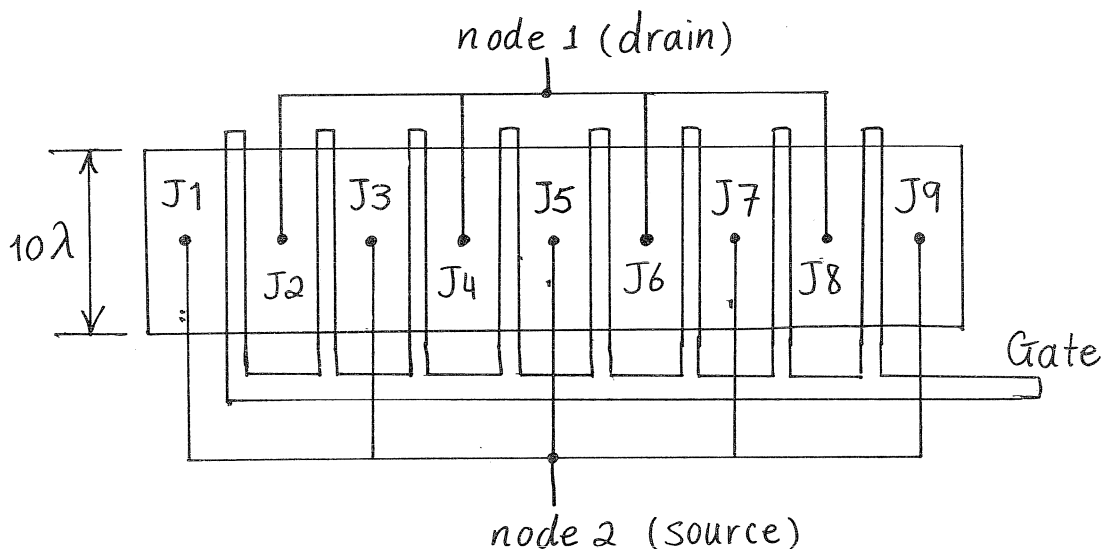


$$A_{3a} = A_{3b} = A_{J_2} = 8 (\mu m)^2$$

$$P_{3a} = P_{3b} = P_{J_2} = 10 \mu m$$

The area is increased by a factor of ≈ 3 , the peri. is increased by a factor of 10.

2.12)



(layout not to scale)

For the drain:

$$A_{J_2} = A_{J_4} = A_{J_6} = A_{J_8} = 6\lambda \times 10\lambda = 60\lambda^2 = \underline{15 (\mu m)^2}$$

$$P_{J_2} = P_{J_4} = P_{J_6} = P_{J_8} = 6\lambda + 6\lambda = 12\lambda = \underline{6 \mu m}$$

(cont.)

2.12) (cont.)

$$\therefore C_{db} = 4(A_{J2}C_j + P_{J2}C_{jsw}) = \underline{0.019 \text{ pF}}$$

For the source:

$$A_{J1} = A_{J9} = 5\lambda \times 10\lambda = 50\lambda^2 = \underline{12.5 (\mu m)^2}$$

$$A_{J3} = A_{J5} = A_{J7} = A_{J2} = \underline{15 (\mu m)^2}$$

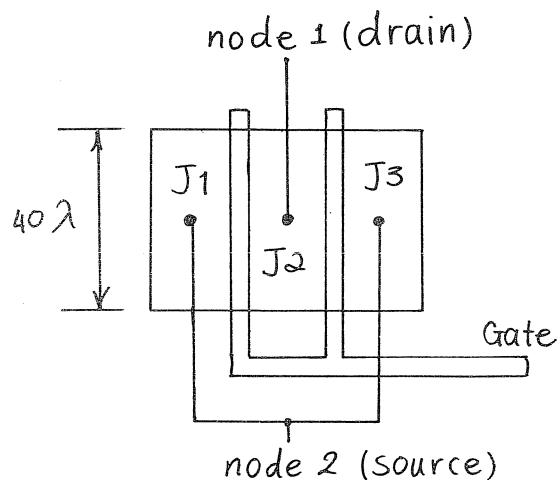
$$P_{J1} = P_{J9} = 5\lambda + 5\lambda + 10\lambda = 20\lambda = \underline{10 \mu m}$$

$$P_{J3} = P_{J5} = P_{J7} = P_{J2} = \underline{6 \mu m}$$

$$\therefore C_{sb} = (A_{J1} + A_{J3} + \dots + A_{J9} + WL)C_j + (P_{J1} + P_{J3} + \dots + P_{J9})C_{jsw} = \underline{0.034 \text{ pF}}$$

Compared to the results of Example 2.2, the changes in capacitances are insignificant.

2.13) (layout not to scale)



For the drain:

$$A_{J2} = 6\lambda \times 40\lambda = \underline{60 (\mu m)^2}$$

$$P_{J2} = 6\lambda + 6\lambda = \underline{6 \mu m}$$

$$\therefore C_{db} = A_{J2}C_j + P_{J2}C_{jsw} = \underline{0.016 \text{ pF}}$$

(cont.)

2.13) (Cont.)

For the source:

$$A_{J1} = A_{J3} = 5\lambda \times 40\lambda = \underline{50 (\mu m)^2}$$

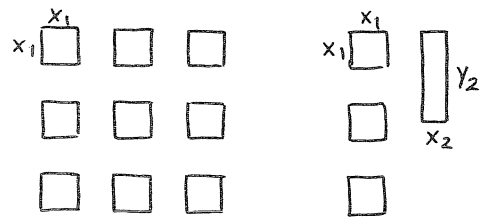
$$P_{J1} = P_{J3} = 5\lambda + 5\lambda + 40\lambda = \underline{25 \mu m}$$

$$\therefore C_{sb} = (A_{J1} + A_{J3} + WL)C_j + (P_{J1} + P_{J3})C_{jsw} = \underline{0.044 pF}$$

2.14) $K = 1.523$

Using (2.26):

$$\frac{Y_2}{X_1} = 2.415, \quad \frac{X_2}{X_1} = 0.631$$



$$\therefore \underline{Y_2 = 2.415 X_1} \quad \& \quad \underline{X_2 = 0.631 X_1}$$

2.15) It is desired to have coeff matching of $\frac{9}{4.523} = 1.9898$

Using $X_1 = 5\mu m$ & the results of problem (2.14):

$$Y_2 = 12.075 \mu m \xrightarrow{\text{rounding}} 12.1 \mu m$$

$$X_2 = 3.155 \mu m \xrightarrow{\text{rounding}} 3.2 \mu m$$

$$\text{actual coeff matching} = \frac{(9 \times 25)}{(3 \times 25) + (12.1 \times 3.2)} = \underline{1.9785}$$

$$\% \text{ matching accuracy error} = \frac{1.9785 - 1.9898}{1.9898} \times 100\% = \underline{-0.6\%}$$

(Cont.)

2.15) (cont.)

If capacitor sizes are doubled to 18 + 9.046

then $k = 1.046$ leading to

$$y_2 = 1.2654 x_1 \quad \& \quad x_2 = 0.82665 x_1$$

With $x_1 = 5 \mu\text{m} \Rightarrow y_2 = 6.33 \mu\text{m} \quad \& \quad x_2 = 4.13 \mu\text{m}$

Rounding results in $y_2' = 6.3 \mu\text{m} \quad \& \quad x_2 = 4.1 \mu\text{m}$

$$\text{actual coeff matching} = \frac{(18 \times 25)}{(8 \times 25) + (6.3 \times 4.1)} = 1.9927$$

$$\% \text{ matching accuracy error} = \frac{1.9927 - 1.9898}{1.9898} \times 100\% = +0.15\%$$

a 4 times improvement!!

2.16) $\frac{1000 \mu}{2 \mu} \times 7 \Omega = \underline{3.5 \text{ k}\Omega}$

2.17) The length of the dashed line is 13 units of $2 \mu\text{m}/\text{unit}$, which is equal to $26 \mu\text{m}$.

$$1000 \mu\text{m} = 38 \times 26 \mu\text{m} + 12 \mu\text{m}$$

$\therefore$ there will be 38 bends,

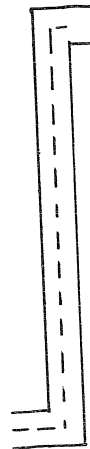
each contributing 0.11 square more than their mid-length.

Therefore, the total resistance will be :

$$3.5 \text{ k}\Omega + (38 \times 0.11 \times 7 \Omega) = \underline{3.529 \text{ k}\Omega}$$

the serpentine's height = $12 \times 2 \mu\text{m} = \underline{24 \mu\text{m}}$

" " width = $38 \times 4 \mu\text{m} = \underline{152 \mu\text{m}}$



Chapter 3

3.14 PROBLEMS

Unless otherwise stated, assume the following:

- npn bipolar transistors:
 $\beta = 100$
 $V_A = 80 \text{ V}$
 $\tau_b = 13 \text{ ps}$
 $\tau_s = 4 \text{ ns}$
 $r_b = 330 \text{ } \Omega$
- n-channel MOS transistors:
 $\mu_n C_{ox} = 92 \text{ } \mu\text{A}/\text{V}^2$
 $V_{tn} = 0.8 \text{ V}$
 $\gamma = 0.5 \text{ V}^{1/2}$
 $r_{ds} (\Omega) = 8,000 L (\mu\text{m}) / I_D (\text{mA})$ in active region
 $C_j = 2.4 \times 10^{-4} \text{ pF}/(\mu\text{m})^2$
 $C_{j\text{-sw}} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$
 $C_{ox} = 1.9 \times 10^{-3} \text{ pF}/(\mu\text{m})^2$
 $C_{gs(\text{overlap})} = C_{gd(\text{overlap})} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$
- p-channel MOS transistors:
 $\mu_p C_{ox} = 30 \text{ } \mu\text{A}/\text{V}^2$
 $V_{tp} = -0.9 \text{ V}$
 $\gamma = 0.8 \text{ V}^{1/2}$

$$r_{ds} (\Omega) = 12,000L (\mu m)/I_D (\text{mA}) \text{ in active region}$$

$$C_j = 4.5 \times 10^{-4} \text{ pF}/(\mu m)^2$$

$$C_{j-sw} = 2.5 \times 10^{-4} \text{ pF}/\mu m$$

$$C_{ox} = 1.9 \times 10^{-3} \text{ pF}/(\mu m)^2$$

$$C_{gs(overlap)} = C_{gd(overlap)} = 2.0 \times 10^{-4} \text{ pF}/\mu m$$

- 3.1 Consider the current mirror shown in Fig. 3.1, where $I_{in} = 80 \mu A$, transistor Q_1 has $W/L = 100 \mu m/1.6 \mu m$, and transistor Q_2 has $W/L = 25 \mu m/1.6 \mu m$. Find the nominal output current as well as the output impedance, r_{out} . Also, find the minimum output voltage such that both transistors remain in the active region.
- 3.2 For the common-source amplifier in Fig. 3.4, derive the relationship between the bias current, I_{bias} , and the dc gain. Assume all transistor sizes are the same.
- 3.3 For the common-source amplifier in Fig. 3.4, assume there is a load capacitor, C_L , that dominates the frequency response. Derive the -3-dB frequency of the amplifier. Find the relationship between the -3-dB frequency and the bias current, I_{bias} . Assume all transistor sizes are the same.
- 3.4 In the high-frequency analysis of a common-source amplifier, show that the transfer function given in equation (3.103) can be derived from nodal equations (3.101) and (3.102).
- 3.5 Find the -3-dB frequency of the common-source amplifier in Fig. 3.4, assuming that all transistors have $W/L = 75 \mu m/1.6 \mu m$ and that $I_{bias} = 75 \mu A$. The first step is to estimate the areas and peripheries of the junctions based on simple layout rules. In this step, assume each transistor is composed of three gate stripes. Based on this geometry, the next step is to estimate the parasitic capacitances of the small-signal model. In this step, assume the dc bias voltage of the output node is halfway between ground and the 5-V power-supply voltage.
- 3.6 Derive the low-frequency output impedance of the source follower shown in Fig. 3.6.
- 3.7 Derive an equation for the impedance looking into the source of a source follower. Use a small-signal model similar to that in Fig. 3.8, but do not take into account C_s and R_{s1} .
- 3.8 Repeat Example 3.9, but assume $C_L = 0.5 \text{ pF}$ when (a) the source is not connected to the substrate, and (b) the source is connected to the substrate.
- 3.9 Repeat Example 3.10, but assume $C_L = 0.5 \text{ pF}$ when (a) the source is not connected to the substrate, and (b) the source is connected to the substrate.
- 3.10 Assume that the common-gate amplifier of Fig. 3.9 has a bias current of 0.1 mA and that all transistors have a W/L of $100 \mu m/1.6 \mu m$. Also assume the following:
 $r_{ds-n} (\Omega) = 8,000L (\mu m)/I_D (\text{mA})$
 $r_{ds-p} (\Omega) = 12,000L (\mu m)/I_D (\text{mA})$

$$R_{in} = 180 \text{ k}\Omega$$

$$C_L = 1 \text{ pF}$$

$$C_{gs} = 0.2 \text{ pF}$$

$$C_{gd} = 15 \text{ fF}$$

$$C_{sb} = 40 \text{ fF}$$

$$C_{db} = 20 \text{ fF}$$

And assume the input source has a 30-fF output capacitance. Find the –3-dB frequency of the amplifier.

- 3.11** Derive the output impedance of the current mirror shown in Fig. P3.11, where a diode-connected transistor has been included in series with the source of the output transistor.

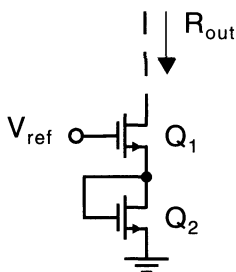


Fig. P3.11

- 3.12** A MOS n-channel cascode current mirror has a bias current of 0.1 mA and transistor sizes given by $W/L = 50 \text{ }\mu\text{m}/1.6 \text{ }\mu\text{m}$. What is the minimum output voltage allowable across the current mirror without any transistors entering the triode region?
- 3.13** Using small-signal analysis, find the output impedance of a MOS cascode current mirror. Include in your analysis the voltage-dependent current source that models the body effect.
- 3.14** Using small-signal analysis, find the equation for the output impedance, r_{out} , of a MOS Wilson current mirror, and show that it is approximately given by

$$r_{out} \cong r_{ds4} \left(\frac{g_{m1} r_{ds1}}{2} \right)$$

Ignore the body effect. Remember to include the effect of the output impedance of the input current source.

- 3.15** Repeat Problem 3.14, but include the voltage-dependent current source that models the body effect.
- 3.16** Repeat Example 3.11, in which a cascode amplifier is analyzed to find its –3-dB frequency, but assume $C_L = 0.5 \text{ pF}$.
- 3.17** Derive an expression for the frequency at which the magnitude of the gain of a cascode amplifier has decreased to unity. What is this value when the parameters of Example 3.11 are used, except that $C_L = 2 \text{ pF}$?

- 3.18** Derive the current gain, $I_{\text{out}}/I_{\text{in}}$, for the bipolar current mirror shown in Fig. 3.22, and show that, for $\beta \gg 1$, this gain is $1 - 2/\beta$. (Neglect finite output impedances.)
- 3.19** Derive the current gain, $I_{\text{out}}/I_{\text{in}}$, for the bipolar current mirror shown in Fig. 3.23, and show that, for $\beta \gg 1$, this gain is $1 - 2/\beta^2$. (Neglect finite output impedances.)
- 3.20** For the bipolar cascode current mirror shown in Fig. 3.25(a), derive the current gain, $I_{\text{out}}/I_{\text{in}}$, and show that, for large β , this gain is given by

$$\frac{I_{\text{out}}}{I_{\text{in}}} \approx 1 - \frac{4}{\beta}$$

- 3.21** For the bipolar Wilson current mirror shown in Fig. 3.25(b), show that the current gain, $I_{\text{out}}/I_{\text{in}}$, is given by

$$\frac{I_{\text{out}}}{I_{\text{in}}} = 1 - \frac{2}{\beta^2 + 2\beta + 2}$$

- 3.22** Derive the output impedance of the simple bipolar current mirror shown in Fig. P3.22. Take the finite output impedance, r_o , into account for both transistors. What is the required value for R , and what is the output impedance if we want the transistors to be biased at 0.2 mA?

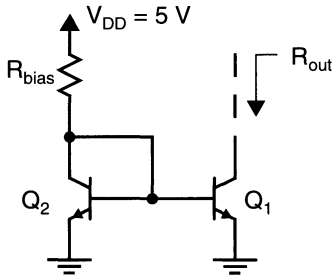


Fig. P3.22

- 3.23** For a common-source amplifier similar to that in Fig. 3.4 but realized using bipolar transistors, derive the relationship between the bias current, I_{bias} , and the dc gain.
- 3.24** Derive the -3 -dB frequency of a bipolar common-emitter amplifier.
- 3.25** Derive the low-frequency gain and output impedance of a bipolar emitter follower that is biased with a current mirror. What are these values for a 0.5-mA bias current?
- 3.26** Derive the low-frequency input impedance and the gain of a bipolar common-base amplifier.
- 3.27** For the circuit shown in Fig. P3.27, show that the output impedance, r_{out} , is approximately given by

$$r_{\text{out}} \cong r_{o2}(1 + g_{m2}(R_e \parallel r_{\pi}))$$

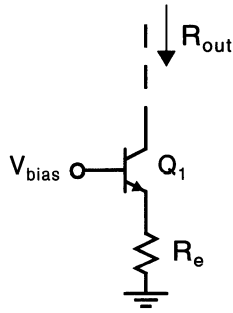


Fig. P3.27

- 3.28** Taking into account the finite current gain of bipolar transistors, but ignoring the finite output impedance, find the large-signal current gains of cascode and Wilson current mirrors.
- 3.29** Assuming $\beta \gg 1$ and $g_m r_o \gg \beta$, show that the output impedance of a bipolar Wilson current mirror is approximately given by

$$r_{out} \cong \frac{\beta r_o}{2}$$

- 3.30** Repeat Problem 3.29 for a bipolar cascode current mirror.
- 3.31** For the differential-input stage in Fig. 3.19, assume that $I_{bias} = 0.1 \text{ mA}$, all transistors have $W/L = 100 \text{ } \mu\text{m}/1.6 \text{ } \mu\text{m}$, and the load capacitance is 100 pF . Find the dc gain and the -3-dB frequency.

Chapter 3 - Problems

3.1) Nominal $I_{out} = \frac{25\mu m}{100\mu m} \times 80\mu A = \underline{20\mu A}$

$$\therefore R_{out} = r_{DS2} = 8000 \times \frac{1.6\mu m}{0.02mA} \\ = \underline{640 k\Omega}$$

Minimum output voltage = V_{eff}

$$= \sqrt{\frac{2 I_D}{\mu_n C_{ox} W/L}} \\ = \sqrt{\frac{2 \times 20 \times 10^{-6}}{92 \times 10^{-6} \times 25/1.6}} \\ = \underline{170 mV}$$

3.2) DC gain, $A_v = -g_{m1} (r_{DS1} // r_{DS2})$

where $r_{DS1} = \frac{8000 L}{I_{bias}}$, $r_{DS2} = \frac{12000 L}{I_{bias}}$

$$\therefore r_{DS1} // r_{DS2} = \frac{96000 L^2}{20000 L I_{bias}} \\ = \underline{\frac{4800 L}{I_{bias}}}$$

Also,

$$g_{m1} = \sqrt{2 \mu_n C_{ox} (W/L) I_{bias}}$$

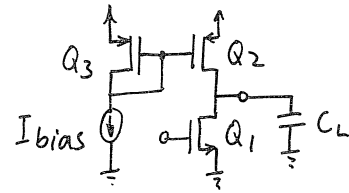
$$\therefore A_v = -4800 \sqrt{\frac{2 \mu_n C_{ox} W L}{I_{bias}}}$$

This result suggests that higher gains are obtained by using smaller bias currents, and/or larger transistor sizes in both width and length.

3.3)

∴ C_L dominates the frequency response,

$$\omega_{3dB} \approx \frac{1}{R_{out} C_L}$$



for

$$R_{out} = r_{ds1} // r_{ds2}$$

$$= \frac{4800 L}{I_{bias}} \text{ from Problem 3.2}$$

$$\therefore \omega_{3dB} = \frac{I_{bias}}{4800 L \times C_L}$$

Increasing the bias current speeds up the circuit.

3.4)

$$V_i (G_{in} + sC_{gs1} + sC_{gd1}) - V_{in} G_{in} - V_{out} sC_{gd1} = 0 \quad (3.101)$$

$$\text{Isolating } V_i \text{ gives } V_i = \frac{V_{in} G_{in} + V_{out} sC_{gd1}}{G_{in} + sC_{gs1} + sC_{gd1}} \quad \textcircled{A}$$

sub ① → (3.102)

$$V_{out} (G_2 + sC_{gd1} + sC_2) - (sC_{gd1} - g_{m1}) \times \frac{V_{in} G_{in} + V_{out} sC_{gd1}}{G_{in} + sC_{gs1} + sC_{gd1}} = 0$$

$$\text{or } \gamma(s) V_{out} = G_{in} (sC_{gd1} - g_{m1}) V_{in}$$

$$\text{where } \gamma(s) = (G_2 + s(C_{gd1} + C_2))(G_{in} + s(C_{gs1} + C_{gd1})) - sC_{gd1} (sC_{gd1} - g_{m1})$$

$$= G_2 G_{in} + s(G_2 C_{gs1} + G_2 C_{gd1} + G_{in} C_{gd1} + G_{in} C_2 + g_{m1} C_{gd1}) + s^2(C_{gd1} C_{gs1} + C_{gd1} C_2 + C_2 C_{gs1} + C_2 C_{gd1} - C_{gd1}^2)$$

$$= G_2 G_{in} [1 + s(R_{in}(C_{gs1} + C_{gd1}(1 + g_{m1} R_2)) + R_2(C_{gd1} + C_2)) + s^2 R_{in} R_2 (C_{gd1} C_{gs1} + C_{gs1} C_2 + C_{gd1} C_2)]$$

$$\therefore \frac{V_{out}}{V_{in}} = \frac{G_{in} (sC_{gd1} - g_{m1})}{\gamma(s)} = \frac{sC_{gd1} - g_{m1}}{G_2 (1 + sa + s^2 b)} = \frac{-g_{m1} R_2 (1 - s \frac{C_{gd1}}{g_{m1}})}{1 + sa + s^2 b}$$

where

$$a = R_{in} [C_{gs1} + C_{gd1} (1 + g_{m1} R_2)] + R_2 (C_{gd1} + C_2)$$

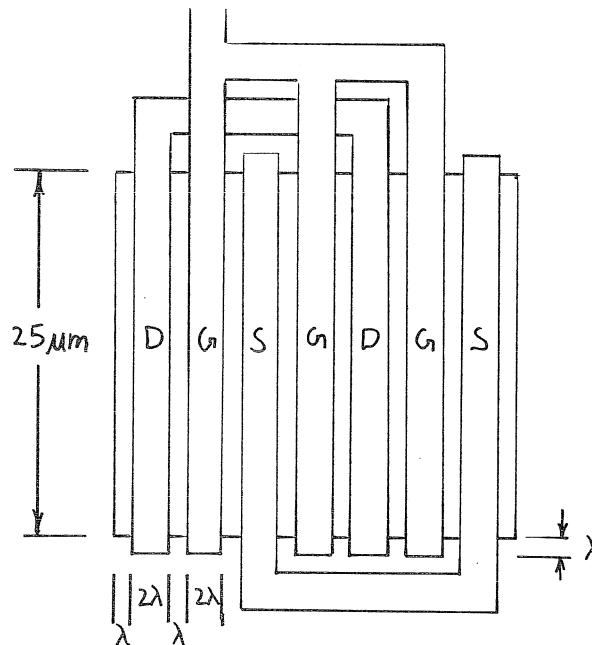
$$b = R_{in} R_2 (C_{gd1} C_{gs1} + C_{gs1} C_2 + C_{gd1} C_2)$$

Q.E.D.

3.5)

* Assume minimum feature size, $\lambda = 0.8 \mu\text{m}$ *

A sample layout of a transistor:



Estimating parasitic capacitances found in the layout above:

$$\begin{aligned} C_{gs1} &= \frac{2}{3} W L C_{ox} + C_{gs-ov} W \\ &= \frac{2}{3} \times 75 \times 1.6 \times 1.9 \times 10^{-15} + 0.2 \times 10^{-15} \times 75 \\ &= \underline{167 \text{ fF}} \end{aligned}$$

$$\begin{aligned} C_{gd1} &= W C_{gd-ov} = 75 \times 0.2 \times 10^{-15} \\ &= \underline{15 \text{ fF}} \end{aligned}$$

For C_{db1} & C_{db2} , we need to first calculate C_{jd} and C_{jsw} .

(assume $V_{SB} = V_{DB} = 5V - 2.5V = 2.5V$)

$$C_{jd1} = \frac{C_{jdo}}{\sqrt{1 + V_{DB}/\Phi_0}} = \frac{2.4 \times 10^{-4}}{\sqrt{1 + 2.5/0.9}} = 1.23 \times 10^{-4} \text{ PF}/\mu\text{m}^2$$

$$C_{jsw1} = \frac{C_{jsw0}}{\sqrt{1 + V_{SB}/\Phi_{sw}}} = \frac{2.0 \times 10^{-4}}{\sqrt{1 + 2.5/0.9}} = 1.03 \times 10^{-4} \text{ PF}/\mu\text{m}$$

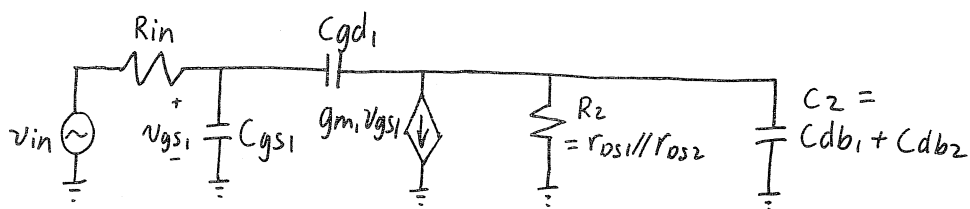
(cont.)

3.5) (cont.) Similarly, $C_{jd2} = 2.3 \times 10^{-4} \text{ pf}/\mu\text{m}^2$ and $C_{jsw2} = 1.29 \times 10^{-4} \frac{\text{pf}}{\mu\text{m}}$

$$\begin{aligned} C_{db1} &= A_{d1} C_{jd1} + P_{d1} C_{jsw1} \\ &= 2 \times 25 \times 4 \times 0.8 \times 1.23 \times 10^{-4} + (25 + 4 \times 4 \times 0.8) \times 1.03 \times 10^{-4} \\ &= \underline{24 \text{ fF}} \end{aligned}$$

$$\begin{aligned} C_{db2} &= A_{d2} C_{jd2} + P_{d2} C_{jsw2} \\ &= 2 \times 25 \times 4 \times 0.8 \times 2.3 \times 10^{-4} + (25 + 4 \times 4 \times 0.8) \times 1.29 \times 10^{-4} \\ &= \underline{42 \text{ fF}} \end{aligned}$$

Small-signal model :



$$\omega_{3dB} \cong \frac{1}{R_{in} [C_{gs1} + C_{gd1} (1 + g_{m1} R_2)] + R_2 (C_{gd1} + C_2)}$$

where

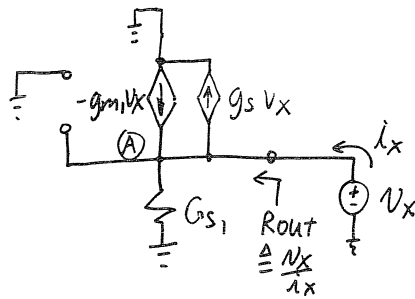
$$\begin{aligned} R_2 &= r_{ds1} || r_{ds2} = \frac{4800 L}{I_{bias}} \text{ from Problem 3.2} \\ &= \frac{4800 \times 1.6}{75 \mu\text{A}} \\ &= \underline{102 \text{ k}\Omega} \end{aligned}$$

$$\begin{aligned} g_{m1} &= \sqrt{2 \mu_n C_{ox} W/L I_{bias}} = \sqrt{2 \times 92 \times 10^{-6} \times 75 / 1.6 \times 75 \times 10^{-6}} \\ &= 0.804 \text{ mA/V} \end{aligned}$$

If we assume $R_{in} \approx R_2 = 102 \text{ k}\Omega$,

$$\begin{aligned} \omega_{3dB} &\cong \frac{1}{102 \times 10^3 [167 + 15 (1 + 0.804 \times 10^{-3} \times 102 \times 10^3)] \times 10^{-15} + 102 \times 10^3 \times 10^{-15} \times (15 + 24 + 42)} \\ &= 6.57 \times 10^6 \text{ rads/sec} = \underline{2\pi \times 1.05 \text{ MHz}} \end{aligned}$$

3.6)



Summing currents at node (A) :

$$i_x - g_{m1} V_x - g_s V_x - N_x G_{s1} = 0$$

$$\therefore G_{out} = 1/R_{out} = i_x / V_x \\ = g_{m1} + g_s + G_{s1}$$

$$\text{where } G_{s1} = g_{ds1} + g_{ds2}$$

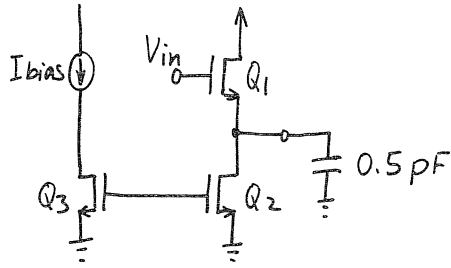
$$\text{or } \underline{R_{out} = [g_{m1} + g_s + \overbrace{g_{ds1} + g_{ds2}}^{G_{s1}}]^{-1}} \cong \frac{1}{g_{m1} + g_s}$$

3.7)

Following the same analysis as in Problem 3.6, this time ignoring R_{s1} (i.e., G_{s1}), the impedance looking into the source is

$$\underline{R_{out} = [g_{m1} + g_s]^{-1}}$$

3.8)



Find ω_0 , Q and zero frequency.

$$a) \quad g_{m1} = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_{bias}} = \sqrt{2 \times 92 \times 10^{-6} \times \frac{100}{1.6} \times 100 \times 10^{-6}} = 1.06 \frac{\text{mA}}{\text{V}}$$

$$r_{ds1} = r_{ds2} = \frac{8000 L}{I_{bias}} = \frac{8000 (1.6 \times 10^{-6})}{100 \times 10^{-6}} \text{ k}\Omega = 128 \text{ k}\Omega$$

$$g_{s1} = \frac{8 g_m}{2 \sqrt{V_{SB} + 12 \Phi_F}} \approx \frac{0.5 g_m}{2 \sqrt{2 + 0.7}} = 0.16 \text{ mA/V}$$

$$C_{in} = [180 \times 10^3]^{-1} = 5.56 \times 10^{-6} \text{ F}$$

$$G_{s1} = g_{s1} + g_{ds1} + g_{ds2} = 0.176 \text{ mA/V}$$

$$C_s = C_L + C_{sb1} = 0.5 + 0.04 \text{ pF} = 0.54 \text{ pF}$$

$$C_{in}' = C_{in} + C_{gd1} = 45 \text{ fF}$$

$$\therefore \omega_0 = \sqrt{\frac{C_{in} (g_{m1} + G_{s1})}{C_{gs1} C_s + C_{in}' (C_{gs1} + C_s)}} \quad \text{where } C_{gs1} = 0.2 \text{ pF}$$

$$= 2.21 \times 10^9 \text{ rads/sec} = \underline{2\pi \times 35 \text{ MHz}}$$

$$Q = \frac{\sqrt{C_{in} (g_{m1} + G_{s1}) [C_{gs1} C_s + C_{in}' (C_{gs1} + C_s)]}}{C_{in} C_s + C_{in}' (g_{m1} + G_{s1}) + C_{gs1} G_{s1}}$$

$$= \frac{3.1149 \times 10^{-17}}{9.382 \times 10^{-17}} = \underline{0.332}$$

$$\omega_z = -\frac{g_{m1}}{C_{gs1}} = 5.3 \times 10^9 \text{ rads/sec} = \underline{2\pi \times 844 \text{ MHz}}$$

(cont.)

3.8)(cont.)

b) Source is connected to the substrate.

Assume C_{gs} , C_{gd1} , and C_{sb1} remain unchanged, and that the only difference is the elimination of transconductance, g_{s1} .

∴ All parameters unchanged except

$$G_{S1} = g_{DS1} + g_{DS2} = 15.6 \mu A/V$$

$$\omega_0 = \sqrt{\frac{5.976 \times 10^{-9}}{1.413 \times 10^{-25}}} = 2.06 \times 10^8 \frac{rads}{sec} = \underline{2\pi \times 33 MHz}$$

$$Q = \frac{2.906 \times 10^{-17}}{5.140 \times 10^{-17}} = \underline{0.57}$$

$$\omega_z = \underline{2\pi \times 844 MHz} \quad (\text{unchanged from part a})$$

As expected, the elimination of the body effect increases Q .

$$\begin{aligned} 3.9) \quad a) \quad C_1 &= \frac{g_{m1} C_{gs1} C_s}{(g_{m1} + G_{S1})(C_{gs1} + C_s)} = \frac{(1.06 \times 10^{-3})(0.2 \times 10^{-12})(0.54 \times 10^{-12})}{(1.06 \times 10^{-3} + 0.176 \times 10^{-3})(0.2 + 0.54) \times 10^{-12}} \\ &= \frac{1.145 \times 10^{-28}}{9.146 \times 10^{-16}} = \underline{0.125 pF} \end{aligned}$$

$$\begin{aligned} R_1 &= \frac{(C_{gs1} + C_s)^2}{C_{gs1} C_s g_{m1}} = \frac{(0.74 \times 10^{-12})^2}{0.2 \times 0.54 \times 10^{-24} \times 1.06 \times 10^{-3}} \\ &= \frac{5.476 \times 10^{-25}}{1.145 \times 10^{-28}} = \underline{4780 \Omega} \end{aligned}$$

$$P_1 \cong \frac{C_{in}}{C_{gs1} + C_{in}} = \underline{2\pi \times 3.61 MHz} \quad (\text{unchanged from Ex. 3.10})$$

$$P_2 = \frac{g_{m1} + G_{S1}}{C_{gs} + C_L} = \frac{1.236 \times 10^{-3}}{0.7 \times 10^{-12}} = \underline{2\pi \times 281 MHz}$$

(cont.)

3.9) b) Now $g_{s1} = g_{ds1} + g_{bs2} = 15.6 \mu A/V$

With all other values unchanged,

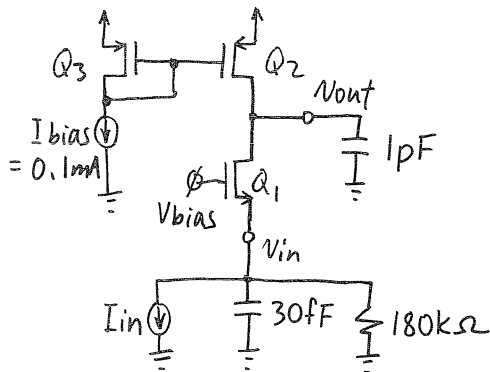
$$C_1 = \frac{1.145 \times 10^{-28}}{1.075 \times 10^{-3} \times 0.74 \times 10^{-12}} = \underline{0.144 \text{ pF}}$$

$$R_1 = \underline{4780 \Omega} \quad (\text{unchanged from part a)})$$

$$p_1 = \underline{2\pi \times 3.61 \text{ MHz}} \quad (\text{same as in part a)})$$

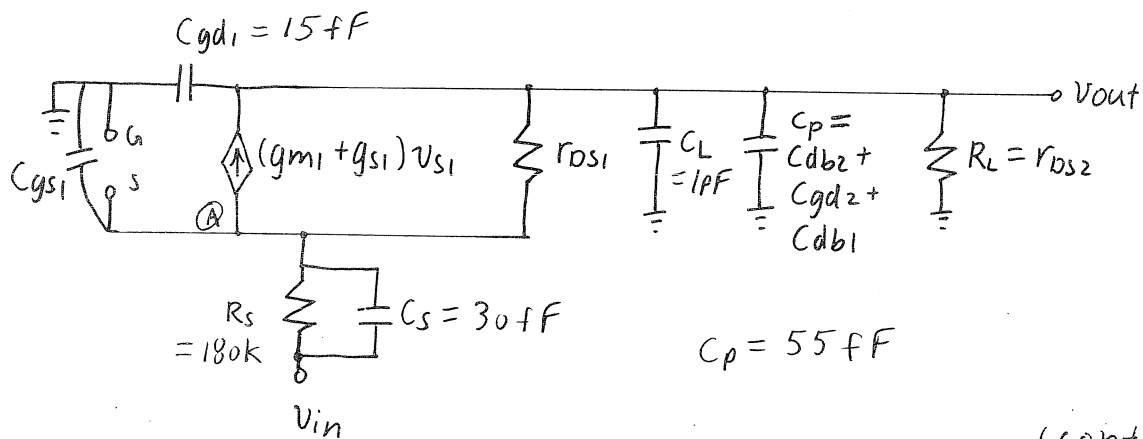
$$p_2 = \frac{1.075 \times 10^{-3}}{0.7 \times 10^{-12}} = \underline{244 \text{ MHz} \times 2\pi}$$

3.10)



Find the -3dB frequency.

Small-signal model:



$$r_{ds1} = \frac{8000 \times 1.6}{0.1} = 128 \text{ k}\Omega$$

$$r_{ds2} = \frac{12,000 \times 1.6}{0.1} = 192 \text{ k}\Omega$$

(cont.)

3.10) (cont.)

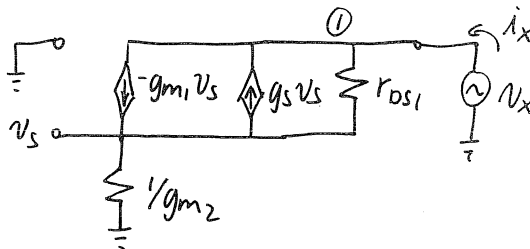
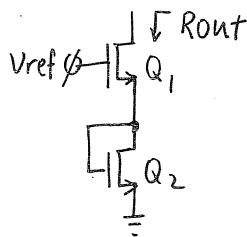
using the open-circuit time-constant approach,
the output node dominates.

$$\omega_{3dB} \cong \frac{1}{[r_{ds2} \parallel (r_{ds1} + R_S)] [C_L + C_P + C_{gd1}]}$$

$$= \frac{1}{(118.3 \text{ k}\Omega)(1.07 \text{ pF})} = 2\pi \times 1.26 \text{ MHz}$$

Roughly $f_{3dB} = \underline{1.3 \text{ MHz}}$

3.11)



Use test voltage, V_x , to find R_{out} .

K.C.L. at node ① :

$$-g_{m1} v_s - g_{s1} v_s - i_x + V_x g_{os1} - V_s g_{os1} = 0$$

$$-v_s (g_{m1} + g_{s1} + g_{os1}) - i_x + V_x g_{os1} = 0 \quad (A)$$

K.C.L. at V_s :

$$V_s g_{m2} + (g_{s1} + g_{m1}) V_s + (V_s - V_x) g_{os1} = 0$$

$$V_s (g_{m1} + g_{m2} + g_{s1} + g_{os1}) - V_x g_{os1} = 0$$

$$\therefore V_s = \frac{g_{os1}}{g_{m1} + g_{m2} + g_{s1} + g_{os1}} V_x \quad (B)$$

(cont.)

3.11 (cont.)

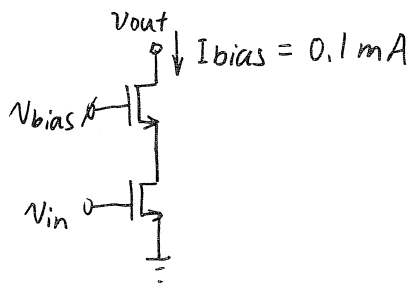
sub ③ → ①

$$\left[-\frac{g_{ds1}(g_{m1} + g_{s1} + g_{ds1})}{(g_{m1} + g_{s1} + g_{ds1} + g_{m2})} + g_{ds1} \right] v_x = i_x$$

$$\begin{aligned} \therefore R_{out} &\triangleq \frac{v_x}{i_x} = \frac{g_{m1} + g_{s1} + g_{ds1} + g_{m2}}{\underbrace{g_{ds1} g_{m2}}} \\ &= r_{ds1} + \frac{g_{m1} + g_{s1}}{g_{m2}} \times r_{ds2} + 1/g_{m2} \end{aligned}$$

$\therefore \underline{R_{out} \approx 2 r_{ds1}}$ which is consistent with the value obtained using the source degeneration formula.

3.12)



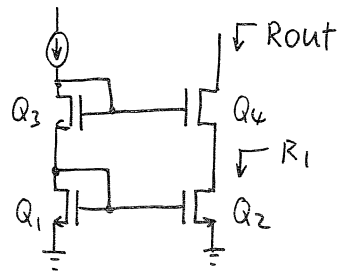
From (3.40),

$$V_{out} > V_{ds2} + V_{eff} = 2 V_{eff1} + V_{tn1}$$

$$\text{where } V_{eff1} = \sqrt{\frac{2 I_{bias}}{\mu_n C_{ox} W/L}} = 0.264 \text{ V}$$

$$\therefore V_{out} \geq 2 \times 0.264 + 0.8 \text{ V} = \underline{1.33 \text{ V}}$$

3.13)

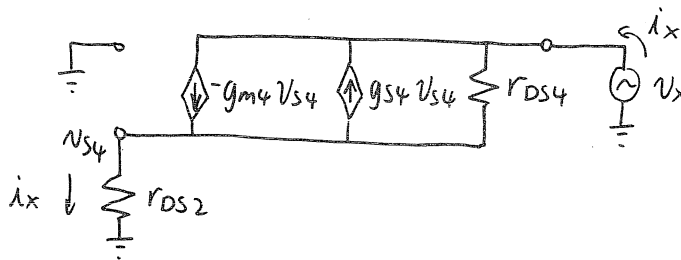


Find R_{out} ,

From section 3.1,

$$R_1 = r_{DS2}$$

Giving us the resulting small signal model:



KCL at output:

$$-(g_{m4} + g_{s4})v_{s4} + g_{DS4}v_x - g_{DS4}v_{s4} - i_x = 0$$

$$\therefore (g_{m4} + g_{s4} + g_{DS4})v_{s4} + i_x = g_{DS4}v_x$$

$$\text{But } v_{s4} = i_x / g_{DS2}$$

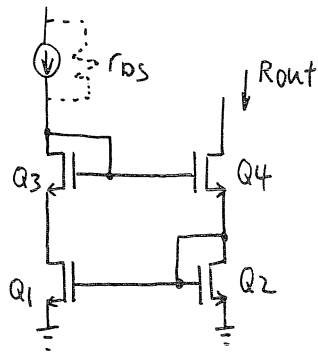
$$\therefore i_x \left[(g_{m4} + g_{s4} + g_{DS4}) / g_{DS2} + 1 \right] = g_{DS4}v_x$$

$$\therefore R_{out} \triangleq \frac{v_x}{i_x} = \frac{1}{g_{DS4}} \left[1 + \frac{1}{g_{DS2}} (g_{m4} + g_{s4} + g_{DS4}) \right]$$

$$\underline{R_{out} = r_{DS4} [1 + r_{DS2} (g_{m4} + g_{s4} + g_{DS4})]}$$

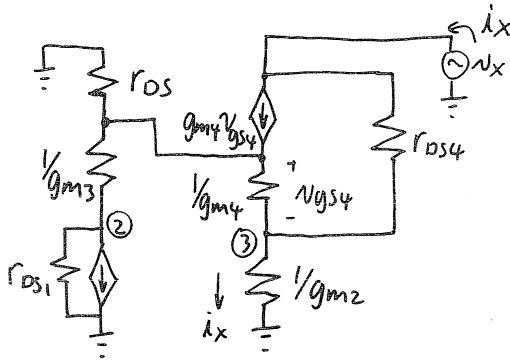
This result is consistent with that obtained using source degeneration equation (3.29).

3.14)



Find R_{out}

Method 1: Nodal Analysis



$$v_{gs4} = v_{g4} - v_{s4} = v_{(2)} \times \frac{r_{DS}}{\frac{1}{g_{m3}} + r_{DS}} - v_{(3)}$$

But $r_{DS} \gg \frac{1}{g_{m3}}$

$$\therefore \frac{r_{DS}}{\frac{1}{g_{m3}} + r_{DS}} \approx 1 \quad \text{and}$$

$$v_{gs4} \approx v_{(2)} - v_{(3)}$$

KCL at output:

$$-i_x + g_{m4} v_{gs4} + g_{DS4} (v_x - v_{(3)}) = 0$$

$$-i_x + g_{m4} (v_{(2)} - v_{(3)}) + g_{DS4} (v_x - v_{(3)}) = 0$$

$$\therefore -i_x - v_{(3)} (g_{m4} + g_{DS4}) + g_{m4} v_{(2)} + g_{DS4} v_x = 0 \quad (A)$$

But $v_{(3)} = i_x / g_{m2}$ and

$$v_{(2)} = -g_{m1} v_{(3)} \times [r_{DS1} \parallel (\frac{1}{g_{m3}} + r_{DS})]$$

$$\approx -\frac{g_{m1} r_{DS1}}{2 g_{m2}} \times i_x \quad (B)$$

(cont.)

3.14 (cont.)

sub (B) $\rightarrow$ (A)

$$-i_x - \frac{1}{g_{m2}} (g_{m4} + g_{DS4}) i_x - \frac{g_{m1} g_{m4} r_{DS1}}{2 g_{m2}} i_x + g_{DS4} v_x = 0$$

$$g_{DS4} v_x = i_x \left(1 + \frac{g_{m4} + g_{DS4}}{g_{m2}} + \frac{g_{m1} g_{m4}}{2 g_{m2}} r_{DS1} \right)$$

$$\therefore R_{out} \triangleq \frac{v_x}{i_x} \approx r_{DS4} \times \left(2 + \frac{g_{m1}}{2} r_{DS1} \right)$$

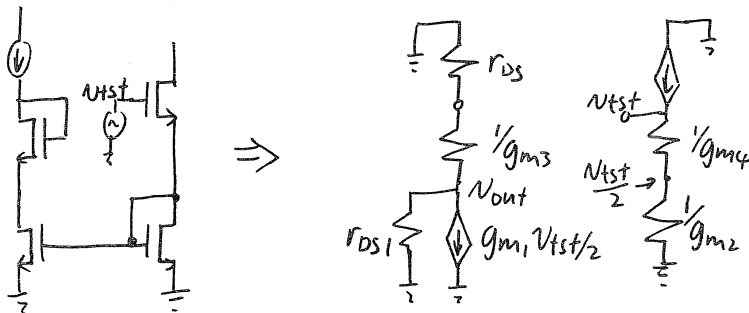
$$\therefore \frac{g_{m1} r_{DS1}}{2} \gg 2$$

$$\therefore \underline{R_{out} \approx r_{DS4} \left(\frac{g_{m1} r_{DS1}}{2} \right)} \quad \text{Q.E.D.}$$

Method 2: Feedback Analysis

$$R_{out} = (1 - A\beta) R_{out} (\text{open loop})$$

where $A\beta$ is the loop gain. To determine $A\beta$, break the feedback loop and apply a test voltage as follows:



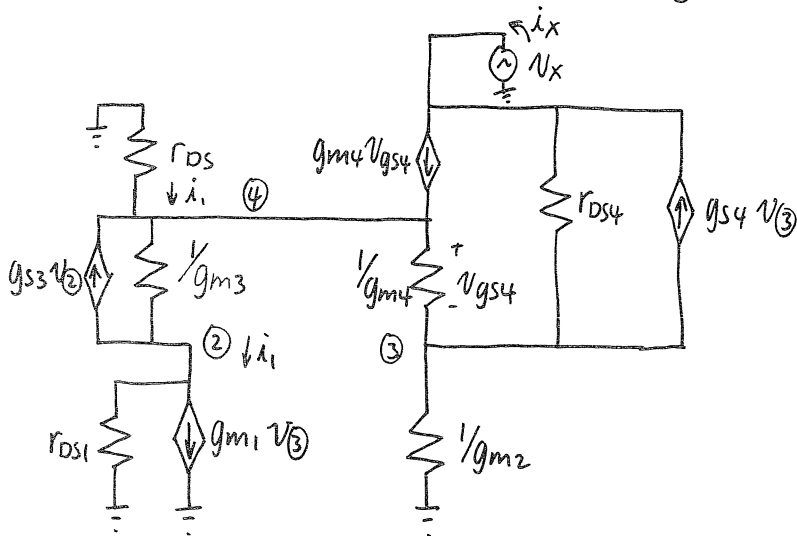
$$v_{out} = -g_{m1} \times \frac{v_{tst}}{2} \times (r_{DS1} \parallel r_{DS})$$

$$\therefore \text{Loop gain} = A\beta = -\frac{g_{m1} r_{DS1}}{4}$$

$$\therefore R_{out} = \left(1 + \frac{g_{m1} r_{DS1}}{4} \right) \times 2 r_{DS4}$$

$$\therefore \underline{R_{out} \approx \left(\frac{g_{m1} r_{DS1}}{2} \right) \times r_{DS4}} \quad \text{Q.E.D.}$$

3.15) Small signal model incorporating the body effect :



$$v_{gs4} = v_{(4)} - v_{(3)}$$

$$v_{(4)} = v_{(2)} + (i_1 + g_{s3} v_{(2)}) / g_{m3} \quad \text{[A]}$$

$$i_1 = -g_{os} v_{(4)} \quad \text{[B]}$$

$$\text{[B]} \rightarrow \text{[A]}$$

$$\therefore v_{(4)} \left(1 + \frac{g_{os}}{g_{m3}} \right) = \left(1 + \frac{g_{s3}}{g_{m3}} \right) v_{(2)}$$

$$v_{(4)} = \frac{g_{m3} + g_{s3}}{g_{m3} + g_{os}} v_{(2)}$$

$$v_{(4)} \approx \left(1 + \frac{g_{s3}}{g_{m3}} \right) v_{(2)}$$

$$\therefore v_{gs4} \approx \left(1 + \frac{g_{s3}}{g_{m3}} \right) v_{(2)} - v_{(3)}$$

new term due to body effect.

KCL at output :

$$-i_x + g_{m4} v_{gs4} + g_{os4} (v_x - v_{(3)}) - g_{s4} v_{(3)} = 0$$

$$-i_x + g_{m4} \left(\left(1 + \frac{g_{s3}}{g_{m3}} \right) v_{(2)} - v_{(3)} \right) + g_{os4} v_x - v_{(3)} (g_{os4} + g_{s4}) \approx 0$$

$$-i_x - v_{(3)} (g_{m4} + g_{os4} + g_{s4}) + g_{m4} \left(1 + \frac{g_{s3}}{g_{m3}} \right) v_{(2)} + g_{os} v_x \approx 0 \quad \text{[C]}$$

new terms

(cont.)

3.15 (cont.)

But $V_3 = i_x / g_{m2}$ [D]

$$\begin{aligned} V_2 &= (i_1 - g_{m1} V_3) r_{DS1} \\ &= -(V_4 g_{DS} + g_{m1} V_3) r_{DS1} \\ &= -[(1 + g_{S3}/g_{m3}) g_{DS} V_2 + g_{m1} V_3] r_{DS1} \end{aligned}$$

$$\therefore V_2 = \frac{-g_{m1} r_{DS1}}{2 + g_{S3}/g_{m3}} V_3 \quad \text{[E]}$$

[D], [E] $\rightarrow$ [C]

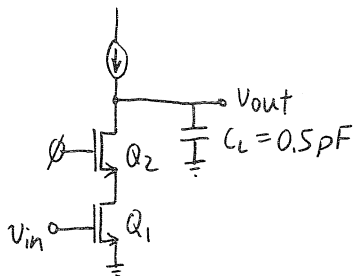
$$-i_x - \frac{i_x}{g_{m2}} \left(g_{m4} \left(1 + \left(1 + \frac{g_{S3}}{g_{m3}} \right) \frac{g_{m1} r_{DS1}}{2 + g_{S3}/g_{m3}} \right) + g_{DS4} + g_{S4} \right) + g_{DS4} V_x \approx 0$$

$$\begin{aligned} \therefore R_{out} \triangleq \frac{V_x}{i_x} &\approx r_{DS4} \left[1 + \frac{g_{m4}}{g_{m2}} \left(1 + \left(1 + \frac{g_{S3}}{g_{m3}} \right) \frac{g_{m1} r_{DS1}}{2 + g_{S3}/g_{m3}} + \frac{g_{S4}}{g_{m2}} + \frac{g_{DS4}}{g_{m2}} \right) \right] \\ &\approx r_{DS4} \left[2 + \left(1 + \frac{g_{S3}}{g_{m3}} \right) \frac{g_{m1} r_{DS1}}{2 + g_{S3}/g_{m3}} + \frac{g_{S4}}{g_{m2}} \right] \end{aligned}$$

$$R_{out} \approx r_{DS4} \left(1 + \frac{g_{S3}}{g_{m3}} \right) \times \frac{g_{m1} r_{DS1}}{2 + g_{S3}/g_{m3}} \rightarrow r_{DS4} \left(\frac{g_{m1} r_{DS1}}{2} \right) \text{ as } g_{S3} \rightarrow 0$$

Q.E.D.

3.16)



$$C_{S2} = 0.26 \text{ pF}$$

$$\begin{aligned} C_{d2} &= C_{gd2} + C_{db2} + C_L + C_{bias} \\ &= 15 + 20 + 500 + 20 \text{ fF} \\ &= 555 \text{ fF} \end{aligned}$$

$$\begin{aligned} \therefore \tau_{Cgs1} &= 36 \text{ nsec} \\ \tau_{Cgd1} &= 75 \text{ nsec} \\ \tau_{CS2} &= 13 \text{ nsec} \end{aligned} \quad \left. \vphantom{\begin{aligned} \tau_{Cgs1} &= 36 \text{ nsec} \\ \tau_{Cgd1} &= 75 \text{ nsec} \\ \tau_{CS2} &= 13 \text{ nsec} \end{aligned}} \right\} \text{unchanged}$$

$$\tau_{cd2} = C_{d2} \frac{g_{m2} r_{DS2}}{2} = 555 \times 10^{-15} \times \frac{10^{-3} \times 10^{10}}{2} = 2.8 \mu\text{sec}$$

$$\therefore \tau_{cd2} \gg \tau_{Cgd1}, \tau_{Cgs1}, \text{ and } \tau_{CS2}$$

$$\therefore \omega_{-3dB} \approx \frac{1}{\tau_{cd2}} = \underline{2\pi \times 57 \text{ kHz}}$$

3.17) Derive ω_{odB}

From (3.161)

$$|A(s)| \approx \frac{g_{m1}}{sC_L} \quad \text{for } s=j\omega, \omega \gg \omega_{-3\text{dB}}$$

for the unity gain frequency

$$|A(s)| \Big|_{s=j\omega_{\text{odB}}} \approx \frac{g_{m1}}{\omega_{\text{odB}} C_L} \equiv 1$$

$$\therefore \underline{\omega_{\text{odB}} = \frac{g_{m1}}{C_L}}$$

Assuming $C_L = 2 \text{ pF}$,

$$\omega_{\text{odB}} = \frac{1 \text{ mA/V}}{2 \text{ pF}} = 5 \times 10^8 \text{ rad/sec} = \underline{2\pi \times 80 \text{ MHz}}$$

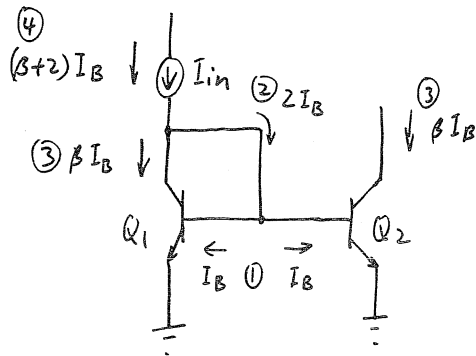
In order to use (3.161), we need to verify that

$\omega_{\text{odB}} \ll \omega_{p2}$. From (3.165),

$$\omega_{p2} \approx \frac{g_{m2}}{C_{S2}} = \frac{1 \text{ mA/V}}{0.26 \text{ pF}} = \underline{2\pi \times 610 \text{ MHz}} \gg \omega_{\text{odB}} \checkmark$$

$\therefore$ assumption is valid.

3.18) Derive current gain. For $\beta \gg 1$, show $\frac{I_{out}}{I_{in}} \approx 1 - \frac{2}{\beta}$



* Numbers indicate process of reasoning or analysis *

① Because $V_{be1} = V_{be2}$
 $\therefore I_{B1} = I_{B2} \triangleq I_B$

From above analysis,

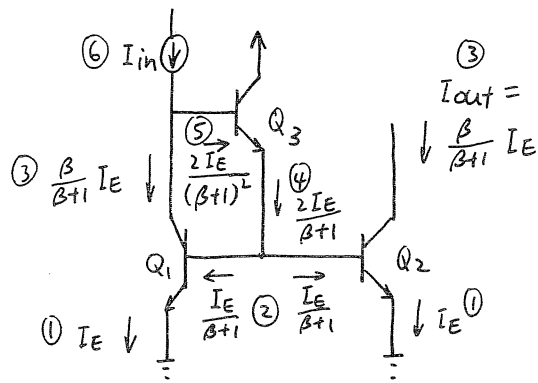
$$I_{out} = \beta I_B$$

$$I_{in} = (\beta + 2) I_B$$

$$\therefore \frac{I_{out}}{I_{in}} = \frac{\beta I_B}{(\beta + 2) I_B} = \frac{\beta + 2}{\beta + 2} - \frac{2}{\beta + 2}$$

$$\therefore \underline{\underline{\frac{I_{out}}{I_{in}} \approx 1 - \frac{2}{\beta} \text{ if } \beta \gg 1 \quad \text{Q.E.D.}}}}$$

3.19)

Show $\frac{I_{out}}{I_{in}} \approx 1 - \frac{2}{\beta^2}$ for $\beta \gg 1$ 

* Numbers indicate process of analysis *

$$\begin{aligned} \textcircled{1} \quad & \because V_{be1} = V_{be2} \\ & \therefore I_{E1} = I_{E2} \triangleq I_E \end{aligned}$$

$$\textcircled{3} \quad I_{out} = \frac{\beta}{\beta+1} I_E$$

$$\textcircled{6} \quad I_{in} = \frac{\beta}{\beta+1} I_E + \frac{2}{(\beta+1)^2} I_E$$

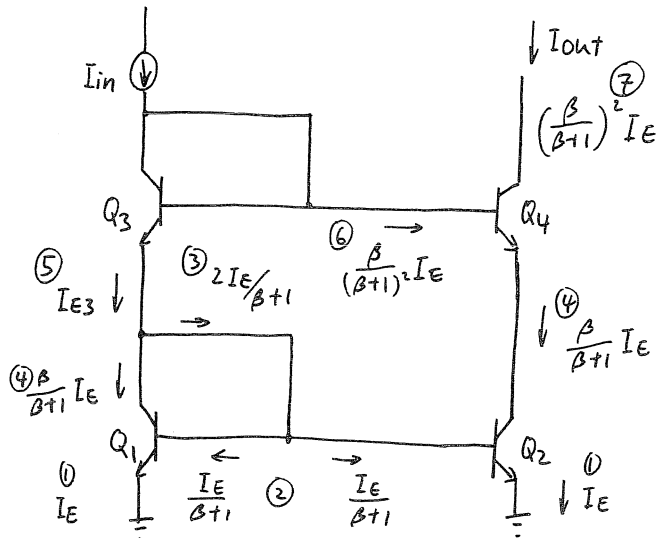
$$\therefore \frac{I_{out}}{I_{in}} = \frac{\frac{\beta}{\beta+1}}{\frac{\beta}{\beta+1} + \frac{2}{(\beta+1)^2}} = \frac{\beta(\beta+1)}{\beta(\beta+1) + 2}$$

$$= \frac{\beta(\beta+1)+2}{\beta(\beta+1)+2} - \frac{2}{\beta(\beta+1)+2} = 1 - \frac{2}{\beta^2 + \beta + 2}$$

$$\therefore \frac{I_{out}}{I_{in}} \approx 1 - \frac{2}{\beta^2} \quad \text{for } \beta \gg 1$$

Q.E.D.

3.20) Show $\frac{I_{out}}{I_{in}} \approx 1 - \frac{4}{\beta}$ for $\beta \gg 1$



① $\therefore V_{be1} = V_{be2},$
 $\therefore I_{E1} = I_{E2} \triangleq I_E$

⑦ $I_{out} = \left(\frac{\beta}{\beta+1}\right)^2 I_E$

⑤ $I_{E3} = \frac{\beta}{\beta+1} I_E + \frac{2}{\beta+1} I_E = \left(\frac{\beta+2}{\beta+1}\right) I_E$

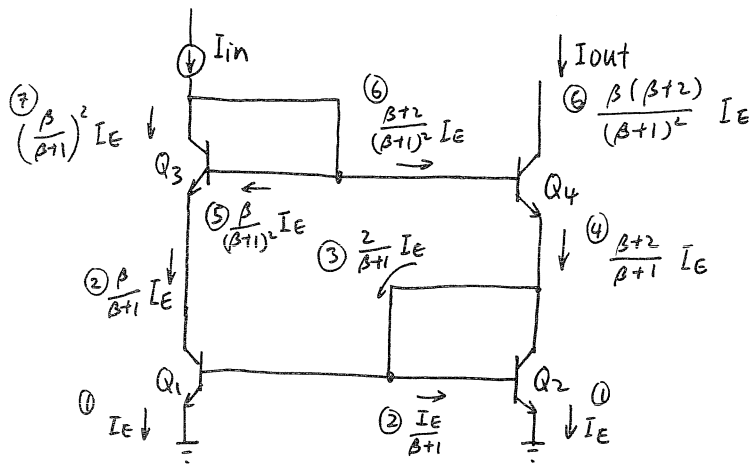
$\therefore I_{B3} = \frac{\beta+2}{(\beta+1)^2} I_E, \quad I_{C3} = \frac{\beta(\beta+2)}{(\beta+1)^2} I_E$

$$\begin{aligned} I_{in} &= I_{C3} + I_{B3} + I_{B4} \\ &= \left[\frac{(\beta+2)\beta}{(\beta+1)^2} + \frac{\beta+2}{(\beta+1)^2} + \frac{\beta}{(\beta+1)^2} \right] I_E \\ &= \frac{1}{(\beta+1)^2} [\beta^2 + 4\beta + 2] I_E \end{aligned}$$

$\therefore \frac{I_{out}}{I_{in}} = \frac{\beta^2}{\beta^2 + 4\beta + 2} = \frac{\beta^2 + 4\beta + 2}{\beta^2 + 4\beta + 2} - \frac{4\beta + 2}{\beta^2 + 4\beta + 2}$

$\therefore \frac{I_{out}}{I_{in}} \approx 1 - \frac{4\beta}{\beta^2} = \underline{1 - \frac{4}{\beta}} \quad \text{Q.E.D.}$

3.21) Show $\frac{I_{out}}{I_{in}} = 1 - \frac{2}{\beta^2 + 2\beta + 2}$



(i) $\therefore V_{be1} = V_{be2}$
 $\therefore I_{E1} = I_{E2} = I_E$

From the above schematic analysis,

$$I_{in} = I_{C3} + I_{B3} + I_{B4} = \left[\left(\frac{\beta}{\beta+1} \right)^2 + \frac{\beta}{(\beta+1)^2} + \frac{\beta+2}{(\beta+1)^2} \right] I_E$$

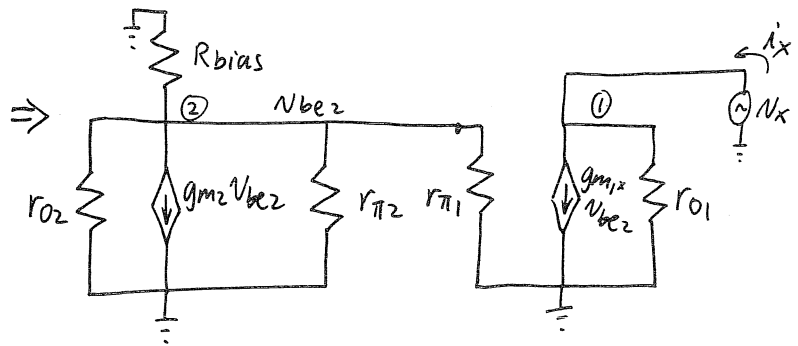
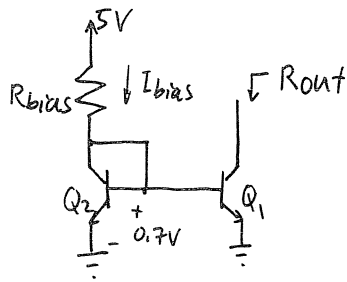
$$I_{out} = \frac{\beta(\beta+2)}{(\beta+1)^2} I_E$$

$$\therefore \frac{I_{out}}{I_{in}} = \frac{\beta(\beta+2)}{\beta^2 + \beta + \beta + 2} = \frac{\beta^2 + 2\beta}{\beta^2 + 2\beta + 2}$$

$$\therefore \frac{I_{out}}{I_{in}} = \frac{\beta^2 + 2\beta + 2}{\beta^2 + 2\beta + 2} - \frac{2}{\beta^2 + 2\beta + 2} = 1 - \frac{2}{\beta^2 + 2\beta + 2}$$

Q.E.D.

3.22)



KCL at ①:

$$-i_x + g_{m1}V_{be2} + N_x/r_{o1} = 0 \quad [A]$$

KCL at ②: Let $R_T \triangleq R_{bias} // r_{o2} // r_{\pi2} // r_{\pi1}$

$$\frac{V_{be2}}{R_T} + g_{m2}V_{be2} = 0 \quad \text{which implies}$$

$$R_T = -1/g_{m2} \quad \text{or} \quad \underline{\underline{V_{be2} = 0}} \quad \checkmark$$

X (impossible as $R_T \geq 0 \Omega$)

$$\therefore [A] \quad -i_x + N_x/r_{o1} = 0$$

$$\Rightarrow \underline{\underline{R_{out} \triangleq \frac{N_x}{i_x} = r_{o1}}}$$

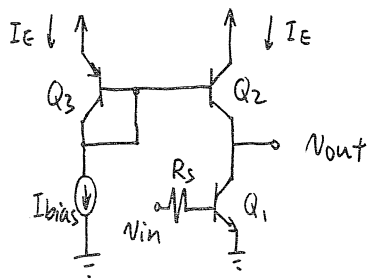
For $I_{bias} = 0.2 \text{ mA}$, assuming $V_{be2} \approx 0.7 \text{ V}$

$$R_{bias} = \frac{5 - 0.7 \text{ V}}{0.2 \text{ mA}} = \underline{\underline{21.5 \text{ k}\Omega}}$$

And

$$R_o = r_{o1} = \frac{V_A}{I_C} = \frac{80 \text{ V}}{0.2 \text{ mA}} = \underline{\underline{400 \text{ k}\Omega}}$$

3.23)



Find A_v

$$I_{bias} = \frac{\beta+2}{\beta+1} I_E \Rightarrow I_E = \frac{\beta+1}{\beta+2} I_{bias}$$

$$I_{C1} = \frac{\beta}{\beta+1} I_E = \frac{\beta}{\beta+2} I_{bias}$$

$$g_{m1} = \frac{I_{C1}}{V_T} = \frac{\beta}{\beta+2} \frac{I_{bias}}{V_T}$$

$$A_v \triangleq \frac{V_{out}}{V_{in}} = -g_{m1} (r_{o1} \parallel r_{o2}) \times \frac{r_{\pi 1}}{r_{\pi 1} + R_S}$$

$$\text{where } r_{\pi 1} = \beta / g_{m1} \\ = (\beta+2) V_T / I_{bias}$$

$$r_{o1} \approx r_{o2} = \frac{V_A}{I_{C2}} = \frac{V_A}{\left(\frac{\beta}{\beta+2}\right) I_{bias}} = \frac{\beta+2}{\beta} \frac{V_A}{I_{bias}}$$

$$\therefore r_{o1} \parallel r_{o2} = \frac{\beta+2}{2\beta} \frac{V_A}{I_{bias}}$$

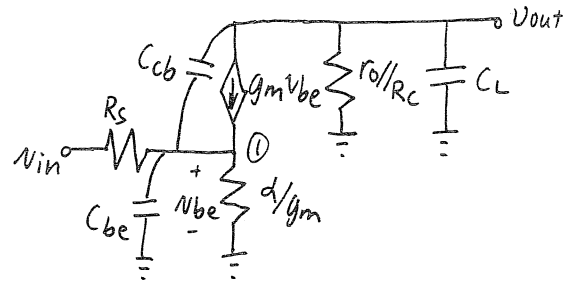
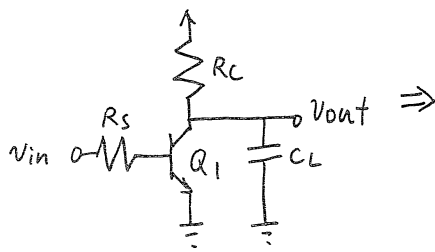
$$\therefore A_v = -g_{m1} (r_{o1} \parallel r_{o2}) \frac{r_{\pi 1}}{r_{\pi 1} + R_S} \\ = -\frac{\beta}{\beta+2} \times \frac{I_{bias}}{V_T} \times \frac{\beta+2}{2\beta} \times \frac{V_A}{I_{bias}} \times \frac{(\beta+2) V_T / I_{bias}}{R_S + (\beta+2) V_T / I_{bias}}$$

$$A_v = -\frac{V_A}{2V_T} \times \frac{1}{1 + \frac{I_{bias}}{(\beta+2)V_T} \times R_S}$$

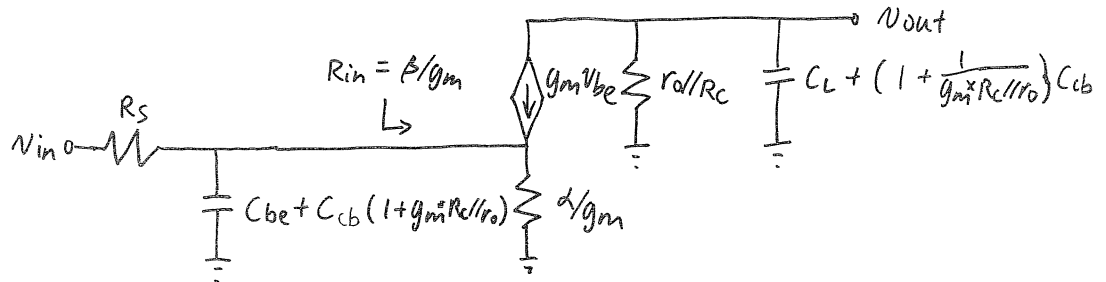
$$\text{IF } R_S \rightarrow 0 \text{ then } A_v = -\frac{V_A}{2V_T}$$

AND is independent of I_{bias} .

3.24)



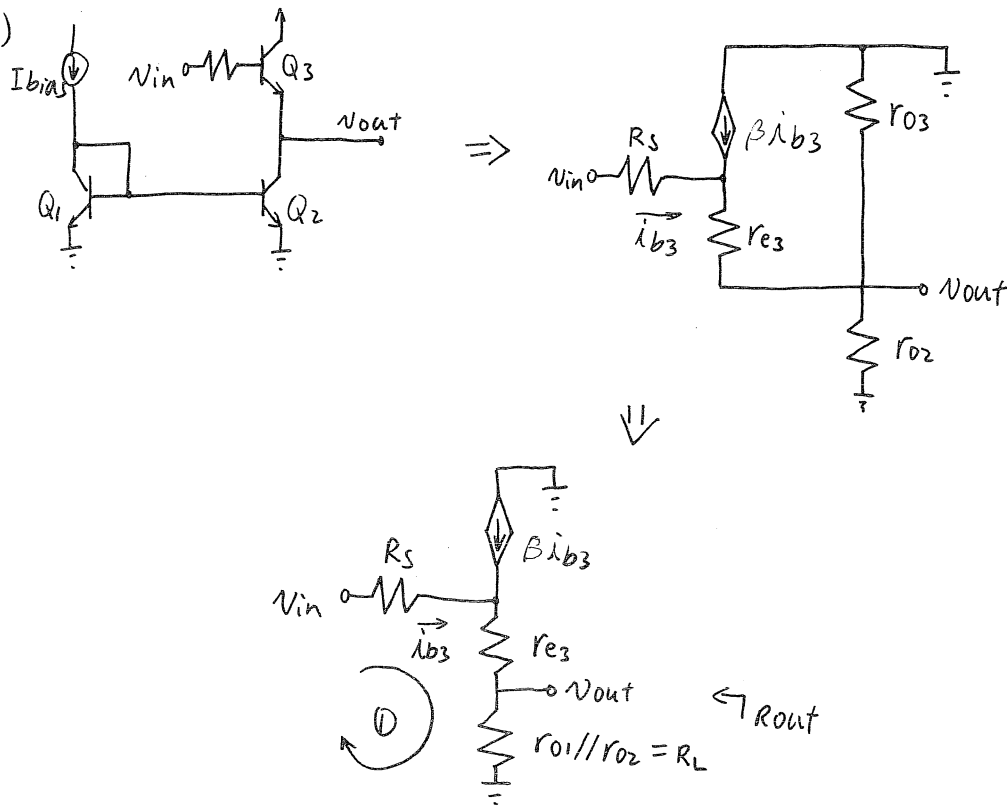
Using Miller's Theorem



Using the method of adding time constants,

$$\left\{ \begin{array}{l} \omega_{-3dB} \approx \frac{1}{\tau_0} + \frac{1}{\tau_{out}} \text{ where} \\ \tau_0 = [R_s \parallel \beta/g_m][C_{be} + C_{cb}(1 + g_m(R_C \parallel r_o))] \\ \tau_{out} = [r_o \parallel R_C][C_L + (1 + \frac{1}{g_m(R_C \parallel r_o)})C_{cb}] \\ \approx [r_o \parallel R_C]C_L \end{array} \right.$$

3.25)



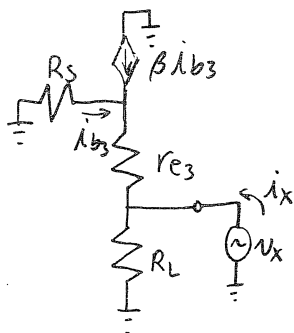
DC gain:

$$\text{KVL } \textcircled{1}: V_{in} = R_s i_{b3} + (\beta + 1) i_{b3} (r_{e3} + R_L)$$

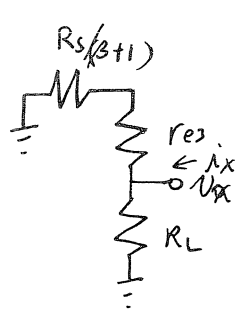
$$\text{and } V_{out} = (\beta + 1) i_{b3} R_L$$

$$\therefore \frac{V_{out}}{V_{in}} = \frac{(\beta + 1) R_L}{(\beta + 1)(r_{e3} + R_L) + R_s} = \frac{R_L}{r_{e3} + R_L + \frac{R_s}{\beta + 1}}$$

Calculating R_{out} :



By source absorption



$$R_{out} \triangleq \frac{V_x}{i_x}$$

$$R_{out} = R_L // \left(r_{e3} + \frac{R_s}{\beta + 1} \right)$$

(cont.)

3.25 (cont.)

When $I_{bias} = 0.5 \text{ mA}$, and assuming $\frac{R_s}{\beta+1} \ll r_{e3}$,

$$I_{c3} \approx I_{c2} = \frac{\beta}{\beta+2} I_{bias} = \frac{100}{102} \times 0.5 \text{ mA} = 0.49 \text{ mA}$$

$$\text{and } r_{o3} \approx r_{o2} = \frac{V_A}{I_{c2}} = \frac{80 \text{ V}}{0.49 \text{ mA}} = 163 \text{ k}\Omega$$

$$\therefore R_L = r_{o1} \parallel r_{o2} = 81 \text{ k}\Omega, \quad r_{e3} = \frac{\beta}{(\beta+1)g_{m3}} = \frac{\beta}{(\beta+1)} \frac{I_{c3}}{V_T}$$

$$\text{and } \frac{V_{out}}{V_{in}} \approx \frac{81 \text{ k}\Omega}{81 \text{ k}\Omega + 53 \Omega} = \frac{100}{101 \times \frac{0.49 \text{ mA}}{26 \text{ mV}}} = \underline{\underline{53 \Omega}}$$

$$= \underline{\underline{0.9993}}$$

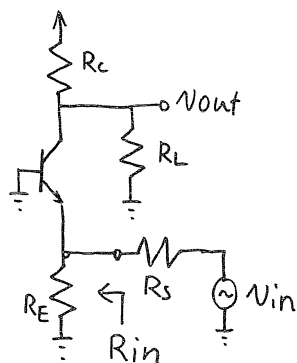
$$R_{out} = R_L \parallel (r_{e3} + \frac{R_s}{\beta+1})$$

$$\approx R_L \parallel r_{e3}$$

$$= 81 \text{ k}\Omega \parallel 53 \Omega$$

$$= \underline{\underline{52 \Omega}}$$

3.26)



Find $\frac{V_{out}}{V_{in}}$ and R_{in}

From the small signal diagram it is clear that $R_{in} = r_e \parallel R_E$.

$$\text{Also, } V_e = \frac{r_e \parallel R_E}{R_s + r_e \parallel R_E} V_{in} \quad (\text{voltage divider})$$

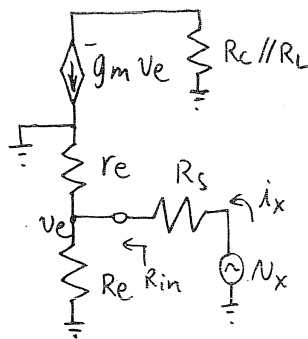
$$V_{out} = +g_m V_e \times R_C \parallel R_L \quad \text{where}$$

$$g_m = \alpha / r_e = \frac{\beta}{(\beta+1) r_e}$$

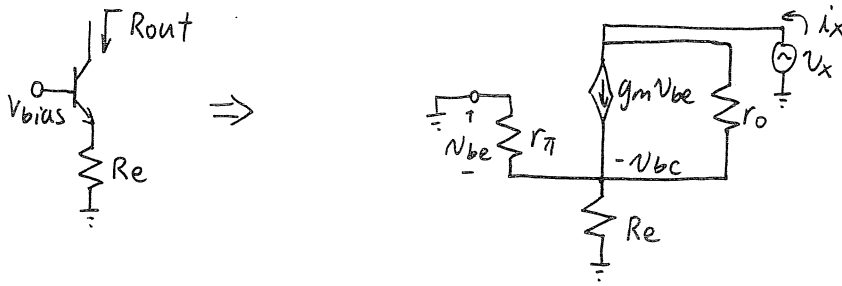
$$\therefore V_{out} = + \frac{\beta}{(\beta+1) r_e} \times \frac{r_e \parallel R_E}{R_s + r_e \parallel R_E} \times R_C \parallel R_L V_{in}$$

$$\frac{V_{out}}{V_{in}} = \frac{\beta}{(\beta+1) r_e} \times \frac{R_E R_e}{r_e + R_E} \times \frac{1}{R_s + \frac{r_e R_E}{r_e + R_E}} \times R_C \parallel R_L$$

$$\underline{\underline{\frac{V_{out}}{V_{in}} = \frac{\beta}{\beta+1} \times \frac{R_E}{r_e R_s + R_E R_s + r_e R_E} \times R_C \parallel R_L}}$$



3.27)



KCL at the output :

$$-i_x + g_m v_{be} + \frac{v_x + v_{be}}{r_o} = 0 \quad [1]$$

$$\text{But } i_x = \frac{-v_{be}}{r_{\pi} \parallel R_e} \Rightarrow v_{be} = -i_x \times (r_{\pi} \parallel R_e) \quad [2]$$

[2] $\rightarrow$ [1]

$$+i_x + g_m i_x \times r_{\pi} \parallel R_e + \frac{i_x}{r_o} (r_{\pi} \parallel R_e) = v_x / r_o$$

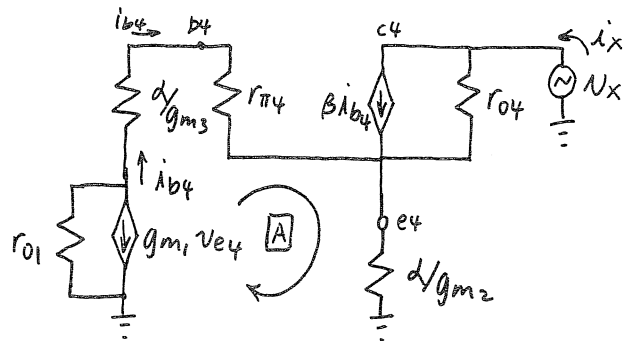
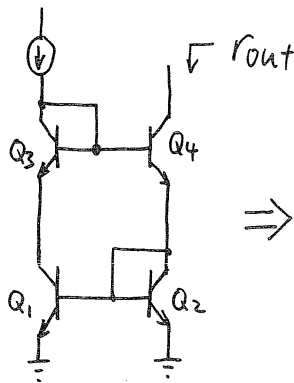
$$\therefore R_{out} \triangleq \frac{v_x}{i_x} = r_o \left(1 + \left(g_m + \frac{1}{r_o} \right) r_{\pi} \parallel R_e \right)$$

$$\because g_m \gg 1/r_o$$

$$\therefore \underline{R_{out} \approx r_o (1 + g_m \times (R_e \parallel r_{\pi}))} \quad \text{Q.E.D.}$$

3.28) * Refer to problems 3.20 and 3.21. *

3.29) Show $r_{out} \approx \frac{\beta r_o}{2}$



KCL at output: $i_x - \beta i_{b4} - (v_x - v_{e4})/r_{o4} = 0$ [1]

KCL at e_4 : $i_x + i_{b4} - v_{e4} g_{m2}/\alpha = 0$ [2]

Now express i_{b4} and v_{e4} in terms of i_x and v_x .

[1] $\times r_{o4}$ $(i_x - \beta i_{b4}) r_{o4} - v_x + v_{e4} = 0$
 $v_{e4} = v_x + (\beta i_{b4} - i_x) r_{o4}$ [3]

[3] $\rightarrow$ [2] $i_x + i_{b4} - [v_x + (\beta i_{b4} - i_x) r_{o4}] g_{m2}/\alpha = 0$
 $i_x + i_{b4} + (i_x - \beta i_{b4}) r_{o4} g_{m2}/\alpha = \frac{g_{m2}}{\alpha} v_x$
 $i_x \left(1 + \frac{r_{o4} g_{m2}}{\alpha}\right) + i_{b4} \left(1 - (\beta+1) r_{o4} g_{m2}\right) = \frac{g_{m2}}{\alpha} v_x$
 $\gg 1$ $\gg 1$

$\therefore r_{o4} g_{m2} i_x - i_{b4} (\beta+1) r_{o4} g_{m2} \approx g_{m2} v_x$

$\Rightarrow i_{b4} \approx \frac{1}{(\beta+1) r_{o4} g_{m2}} [r_{o4} g_{m2} i_x - g_{m2} v_x]$

$i_{b4} \approx \frac{i_x}{\beta+1} - \frac{v_x}{(\beta+1) r_{o4}}$ [4]

[4] $\rightarrow$ [2] $i_x + \frac{i_x}{\beta+1} - \frac{v_x}{(\beta+1) r_{o4}} - v_{e4} \frac{g_{m2}}{\alpha} = 0$

$i_x \left(1 + \frac{1}{\beta+1}\right) - \frac{v_x}{(\beta+1) r_{o4}} = v_{e4} \frac{g_{m2}}{\alpha}$
 $\ll 1$

$v_{e4} \approx \frac{\alpha}{g_{m2}} \left[i_x - \frac{v_x}{(\beta+1) r_{o4}}\right]$

[5]

(cont.)

3.29 (cont.)

KVL around bias network loop [A] :

$$(g_m v_{e4} + i_{b4}) r_{o1} + (\alpha/g_{m3} + r_{\pi4}) i_{b4} + v_{e4} = 0$$

$$\boxed{4} \quad v_{e4} (\underbrace{g_{m1} r_{o1} + 1}_{\gg 1}) + i_{b4} (\underbrace{r_{o1} + r_{\pi4} + \alpha/g_{m3}}_{\ll r_{o1}}) = 0 \quad \boxed{6}$$

$\boxed{5} \rightarrow \boxed{6}$

$$\frac{\alpha}{g_{m2}} \left[i_x - \frac{N_x}{(\beta+1)r_{o4}} \right] g_{m1} r_{o1} + \left(\frac{i_x}{\beta+1} - \frac{N_x}{(\beta+1)r_{o4}} \right) r_{o1} \approx 0$$

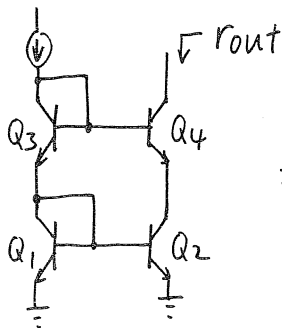
$$r_{o1} \left[i_x - \frac{N_x}{(\beta+1)r_{o4}} \right] + \frac{i_x r_{o1}}{\beta+1} - \frac{N_x}{\beta+1} \approx 0$$

$$i_x r_{o1} \left(1 + \underbrace{\frac{1}{\beta+1}}_{\ll 1} \right) - N_x \times \frac{2}{\beta+1} \approx 0$$

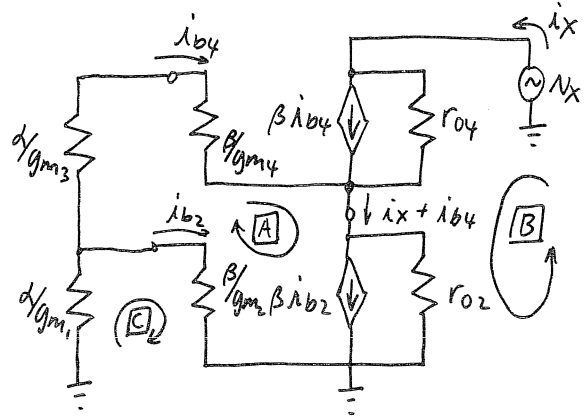
$$\therefore r_{out} \triangleq \frac{v_x}{i_x} \approx \frac{\beta+1}{2} r_{o1} \approx \frac{\beta}{2} r_{o1}$$

$$\therefore \underline{r_{out} \approx \frac{\beta}{2} r_{o1}} \quad \text{Q.E.D.}$$

3.30) Show $r_{out} \approx \beta r_o/2$



$\Rightarrow$



KVL about loop [B]:

$$-v_x + (i_x - \beta i_{b4})r_o + (i_x + i_{b4} - \beta i_{b2})r_o = 0$$

$$2r_o i_x + (1 - \beta)r_o i_{b4} - \beta r_o i_{b2} = v_x \quad [1]$$

KVL about loop [A]:

$$(\alpha/g_{m3} + \beta/g_{m4})i_{b4} + (i_x + i_{b4} - \beta i_{b2})r_o - i_{b2}\beta/g_{m2} = 0$$

$$(\alpha/g_{m3} + \beta/g_{m4} + r_o)i_{b4} + r_o i_x - (\beta r_o + \beta/g_{m2})i_{b2} = 0 \quad [2]$$

KVL about loop [C]:

$$(i_{b4} + i_{b2})\alpha/g_{m1} + i_{b2}\beta/g_{m2} = 0$$

$$\alpha/g_{m1} \cdot i_{b4} + (\beta/g_{m2} + \alpha/g_{m1})i_{b2} = 0$$

$$i_{b4} = -\frac{g_{m1}}{\alpha} \left(\frac{\beta}{g_{m2}} + \frac{\alpha}{g_{m1}} \right) i_{b2}$$

$$= -\left((\beta+1) \frac{g_{m1}}{g_{m2}} + 1 \right) i_{b2}$$

$$\underline{i_{b4} \approx -(\beta+2)i_{b2}} \quad [3]$$

[3] $\rightarrow$ [1]

$$2r_o i_x + (\beta-1)(\beta+2)r_o i_{b2} - \beta r_o i_{b2} = v_x$$

$$\therefore i_{b2} = \frac{2r_o i_x - v_x}{r_o(-\beta^2 - 2\beta + 2)} \quad [4]$$

[3], [4] $\rightarrow$ [2]

$$\underbrace{(\alpha/g_{m3} + \beta/g_{m4} + r_o)}_{\approx \beta/g_{m4} + r_o} \cdot \underbrace{-(\beta+2)}_{\approx \beta} \frac{2r_o i_x - v_x}{r_o(-\beta^2 - 2\beta + 2)} + i_x r_o - \underbrace{(\beta r_o + \beta/g_{m2})}_{\approx \beta r_o} \frac{2r_o i_x - v_x}{r_o(-\beta^2 - 2\beta + 2)} = 0$$

((cont.))

3.30 (cont.)

$$\frac{2r_o i_x - N_x}{r_o(-\beta^2 - \beta + 2)} \left[\left(\beta/g_{m4} + r_o \right)(-\beta) - \beta r_o \right] + i_x r_o \approx 0$$

$$\frac{2r_o i_x - N_x}{r_o(+\beta^2)} \left[2\beta r_o + \beta^2/g_{m4} \right] + i_x r_o \approx 0$$

$$i_x \left(\underbrace{4r_o/\beta + 2/g_{m4} + r_o}_{\approx r_o} \right) - N_x \left[\underbrace{2/\beta + 1/(r_o g_m)}_{2/\beta} \right] \approx 0$$

$$\therefore r_{out} \triangleq \frac{N_x}{i_x} \approx \frac{\beta}{2} r_o \quad \text{Q.E.D.}$$

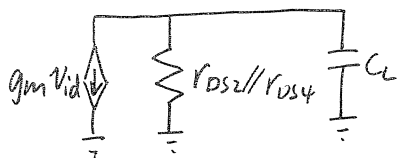
3.31) Find DC gain and ω -3dB

$$\text{DC gain} \triangleq \frac{V_{out}}{V_{in}} = g_{m1} \times r_{DS2} // r_{DS4}$$

$$g_{m1} = \sqrt{2 \mu_n C_{ox} W/L \cdot I_{bias}/2} = \sqrt{2 \times 92 \times 10^{-6} \times \frac{100}{1.6} \times 0.05 \times 10^{-3}} \\ = \underline{0.758 \text{ mA/V}}$$

$$\left. \begin{aligned} r_{DS2} &= \frac{8000L}{I_D} = \frac{8000 \times 1.6}{0.05 \text{ mA}} = 256 \text{ k}\Omega \\ r_{DS4} &= \frac{12000L}{I_D} = \frac{12000 \times 1.6}{0.05 \text{ mA}} = 384 \text{ k}\Omega \end{aligned} \right\} r_{DS2} // r_{DS4} = 154 \text{ k}\Omega$$

$$\therefore \text{DC gain} = 0.758 \times 10^{-3} \times 154 \times 10^3 = \underline{116}$$



$$\omega\text{-3dB} = \frac{1}{r_{DS2} // r_{DS4} \times C_L} \\ = \frac{1}{154 \times 10^3 \times 100 \times 10^{-12}} \\ = \underline{2\pi \times 10 \text{ kHz}}$$

Chapter 4

4.6 PROBLEMS

Unless otherwise stated, assume dBm values are referenced to $50\ \Omega$.

- 4.1 If a signal is measured to have $V_{(rms)}$ volts, what is the difference in db if it is expressed in dBm referenced to $50\ \Omega$ as opposed to being referenced to $75\ \Omega$?
- 4.2 Consider the sum of two noise sources of values $-20\ \text{dBm}$ and $-23\ \text{dBm}$. Find the total noise power in dBm for the cases in which the two noise sources are (a) uncorrelated, (b) $C = 0.3$, (c) $C = +1$, and (d) $C = -1$.
- 4.3 The output noise of a circuit is measured to be $-40\ \text{dBm}$ around $100\ \text{kHz}$ when a resolution bandwidth of $30\ \text{Hz}$ is used. What is the expected dBm measurement if a resolution bandwidth of $10\ \text{Hz}$ is used? Find the root spectral density in $\text{V}/\sqrt{\text{Hz}}$.
- 4.4 At $0.1\ \text{Hz}$, a low-frequency measurement has a noise value of $-60\ \text{dBm}$ when a resolution bandwidth of $1\ \text{MHz}$ is used. Assuming $1/f$ noise dominates, what would be the expected noise value (in dBm) over the band from $1\ \text{MHz}$ to $1\ \text{Hz}$?
- 4.5 Show that, when two resistors of values R_1 and R_2 are in series, their noise model is the same as a single resistance of value $R_1 + R_2$. Repeat the problem for parallel resistances.
- 4.6 Sketch the spectral density of voltage noise across a 100-pF capacitor when it is in parallel with a $1\text{-k}\Omega$ resistor. Make another sketch for the same capacitor but with a $1\text{-M}\Omega$ resistance in parallel. What can you say about the area under the curves of the two sketches?
- 4.7 Consider the circuit shown in Fig. P4.7 on p. 218, where the opamp has a unity-gain frequency of $1\ \text{MHz}$ and equivalent input voltage and current noise sources of $V_n(f) = 20\ \text{nV}/\sqrt{\text{Hz}}$ and $I_n(f) = 10\ \text{pA}/\sqrt{\text{Hz}}$, respectively. Estimate the total rms noise voltage at V_o .
- 4.8 Assuming an ideal noiseless opamp, find the noise level at V_x and V_o for the opamp circuit shown in Fig. P4.8. By what factor is the noise value at V_o larger (or smaller) than $kT/1\ \text{nF}$? How do you account for this increase (or decrease)? Also explain why the noise value at V_x is smaller than $kT/1\ \text{nF}$.

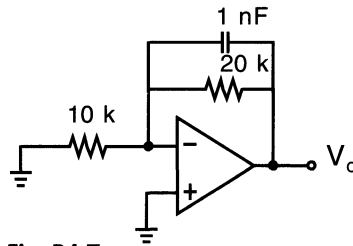


Fig. P4.7

Opamp

$$V_n(f) = 20 \text{ nV}/\sqrt{\text{Hz}}$$

$$I_{n+}(f) = I_{n-}(f) = 10 \text{ pA}/\sqrt{\text{Hz}}$$

$$f_t = 1 \text{ MHz}$$

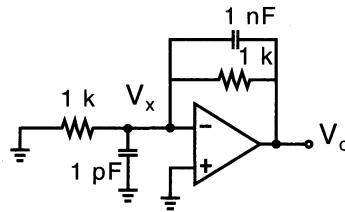


Fig. P4.8

Ideal noiseless opamp

- 4.9 Consider an inductor of value L and an arbitrary resistor in parallel, as shown in Fig. P4.9. Show that the current noise, $i_{no}(t)$, has a noise value given by

$$I_{no(rms)}^2 = \frac{kT}{L}$$

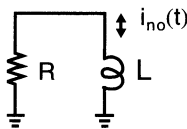


Fig. P4.9

- 4.10 The two circuits shown in Fig. P4.10 realize a first-order, low-pass filter. Assume the opamps are ideal and noiseless.
- Show that the two circuits have the same input-output transfer function.
 - Estimate the total output noise for circuit I, using only dominant noise sources.
 - Repeat (b) for circuit II.
- 4.11 Modify circuit I in Fig. P4.10 such that the new circuit has the same transfer function but uses an 80-pF capacitor instead of an 80-nF capacitor. If the opamp is ideal and noiseless, what is the new total output noise?

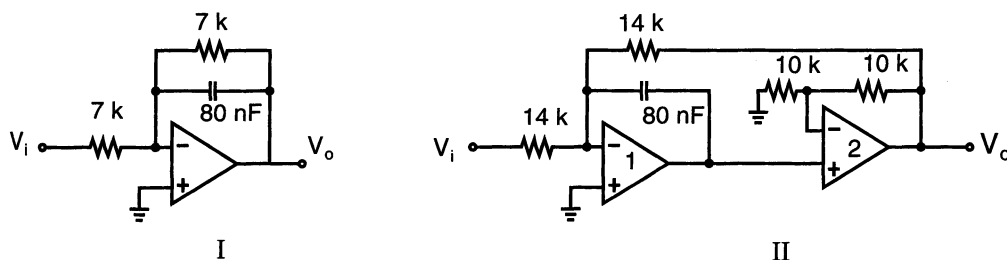


Fig. P4.10

- 4.12** Using the $1/f$ tangent principle, estimate the total noise above 0.1 Hz for the spectral density shown in Fig. 4.3.
- 4.13** Consider a bandpass amplifier that has equivalent input noise root spectral density and amplifier response, as shown in Fig. P4.13. Sketch the root spectral density for the output signal. Estimate the total output rms noise value by applying the $1/f$ tangent principle.

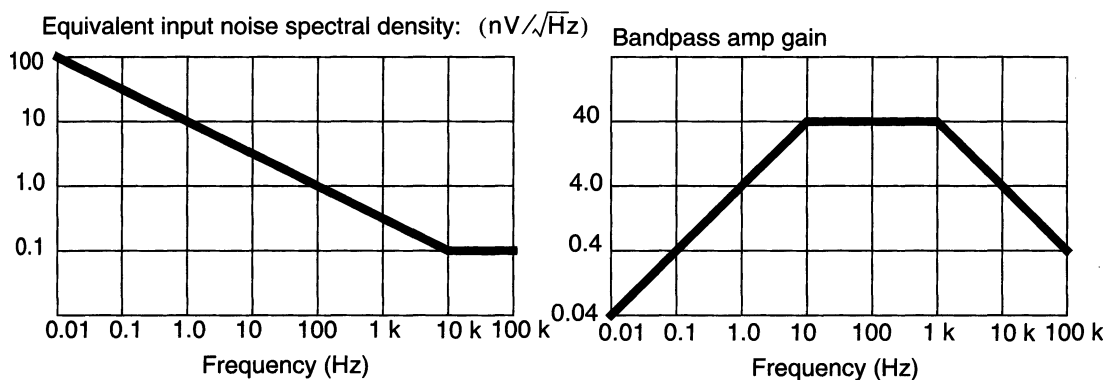


Fig. P4.13

- 4.14** Consider the noise root spectral density of the signal shown in Fig. P4.14. Find the total rms noise value using a graphical approach for 0.01 to ∞ Hz. Compare your result with that obtained when using the $1/f$ tangent principle.
- 4.15** We saw on page 193, Eq. (4.36), that the noise bandwidth of a first-order, low-pass filter is $(\pi/2)f_0$. Show that the noise bandwidth of a second-order, low-pass filter given by

$$A(s) = \frac{A_o}{\left(1 + \frac{s}{2\pi f_0}\right)^2}$$

is equal to $(\pi/4)f_0$.

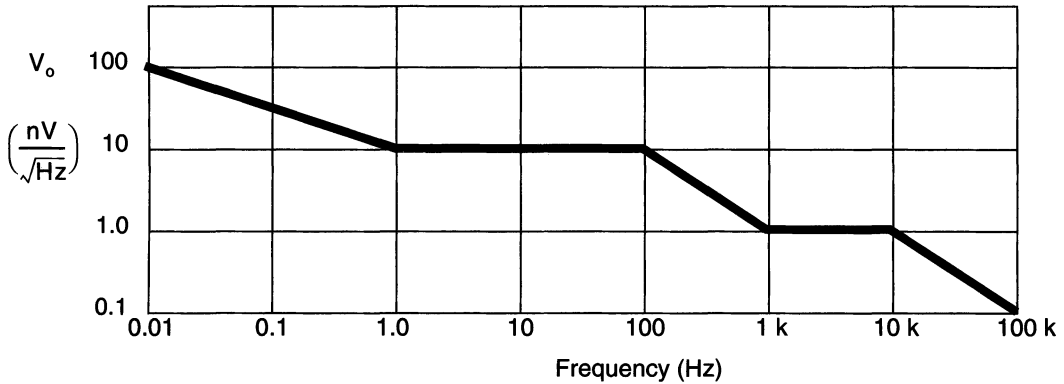


Fig. P4.14

- 4.16** Consider the two bipolar current mirrors shown in Fig. P4.16, where I_{in} is a 1-mA bias current plus a 100- μ A(rms) signal current. Assuming that the base resistance for each transistor is $r_b = 330 \, \Omega$ and dominates the output noise, estimate the resulting SNR (in dB) for the two current mirrors over a 50-MHz bandwidth (also assume the output noise is white).

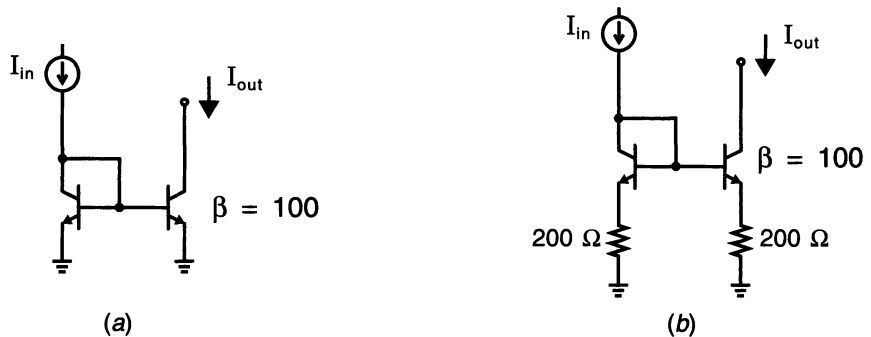


Fig. P4.16

- 4.17** Estimate the total output noise rms value for a low-pass filter, as shown in Fig. 4.15, when $C_f = 1 \, \text{nF}$, $R_f = 16 \, \text{k}$, $R_1 = 1.6 \, \text{k}$, and $R_2 = 0$. Also, find the SNR for an input signal equal to 100 mV rms. Assume that the noise voltage of the opamp is given by $V_n(f) = 20 \, \text{nV}/\sqrt{\text{Hz}}$, that both its noise currents are $I_n(f) = 0.6 \, \text{pA}/\sqrt{\text{Hz}}$, and that its unity-gain frequency equals 2 MHz.
- 4.18** Consider the CMOS differential input stage shown in Fig. 4.17, where Q_5 supplies a bias current of 100 μ A, resulting in $g_{m1} = g_{m2} = 1 \, \text{mA/V}$ and $g_{m3} = g_{m4} = 0.5 \, \text{mA/V}$. Find the equivalent input noise spectral density associated with thermal noise. If the bias current is doubled, how does the equivalent input noise density change?

Chapter 4 - Problems

$$4.1) \text{ dBm}_{(50\Omega)} = 10 \log \frac{V_{\text{rms}}^2}{50 \text{ m}}$$

$$\text{dBm}_{(75\Omega)} = 10 \log \frac{V_{\text{rms}}^2}{75 \text{ m}}$$

$$\therefore \text{dBm}_{(50\Omega)} - \text{dBm}_{(75\Omega)} = 10 \log \frac{75}{50} \approx \underline{1.76}$$

$$\therefore \underline{\text{dBm}_{(50\Omega)} = \text{dBm}_{(75\Omega)} + 1.76}$$

$$4.2) 10 \log \frac{P_1}{1 \text{ mW}} = -20 \Rightarrow P_1 = 10 \mu\text{W}$$

$$10 \log \frac{P_2}{1 \text{ mW}} = -23 \Rightarrow P_2 = 5 \mu\text{W}$$

$$a) P_{\text{total}} = P_1 + P_2 = 15 \mu\text{W} \Rightarrow \underline{P_{\text{total}} = -18.24 \text{ dBm}}$$

$$b) P_{\text{total}} = P_1 + P_2 + 2C\sqrt{P_1 P_2} = 19.24 \mu\text{W} \\ \Rightarrow \underline{P_{\text{total}} = -17.16 \text{ dBm}}$$

$$c) \underline{P_{\text{total}} = -15.35 \text{ dBm}}$$

$$d) \underline{P_{\text{total}} = -30.67 \text{ dBm}}$$

$$4.3) \text{ Expected dBm for 10 Hz bandwidth} = -40 - 10 \log \frac{30}{10} \\ = \underline{-44.77 \text{ dBm}}$$

$$-40 = 10 \log \frac{P}{1 \text{ mW}} \Rightarrow \underline{P = 0.1 \mu\text{W}}$$

(cont.)

$$4.3) \text{ (cont.) } \text{Spectral density} = \frac{0.1 \mu\text{W}}{30 \text{ Hz}} = \frac{3.3}{1000} \frac{(\text{mV})^2}{\text{Hz}}$$

$$\Rightarrow \text{root spectral density} = \underline{0.058 \frac{\text{mV}}{\sqrt{\text{Hz}}}}$$

$$4.4) 10 \log \frac{P}{1 \text{ mW}} = -60 \Rightarrow P = 1 \text{ nW}$$

$$V_n^2(f) \Big|_{f=0.1 \text{ Hz}} = \frac{1 \text{ nW}}{1 \text{ mHz}} = 1 \frac{(\text{mV})^2}{\text{Hz}}$$

$$\text{Assuming } V_n^2(f) = \frac{K_v^2}{f} \text{ and } V_n^2(f) \Big|_{f=0.1 \text{ Hz}} = 1 \frac{(\text{mV})^2}{\text{Hz}},$$

$$V_n^2(f) = \frac{0.1 (\text{mV})^2}{f}$$

Total noise power in (1 mHz, 1 Hz) is :

$$\int_{1 \text{ m}}^1 V_n^2(f) df = 0.1 (\text{mV})^2 [\ln 1 - \ln 1 \text{ m}] = 0.69 \mu\text{W}$$

$$10 \log \frac{0.69 \mu}{1 \text{ m}} = \underline{-31.6 \text{ dBm}}$$

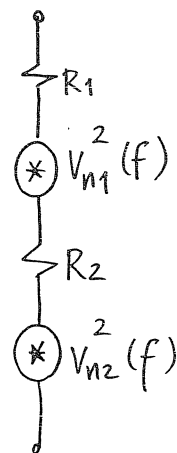
$$4.5) V_{n1}^2(f) = 4KT R_1 \text{ \& } V_{n2}^2(f) = 4KT R_2$$

The two noise sources are uncorrelated.

$$\therefore V_n^2(f) = V_{n1}^2(f) + V_{n2}^2(f) = 4KT R_1 + 4KT R_2$$

$$= 4KT (R_1 + R_2)$$

which is equal to the noise spectral density of the series combination of R_1 \& R_2 .



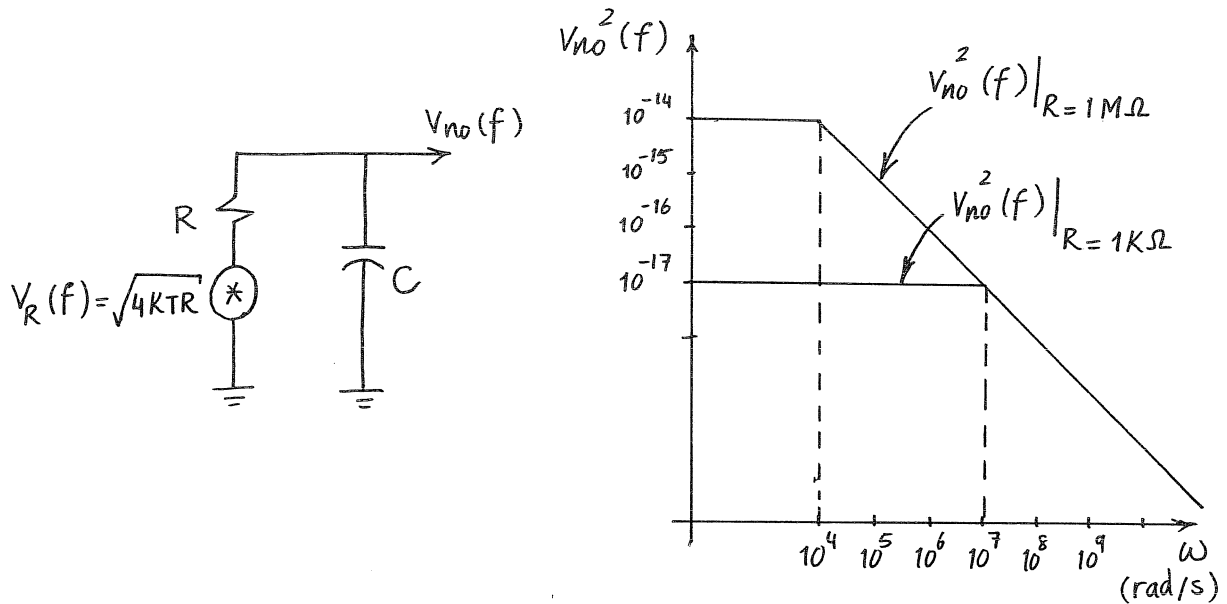
$$4.6) \quad V_{no}^2(f) = \frac{4KTR}{1 + \left(\frac{\omega}{\omega_{3dB}}\right)^2} \quad \text{where } \omega_{3dB} = \frac{1}{RC}$$

For $C = 100 \text{ pF}$ & $R = 1 \text{ K}\Omega$, $\omega_{3dB} = 10^7 \text{ rad/sec}$.

$$\text{and } V_{no}^2(0) = (4.08 \text{ nV})^2 = 1.66 \times 10^{-17} \text{ V}^2$$

For $C = 100 \text{ pF}$ & $R = 1 \text{ M}\Omega$, $\omega_{3dB} = 10^4 \text{ rad/sec}$

$$\text{and } V_{no}^2(0) = (129 \text{ nV})^2 = 1.66 \times 10^{-14} \text{ V}^2$$



The area underneath both curves are the same and

$$\text{is equal to } \frac{KT}{C} = (6.45 \mu\text{V})^2$$

$$4.7) \quad I_{nf}^2 = 8.28 \times 10^{-25} \left(\frac{\text{A}^2}{\text{Hz}} \right), \quad I_{n1}^2 = 1.656 \times 10^{-24} \left(\frac{\text{A}^2}{\text{Hz}} \right)$$

$$I_{n-}^2(f) = 100 \times 10^{-24} \left(\frac{\text{A}^2}{\text{Hz}} \right)$$

$$V_{no1}^2(0) = (I_{nf}^2(0) + I_{n1}^2(0) + I_{n-}^2(0)) R_f^2$$

$$\approx I_{n-}^2(0) R_f^2 = (0.2 \frac{\mu\text{V}}{\sqrt{\text{Hz}}})^2$$

(cont.)

$$4.7) \text{ (cont.) } V_{no1}^2(\text{rms}) = (0.2 \frac{\mu V}{\sqrt{\text{Hz}}}) \times \frac{1}{4(20\text{K})(1\text{n})} = \underline{(22.4 \mu V)^2}$$

$$V_{no2}^2(0) = V_n^2(f) (1 + \frac{R_f}{R_1}) = (60 \frac{\text{nV}}{\sqrt{\text{Hz}}})^2$$

The noise is lowpass filtered at $f_0 = \frac{1}{2\pi R_f C_f} \approx 8 \text{ KHz}$
until $f_1 = \frac{R_f}{R_1} f_0 = 16 \text{ KHz}$.

$$\therefore V_{no2}^2(\text{rms}) = (60 \times 10^{-9})^2 \frac{\pi \times 8\text{K}}{2} + (20 \times 10^{-9})^2 \frac{\pi}{2} (1\text{M} - 16\text{K})$$

$$\approx \underline{(25.9 \mu V)^2}$$

The total output noise is estimated to be:

$$V_{no}^2(\text{rms}) = V_{no1}^2(\text{rms}) + V_{no2}^2(\text{rms}) = (34.2 \mu V)^2$$

$$\Rightarrow \underline{V_{no}(\text{rms}) = 34.2 \mu V \text{ rms}}$$

4.8) Since the opamp is assumed to be ideal and noiseless, we have $V_x = 0$ at all times.

$$\therefore V_{nx}^2(f) = 0 \Rightarrow \underline{V_{nx}^2(\text{rms}) = 0}$$

This noise is smaller than $\frac{KT}{1\text{pF}}$ since the voltage across the capacitor is fixed by another device (i.e. opamp).

For the noise at V_o , we have:

$$V_{no}^2(f) = (i_R^2(f) + i_{R_f}^2(f)) \frac{R_f^2}{1 + \omega^2 R_f^2 C^2}$$

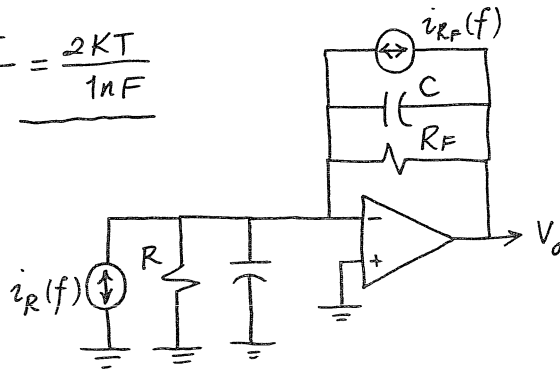
$$= \left(\frac{4KT}{R} + \frac{4KT}{R_f} \right) R_f^2 \times \frac{\pi}{2} \frac{1}{2\pi R_f C}$$

(cont.)

4.8) (Cont.)

$$\therefore V_{no}^2(f) = \frac{2KT}{C} = \frac{2KT}{1nF}$$

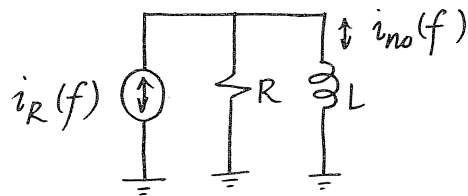
The noise at V_o
is twice as
much as $\frac{KT}{1nF}$.



The reason for this increase is the additional noise current source $i_R(f)$. Since the voltage across this current source is fixed (i.e. it is equal to zero), it does not affect the ω_{3dB} of the filter and acts as an ideal noise current source!

$$4.9) \quad I_{no}^2(f) = \left| \frac{R}{R + j\omega L} \right|^2 I_R^2(f)$$

$$= \frac{1}{1 + \frac{L^2}{R^2} \omega^2} \frac{4KT}{R}$$



$$I_{no(rms)}^2 = \int_0^\infty \frac{4KT/R}{1 + \frac{L^2}{R^2} \omega^2} df$$

$$= \frac{4KT}{R} \frac{1}{2\pi} \frac{R}{L} \int_0^\infty \frac{d\omega}{1 + \omega^2} = \frac{KT}{L}$$

4.10) a) For circuit (I) :

$$\underline{\underline{\frac{V_o}{V_i} = \frac{-1}{1 + j\omega(7K)(80n)}}} \quad (1)$$

For circuit (II) :

$$\begin{aligned} \left(\frac{V_o}{2} (j\omega(80n)) + \frac{V_i}{14K} \right) 14K &= -V_o \\ \Rightarrow \underline{\underline{\frac{V_o}{V_i} = \frac{-1}{1 + j\omega(80n)(7K)}}} &\quad (2) \end{aligned}$$

(1) & (2) show identical transfer functions.

$$b) V_{no}^2(0) = [I_{n1}^2(f) + I_{nf}^2(f)] R_f^2 = \left(15.2 \frac{nV}{\sqrt{Hz}} \right)^2$$

$$\therefore V_{no(rms)}^2 = V_{no}^2(0) \frac{1}{4R_F C_F} = \underline{\underline{(0.32 \mu V)^2}}$$

c) the transfer function for current source entering the negative terminal of opamp 1 is :

$$\begin{aligned} \frac{V_o}{I_{i1}} &= \frac{-R_F}{1 + j\omega \frac{C_F R_F}{2}} \\ \Rightarrow V_{no1}^2(f) &= [I_{n1}^2(f) + I_{nf}^2(f)] \frac{R_F^2}{1 + \omega^2 \frac{C_F^2 R_F^2}{4}} \\ V_{no1}^2(0) &= [I_{n1}^2(f) + I_{nf}^2(f)] R_F^2 = \left(21.5 \frac{nV}{\sqrt{Hz}} \right)^2 \\ V_{no1(rms)}^2 &= V_{no1}^2(0) \frac{1}{4 \frac{R_F C_F}{2}} = \underline{\underline{(0.45 \mu V)^2}} \end{aligned}$$

the transfer function for the current source entering the negative terminal of opamp 2 is :

$$\frac{V_o}{I_{i2}} = \frac{-R_2}{1 + \frac{1}{j\omega C_F R_F/2}} = \frac{-j\omega R_2 R_F C_F/2}{1 + j\omega C_F R_F/2}$$

(cont.)

4.10) (cont.) This is the transfer function of a high-pass filter, which means, ideally, the total output noise due to any current noise at the negative terminal of opamp 2 is ∞ .

$$4.11) \quad \frac{V_o}{V_i} = \frac{-R_F/R_1}{1+j\omega C_F R_F}$$

C_F is reduced by a factor of 1000. Therefore, to keep the same transfer function, R_F & R_1 must be increased by the same factor. The parameters of the new circuit are:

$$R_F = 7\text{M}\Omega, \quad R_1 = 7\text{M}\Omega, \quad C_F = 80\text{ pF}$$

$$\Rightarrow V_{no}^2(0) = [I_{n1}^2(f) + I_{nf}^2(f)] R_F^2 = 8KT R_F$$

$$V_{no(rms)}^2 = 8KT R_F \frac{1}{4R_F C_F} = \frac{2KT}{C_F}$$

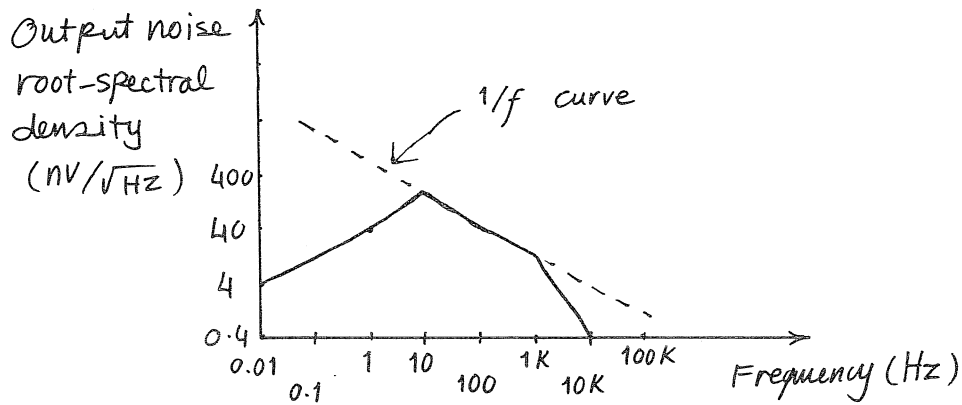
$\therefore$ the total noise power is just a function of C_F (for the case $R_F = R_1$) and increases by decreasing C_F :

$$V_{no(rms)}^2 = \underline{(10.2 \mu\text{V})^2}$$

4.12) The $\frac{1}{f}$ tangent line touches the spectral density of Fig. 4.3 at 1000 Hz. The noise power can be estimated

$$\text{as: } V_n(rms)^2 = 10 \frac{(\mu\text{V})^2}{\text{Hz}} \times \frac{\pi \times 1000}{2} = \underline{(125 \mu\text{V})^2}$$

4.13)



Estimate output noise by integrating from 1 to 10KHz

$$V_{ho(rms)}^2 \cong \int_1^{10} 40^2 f df + \int_{10}^{1k} \frac{400^2}{f} df + \int_{1k}^{10k} \frac{0.4^2 (10k)^2}{f^2} df$$

$$= 7.92 \times 10^4 (mV)^2 + 7.37 \times 10^5 (mV)^2 + 1.44 \times 10^4 (mV)^2$$

$$= 8.31 \times 10^5 (mV)^2 \cong (1 \mu V)^2 \quad \text{Roughly } 1 \mu V \text{ RMS}$$

4.14) A: Graphical Approach

Region 1 : $0.01 < f < 1$ $V_n^2(f) = \left| \frac{10n}{\sqrt{f}} \frac{V}{\sqrt{Hz}} \right|^2 = \frac{10^{-16}}{f} \frac{V^2}{Hz}$

$$\Rightarrow V_{n1}^2 = \int_{0.01}^1 V_n^2(f) df = 4.6 \times 10^{-16} V^2$$

Region 2 : $1 < f < 100$ $V_n^2(f) = 10^{-16} \frac{V^2}{Hz}$

$$\Rightarrow V_{n2}^2 = 99 \times 10^{-16} V^2$$

Region 3 : $100 < f < 1K$ $V_n^2(f) = \frac{10^{-12}}{f^2} \frac{V^2}{Hz}$

$$\Rightarrow V_{n3}^2 = \int_{100}^{1K} \frac{10^{-12}}{f^2} df = 90 \times 10^{-16} V^2$$

Region 4 : $1K < f < \infty$ $V_{n4}^2 = 10^{-18} \times \frac{\pi}{2} \times 9000 = 14.1 \times 10^{-15} V^2$

$$\therefore V_{n(rms)}^2 = V_{n1}^2 + V_{n2}^2 + V_{n3}^2 + V_{n4}^2 = \underline{(0.18 \mu V)^2}$$

(cont.)

4.14) (cont.) B: Using $\frac{1}{f}$ tangent line

The $\frac{1}{f}$ line touches the curve at 100 Hz & 10 KHz simultaneously. If we approximate the noise at 100Hz and 10KHz with two lowpass-filtered noise, we will

$$\text{have : } V_n^2 = 10^{-16} \times \frac{\pi}{2} \times 100 + 10^{-18} \times \frac{\pi}{2} \times 9000 = \underline{(0.17 \mu V)^2}$$

$$4.15) \quad A(s) = \frac{A_0}{\left(1 + \frac{s}{2\pi f_0}\right)^2} \Rightarrow A(j\omega) = \frac{A_0}{\left(1 + j \frac{f}{f_0}\right)^2}$$

$$\Rightarrow \text{equivalent noise bandwidth} = \frac{1}{A_0^2} \int_0^\infty |A(j\omega)|^2 df$$

$$= f_0 \int_0^\infty \frac{df}{(1+f^2)^2} = \underline{f_0 \frac{\pi}{4}}$$

4.16) The equivalent circuits for noise calculations are shown below:

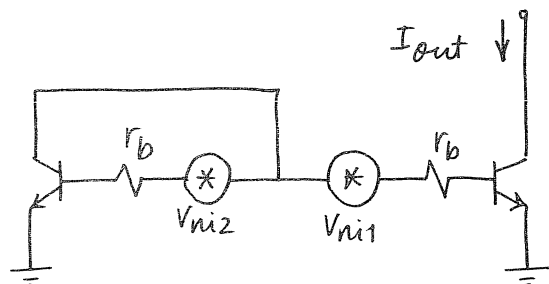


Fig. P4.16(a)

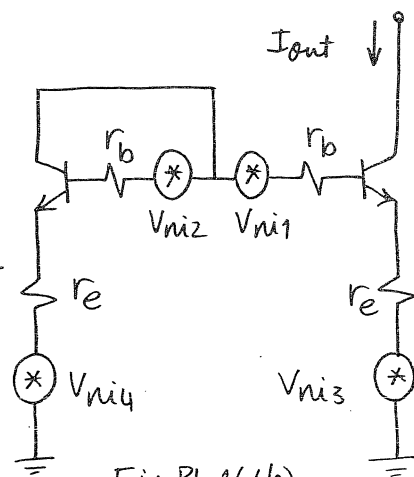


Fig. P4.16(b)

(cont.)

4.16) (cont.) SNR for the current mirror of Fig. P4.16(a):

Using superposition for V_{ni1} & V_{ni2} , and approximating

$\beta + 1 \approx \beta = 100$, we have:

$$I_{no}^2(f) = [V_{ni1}^2(f) + V_{ni2}^2(f)] \frac{\beta^2}{(r_b + r_{\pi})^2}$$

$$\text{where } V_{ni1}^2(f) = V_{ni2}^2(f) = 4KT r_b = 5.46 \frac{(nV)^2}{Hz}$$

$$\text{and } r_{\pi} = \frac{\beta}{g_m} = \frac{\beta V_T}{I_C} = 2.6 K\Omega$$

$$\therefore I_{no}^2(f) = 0.0127 \frac{(nA)^2}{Hz}$$

$$\therefore I_{no(rms)}^2 = 50 MHz \times I_{no}^2(f) = (0.8 \mu A)^2$$

$$\therefore SNR = 20 \log \frac{100 \mu}{0.8 \mu} = \underline{41.94 \text{ dB}}$$

SNR for the current mirror of Fig. P4.16(b):

$$I_{no}^2(f) = [V_{ni1}^2(f) + V_{ni2}^2(f) + V_{ni3}^2(f) + V_{ni4}^2(f)] \frac{\beta^2}{(r_b + r_{\pi} + \beta r_e)^2}$$

$$\text{where } V_{ni3}^2(f) = V_{ni4}^2(f) = 4KT r_e = 3.31 \frac{(nV)^2}{Hz}$$

$$V_{ni1}^2(f) = V_{ni2}^2(f) = 4KT r_b = 5.46 \frac{(nV)^2}{Hz}$$

$$r_{\pi} = \frac{\beta}{g_m} = \frac{\beta V_T}{I_C} = 2.6 K\Omega$$

$$\therefore I_{no}^2(f) = 0.0003336 \frac{(nA)^2}{Hz}$$

$$\therefore I_{no(rms)}^2 = 50 MHz \times I_{no}^2(f) = (0.13 \mu A)^2$$

$$\therefore SNR = 20 \log \frac{100 \mu}{0.13 \mu} = \underline{57.7 \text{ dB}}$$

4.17) Using (4.61): $V_{no1}^2(0) = (I_{n1}^2(0) + I_{nf}^2(0) + I_{n-}^2(0)) R_F^2$

where $I_{nf}^2(f) = \frac{4KT}{R_F} = 1.035 \left(\frac{pA}{\sqrt{Hz}} \right)^2$

∴ $V_{no1}^2(0) = (2 \times 0.6^2 + 1.035) (16K)^2 \left(\frac{pV}{\sqrt{Hz}} \right)^2 = (21.2 \frac{nV}{\sqrt{Hz}})^2$

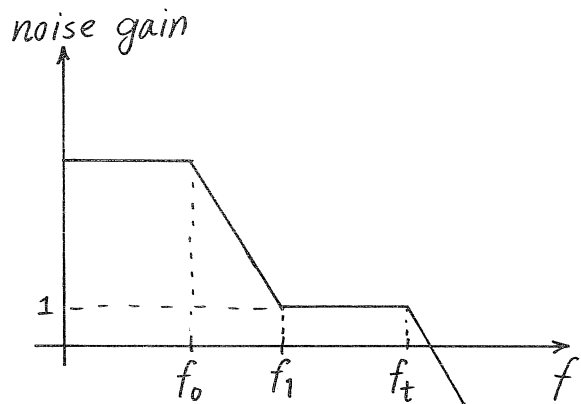
$V_{no1}^2(rms) = V_{no1}^2(0) \times \frac{1}{4R_F C_F} = (2.65 \mu V)^2$

Using (4.62): $V_{no2}^2(0) = V_n^2(f) \left(1 + \frac{R_F}{R_1} \right)^2 = (220 \frac{nV}{\sqrt{Hz}})^2$

this noise is lowpass
filtered at $f_0 = \frac{1}{2\pi R_F C_F}$

until $f_1 = \frac{R_F}{R_1} f_0$

where the noise gain
reaches unity.



$$V_{no2}^2(rms) = (220 \times 10^{-9})^2 \frac{1}{4R_F C_F} + (20 \times 10^{-9})^2 \frac{\pi}{2} (f_t - f_1)$$

$$= (44.16 \mu V)^2$$

Using (4.64): $V_{no}^2(rms) = (44.24 \mu V)^2$

SNR for a 100 mV signal = $20 \log \frac{100 m}{44.24 \mu}$

$= 67.08 dB$

4.18) Using (4.97):

$$\begin{aligned} V_{neg}^2(f) &= \left(\frac{16}{3}\right) KT \frac{1}{g_{m1}} + \frac{16}{3} KT \left(\frac{g_{m3}}{g_{m1}}\right)^2 \frac{1}{g_{m3}} \\ &= \underline{\underline{\left(5.75 \frac{nV}{\sqrt{Hz}}\right)^2}} \end{aligned}$$

Since $g_m \propto \sqrt{I_{BIAS}}$, if I_{BIAS} is doubled, g_{m1} & g_{m3} will be increased by $\sqrt{2}$. Therefore, $V_{neg}^2(f)$ will be reduced by $\sqrt{2}$, and we have:

$$V_{neg}^2(f) = \frac{1}{\sqrt{2}} \left(5.75 \frac{nV}{\sqrt{Hz}}\right)^2 = \underline{\underline{\left(4.84 \frac{nV}{\sqrt{Hz}}\right)^2}}$$

Chapter 5

5.5 PROBLEMS

Unless otherwise stated, assume the following:

- n-channel MOS transistors:

$$\mu_n C_{ox} = 92 \mu A/V^2$$

$$V_{tn} = 0.8 V$$

$$\gamma = 0.5 V^{1/2}$$

$$r_{ds}(\Omega) = 8,000 L(\mu m)/I_D(mA) \text{ in active region}$$

$$C_j = 2.4 \times 10^{-4} pF/(\mu m)^2$$

$$C_{j-sw} = 2.0 \times 10^{-4} pF/\mu m$$

$$C_{ox} = 1.9 \times 10^{-3} pF/(\mu m)^2$$

$$C_{gs(overlap)} = C_{gd(overlap)} = 2.0 \times 10^{-4} pF/\mu m$$
- p-channel MOS transistors:

$$\mu_p C_{ox} = 30 \mu A/V^2$$

$$V_{tp} = -0.9 V$$

$$\gamma = 0.8 V^{1/2}$$

$$r_{ds}(\Omega) = 12,000 L(\mu m)/I_D(mA) \text{ in active region,}$$

$$C_j = 4.5 \times 10^{-4} pF/(\mu m)^2$$

$$C_{j-sw} = 2.5 \times 10^{-4} pF/\mu m$$

$$C_{ox} = 1.9 \times 10^{-3} pF/(\mu m)^2$$

$$C_{gs(overlap)} = C_{gd(overlap)} = 2.0 \times 10^{-4} pF/\mu m$$

- 5.1 Assume that $I_{D5} = 100 \mu A$, all transistor lengths are $1.2 \mu m$, and $C_C = 10 pF$. For the two-stage opamp of Fig. 5.2,
 - a. Estimate the -3-dB frequency of the first stage.
 - b. Estimate the unity-gain frequency of the opamp.
 - c. Find the slew rate.
- 5.2 Repeat Example 5.3, but let $C_C = 4 pF$. What circuit changes could be made to double the slew rate while keeping ω_t and C_C unchanged?
- 5.3 How should the sizes of Q_6 and Q_7 change in Problem 5.2 to maintain zero systematic offset voltage?
- 5.4 Assume Q_3 and Q_4 of the two-stage opamp of Fig. 5.4 are both $50 \mu m$ wide, all transistor lengths are $1.6 \mu m$, and $I_{D5} = 100 \mu A$. Estimate the inherent input-offset voltage.
- 5.5 Ignoring the body effect, what is the output voltage range of the opamp of Fig. 5.2? What is the range of the common-mode input voltage, assuming $\pm 5 V$ power supplies are used? Assume $I_{D5} = 100 \mu A$.

- 5.6** Consider the opamp shown in Fig. 5.2, where Q_{16} is placed on the output side of C_C (rather than where it is shown). Explain why this circuit would then oscillate for large positive output values.
- 5.7** Repeat Problem 5.5, but take the body effect into account and find approximate answers.
- 5.8** Consider a system response, $H(s)$, with one dominant pole, other real-axis poles, and zeros such that

$$H(s) = \frac{A_0 \times \Pi \left(1 + \frac{s}{\omega_{zi}} \right)}{\left(1 + \frac{s}{\omega_{p1}} \right) \Pi \left(1 + \frac{s}{\omega_{pi}} \right)}$$

Also consider another approximating function, $H_{app}(s)$, given by

$$H_{app}(s) = \frac{A_0}{\left(1 + \frac{s}{\omega_{p1}} \right) \left(1 + \frac{s}{\omega_{eq}} \right)}$$

Show that, when $H(s)$ is approximated by $H_{app}(s)$ such that the phase of the two systems are approximately equal at ω_t of $H(s)$, ω_{eq} should be equal to

$$\frac{1}{\omega_{eq}} = \sum \frac{1}{\omega_{pi}} - \sum \frac{1}{\omega_{zi}}$$

- 5.9** An opamp has its first pole at 3 kHz and has high-frequency poles at 130 MHz, 160 MHz, and 180 MHz. Using iteration, find the frequency where the phase shift is -135° and, therefore, find the equivalent time constant that models the high-frequency poles. How does this compare to the estimate given by equation (5.45)?
- 5.10** A two-stage opamp has a compensation capacitor connected between the input and the output of the second stage. It has an equivalent second-pole frequency of 60 MHz. Assume the input transistors of the first stage have a transconductance of 0.775 mA/V and the gain of the output buffer is exactly unity. What is the required size of the compensation capacitor if the phase margin is to be 55° for the feedback configuration shown in Fig. P5.10?

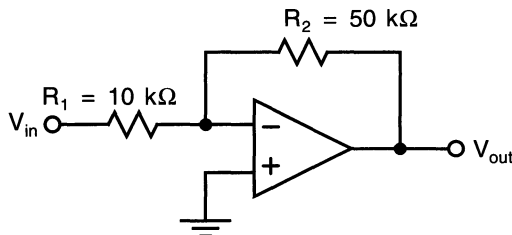


Fig. P5.10

- 5.11** For the result of Problem 5.10, what is a near-optimum size for a capacitor to be placed in parallel with R_2 in order to obtain lead compensation?
- 5.12** An opamp has an open-loop transfer function given by

$$H(s) = \frac{A_0}{\left(1 + \frac{s}{\omega_1}\right)\left(1 + \frac{s}{\omega_2}\right)}$$

Assume $A_0 = 10^4$ and $\omega_2 = 10^8$ rad/s. A zero, ω_z , is added such that the frequency of the zero is the same as the resulting open-loop unity-gain frequency, ω_t . Find ω_1 and ω_t so that the phase margin is 80° .

- 5.13** An opamp has an open-loop transfer function given by

$$A(s) \cong \frac{A_0(1 + s\tau_z)}{s\tau_1(1 + s\tau_2)}$$

Find the transfer function of the closed-loop amplifier, assuming a feedback factor β exists. Find approximate equations for the resonant frequency and the Q factor of the denominator of the transfer function of the closed-loop amplifier.

- 5.14** Prove that the frequencies of the first and second poles of the amplifier of Fig. 5.2 are still given by (5.70) and (5.71), even when lead compensation is used (i.e., when R_C is included in Fig. 5.8).
- 5.15** Design the bias circuit of Fig. 5.10 to have $V_{\text{eff}} = 0.25$ V for all transistors. For $Q_{12} - Q_{14}$ having sizes of $10 \mu\text{m}/1.2 \mu\text{m}$ and Q_{15} having a size of $40 \mu\text{m}/1.2 \mu\text{m}$, what is the required size of R ? What would the effective gate-source voltage be at 70°C ? Confirm your answer using SPICE.

Chapter 5 - Problems

5.1)

a) Find $\omega_{-3dB} = \frac{1}{R_{out} C_{eq}} = \frac{1}{r_{DS2} // r_{DS4} \times C_L (1 + A_2)}$

where $r_{DS4} = \frac{8000 L}{I_{D4}} = \frac{8000 \times 1.2}{0.05} = 192 \text{ k}\Omega$

$$r_{DS2} = \frac{12000 L}{I_{D2}} = \frac{12000 \times 1.2}{0.05} = 288 \text{ k}\Omega$$

$$\therefore r_{DS4} // r_{DS2} = 115 \text{ k}\Omega$$

$$A_2 = g_{m7} (r_{DS6} // r_{DS7})$$

$$r_{DS6} = \frac{12000 \times 1.2}{0.1} = 144 \text{ k}\Omega$$

$$r_{DS7} = \frac{8000 \times 1.2}{0.1} = 96 \text{ k}\Omega$$

$$\therefore r_{DS6} // r_{DS7} = 58 \text{ k}\Omega$$

$$g_{m7} = \sqrt{2 \mu_n C_{ox} W/L I_{D5}} = (2 \times 92 \times 10^{-6} \times 300 / 1.2 \times 0.1 \times 10^{-3})^{\frac{1}{2}}$$
$$= 2.15 \text{ mA/V}$$

$$\therefore A_2 = 2.15 \times 10^{-3} \times 58 \times 10^3 = 124$$

$$\therefore \omega_{-3dB} = \frac{1}{115 \times 10^3 \times 10 \times 10^{-12} \times 125} = \underline{\underline{2\pi \times 1.1 \text{ kHz}}}$$

b) Find unity-gain frequency, ω_t .

$$\omega_t = g_{m1} / c_c \quad \text{where} \quad g_{m1} = \sqrt{2 \times 30 \times 10^{-6} \times \frac{300}{1.2} \times 50 \times 10^{-6}}$$
$$= 0.866 \text{ mA/V}$$

$$\therefore \omega_t = \frac{0.866 \times 10^{-3}}{10 \times 10^{-12}} = \underline{\underline{2\pi \times 14 \text{ MHz}}}$$

c) Find slew rate.

$$SR = \frac{2 I_{D1}}{C_c} = \frac{2 \times 50 \mu\text{A}}{10 \text{ pF}} = \underline{\underline{10 \text{ V}/\mu\text{sec}}}$$

5.2) With $C_c = 4\text{pF}$,

$$SR = \frac{2I_{D1}}{C_c} = \frac{2 \times 50\mu\text{A}}{4\text{pF}} = \underline{25\text{ V}/\mu\text{sec}}$$

To double the slew rate while maintaining C_c , we need to double I_{D1} by doubling the width of Q_5 .

In order to prevent this from changing g_{m1} and g_{m2} and hence ω_t , we need to reduce the widths of Q_1 and Q_2 by half.

5.3) Referring to Problem 5.2, if we scale the widths of Q_1 and Q_2 by half, we need to maintain the equality

$$\frac{W/L_7}{W/L_4} = 2 \frac{W/L_6}{W/L_5} \quad (5.28)$$

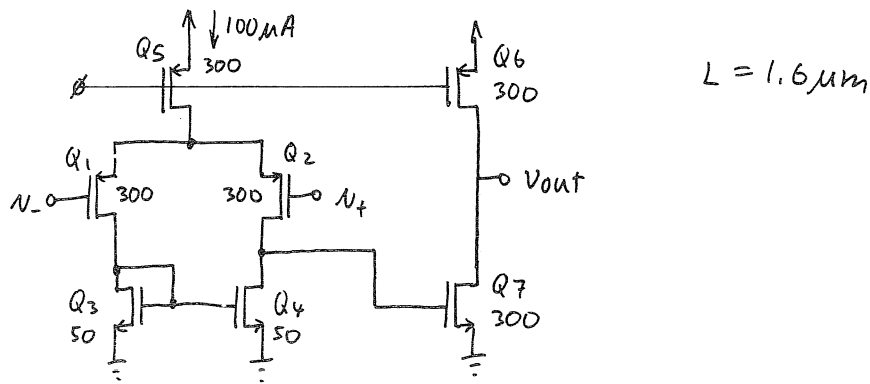
$$\therefore \frac{(W/L)_7}{(W/L)_6} = \frac{2(W/L)_4}{(W/L)_5} = \frac{2(150/1.2)}{(600/1.2)} = \underline{\frac{1}{2}}$$

2 POSSIBILITIES ARE :

1) IF bias current at output stage remains unchanged $\Rightarrow (W/L)_6 = \frac{300}{1.2}$ & $(W/L)_7 = \frac{150}{1.2}$

2) IF bias current at output stage is doubled $\Rightarrow (W/L)_6 = \frac{600}{1.2}$ & $(W/L)_7 = \frac{300}{1.2}$

5.4)



To remove the inherent, systematic offset, the widths of Q_3 and Q_4 should become $150\mu\text{m}$. In that case,

$$V_{eff7} = V_{eff4} = \sqrt{\frac{2I_{D7}}{\mu_n C_{ox} W/L_7}} = \sqrt{\frac{2 \times 100 \times 10^{-6}}{92 \times 10^{-6} \times 300/1.6}} = \underline{0.1077\text{V}}$$

However, with the current circuit,

$$V_{eff4} = \sqrt{\frac{2 \times 50 \times 10^{-6}}{92 \times 10^{-6} \times 50/1.6}} = 0.1865\text{V}$$

$\therefore$ An input offset voltage will have to be applied in order to decrease V_{gs7} by

$$\Delta V_{gs7} = 0.1865 - 0.1077 = 78.8\text{mV}$$

For the input-referred offset voltage V_{ioff} ,

$$V_{ioff} = \Delta V_{gs7} / A_1$$

where A_1 is the first stage's voltage gain

$$A_1 = g_{m1} \times r_{DS2} // r_{DS4}$$

$$g_{m1} = \sqrt{2 \times 30 \times 10^{-6} \times \frac{300}{1.6} \times 50 \times 10^{-6}} = 0.75\text{mA/V}$$

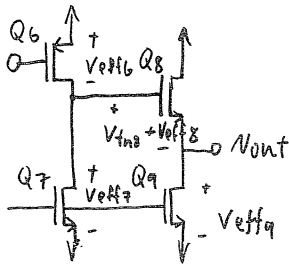
$$r_{DS4} = 8000 \times 1.6 / 0.05\text{mA} = 256\text{k}\Omega$$

$$r_{DS2} = 12000 \times 1.6 / 0.05\text{mA} = 384\text{k}\Omega$$

$$\therefore A_1 = 0.75 \times (256 // 384) = 115$$

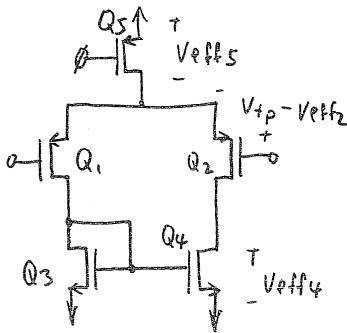
And $\underline{V_{ioff} = \frac{78.8\text{mV}}{115} = 0.7\text{mV}}$ (applied to v_+ input).

5.5) Find max. and min v_{out} and V_{in} (common mode)



$$V_{out-max} = V_{DD} - V_{eff6} - V_{th8} - V_{eff8}$$

$$V_{out-min} = V_{SS} + V_{eff9}$$



$$V_{in-cm-max} = V_{DD} - V_{eff5} - V_{eff1} + V_{tp1}$$

$$V_{in-cm-min} = V_{SS} + V_{eff3} + V_{th3} + V_{tp1}$$

Calculating all values,

$$V_{eff5} = V_{eff6} = \sqrt{\frac{2I_{D5}}{\mu_n C_{ox} W/L}} = \sqrt{\frac{2 \times 100 \times 10^{-6}}{30 \times 10^{-6} \times 300/1.6}} = 0.189V$$

similarly,

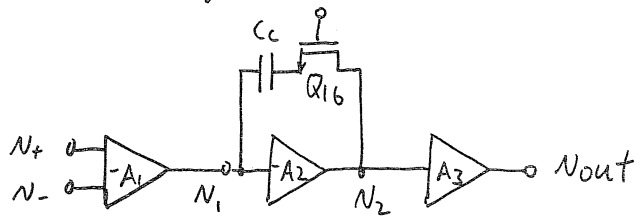
$$V_{eff1} = V_{eff2} = 0.133V$$

$$V_{eff3} = V_{eff4} = 0.108V$$

$$V_{eff8} = V_{eff9} = 0.108V$$

$$\left\{ \begin{array}{l} V_{out-max} = 5 - 0.189 - 0.8 - 0.108 = 3.9V \\ V_{out-min} = -5 + 0.108 = -4.9V \\ V_{in-cm-max} = 5 - 0.189 - 0.133 - 0.9 = 3.8V \\ V_{in-cm-min} = -5 + 0.108 + 0.8 - 0.9 = -4.99V \end{array} \right.$$

5.6) Bad design of compensation network:



In order to understand why oscillations occur on account of this arrangement, we have to keep in mind our use of Q_{16} , operated in the triode region, as a substitute for a feed-forward resistor. When v_{out} is large and positive, v_2 is similarly so. Should the voltage across C_c happen to be small, a large voltage would appear across the drain and source of Q_{16} . If this voltage were large enough to put Q_{16} into the active mode, capacitor C_c would effectively become disconnected from the output. The result would be an uncompensated op amp that is prone to oscillations.

By keeping Q_{16} connected to the input of the second stage, most of the large signal swings will appear across the capacitor instead. As such, Q_{16} will more reliably remain in the triode region of operation.

- 5.7) The body effect changes the threshold voltages of Q_8 , Q_1 and Q_2 . The change in V_{tn8} affects only $V_{out-max}$ while the changes in V_{tp1} and V_{tp2} affect both $V_{in-cm-max}$ and $V_{in-cm-min}$.

From Problem 5.5, $V_{out-min} = -4.9V$

The other values must be solved iteratively as the threshold voltages are dependent on V_{SB} 's.

For $V_{out-max}$:

Assume $V_{tn8} = 0.8V$ for first iteration

$$\begin{aligned} \therefore V_{SB8} &= V_{out-max} - V_{SS} = V_{DD} - V_{eff6} - V_{tn8} - V_{eff8} - V_{SS} \\ &= 5 - 0.189 - 0.8 - 0.108 - (-5V) \\ &= 8.9V \end{aligned}$$

$$\begin{aligned} \therefore V_{tn8} &= V_{tn0} + \gamma (\sqrt{V_{SB8} + 2\phi_F} - \sqrt{2\phi_F}) \\ &= 0.8V + 0.5 (\sqrt{8.9 + 0.7} - \sqrt{0.7}) \\ &= 1.93V \end{aligned}$$

1st iteration

$$\begin{aligned} \therefore V_{SB8} &= 5 - 0.189 - 1.93 - 0.108 - (-5V) \\ &= 7.77V \end{aligned}$$

$$V_{tn8} = 0.8 + 0.5 (\sqrt{7.77 + 0.7} - \sqrt{0.7})$$

$$\underline{V_{tn8} = 1.84V}$$

2nd iteration

$$\therefore V_{out-max} = V_{DD} - V_{eff6} - V_{tn8} - V_{eff8} = 5 - 0.189 - 1.84 - 0.108$$

$V_{out-max} = 2.9V$ which is almost 1 volt lower than our original result in PS.5.

For $V_{in-cm-max}$:

$$V_{SB1} = V_{eff5} = 0.189V \quad (\text{no iterations required})$$

$$\begin{aligned} V_{tp1} &= V_{tp0} - \gamma (\sqrt{V_{SB1} + 2\phi_F} - \sqrt{2\phi_F}) \\ &= -0.9 - 0.8 (\sqrt{0.189 + 0.7} - \sqrt{0.7}) \\ &= -0.98V \end{aligned}$$

$$\therefore V_{in-cm-max} = V_{DD} - V_{eff5} - V_{eff1} + V_{tp1} = 5 - 0.189 - 0.133 - 0.98$$

$$\underline{V_{in-cm-max} = 3.7V}$$

(cont.)

5.7) (cont.)

For $V_{in,cm-min}$:

Assume $V_{tp1} = -0.9V$

$$\begin{aligned} V_{S1} &= V_{DD} - (V_{in,cm-min} + V_{eff1} - V_{tp1}) \\ &= V_{DD} - (V_{SS} + V_{eff3} + V_{tn3} + V_{tp1} + V_{eff1} - V_{tp1}) \\ &= V_{DD} - V_{SS} - V_{eff3} - V_{tn3} - V_{eff1} = 5 - (-5) - 0.108 - 0.8 - 0.133 \\ &= 8.96V \end{aligned}$$

$$\begin{aligned} \therefore V_{tp1} &= V_{tp0} - \gamma (\sqrt{V_{S1} + 2\phi_F} - \sqrt{2\phi_F}) \\ &= -0.9 - 0.8 (\sqrt{8.96 + 0.7} - \sqrt{0.7}) \\ &= \underline{\underline{-2.7V}} \end{aligned}$$

$$\begin{aligned} \therefore V_{in,cm-min} &= V_{SS} + V_{eff3} + V_{tn3} + V_{tp1} = -5 + 0.108 + 0.8 - 2.7 \\ &= \underline{\underline{-6.8V}} \end{aligned}$$

Note that the min common-mode input voltage is lower than the $-5V$ supply.

5.8) Show that $1/\omega_{eq} = \sum \frac{1}{\omega_{pi}} - \sum \frac{1}{\omega_{zi}}$.

We are given that

$$\angle[H(j\omega)] \approx \angle[H_{app}(j\omega)]$$

$$\begin{aligned} LS = \angle[H(j\omega)] &\approx \angle\left[\frac{\pi(1+j\omega/\omega_{zi})}{\pi(1+j\omega/\omega_{pi})}\right] \\ &= \tan^{-1}\left(\frac{\omega/\omega_{zi}}{1}\right) + \tan^{-1}\left(\frac{\omega/\omega_{z2}}{1}\right) + \dots + \tan^{-1}\left(\frac{\omega/\omega_{zn}}{1}\right) - \tan^{-1}\left(\frac{\omega/\omega_{p1}}{1}\right) \\ &\quad - \dots - \tan^{-1}\left(\frac{\omega/\omega_{pn}}{1}\right) \end{aligned}$$

But the Taylor Series expansion for $\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$
 $\approx x$ for $x \ll 1$

$\therefore$ If the op amp is compensated such that the unity gain frequency, ω_t , is much lower than all higher order poles and zeros but much higher than ω_p ,

$$LS \approx \frac{\omega_t}{\omega_{z1}} + \dots + \frac{\omega_t}{\omega_{zn}} - 90^\circ - \frac{\omega_t}{\omega_{p2}} - \dots - \frac{\omega_t}{\omega_{pn}} = \omega_t \left(\sum \frac{1}{\omega_{zi}} - \sum \frac{1}{\omega_{pi}} \right) - 90^\circ$$

$$\text{Similarly, } RS = \angle[H_{app}(j\omega)] = \tan^{-1}\left(\frac{\omega_t/\omega_{eq}}{1}\right) - 90^\circ \approx \frac{\omega_t}{\omega_{eq}} - 90^\circ$$

$$\begin{aligned} \therefore \omega_t \left(\sum \frac{1}{\omega_{zi}} - \sum \frac{1}{\omega_{pi}} \right) &\approx -\omega_t/\omega_{eq} \\ \text{or } \underline{\underline{\frac{1}{\omega_{eq}} = \sum \frac{1}{\omega_{pi}} - \sum \frac{1}{\omega_{zi}}}} &\quad \text{Q.E.D.} \end{aligned}$$

5.9) Given:

$$H(s) = \frac{K}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})(1 + s/\omega_{p3})(1 + s/\omega_{p4})}$$

where $\omega_{p1} = 2\pi \times 3 \text{ kHz}$, $\omega_{p2} = 2\pi \times 130 \text{ MHz}$
 $\omega_{p3} = 2\pi \times 160 \text{ MHz}$, $\omega_{p4} = 2\pi \times 180 \text{ MHz}$

$$\angle H(j\omega) = -90^\circ - \tan^{-1}\left(\frac{\omega}{130}\right) - \tan^{-1}\left(\frac{\omega}{160}\right) - \tan^{-1}\left(\frac{\omega}{180}\right)$$

* Given that $\angle H(j\omega_{eq}) = -135^\circ$ then trial and error gives $\omega_{eq} \cong 2\pi \times 41.3 \text{ MHz}$

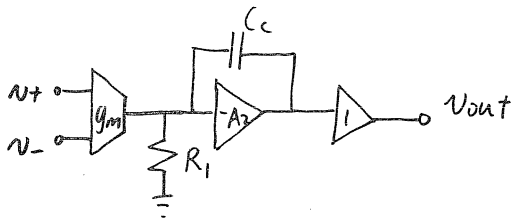
Using Eq. (5.45),

$$\frac{1}{\omega_{eq}} \cong \frac{1}{\omega_{p2}} + \frac{1}{\omega_{p3}} + \frac{1}{\omega_{p4}} = \frac{1}{2\pi} \left(\frac{1}{130 \times 10^6} + \frac{1}{160 \times 10^6} + \frac{1}{180 \times 10^6} \right)$$

$$\therefore \omega_{eq} \cong 2\pi \times 51.3 \text{ MHz}$$

This estimate is about 24% above the true value.

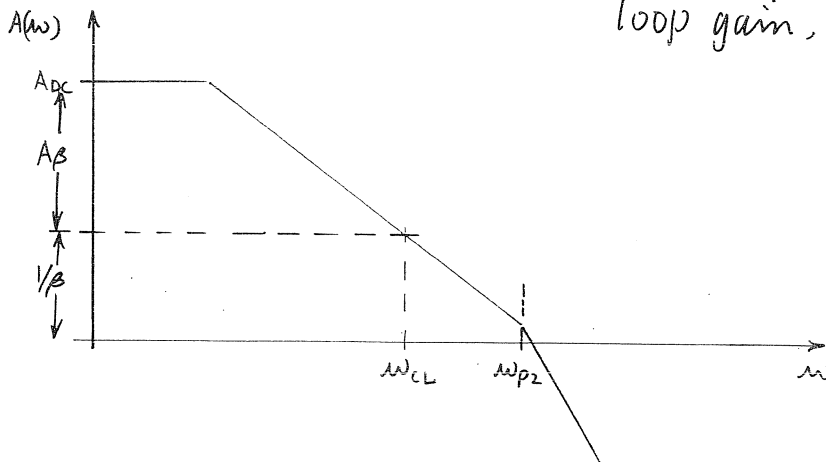
5.10) Two stage opamp:



$$g_m = 0.775 \text{ mA/V}$$

$$\omega_{p2} = 60 \text{ MHz}$$

ω_{CL} is defined as the frequency at which the loop gain, $A\beta$, is unity.



(cont.)

5.10 (cont.)

For a closed loop phase margin of 55°
eqn (5.53) states

$$\omega_t = \tan(90^\circ - PM) \omega_{eq}$$

$$= \tan(90^\circ - 55^\circ) \omega_{eq}$$

$$\omega_t = 0.7 \omega_{eq}$$

For $\omega_{eq} = 2\pi \times 60 \text{ MHz}$, $\omega_t = 2\pi \times 42 \text{ MHz}$

Now including a feedback factor β into
eqn (5.46), we have

$$A(s) = \frac{\omega_t a \beta}{s(1 + s/\omega_{eq})} = \frac{\frac{g_m \beta}{C_c}}{s(1 + s/\omega_{eq})}$$

$$\text{Here } \beta = \frac{R_1}{R_1 + R_2} = \frac{1}{6}$$

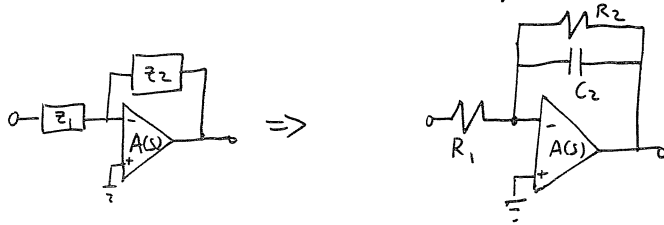
We require that $|A(j\omega_t)| = 1$ leading to

$$|A(j\omega_t)| = 1 = \frac{\frac{g_m \beta}{C_c}}{|\omega_t| \sqrt{1 + \left(\frac{\omega_t}{\omega_{eq}}\right)^2}}$$

$$\therefore C_c = \frac{g_m \beta}{\omega_t \sqrt{1 + 0.7^2}} = \frac{0.775 \times 10^{-3} \times \frac{1}{6}}{2\pi \times 42 \times 10^6 \times 1.221}$$

$$C_c = \underline{0.4 \text{ pF}}$$

5.11) Find C_2 that provides compensation.



In Problem 5.10, we found that

$$\beta = \frac{R_1}{R_1 + R_2}.$$

This can be generalized to $\beta = \frac{Z_1}{Z_1 + Z_2}$. We now wish to use the feedback network for compensation.

For $Z_1 = R_1$, $Z_2 = R_2 \parallel \frac{1}{sC_2} = \frac{R_2}{1 + sR_2C_2}$

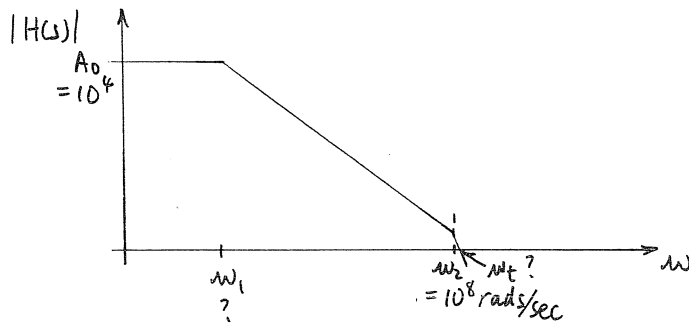
$$\beta = \frac{R_1}{R_1 + \frac{R_2}{1 + sR_2C_2}} = \frac{R_1(1 + sR_2C_2)}{sR_1R_2C_2 + R_1 + R_2} \quad \leftarrow \text{new zero added}$$

From Section 5.2, lead compensation is achieved by placing a zero at 1.2 times the unity loop gain frequency, ω_t .

$$\therefore \omega_z = \frac{1}{R_2C_2} \equiv 1.2 \times \omega_t$$

$$\therefore C_2 = \frac{1}{1.2 \times 0.7 \omega_{p2} \times R_2} = \underline{\underline{63 \text{ fF}}}$$

5.12) Find ω_1 and ω_t .



$$\therefore \omega_z = \omega_t$$

$$\therefore H(s) = \frac{A_0 (1 + s/\omega_t)}{(1 + s/\omega_1)(1 + s/\omega_2)}$$

$$\text{and } \angle H(j\omega_t) = 180^\circ - \text{PM} = -100^\circ = +45^\circ - \tan^{-1}\left(\frac{\omega_t}{\omega_1}\right) - \tan^{-1}\left(\frac{\omega_t}{\omega_2}\right)$$

from zero

$$\therefore A_0 \gg 1$$

$$\therefore \omega_t \gg \omega_1 \text{ by the constant gain bandwidth}$$

$$\therefore \tan^{-1}\left(\frac{\omega_t}{\omega_1}\right) \approx 90^\circ \text{ product}$$

$$\therefore -100^\circ \approx +45^\circ - 90^\circ - \tan^{-1}\left(\frac{\omega_t}{\omega_2}\right)$$

$$\frac{\omega_t}{\omega_2} \approx \tan(55^\circ)$$

$$\therefore \underline{\omega_t \approx 1.43 \omega_2 = 1.43 \times 10^8 \text{ rad/sec}}$$

$$|H(j\omega_t)| = A_0 \sqrt{2} \times \frac{1}{\sqrt{1 + \left(\frac{\omega_t}{\omega_1}\right)^2} \times \sqrt{1 + \left(\frac{\omega_t}{\omega_2}\right)^2}} \equiv 1$$

$$\left[1 + \left(\frac{\omega_t}{\omega_1}\right)^2\right] \left[1 + \left(\frac{\omega_t}{\omega_2}\right)^2\right] = 2 A_0^2$$

$$1 + \left(\frac{\omega_t}{\omega_1}\right)^2 = \frac{2 \times 10^8}{1 + 1.43^2}$$

$$\omega_1 = \frac{1.43 \times 10^8 \text{ rad/sec}}{8.1 \times 10^3}$$

$$\underline{\omega_1 = 1.8 \times 10^4 \text{ rad/sec}}$$

5.13) Given $A(s) \approx \frac{A_0(1+sT_z)}{sT_1(1+sT_2)}$, the closed loop transfer function is given by

$$A_{CL}(s) = \frac{A(s)}{1+A(s)\beta} \approx \frac{A_0(1+sT_z)/(sT_1(1+sT_2))}{1+A_0\beta(1+sT_z)/(sT_1(1+sT_2))}$$

$$= \frac{A_0(1+sT_z)}{sT_1(1+sT_2)+A_0\beta(1+sT_z)}$$

$$\therefore A_{CL}(s) = \frac{A_0(1+sT_z)}{T_1T_2[s^2+s(\frac{1}{T_2}+A_0\beta\frac{T_z}{T_1T_2})+\frac{A_0\beta}{T_1T_2}]}$$

Equating the coefficients of the denominator polynomial to the standard second-order pole polynomial,

$$s^2 + \frac{\omega_0}{Q}s + \omega_0^2,$$

we find that

$$\frac{\omega_0}{Q} = \frac{1}{T_2} + A_0\beta\frac{T_z}{T_1T_2}, \quad \omega_0^2 = \frac{A_0\beta}{T_1T_2}$$

$$\therefore \omega_0 = \sqrt{\frac{A_0\beta}{T_1T_2}}$$

$$\therefore Q = \frac{\omega_0}{\frac{1}{T_2} + A_0\beta\frac{T_z}{T_1T_2}}$$

$$= \frac{\omega_0 T_2}{1 + A_0\beta T_z/T_1}$$

Since T_z is roughly T_1/A_0 we cannot make any approximations here.

5.14) Given $R_c = 0$, Equations (5.66) ~ (5.68) describe the denominator polynomial, $D(s)$, as

$$D(s) = 1 + sa + s^2b \quad \text{where}$$

$$a = (C_2 + C_c)R_2 + (C_1 + C_c)R_1 + g_{m7}R_1R_2C_c$$

$$b = R_1R_2(C_1C_2 + C_1C_c + C_2C_c)$$

When $R_c \neq 0$, the admittance sC_c becomes $\frac{sC_c}{1+sR_cC_c}$. Thus, we can obtain the new denominator polynomial, $D'(s)$, by simply substituting C_c with $\frac{C_c}{1+sR_cC_c}$.

$$\begin{aligned} \therefore a' &= (C_2 + \frac{C_c}{1+sR_cC_c})R_2 + (C_1 + \frac{C_c}{1+sR_cC_c})R_1 + g_{m7}R_1R_2\frac{C_c}{1+sR_cC_c} \\ &= \frac{1}{1+sR_cC_c} [(C_2(1+sR_cC_c) + C_c)R_2 + (C_1(1+sR_cC_c) + C_c)R_1 \\ &\quad + g_{m7}R_1R_2C_c] \end{aligned}$$

$$a' = \frac{1}{1+sR_cC_c} [a + s(R_2C_2R_cC_c + R_1C_1R_cC_c)]$$

$$b' = R_1R_2 [C_1C_2 + C_1\frac{C_c}{1+sR_cC_c} + C_2\frac{C_c}{1+sR_cC_c}]$$

$$b' = \frac{1}{1+sR_cC_c} [b + sR_1C_1R_2C_2R_cC_c]$$

$$\therefore D'(s) = 1 + sa' + s^2b' = \underbrace{\left(\frac{1}{1+sR_cC_c} \right)}_{\text{this term is moved to the numerator}} \underbrace{[(1+sR_cC_c) + s(a + s(R_2C_2R_cC_c + R_1C_1R_cC_c)) + s^2(b + sR_1C_1R_2C_2R_cC_c)]}_{\triangleq D''(s)}$$

The poles of the system are determined by the roots of $D''(s)$.

$$D''(s) = 1 + s(a + R_cC_c) + s^2(b + R_2C_2R_cC_c + R_1C_1R_cC_c) + s^3R_1C_1R_2C_2R_cC_c$$

$$\therefore R_c \ll R_1 \text{ or } R_2$$

$\therefore$ any time constant with R_c is much less than time constants with R_1 or R_2

$$\therefore a \gg R_cC_c \text{ and } b \gg R_2C_2R_cC_c + R_1C_1R_cC_c$$

(cont.)

5.14 (cont.)

$$\therefore D''(s) \approx 1 + sa + s^2b + s^3 R_1 C_1 R_2 C_2 R_C C_C$$

$\therefore$ system poles are spaced far apart with $\omega_{p1} \ll \omega_{p2} \ll \omega_{p3}$

$$\therefore (1 + \frac{s}{\omega_{p1}})(1 + \frac{s}{\omega_{p2}})(1 + \frac{s}{\omega_{p3}}) \approx 1 + \frac{s}{\omega_{p1}} + \frac{s^2}{\omega_{p1}\omega_{p2}} + \frac{s^3}{\omega_{p1}\omega_{p2}\omega_{p3}}$$

Equating these coefficients, we have the original

$$\omega_{p1} \approx 1/a$$

Approximations,

$$\omega_{p2} \approx a/b$$

$\therefore$ Equations (5.70) and (5.71) still hold true.

Q.E.D.

5.15) Find R_B and V_{eff} at 70°C .

From Eq. (5.108)

$$R_B = 1/g_{m13} = 1/(\mu_n C_{ox} W/L V_{eff}) = 1/(92 \times 10^{-6} \times 10/1.2 \times 0.25)$$

$$\underline{R_B = 5.2 \text{ k}\Omega}$$

To determine the effect of temperature on V_{eff} , note that

$$\mu_n \propto T^{-3/2} \text{ (see pg 250)}$$

$$\text{Also } R_B \propto \frac{1}{\mu_n V_{eff}} \propto \frac{1}{T^{-3/2} V_{eff}} \text{ assuming } R_B \text{ is a constant}$$

$$\therefore T_1^{-3/2} V_{eff1} = T_2^{-3/2} V_{eff2}$$

$$\text{Let } T_1 = 300\text{K} \text{ (27}^\circ\text{C)}, T_2 = 343\text{K} \text{ (70}^\circ\text{C)}, V_{eff1} = 0.25\text{V}$$

$$\therefore V_{eff2} = 0.25 \times \left(\frac{300}{343}\right)^{-3/2} = \underline{0.31\text{V}}$$

Results from HSPICE:

$$\text{at } 27^\circ\text{C: } V_{eff} = V_{osat} = \underline{0.26\text{V}}$$

$$\text{at } 70^\circ\text{C: } V_{eff} = V_{osat} = \underline{0.31\text{V}}$$

$\therefore$ Results are consistent with our calculations.

Chapter 6

6.10 PROBLEMS

Unless otherwise stated, assume the following:

- n-channel MOS transistors:
 $\mu_n C_{ox} = 92 \mu\text{A}/\text{V}^2$
 $V_{tn} = 0.8 \text{ V}$
 $\gamma = 0.5 \text{ V}^{1/2}$
 $r_{ds}(\Omega) = 8,000L(\mu\text{m})/I_D(\text{mA})$ in active region
 $C_j = 2.4 \times 10^{-4} \text{ pF}/(\mu\text{m})^2$
 $C_{j-sw} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$
 $C_{ox} = 1.9 \times 10^{-3} \text{ pF}/(\mu\text{m})^2$
 $C_{gs(\text{overlap})} = C_{gd(\text{overlap})} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$
- p-channel MOS transistors:
 $\mu_p C_{ox} = 30 \mu\text{A}/\text{V}^2$
 $V_{tp} = -0.9 \text{ V}$
 $\gamma = 0.8 \text{ V}^{1/2}$
 $r_{ds}(\Omega) = 12,000L(\mu\text{m})/I_D(\text{mA})$ in active region
 $C_j = 4.5 \times 10^{-4} \text{ pF}/(\mu\text{m})^2$
 $C_{j-sw} = 2.5 \times 10^{-4} \text{ pF}/\mu\text{m}$
 $C_{ox} = 1.9 \times 10^{-3} \text{ pF}/(\mu\text{m})^2$
 $C_{gs(\text{overlap})} = C_{gd(\text{overlap})} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$

- 6.1 Calculate the output impedance of the two-transistor diode-connected circuit shown in Fig. P6.1 using small-signal analysis. Assume both transistors are in the active region, ignore the body effect, and assume $g_{m1} = g_{m2}$, $r_{ds1} = r_{ds2}$, and $g_m r_{ds} \gg 1$.
- 6.2 Assume n is chosen to be 1 in Fig. 6.1 and all transistors are taken to be equal sizes, except $(W/L)_5$ is chosen so that $V_{DS3} = V_{DS2} = V_{eff3} + 0.15 \text{ V}$, where V_{eff} is chosen to be 0.2 V for all transistors except Q_5 . Ignoring the body effect, find the required size for all transistors assuming they all have lengths equal to 1.6 μm . Assume that $I_{bias} = 50 \mu\text{A}$.
- 6.3 What are the drain-source voltages of Q_3 and Q_4 from Problem 6.2 if the lengths of Q_3 and Q_4 are decreased to 1 μm to maximize speed?

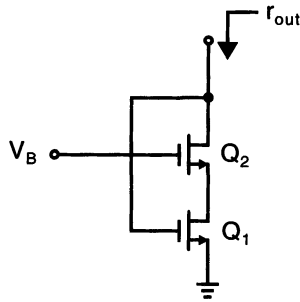


Fig. P6.1

- 6.4 What is the value required for R_B in Fig. 6.2 to give $V_{\text{eff}3} = 0.2 \text{ V}$?
- 6.5 Assuming that μ_n is proportional to $T^{-3/2}$, what would $V_{\text{eff}3}$ be at 100°C if it was 0.2 V at 20°C ?
- 6.6 Assume that R_B from Fig. 6.2 has a temperature dependence of $+0.3\%/^\circ\text{C}$. What would $V_{\text{eff}3}$ be at 100°C if it was 0.2 V at 20°C ? Assume μ_n varies as in Problem 6.5.
- 6.7 Using small-signal analysis and ignoring the body effect, show that the circuit of Fig. 6.3 has an output impedance given by $r_{\text{out}} \equiv g_{m1}r_{ds1}r_{ds2}(1 + A)$.
- 6.8 For the circuit shown in Fig. P6.8, using small-signal analysis, find the output impedance. What is an upper limit on the output impedance assuming $R_E \gg r_\pi$? Compare this to the upper limit on the output impedance of a simple cascode mirror (i.e., $A = 0$) for the same case.

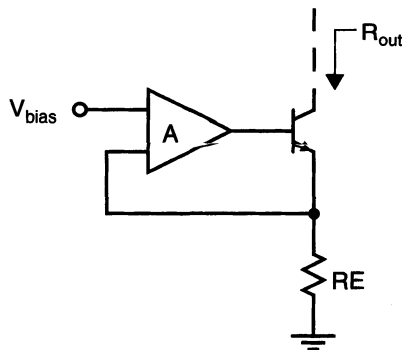


Fig. P6.8

- 6.9 For the circuit shown in Fig. 6.6 with the W/L ratios shown and assuming all lengths are $1.6 \mu\text{m}$, estimate the output impedance. Assume that all current sources are ideal and that $I_{\text{in}} = 7I_{\text{bias}}$. Compare this to the output impedance of a wide-swing cascode current mirror where the gate of Q_1 was connected to $V_{\text{casc-n}}$ from the bias-generation circuitry of Fig. 6.2. Assume $I_{\text{bias}} = 50 \mu\text{A}$.
- 6.10 For the folded-cascode amplifier shown in Fig. 6.9, with the transistor sizes given in Table 6.1, find the unity-gain frequency and the slew rate. Assume

± 2 V power supplies are used, that C_L is 10 pF, and that the power dissipation is 1 mW.

- 6.11 For the folded-cascode amplifier shown in Fig. 6.9, with the transistor sizes given in Table 6.1, estimate the approximate frequency of the second pole caused by the parasitic capacitances at the drains of Q_1 and Q_2 . Assume ± 2 V power supplies are used, that C_L is 10 pF, and that the power dissipation is 1 mW. To simplify matters, junction capacitances can be ignored. What should C_L be to achieve 70° phase margin? What would the corresponding unity-gain frequency and slew rate be?
- 6.12 For the amplifier of Problem 6.11 with lead compensation, find the load capacitance, C_L , for a 70° phase margin. What size resistor should be used? What is the final unity-gain frequency and slew rate? Assume lead compensation with $\omega_z = 1.2\omega_1$ improves the initial unity-gain frequency by 20 percent and the phase margin by 30 degrees.
- 6.13 For the folded-cascode amplifier shown in Fig. 6.9, assume the bias currents of Q_1 and Q_2 are K times greater than the bias currents in Q_5 and Q_6 . Assume the total bias current in the opamp, I_{total} , is determined by a specified power dissipation. Derive an equation for the unity-gain frequency in terms of K and I_{total} . Then show that the unity-gain frequency is maximized by taking K large.
- 6.14 Repeat Problem 6.13, but this time show that the dc gain is optimized by taking K large.
- 6.15 Derive an equation for the ratio of the unity-gain frequency of the folded-cascode amplifier of Fig. 6.9 to the unity-gain frequency of the current-mirror opamp of Fig. 6.11 in terms of K and I_{total} assuming both amplifiers have the same size input transistors, total power dissipation, and load capacitances. For the folded-cascode opamp, define K to be I_{D2}/I_{D7} . What is this ratio for K equal to 1, 2, and 4?
- 6.16 Consider the current-feedback opamp described in Example 6.13, in a feedback configuration shown in Fig. 6.12, with $C_1 = 1$ pF and $C_2 = C_C = C_{\text{load}} = 5$ pF. What is the linear settling time for 1 percent accuracy?
- 6.17 For the case described in Example 6.4, and assuming a current-mirror opamp described in Example 6.3, also assume the input voltage is a 1-V step change. Would the opamp slew rate limit? If so, how close must the output voltage come to its final value before its output voltage rate of change is less than the slew rate? How long would this take? How much longer would be required for linear settling to 1 percent of the total voltage change at the output?
- 6.18 What are the single-ended slew rates of the fully differential folded-cascode and current-mirror opamps, assuming $K = 2$ and the current densities are the same as those of Example 6.2 and Example 6.3, respectively? Calculate the slew rates in both the positive and the negative directions. Assume the load capacitances are 10 pF between each output and ground. For the folded-

cascode design, assume the clamp transistors Q_{11} and Q_{12} of Fig. 6.13 have not been included. Also, assume the current sources biasing the output stages do not change during transients.

- 6.19** Derive the slew rate of the fully differential opamp of Fig. 6.15 in terms of C_L , K , and I_{bias} .
- 6.20** Derive an equation for the unity-gain frequency of the fully differential opamp shown in Fig. 6.16 in terms of K and I_{total} .
- 6.21** Assume the CMFB circuit of Fig. 6.19 has ± 2.5 V power supply voltages, and that the current sources require 0.5 V across them in order to have all transistors remain in the active regions. Ignoring the body effect, what V_{eff} bias voltage should be used for the p-channel transistors to maximize signal swing? What is the maximum single-ended signal swing before the gain of the common-mode feedback circuitry goes to zero and why?
- 6.22** Repeat Problem 6.21, but do not ignore the body effect. Assume the n-wells of the p-channel transistors are connected to V_{DD} .
- 6.23** Derive the low-frequency gain of the current-feedback amplifier of Fig. 6.23 without assuming that the impedance looking into the emitters of Q_1 and Q_2 is zero.
- 6.24** Consider the circuit shown in Fig. P6.24, where a high-output-impedance opamp is used. Capacitor C_p is the parasitic capacitance at the opamp input, while C_o is the output capacitance. Assuming linear settling, show that the optimal value for minimizing the time constant of this circuit is given by

$$C_{1,opt} = \sqrt{MC_oC_p}$$

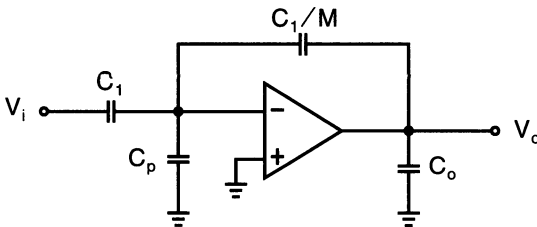


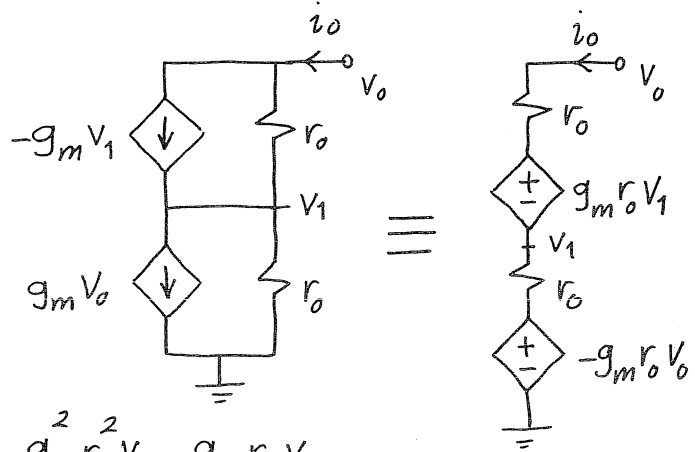
Fig. P6.24

- 6.25** Based on the results of Problem 6.24, derive τ_{min} for $M = 1$, $C_o = 1$ pF, and $C_p = 0.05$ pF. Sketch τ for other values of C_1 between 0.1 pF and 1 pF.

Chapter 6 – Problems

6.1) KVL:

$$\begin{cases} V_o = r_o i_o + g_m r_o V_1 + i_o r_o + (-g_m r_o V_o) \\ V_1 = r_o i_o - g_m r_o V_o \end{cases}$$



$$\therefore V_o = 2r_o i_o + g_m r_o^2 i_o - g_m^2 r_o^2 V_o - g_m r_o V_o$$

$$\Rightarrow r_{out} = \frac{V_o}{i_o} = \frac{r_o (2 + g_m r_o)}{g_m^2 r_o^2 + g_m r_o + 1}$$

Assuming $g_m r_o \gg 1 \Rightarrow g_m^2 r_o^2 \gg g_m r_o$

$$\Rightarrow r_{out} \approx \frac{r_o (g_m r_o)}{g_m^2 r_o^2} \approx \frac{1}{g_m}$$

6.2) For transistors Q_1 to Q_4 :

$$50 \mu A = \frac{\mu_n C_{ox}}{2} \left(\frac{W}{L} \right)_i (0.2)^2 \quad \text{for } i=1 \text{ to } 4$$

$$\Rightarrow \left(\frac{W}{L} \right)_i = 27.2 \quad \text{for } i=1 \text{ to } 4$$

Also, $V_{GS5} = V_{GS4} + V_{DS3} = V_{eff4} + V_{tn} + V_{eff3} + 0.15 = 1.35 V$

$$\Rightarrow V_{eff5} = V_{GS5} - V_{tn} = 0.55 V$$

Using $I_{BIAS} = 50 \mu A \Rightarrow$

$$50 \mu A = \frac{\mu_n C_{ox}}{2} \left(\frac{W}{L} \right)_5 (0.55)^2 \Rightarrow \left(\frac{W}{L} \right)_5 = 3.6$$

6.3) Reducing L by 1.6 will decrease V_{eff} by a factor of $\sqrt{1.6}$. Therefore,

$$\begin{aligned} V_{DS3} &= V_{GS5} - V_{GS4} = 1.35 - (V_{tn} + V_{eff}) \\ &= 1.35 - (0.8 + 0.16) = \underline{0.39 \text{ V}} \end{aligned}$$

$$\begin{aligned} V_{DS4} &= V_{D4} - V_{DS3} = V_{GS3} - V_{DS3} = (V_{tn} + V_{eff}) - 0.39 \\ &= (0.8 + 0.16) - 0.39 = \underline{0.57 \text{ V}} \end{aligned}$$

$$6.4) \quad I_{D3} = I_{D2} \Rightarrow \left(\frac{W}{L}\right)_2 V_{eff2}^2 = \left(\frac{W}{L}\right)_3 V_{eff3}^2 \Rightarrow \underline{V_{eff3} = 2 V_{eff2}}$$

$$\text{Also, } V_{GS3} = V_{GS2} + R_B I$$

$$\Rightarrow V_{eff3} = \frac{V_{eff3}}{2} + R_B \frac{\mu_n C_{ox}}{2} \left(\frac{W}{L}\right)_3 V_{eff3}^2$$

$$\Rightarrow \underline{10 \mu_n C_{ox} R_B V_{eff3} = 1}$$

$$\text{Using } V_{eff3} = 0.2 \Rightarrow \underline{R_B = 5.43 \text{ K}\Omega}$$

6.5) Using the result of Problem 6.4, we have:

$$10 \mu_n C_{ox} R_B V_{eff3} = 1$$

$$\text{Also, } \frac{\mu_n(100^\circ\text{C})}{\mu_n(20^\circ\text{C})} = \left(\frac{373}{293}\right)^{-1.5} = \underline{0.7}$$

Therefore, V_{eff3} will be increased by $\frac{1}{0.7}$.

$$\text{Equivalently, } V_{eff3} = \frac{1}{0.7} \times 0.2 = \underline{0.29 \text{ V}}$$

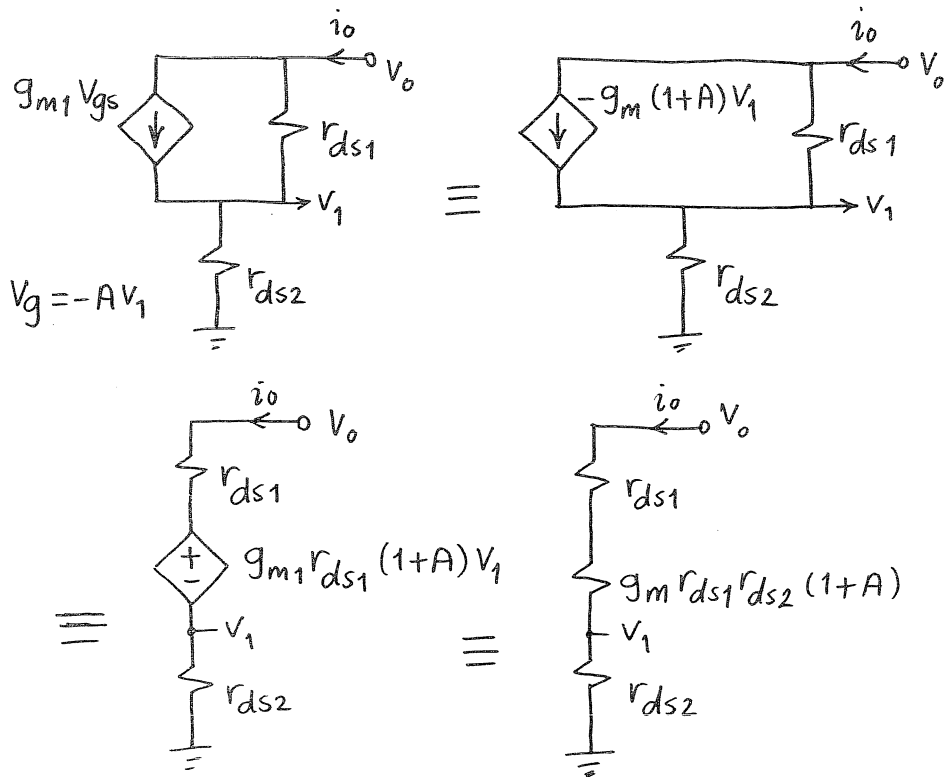
$$6.6) \quad R_B(100^\circ\text{C}) = (1 + 80 \times \frac{0.3}{100}) R_B(20^\circ\text{C}) = \underline{6.73 \text{ K}\Omega}$$

Using the result of Problem 6.5, $\mu_n R_B$ is decreased at 100°C by a factor of $0.7 \times 1.24 = 0.868$.

Therefore, $V_{\text{eff}3}$ will increase by $\frac{1}{0.868}$.

$$\therefore V_{\text{eff}3} = \frac{1}{0.868} \times 0.2 = \underline{0.23 \text{ V}}$$

6.7) The equivalent circuit of Fig. 6.3 can be simplified:

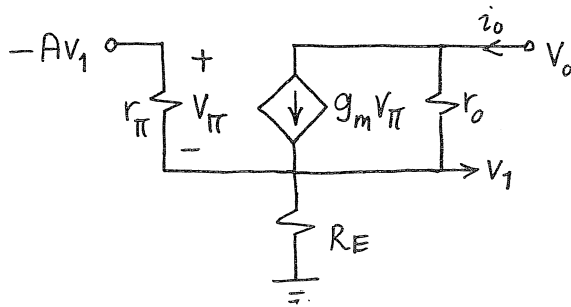


$$\Rightarrow r_o = \frac{V_o}{i_o} = r_{ds1} + r_{ds2} + (1+A)g_{m1}r_{ds1}r_{ds2}$$

ignoring the first two terms:

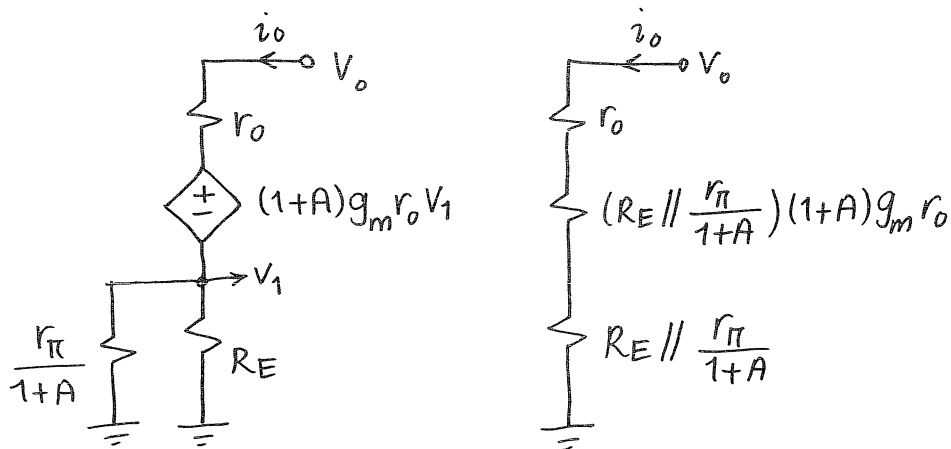
$$\underline{r_o \approx (1+A)g_{m1}r_{ds1}r_{ds2}}$$

6.8) The equivalent circuit of Fig. P6.8 can be simplified as:



Noting that $-AV_1 = V_\pi + V_1 \Rightarrow V_\pi = -(1+A)V_1$

The circuit can be simplified further as :



$$\Rightarrow R_{out} = \frac{V_o}{i_o} = \underline{r_o + (R_E \parallel \frac{r_\pi}{1+A}) (1 + (1+A)g_m r_o)}$$

$$\text{Assuming } R_E \gg r_\pi \Rightarrow R_E \parallel \frac{r_\pi}{1+A} \approx \frac{r_\pi}{1+A}$$

$$\Rightarrow R_{out} \approx r_o + \frac{r_\pi}{1+A} (1+A)g_m r_o = r_o (1 + r_\pi g_m) = \underline{r_o (1 + \beta)}$$

This result is independent of A !

6.9) Using (6.10): $R_{out} = g_{m1} r_{ds1} r_{ds2} (1 + A)$

where $A = g_{m3} r_{ds3}$ for the circuit of Fig. 6.6.

∴ $R_{out} = g_{m1} r_{ds1} r_{ds2} (1 + g_{m3} r_{ds3})$

$I_{BIAS} = 50 \mu A \Rightarrow I_{D1} = I_{D2} = 350 \mu A \text{ \& } I_{D3} = 200 \mu A$

$\Rightarrow r_{ds1} = r_{ds2} = \frac{8K \times 1.6}{0.35} = \underline{36.57 K\Omega}$

$r_{ds3} = \frac{8K \times 1.6}{0.2} = \underline{64 K\Omega}$

$g_{m1} = \sqrt{2 I_{D1} \mu_n C_{ox} \frac{70}{1.6}} = 1.68 \text{ mA/V}$

$g_{m3} = \sqrt{2 I_{D3} \mu_n C_{ox} \frac{10}{1.6}} = 0.48 \text{ mA/V}$

$\Rightarrow R_{out} = 2.27 M\Omega (1 + 30.72) = \underline{7.13 \times 10^7 \Omega}$
enhanced output

the impedance is 32 times larger than that of a wide-swing cascode current mirror given by

$R'_{out} = g_{m1} r_{ds1} r_{ds2} = \underline{2.25 M\Omega}$

6.10) $2 I_{D3} = \frac{1mW}{4V} = 250 \mu A \Rightarrow I_{D3} = I_{D4} = 125 \mu A$

Also, $5 I_{D5} = 125 \mu A \Rightarrow I_{D5} = I_{D6} = 25 \mu A$
& $I_{D1} = I_{D2} = 100 \mu A$

$g_{m1} = \sqrt{2 I_{D1} \mu_n C_{ox} (W/L)_1} = 1.9 \text{ mA/V}$

$\omega_t = g_{m1} / C_L = \frac{1.9 \text{ mA/V}}{10 \text{ pF}} = 1.9 \times 10^8 \text{ rad/s} \Rightarrow \underline{f_t = 30.2 \text{ MHz}}$
(cont.)

6.10) (cont.) the slew rate without the clamp transistors:

$$SR = \frac{I_{D4}}{C_L} = \underline{12.5 \text{ V}/\mu\text{s}}$$

With the clamp transistors:

$$I_{D11} = \frac{I_{bias2} + \frac{125\mu}{30}}{31} = \frac{200\mu + \frac{125\mu}{30}}{31} = 6.6 \mu\text{A}$$

$$\Rightarrow I_{D3} = 30 I_{D11} = 198 \mu\text{A}$$

$$\Rightarrow SR = \frac{198 \mu\text{A}}{10 \text{ pF}} = \underline{19.8 \text{ V}/\mu\text{s}}$$

6.11) Ignoring the junction capacitances, the total capacitance

at the drain of Q_2 can be calculated as:
(also ignore Q_{13} since it is small)

$$\begin{aligned} C_{p2} &= C_{dQ2} + C_{dQ4} + C_{sQ5} = C_{gd(\text{overlap})} (W_2 + W_4 + W_5) + \frac{2}{3} W_5 L C_{ox} \\ &= 0.2 \text{ fF}/\mu\text{m} (300 + 300 + 60) + \frac{2}{3} \times 60 \times 1.6 \times 1.9 \text{ fF} = \underline{254 \text{ fF}} \end{aligned}$$

the total conductance at this node is dominated by

g_{m5} . Using the result of problem 6.10, we have:

$$g_{m5} = \sqrt{2 I_{D5} \mu_p C_{ox} (W/L)_5} = 0.237 \text{ mS}$$

The second pole (half-circuit concept):

$$\omega_2 = \frac{g_{m5}}{C_{p2}} = 9.33 \times 10^8 \text{ rad/s}$$

$$\Rightarrow f_2 = \frac{\omega_2}{2\pi} = \underline{148.5 \text{ MHz}}$$

(cont.)

6.11) (cont.) Using (5.52), for a 70° phase margin, we must

have: $\frac{f'_t}{f_2} = \tan 20^\circ \Rightarrow \underline{f'_t = 54.05 \text{ MHz}}$

Using (6.30): $C'_L = \frac{g_{m1}}{\omega'_t} = \frac{1.9 \text{ m}}{2\pi \times f'_t} = \underline{5.59 \text{ pF}}$

Finally, using (16.32):

$$SR' = \frac{I_{D4}}{C'_L} = \frac{125 \mu}{5.59 \text{ p}} = \underline{22.3 \text{ V}/\mu\text{s}}$$

With the clamp transistors: $SR' = \frac{198 \mu}{5.59 \text{ p}} = \underline{35.4 \text{ V}/\mu\text{s}}$

6.12) This is equivalent to a 40° phase margin in the original design. Using (5.52), we have:

$$\frac{f'_t}{f_{p2}} = \tan 50^\circ \Rightarrow f'_t = 1.19 f_2 \approx \underline{176.7 \text{ MHz}}$$

$$f_z = 1.2 f'_t = \underline{212 \text{ MHz}}$$

The final unity-gain frequency: $f_t = 1.2 f'_t = \underline{212 \text{ MHz}}$

$$\Rightarrow C_L = \frac{g_{m1}}{\omega_t} = \underline{1.43 \text{ pF}}$$

$$\omega_z = \frac{1}{R_c C_L} \Rightarrow R_c = \frac{1}{\omega_z C_L} = \underline{525 \Omega}, \quad SR = \frac{I_{D4}}{C_L} = \underline{87 \text{ V}/\mu\text{s}}$$

With the clamp xtors: $SR = \frac{198 \mu}{1.43 \text{ p}} = \underline{138 \text{ V}/\mu\text{s}}$

$$6.13) \quad I_{D1} = I_{D2} = K I_{D5} = K I_{D6}$$

$$I_{\text{total}} = 2(I_{D1} + I_{D5}) = 2(1+K)I_{D5}$$

$$\Rightarrow I_{D5} = \frac{I_{\text{total}}}{2(K+1)}, \quad I_{D1} = \frac{K}{K+1} \cdot \frac{I_{\text{total}}}{2}$$

$$g_{m1} = \frac{2I_{D1}}{V_{\text{eff}1}} = \frac{K}{K+1} \frac{I_{\text{total}}}{V_{\text{eff}1}}, \quad \omega_t = \frac{K}{K+1} \frac{I_{\text{total}}}{V_{\text{eff}1} \cdot C_L}$$

Assuming a constant I_{total} & $V_{\text{eff}1}$, both g_{m1} & ω_t increase with increasing K .

$$6.14) \quad \text{The dc gain is given by } A_{V_o} = g_{m1} r_{\text{out}}$$

$$\text{where, approximately: } r_{\text{out}} \approx g_{m8} \frac{r_{ds8}^2}{2}$$

$$\text{and } g_{m8} = \frac{2I_{D8}}{V_{\text{eff}8}}, \quad r_{ds8} = \frac{\alpha L}{I_{D8}}, \quad \alpha \text{ is constant!}$$

$$\Rightarrow r_{\text{out}} \approx \frac{\alpha^2 L^2}{V_{\text{eff}8} I_{D8}}$$

$$\Rightarrow A_{V_o} = \frac{2I_{D1}}{V_{\text{eff}1}} \frac{\alpha^2 L^2}{V_{\text{eff}8} I_{D8}} = \left(\frac{I_{D1}}{I_{D8}} \right) \left(\frac{2 \alpha^2 L^2}{V_{\text{eff}1} V_{\text{eff}8}} \right)$$

Noting $I_{D8} = I_{D5}$ and $I_{D1} = K I_{D5}$ results in:

$$A_{V_o} = K \left(\frac{2 \alpha^2 L^2}{V_{\text{eff}1} V_{\text{eff}8}} \right)$$

A_{V_o} increases with K !

6.15) For the folded-cascode amplifier of Fig. 6.9, we have:

$$\omega_t = \frac{g_{m1}}{C_L} = \frac{\sqrt{2I_{D1}\mu_n C_{ox}(W/L)_1}}{C_L}$$

$$\text{Also, } 2I_{D1} + \frac{2}{K} I_{D1} = I_{\text{total}} \Rightarrow I_{D1} = \frac{K}{2(K+1)} I_{\text{total}}$$

$$\Rightarrow \omega_t (\text{folded-cascode}) = \frac{1}{C_L} \sqrt{\frac{\mu_n C_{ox} I_{\text{total}} (W/L)_1 K}{K+1}}$$

For the current-mirror opamp, ω_t is given by (6.48).

Therefore,

$$\frac{\omega_t (\text{folded-cas.})}{\omega_t (\text{current-mir.})} = \sqrt{\frac{K+3}{2K(K+1)}}$$

K	1	2	4
ω_t Ratio	1	0.645	0.418

6.16) From example 6.4 : $C_p = 0.46 \text{ pF}$

$$C_L = 5 + 5 + \frac{5(1+0.46)}{1+0.46+5} = 11.13 \text{ pF}$$

$$\omega_t = \frac{K g_{m1}}{C_L} = \frac{2 \times 1.7 \text{ mA/V}}{11.13 \text{ pF}} = 3.05 \times 10^8 \text{ rad/s or } f_t = 48.6 \text{ MHz}$$

$$\beta = \frac{C_2}{C_1 + C_p + C_2} = \frac{5}{1+0.46+5} = 0.77$$

$$\Rightarrow \tau = \frac{1}{\beta \omega_t} = 4.24 \text{ nsec}$$

For a 1% accuracy, we need 4.6τ or 19.5 nsec.

6.17) Using (6.53) & (6.68): $SR = 25.4 \text{ V}/\mu\text{s}$.

Assuming a high-gain opamp:

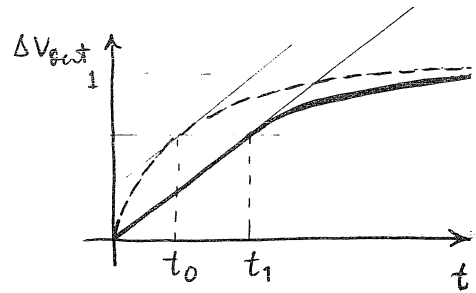
$$A_V = \frac{\Delta V_o}{\Delta V_{in}} = \frac{-C_1}{C_2} = \frac{-5 \text{ pF}}{5 \text{ pF}} = -1 \Rightarrow \Delta V_o = -1 \text{ V}.$$

$\Rightarrow$ the output voltage rate of change would be

$$\left. \frac{dV_o}{dt} \right|_{\max} = \frac{\Delta V_o}{\tau} = \frac{1}{7.8 \text{ ns}} = 128 \text{ V}/\mu\text{s} > 25.4 \text{ V}/\mu\text{s}$$

Therefore, the output will be limited by the slew rate!

The output voltage will ramp up with $25.4 \text{ V}/\mu\text{s}$ until the exponential-curve derivative is equal to the slew rate:



At time t_0 , we have: $\frac{1}{\tau} e^{-t_0/\tau} = 25.4 \text{ V}/\mu\text{s}$

$$\Rightarrow \underline{t_0 = 12.63 \text{ ns}} \quad \& \quad (1 - e^{-t_0/\tau}) = 0.8 \text{ V}$$

$\Rightarrow$ at time $t = t_1$, the output has reached to 80% of its final value. ($t_1 = 31.5 \text{ ns}$)

$$\text{For } t \geq t_1 : \Delta V_o(t) = 0.8 + 0.2(-e^{-(t-t_1)/\tau} + 1)$$

$$\text{For } \Delta V_o(t) = 0.99 \Rightarrow \underline{t - t_1 = 23.37 \text{ ns}}$$

This is the time required after t_1 for the output to settle to 1% of its final value.

6.18) Fully dif. folded-cas.:

$$\text{positive SR} = \frac{I_3 - I_q}{C_L} = \frac{(K+1)I_q - I_q}{C_L} = \frac{K I_q}{C_L}$$

$$\text{negative SR} = \frac{I_q}{C_L}$$

∴ for $K=2$ and $I_q = 40 \mu A$:

$$\underline{\text{positive SR} = 8 \text{ V}/\mu\text{s}} \quad \underline{\text{negative SR} = 4 \text{ V}/\mu\text{s}}$$

Fully dif. current-mir.:

$$\text{positive SR} = \text{negative SR} = \frac{K I_{BIAS}}{2 C_L}$$

∴ for $K=2$ & $I_{BIAS} = 160 \mu A$, we have:

$$\underline{\text{positive SR} = \text{negative SR} = 16 \text{ V}/\mu\text{s}}$$

$$6.19) \quad \left. \frac{dV_{out+}}{dt} \right|_{\max} = \left. \frac{dV_{out-}}{dt} \right|_{\max} = \underline{\frac{K I_{BIAS}}{C_L}}$$

Note that both maximums occur simultaneously.

$$6.20) \quad \omega_t = \frac{K g_{m4}}{C_L} \quad \text{where} \quad g_{m4} = \sqrt{2 I_{D4} \mu_n C_{ox} (W/L)_4}$$

$$\text{Also, } I_{\text{total}} = (2+K) I_{D4} \Rightarrow I_{D4} = \frac{I_{\text{total}}}{2+K}$$

$$\underline{\omega_t = \frac{K}{C_L} \sqrt{\frac{2}{2+K} I_{\text{total}} \mu_n C_{ox} (W/L)_4}}$$

6.21) The maximum value of V_{eff} without the I_B -transistor going into the triode region is $2 - |V_{TP}| = 1.1 \text{ V}$. At this voltage, all the bias current will flow through either Q_1 or Q_2 .

Therefore, $I = K(1.1)^2$

For the bias condition ($V_{out+} = 0$), we have:

$$I/2 = K V_{eff}^2 \Rightarrow \underline{V_{eff} = \frac{1.1}{\sqrt{2}} = 0.78 \text{ V}}$$

Transistor Q_1 will shut off when $V_{S_{G1}} = |V_{TP}| = 0.9 \text{ V}$. At this point, $V_{S1} = 2 \text{ V}$ and I_B flows through Q_2 .

$$\therefore \underline{V_{out+}|_{\max} = 2 - 0.9 = 1.1 \text{ V}}$$

When V_{out+} goes below zero, V_{S1} starts to fall off from its bias value (i.e. 1.68 V) until it reaches 0.9 V . At this point, Q_2 shuts off and I_B will pass through Q_1 .

$$\therefore \underline{V_{out+}|_{\min} = 0.9 - 2 = -1.1 \text{ V}}$$

Note that a V_{eff} (bias) that is higher than the optimum value (i.e. 0.78 V) will cause I_B -transistor to enter the triode region at a lower V_{out+} voltage. Also, a V_{eff} (bias) that is lower than the optimum will shut off Q_1 at a lower V_{out+} voltage. In both cases, the voltage range of V_{out+} for linear operation is reduced.

$$6.22) |V_{tp}| = |V_{tp0}| + \gamma (\sqrt{|V_{SB}| + |2\phi_F|} - \sqrt{|2\phi_F|})$$

$$\text{For } V_S = 2 \text{ V and } V_B = 2.5 \text{ V, } |2\phi_F| \approx 0.7 \text{ V}$$

$$\Rightarrow |V_{tp}| = 1.1 \text{ V} \Rightarrow I = K(2 - 1.1)^2$$

For the bias condition ($V_{out+} = 0$), we have

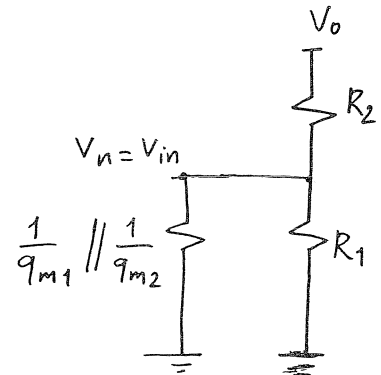
$$I/2 = K(V_{eff})^2 \Rightarrow V_{eff} = \frac{0.9}{\sqrt{2}} = \underline{0.63 \text{ V}}$$

$$V_{out+}|_{\max} = 2 - 1.1 = \underline{0.9 \text{ V}}$$

$$V_{out+}|_{\min} = |V_{tp}| - V_{SG1} = |V_{tp}| - (|V_{tp}| + 0.9) = \underline{-0.9 \text{ V}}$$

$$6.23) V_{in} = V_o \frac{R_1 \parallel \frac{1}{g_{m1}} \parallel \frac{1}{g_{m2}}}{R_2 + R_1 \parallel \frac{1}{g_{m1}} \parallel \frac{1}{g_{m2}}}$$

$$\Rightarrow \frac{V_o}{V_{in}} = \frac{R_1 + R_2}{R_1} + \underline{R_2 (g_{m1} + g_{m2})}$$



The first term can be recognized as the voltage gain when $g_{m1} = g_{m2} = 0$

6.24) Using Eq. (6.62), (6.64), (6.65), and (6.66):

$$\tau = \frac{1}{\omega_{3dB}} = \frac{1}{\beta \omega_t} \quad \text{where} \quad \beta = \frac{C_1/M}{C_1/M + C_1 + C_p} \quad \& \quad \omega_t = \frac{K g_m}{C_0 + \frac{C_1/M (C_1 + C_p)}{C_1/M + C_1 + C_p}}$$

$$\Rightarrow \tau = \frac{1}{K g_m} \left[(M+1)C_0 + C_p + \frac{M C_0 C_p}{C_1} + C_1 \right]$$

$$\frac{\partial \tau}{\partial C_1} = 0 \Rightarrow -\frac{C_0 C_p}{C_1^2} + \frac{1}{M} = 0 \Rightarrow \underline{C_{1,opt} = \sqrt{M C_0 C_p}}$$

6.25) $\tau = \frac{1}{K g_m} \left[2C_0 + C_p + \frac{C_0 C_p}{C_1} + C_1 \right] = \frac{1}{K g_m} \left[2.05 + \frac{0.05}{C_1} + C_1 \right] \text{ ps}$

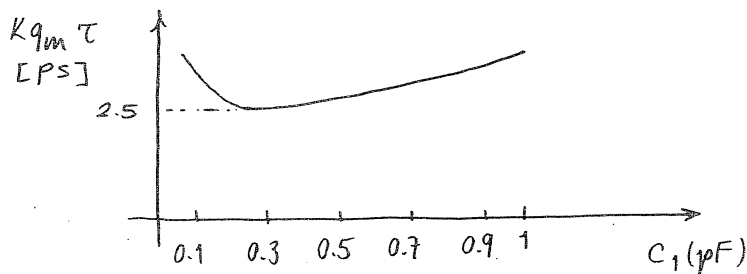
where C_1 must be expressed in pF unit.

$$C_{1,opt} = \sqrt{C_0 C_p} = \underline{0.22 \text{ pF}}$$

$$\Rightarrow K g_m \tau \Big|_{\min} = \underline{2.5 \text{ ps}}$$

The following table shows $K g_m \tau$ for some other values of C_1

C_1 (pF)	0.1	0.3	0.5	0.7	0.9	1
$K g_m \tau$ [ps]	2.65	2.52	2.65	2.82	3	3.1



Chapter 7

7.7 PROBLEMS

Unless otherwise stated, assume the following:

- n-channel MOS transistors:
 - $\mu_n C_{ox} = 92 \mu A/V^2$
 - $V_{tn} = 0.8 V$
 - $\gamma = 0.5 V^{1/2}$
 - $r_{ds} (\Omega) = 8000 L (\mu m)/I_D (mA)$ in active region
 - $C_j = 2.4 \times 10^{-4} pF/(\mu m)^2$
 - $C_{j-sw} = 2.0 \times 10^{-4} pF/\mu m$
 - $C_{ox} = 1.9 \times 10^{-3} pF/(\mu m)^2$
 - $C_{gs(overlap)} = C_{gd(overlap)} = 2.0 \times 10^{-4} pF/\mu m$
- p-channel MOS transistors:
 - $\mu_p C_{ox} = 30 \mu A/V^2$
 - $V_{tp} = -0.9 V$
 - $\gamma = 0.8 V^{1/2}$
 - $r_{ds} (\Omega) = 12,000 L (\mu m)/I_D (mA)$ in active region
 - $C_j = 4.5 \times 10^{-4} pF/(\mu m)^2$
 - $C_{j-sw} = 2.5 \times 10^{-4} pF/\mu m$
 - $C_{ox} = 1.9 \times 10^{-3} pF/(\mu m)^2$
 - $C_{gs(overlap)} = C_{gd(overlap)} = 2.0 \times 10^{-4} pF/\mu m$

- 7.1** A simple CMOS inverter is used as a comparator, as shown in Fig. P7.1. Assume ϕ_{1a} drops before ϕ_1 drops, so that only the charge injection from Q_1 need be considered. What is the change in V_{out} due to this charge injection?

in track mode and a loop gain of 4 when in latch mode. Assume the input transistors are $25\text{-}\mu\text{m}$ wide and that the bias current of the first stage is 0.1 mA .

- 7.8** Assume the comparator of Fig. 7.16 has the device sizes found in Problem 7.7. Estimate the equivalent time constant when the comparator is in track mode. If the input signal changes from 0 to 5 mV at time 0 , how long would it take the differential output voltage to change by 50 mV ?
- 7.9** Consider the comparator in Fig. 7.16 with the device sizes found in Problem 7.7. Estimate the equivalent time constant of the latch-and-track stage when the comparator is in latch mode. If the differential output voltage is 50 mV at the beginning of the latch phase, how long would it take for the output voltage to change to a differential voltage of 2 V ?
- 7.10** What is the minimum power-supply voltage that can be used for the comparator in Fig. 7.25 if the current source and the resistive loads both require 0.4 V across them? You may assume that for the bipolar transistors $V_{\text{CE-sat}} = 0.3\text{ V}$.
- 7.11** The CMOS transmission gate shown has an input voltage of 2.5 V when it turns off. The W/L of the n-channel is $5\text{ }\mu\text{m}/0.8\text{ }\mu\text{m}$ and the W/L of the p-channel is $15\text{ }\mu\text{m}/0.8\text{ }\mu\text{m}$. Assume the total parasitic capacitance between the output node and ground is 50 fF , $V_{\text{DD}} = 5\text{ V}$, and that the clock signals change very fast. Also ignore changes due to overlap capacitance. Estimate the change in output voltage due to charge injection. What will the final output voltage be?

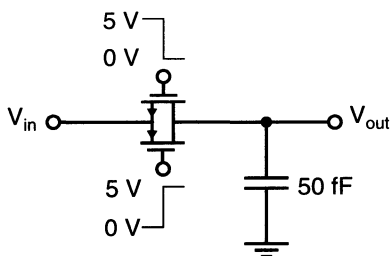
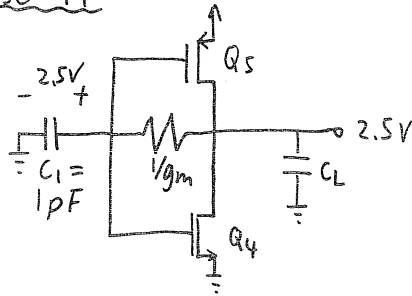


Fig. P7.11

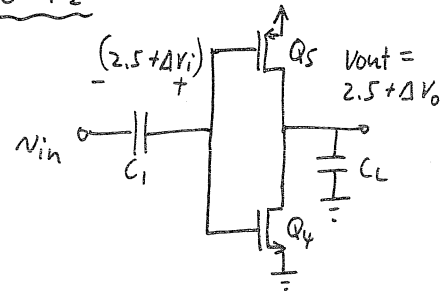
Chapter 7 - Problems

7.1) Find offset due to charge injection

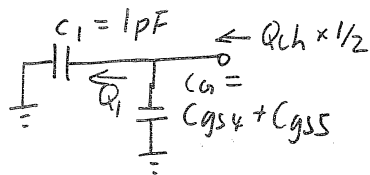
Phase ϕ_1 :



Phase ϕ_2 :



Channel charge stored in Q_1 is partially* injected into C_1 at the end of ϕ_{1a} .



* Assume half is injected into the node attached to the source of Q_1

$$Q_{ch} = V_{eff,1} C_{ox} W_1 L_1$$

$$V_{eff,1} = V_{GS,1} - V_{tn}$$

$$\begin{aligned} V_{tn} &= V_{tn0} + \gamma (\sqrt{V_{SB} + 2\phi_F} - \sqrt{2\phi_F}) \\ &= 0.8 + 0.5 (\sqrt{2.5 + 0.7} - \sqrt{0.7}) \\ &= \underline{1.28 V} \end{aligned}$$

$$\therefore V_{eff,1} = (5 - 2.5) - 1.28 = 1.22 V$$

$$\therefore Q_{ch} = 1.22 V \times 1.9 \times 10^{-3} \frac{PF}{\mu m^2} \times 5 \mu m \times 0.8 \mu m = 9.27 \times 10^{-15} C$$

$$\begin{aligned} \text{Now } C_{gs5} &= \frac{2}{3} C_{ox} W_5 L_5 \quad (\text{neglect overlap capacitance}) \\ &= \frac{2}{3} \times 1.9 \times 10^{-3} \times 92 \times 0.8 \\ &= 0.093 pF \end{aligned}$$

$$\begin{aligned} C_{gs4} &= \frac{2}{3} C_{ox} W_4 L_4 \\ &= 0.030 pF \end{aligned}$$

$$\therefore C_n = C_{gs5} + C_{gs4} = 0.123 pF$$

(Cont.)

7.1 (cont.)

$$\therefore Q = -CV$$

$$\therefore \Delta V_i = \frac{-\Delta Q}{C} = -\frac{Q_{ch}}{2(C_1 + C_2)} = \frac{-9.27 \times 10^{-15}}{2(1\text{pF} + 0.123\text{pF})} = -4.1\text{mV}$$

Since the inverters gain is given as -24

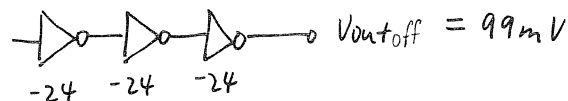
$$\Delta V_o = -24 \times \Delta V_i$$

$$\therefore \Delta V_o = -24 \times \Delta V_i = \underline{+99\text{mV}}$$

$\therefore$ V_{out} shifts up by 99mV due to charge injection.

7.2) Find the input-referred offset voltage.

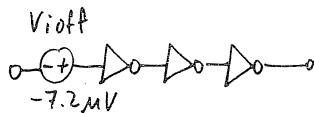
From Problem 7.1, we found the output offset voltage to be $+99\text{mV}$.



To refer this back to the input, we divide by the total gain

$$\text{i.e., } V_{ioff} = V_{outoff} / (-24)^3$$

$$\underline{V_{ioff} = -7.2\text{mV}}$$

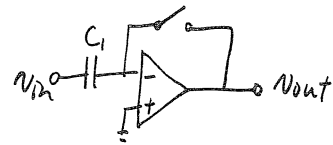
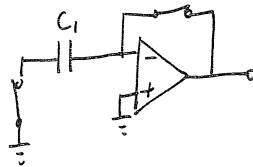


7.3) Estimate time constants :

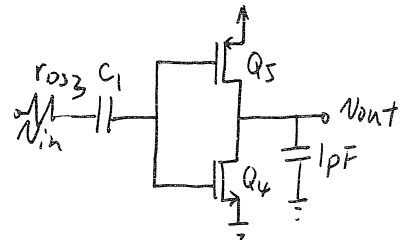
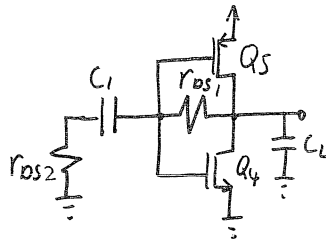
CONCEPT

Reset Phase (ϕ_1)

Comparison Phase (ϕ_2)

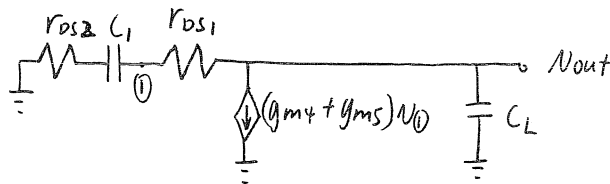


CIRCUIT

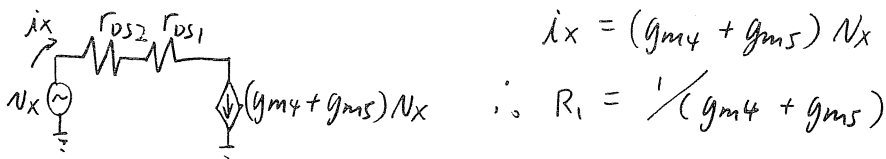


Reset Phase (ϕ_1) :

To find system time constants, calculate the resistances associated with each capacitor.



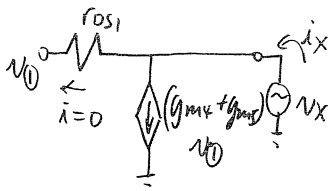
for C_1 :



$$i_x = (g_{m4} + g_{m5}) v_x$$

$$\therefore R_1 = 1 / (g_{m4} + g_{m5})$$

for C_2 :



Again,

$$i_x = (g_{m4} + g_{m5}) v_0 = (g_{m4} + g_{m5}) v_x$$

$$\therefore R_2 = v_x / i_x = 1 / (g_{m4} + g_{m5})$$

$$\therefore \tau_{\phi_1} = \tau_1 + \tau_2 = R_1 C_1 + R_2 C_2$$

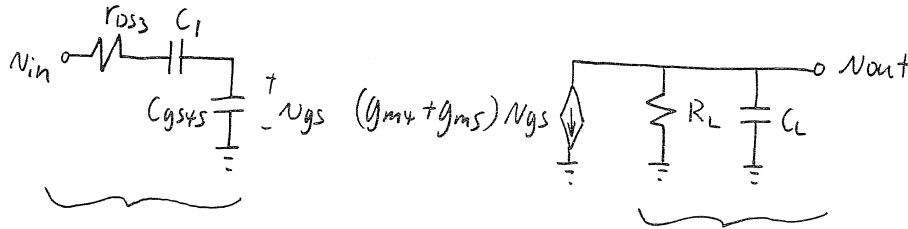
$$\tau_{\phi_1} = \frac{(C_1 + C_2)}{g_{m4} + g_{m5}}$$

for the reset phase

(cont.)

7.3) (cont.)

Comparison Phase (ϕ_2):



$$\tau_{in} = (C_1 \text{ in series w. } r_{DS3}) \times r_{DS3}$$

$$\approx C_{gs45} r_{DS3}$$

$$\tau_{out} = R_L C_L$$

$$= (r_{DS4} // r_{DS5}) C_L$$

$$\tau_{\phi_2} = \tau_{in} + \tau_{out} = r_{DS3} C_{gs45} + r_{DS4} // r_{DS5} \times C_L \approx r_{DS4} // r_{DS5} \times C_L$$

Calculating values:

$$g_{m4} = \mu_n C_{ox} \frac{W}{L} V_{eff4} = 92 \times 10^{-6} \times 30 / 0.8 \times (2.5 - 0.8)$$

$$= 5.9 \text{ mA/V}$$

$$g_{m5} = \mu_p C_{ox} \frac{W}{L} V_{eff5} = 30 \times 10^{-6} \times 92 / 0.8 \times (2.5 - 0.8)$$

$$= 5.9 \text{ mA/V}$$

$$r_{DS1} = \frac{1}{\mu_n C_{ox} \frac{W}{L} V_{eff1}} = \frac{1}{92 \times 10^{-6} \times 5 / 0.8 \times (2.5 - V_{tn1})} \quad \text{where}$$

$$V_{tn1} = V_{tn0} + \gamma (\sqrt{V_{SB1} + 2\phi_F} - \sqrt{2\phi_F})$$

$$= 0.8 + 0.5 (\sqrt{2.5 + 0.7} - \sqrt{0.7}) = 1.28 \text{ V}$$

$$\therefore r_{DS1} = 1.4 \text{ k}\Omega$$

$$r_{DS3} = \frac{1}{92 \times 10^{-6} \times 5 / 0.8 \times (2.5 - 0.8)} = 1 \text{ k}\Omega$$

$$\therefore \tau_{\phi_1} = \frac{C_1 + C_2}{g_{m4} + g_{m5}} = \frac{2 \text{ pF}}{2 \times 5.9 \text{ mA/V}} = \underline{0.17 \text{ nsec}}$$

$$\tau_{\phi_2} = r_{DS4} // r_{DS5} \times C_L \quad \text{where}$$

$$I_{D4} = I_{D5} = \frac{\mu_n C_{ox}}{2} \frac{W}{L} V_{eff}^2 = \frac{92 \times 10^{-6}}{2} \times \frac{30}{0.8} (2.5 - 0.8)^2$$

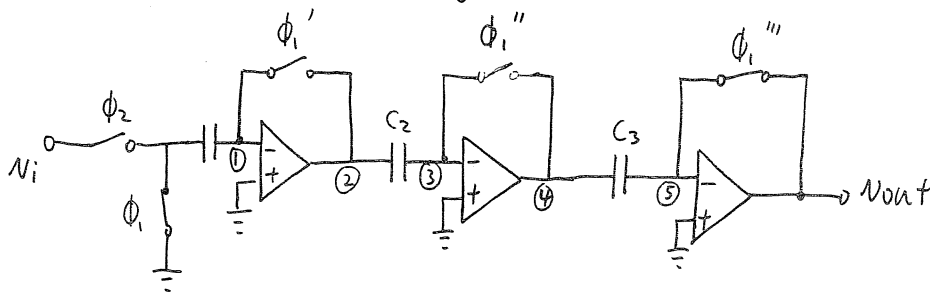
$$= 5 \text{ mA}$$

$$\therefore r_{DS4} = \frac{8000 \times 0.8}{5 \text{ mA}} = 1.3 \text{ k}\Omega, \quad r_{DS5} = \frac{12000 \times 0.8}{5 \text{ mA}} = 1.9 \text{ k}\Omega$$

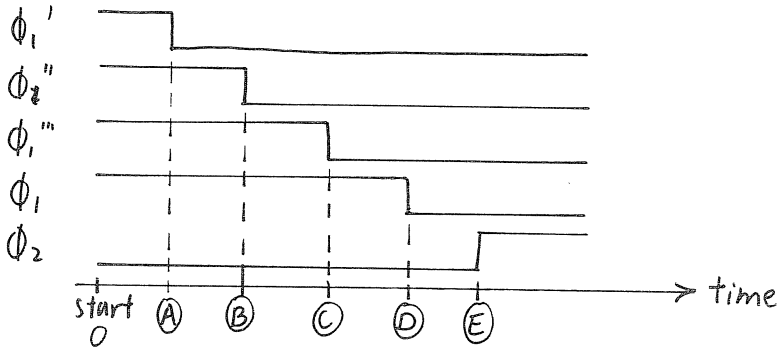
$$\therefore \tau_{\phi_2} = 1.3 \text{ k}\Omega // 1.9 \text{ k}\Omega \times 1 \text{ pF} = \underline{0.77 \text{ nsec}}$$

$\therefore$ The time constants during the reset and comparison phases are 0.17 nsec and 0.77 nsec respectively.

7.4) Show clock feedthrough is $V_{err-3}/(A_1 A_2)$



Clocking diagram :



* Switches are closed with waveforms high *

Time	$V_{①}$	$V_{②}$	$V_{③}$	$V_{④}$	$V_{⑤}$	V_{out}
0	0V	0V	0V	0V	0V	0V
A	V_{err1}	$-A_1 V_{err1}$	0V	0V	0V	0V
B	V_{err1}	$-A_1 V_{err1}$	V_{err2}	$-A_2 V_{err2}$	0V	0V
C	V_{err1}	$-A_1 V_{err1}$	V_{err2}	$-A_2 V_{err2}$	V_{err3}	$-A_3 V_{err3}$
D	"	"	"	"	"	"
E	$V_i + V_{err1}$	$-A_1 (V_i + V_{err1})$	$-A_1 V_i + V_{err2}$	$(A_1 A_2 V_i - A_2 V_{err2})$	$(-A_1 A_2 A_3 V_i - A_3 V_{err3})$	$(A_1 A_2 V_i + V_{err3})$

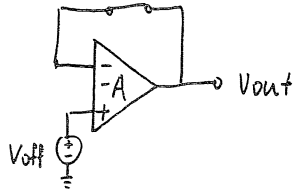
∴ at the end of ϕ_2 ,

$$V_{out} = -(A_1 A_2 A_3 V_i + A_3 V_{err3})$$

$$= -A_1 A_2 A_3 (V_i + V_{ioff}) \text{ where } V_{ioff} = \frac{V_{err3}}{A_1 A_2}$$

∴ the input-referred offset is $\frac{V_{err3}}{A_1 A_2}$. Q.E.D.

7.5) Repeat Problem 7.4, but now each opamp has input offset, V_{offi} .



$$V_{out} = -A(V_{off} - V_{out})$$

$$\therefore \underline{V_{out} = \frac{-A}{1-A} V_{off}}$$

Time	V_D	V_2	V_3	V_4	V_5	V_{out}
0	$\underbrace{-\frac{A_1}{1-A_1} V_{off1}}_{\equiv K_1}$	$\underbrace{-\frac{A_1}{1-A_1} V_{off1}}_{\equiv K_1}$	$\underbrace{-\frac{A_2}{1-A_2} V_{off2}}_{\equiv K_2}$	$\underbrace{-\frac{A_2}{1-A_2} V_{off2}}_{\equiv K_2}$	$\underbrace{-\frac{A_3}{1-A_3} V_{off3}}_{\equiv K_3}$	$\underbrace{-\frac{A_3}{1-A_3} V_{off3}}_{\equiv K_3}$
(A)	$K_1 + V_{err1}$	$K_1 - A_1 V_{err1}$	K_2	K_2	K_3	K_3
(B)	"	"	$K_2 + V_{err2}$	$K_2 - A_2 V_{err2}$	K_3	K_3
(C)	"	"	"	"	$K_3 + V_{err3}$	$K_3 - A_3 V_{err3}$
(D)	"	"	"	"	"	"
(E)	$(V_i + V_{err1} + K_1)$	$(-A_1 V_i + V_{err2} + K_2)$	$(A_1 A_2 V_i + V_{err3} + K_3)$			
	$(-A_1(V_i + V_{err1}) - K_1)$	$(A_1 A_2 V_i - A_2 V_{err2} + K_2)$				

$$V_{out} = -A_1 A_2 A_3 V_i - A_3 V_{err3} + K_3$$

$$\therefore V_{out} = -\left(A_1 A_2 A_3 V_i + A_3 V_{err3} + \frac{A_3}{1-A_3} V_{off3}\right)$$

$$= -A_1 A_2 A_3 (V_i + V_{ioff})$$

$$\text{where } V_{ioff} = \frac{V_{err3}}{A_1 A_2} + \frac{1}{A_1 A_2 (1-A_3)} V_{off3}$$

$\therefore$ the input-referred offset voltage is

$$\underline{V_{ioff} = \left(V_{err3} + \frac{1}{1-A_3} V_{off3}\right) / (A_1 A_2)}$$

7.6)

$$\omega_u = g_m / C_{gs}$$

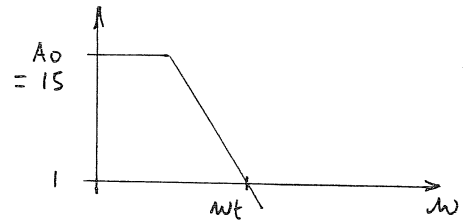
$$\omega_t = \omega_u / 2 = g_m / 2 C_{gs}$$

$$\text{But } C_{gs} = \frac{2}{3} C_{ox} W L$$

$$\text{and } g_m = \mu_n C_{ox} W / L V_{eff}$$

$$\therefore \omega_t = \frac{\mu_n C_{ox} W / L V_{eff}}{\frac{4}{3} C_{ox} W L}$$

$$\omega_t = \frac{3 \mu_n V_{eff}}{4 L^2}$$



The time constant, τ , for a single stage is

$$\tau = \frac{1}{\omega_{p1}}$$

The gain-bandwidth product = $\omega_t = A_0 \omega_{p1}$

$$\therefore \omega_{p1} = \frac{\omega_t}{A_0} \text{ and } \tau = A_0 / \omega_t$$

$$\tau_{comparator} = 3 \times \tau = 3 A_0 / \omega_t$$

$$= 3 A_0 \times \frac{4 L^2}{3 \mu_n V_{eff}}$$

$$= \frac{4 A_0 L^2}{\mu_n V_{eff}}$$

Assuming $V_{eff} = 0.25V$, $\mu_n = 0.05 \text{ m}^2/\text{V}\cdot\text{s}$ and a settling time of $3 \tau_{comparator}$,

$$\tau_{comparator} = \frac{4(15)(0.8 \times 10^{-6})^2}{0.05 \times 0.25} = 3.07 \text{ nsec}$$

$$\therefore f_{max} = \frac{1}{6 \tau_{comparator}} = \frac{1}{6 \times 3.07 \times 10^{-9}}$$

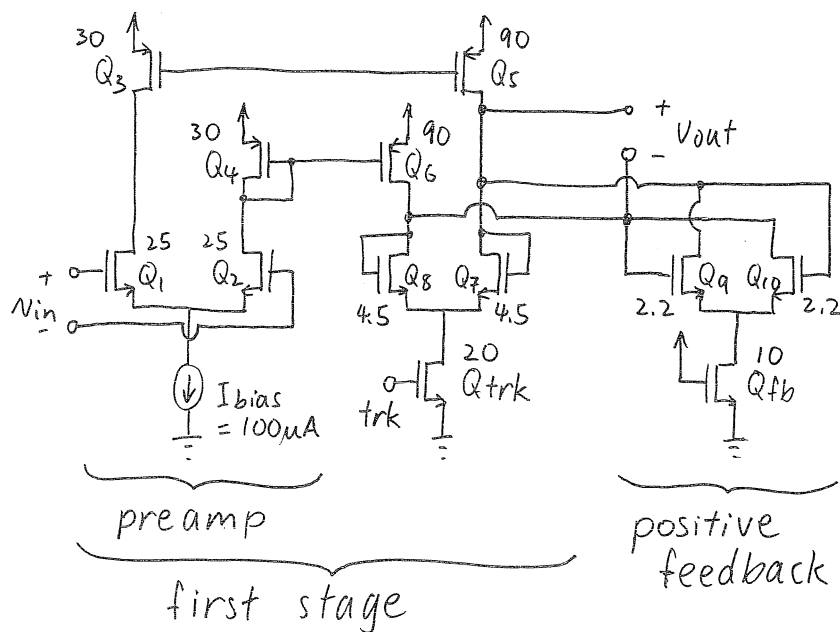
$\therefore \underline{f_{max} = 54 \text{ MHz}}$ is the maximum frequency of operation

$$\text{resolution} = A_0^3 = 15^3 = 3375$$

$$= \underline{71 \text{ dB}}$$

7.7) Determine the transistor sizes.

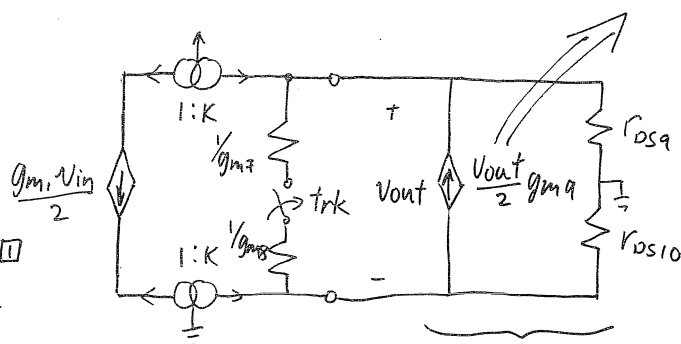
* Devices are all of length $L = 0.8 \mu\text{m}$ *



ASIDE:
by source
absorption

Simplified
small signal
model

$$K = \frac{W_{5,6}}{W_{3,4}} = \frac{2 I_{D5,6}}{I_{bias}} \quad \square$$



$$R = -\frac{2}{g_{m9}}$$

KCL at $V_{out} +$:

$$K g_{m1} \frac{V_{in}}{2} + \frac{V_{out}}{2} g_{m9} - V_{out} / \left(\frac{1}{g_{m7}} + \frac{1}{g_{m8}} \right) - \cancel{V_{out} / (r_{os9} + r_{os10})} = 0$$

$$\therefore V_{in} K \frac{g_{m1}}{2} + V_{out} \left(\frac{g_{m9}}{2} - \frac{g_{m7} g_{m8}}{g_{m7} + g_{m8}} \right) \approx 0$$

$$\therefore \frac{V_{out}}{V_{in}} \approx \frac{-K g_{m1} / 2 \times (2 (g_{m7} + g_{m8}))}{g_{m9} (g_{m7} + g_{m8}) - 2 g_{m7} g_{m8}} = \frac{-K g_{m1} (g_{m7} + g_{m8})}{g_{m9} (g_{m7} + g_{m8}) - 2 g_{m7} g_{m8}}$$

Assuming Q_7 and Q_8 are matched so that $g_{m7} = g_{m8}$

$$\therefore \frac{V_{out}}{V_{in}} = \frac{K g_{m1}}{g_{m7} - g_{m9}} \quad \square \quad \text{Tracking gain}$$

(cont.)

7.7 (cont.)

The gain of the first stage is determined by ignoring the positive feedback circuit.

∴ Let $g_{mq} \equiv 0$

$$\text{and } \frac{v_{out}}{v_{in}} = K \frac{g_{m1}}{g_{m7}} \equiv 5$$

$$\therefore \underline{g_{m7} = K/5 \times g_{m1}} \quad [B]$$

Setting the tracking gain to 10 gives

$$\frac{K g_{m1}}{g_{m7} - g_{mq}} \equiv 10 \quad [C]$$

$$\text{sub } [B] \rightarrow [C] \quad \frac{K g_{m1}}{K/5 \times g_{m1} - g_{mq}} = 10$$

$$\underline{g_{mq} = \frac{K g_{m1}}{10}} \quad [D] \quad \text{or} \quad \underline{g_{mq} = 1/2 g_{m7}} \quad [E]$$

Under ideal bias conditions, $v_{out+} = v_{out-}$.

Thus, even though Q_9 and Q_{10} are cross-coupled unlike Q_7 and Q_8 , their terminals are biased at the same voltages as Q_9 and Q_{10} .

∴ To satisfy equation [E], we simply scale device widths such that

$$\underline{W_{9,10} = 1/2 W_{7,8}} \quad [F]$$

Note that the drain currents are also scaled

$$\text{i.e., } \underline{I_{D9,10} = 1/2 I_{D7,8}} \quad [G]$$

Now find $W_{7,8}$ in terms of $W_{1,2} = 25 \mu\text{m}$:

From [B], [I]

$$g_{m7} = K/5 g_{m1} = \frac{2}{5} \times \frac{I_{D5,6}}{I_{bias}} \times g_{m1}$$

$$\text{where } I_{D5,6} = I_{D7,8} + I_{D9,10} \quad [H]$$

$$[G] \rightarrow [H] \quad = 3/2 I_{D7,8}$$

(cont.)

7.7) (cont.)

$$\therefore g_{m7}^2 = \left(\frac{2}{5} \times \frac{3}{2} \frac{I_{D7,8}}{I_{bias}} \times g_{m1} \right)^2$$

$$\therefore \frac{2 \cancel{\mu_n C_{ox}}}{\cancel{L}} W_{7,8} \cancel{I_{D7,8}} = \frac{9}{25} \times \frac{I_{D7,8}}{I_{bias}} \times \frac{2 \cancel{\mu_n C_{ox}}}{\cancel{L}} W_{1,2} \frac{I_{bias}}{2}$$
$$\frac{W_{7,8}}{W_{1,2}} = \frac{9}{50} \frac{I_{D7,8}}{I_{bias}} \quad \text{for } W_{1,2} = 25 \mu\text{m}$$

$$\therefore W_{7,8} = \frac{9}{2} \frac{I_{D7,8}}{I_{bias}} \mu\text{m}$$

If we let $I_{D7,8} = I_{bias}$,

$$\therefore \underline{W_{7,8} = \frac{9}{2} \mu\text{m} = 4.5 \mu\text{m}}$$

$$\therefore \underline{W_{9,10} = \frac{1}{2} W_{7,8} = 2.2 \mu\text{m}}$$

$$\therefore K = \frac{2 I_{D5,6}}{I_{bias}} = \frac{2 \left(\frac{3}{2} I_{D7,8} \right)}{I_{bias}} = \underline{3 = \frac{W_{5,6}}{W_{3,4}}}$$

If we let $W_{3,4} = 30 \mu\text{m}$

then $W_{5,6} = 90 \mu\text{m}$

Since Q_{trk} and Q_{fb} operate as switches in the triode region, their size is not critical.

If we set

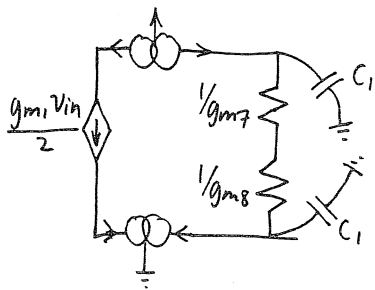
$$\underline{W_{trk} = 20 \mu\text{m}},$$

then to match voltage levels as closely as possible choose

$$\underline{W_{fb} = \frac{1}{2} W_{trk} = 10 \mu\text{m}}$$

Final device widths are marked on the original schematic diagram.

7.8) Estimate time constant in track mode, $\tau_{trk} \approx \tau_{gain} + \tau_{pre}$



Gain stage

$$\tau_{gain} = \frac{C_L}{g_m7}$$

where $C_L = C_{db5} + C_{db7} + C_{gs10} + C_{db9}$
 $\approx C_{db5}$ (device Q_5 is much larger than devices Q_7, Q_9, Q_{10})

$$= A_{d5} C_{d5} + P_{dsw5} \times C_{ov}$$

$$= 30 \times 4 \times 0.8 \times 4.5 \times 10^{-4} \text{ pF}/\mu\text{m}^2$$

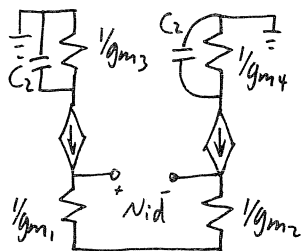
$$+ 2 \times 4 \times 0.8 \times 2.5 \times 10^{-4} \text{ pF}/\mu\text{m}^2$$

$$C_L \approx 52 \text{ fF}$$

and $g_m7 = 0.107 \text{ mA/V}$ from Problem 7.7.

$$\therefore \tau_{gain} = \frac{52 \times 10^{-15} \text{ F}}{0.107 \times 10^{-3} \text{ A/V}} = 0.47 \text{ nsec}$$

Now look at time constant associated with the preamp, τ_{pre} .



Preamplifier stage

$$\tau_{pre} = \frac{C_2}{g_m3} \text{ where}$$

$$C_2 = C_{db3} + C_{gs3} + C_{gs5} + C_{db1} \quad \text{negligible } \because Q_1 \text{ small}$$

$$C_{db3} = A_{d3} C_{d3} + P_{dsw3} \times C_{ov}$$

$$= 100 \times 0.8 \times 4 \times 4.5 \times 10^{-4} +$$

$$(2 \times 4 \times 0.8 + 100) \times 2.5 \times 10^{-4}$$

$$= 170 \text{ fF}$$

$$C_{gs5} = \frac{2}{3} W L C_{ox} + W C_{ov}$$

$$= \frac{2}{3} \times 30 \times 0.8 \times 1.9 \times 10^{-3}$$

$$+ 30 \times 2 \times 10^{-4}$$

$$= 40 \text{ fF}$$

$$C_{gs3} = \frac{2}{3} W L C_{ox} + W C_{ov}$$

$$= \frac{2}{3} \times 100 \times 0.8 \times 1.9 \times 10^{-3} + 100 \times 2 \times 10^{-4}$$

$$= 120 \text{ fF}$$

$$\therefore C_2 = 170 + 120 + 40 \text{ fF} = 330 \text{ fF}$$

$$\text{and } g_m3 = \sqrt{2 M_n C_{ox} W/L I_{D3}} = \sqrt{2 \times 30 \times 10^{-6} \times 100 \times 0.8 \times 50 \times 10^{-6}}$$

$$= 0.612 \text{ mA/V}$$

$$\therefore \tau_{pre} = \frac{330 \times 10^{-15}}{0.612 \times 10^{-3}} = 0.54 \text{ nsec}$$

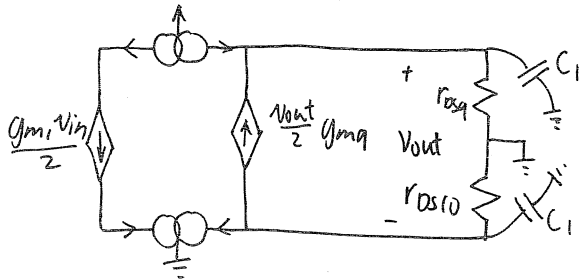
$$\therefore \tau_{trk} \approx \tau_{gain} + \tau_{pre} = 1.01 \text{ nsec}$$

(cont.)

7.8 (cont.)

Assuming it takes three time constants to settle from a step change of 50 mV, it would take 3 nsec.

7.9) Determine time constant in latch mode.



From Problem 7.8,

$$C_1 = 52 \text{ fF}$$

$$\begin{aligned} g_{mq} &= \sqrt{2 \mu_n C_{ox} W/L I_{DQ}} \\ &= \sqrt{2 \times 92 \times 10^{-6} \times 2.4/0.8 \times 5 \times 10^{-6}} \\ &= 53 \text{ mA/V} \end{aligned}$$

KCL at V_{out+} :

$$\frac{g_{m1}}{2} v_{in} + \frac{V_{out}}{2} g_{mq} - \frac{V_{out}}{r_{oq} + r_{o10}} - V_{out} s \frac{C_1}{2} = 0$$

$$\begin{aligned} \therefore \frac{V_{out}}{v_{in}} &\approx \frac{-g_{m1}/2}{g_{mq}/2 - \frac{1}{2(r_{oq})} - sC_1/2} \\ &= \frac{g_{m1}}{C_1 (s - g_{mq}/C_1 + \frac{1}{C_1 r_{oq}})} \approx \frac{g_{m1}}{C_1} \times \frac{1}{s - g_{mq}/C_1} \end{aligned}$$

$\therefore$ There is an unstable pole at

$$\omega = g_{mq}/C_1$$

and the associated time constant is

$$\tau = C_1/g_{mq} = 52 \text{ fF}/0.53 \text{ mA/V}$$

$$\tau = 0.99 \text{ nsec}$$

For a differential output of 2 V,

$$V_{out}(t) = V_{out0} e^{t/\tau}$$

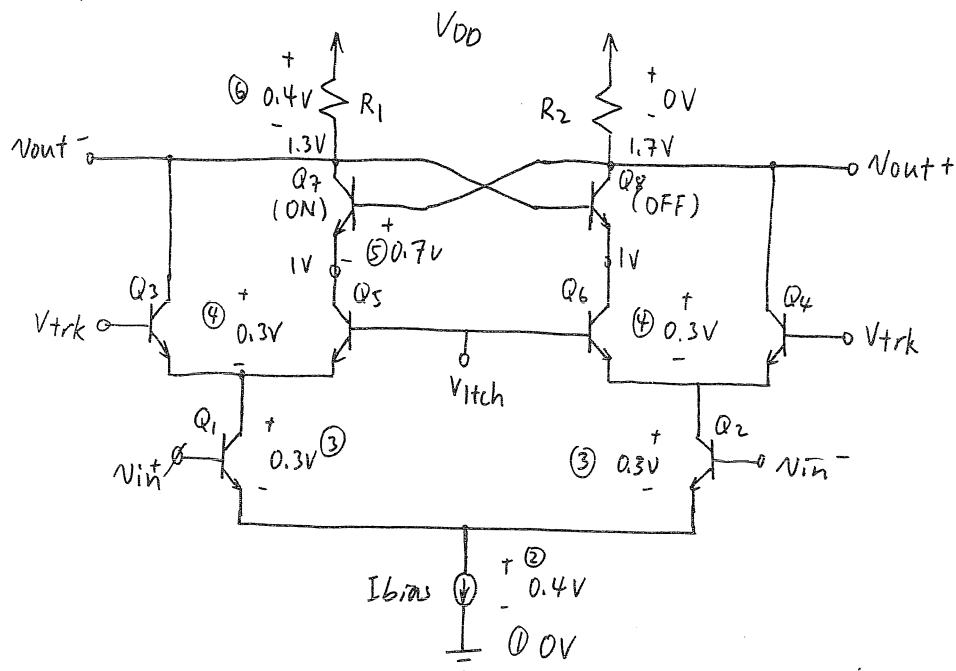
$$\therefore 2 \text{ V} = 0.05 e^{t/0.99 \times 10^{-9}}$$

$$t = 0.99 \times 10^{-9} \ln 40$$

$$= 3.65 \text{ nsec}$$

$\therefore$ it takes 3.65 nsec.

7.10) Find ^{the} minimum power supply voltage.

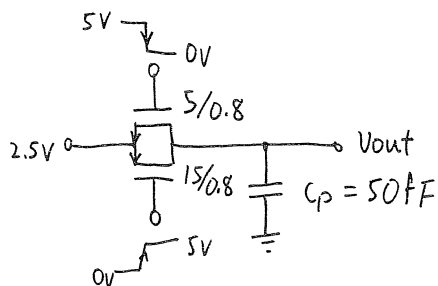


From the above diagram we see that the ^{minimum} voltage drops from R_2 through devices Q_7 , Q_5 , Q_1 , & I_{bias} to ground add up to 1.7V. Assuming $V_{DD} = 1.7V$, we see that V_{out-} is at $1.7V - 0.4V = 1.3V$.

∴ we have correctly assumed that Q_8 is off.

∴ this circuit requires a minimum supply voltage of 1.7V.

7.11)



Determine the voltage change due to each device.

N-MOS:

$$\Delta V_N = - \frac{V_{effn} C_{ox} W_n L_n}{2 C_p} \quad \text{where}$$

$$V_{effn} = 5V - 2.5V - V_{tn}$$

$$V_{tn} = V_{tn0} + \gamma (\sqrt{V_{SB} + 2\phi_F} - \sqrt{2\phi_F})$$

$$= 0.8 + 0.5 (\sqrt{2.5 + 0.7} - \sqrt{0.7})$$

$$V_{tn} = 1.28V$$

$$\therefore V_{effn} = 1.22V$$

$$\therefore \Delta V_N = - \frac{1.22V \times 1.9 \times 10^{-3} \text{ PF}/\mu\text{m}^2 \times 5 \times 0.8}{2 \times 50 \times 10^{-15}} = \underline{\underline{-93 \text{ mV}}}$$

PMOS: $\Delta V_p = \frac{V_{effp} C_{ox} W_p L_p}{2 C_p} \quad \text{where}$

$$V_{effp} = 2.5V - 0 - V_{tp}$$

$$V_{tp} = V_{tp0} + \gamma (\sqrt{V_{BS} - 2\phi_F} - \sqrt{2\phi_F})$$

To determine ϕ_F , use

$$\gamma = \frac{\sqrt{2qK_{si} \epsilon_0 N_A}}{C_{ox}}$$

$$\therefore 0.8 = \frac{\sqrt{2 \times 1.602 \times 10^{-19} \times 8.85 \times 10^{-12} \times 11.8 \times N_A}}{1.9 \times 10^{-15}}$$

$$\therefore \underline{N_A = 6.9 \times 10^{22} \text{ m}^{-3}}$$

$$\Rightarrow \phi_F = \frac{kT}{q} \ln \left(\frac{N_A}{n_i} \right) = \frac{1.38 \times 10^{-23} \times 300}{1.602 \times 10^{-19}} \ln \left(\frac{6.9 \times 10^{22}}{1.1 \times 10^{16}} \right)$$

$$= 0.40V$$

(cont.)

7.11 (cont.)

$$\therefore V_{tp} = -0.9 - 0.8(\sqrt{2.5+0.8} - \sqrt{0.8}) \\ = 1.64V$$

$$\therefore V_{thp} = 2.5 - 1.64V = 0.86V$$

$$\therefore \Delta V_p = \frac{0.86V \times 1.9 \times 10^{-3} pF \times 15 \times 0.8}{2 \times 50 \times 10^{-3} pF} = +196 mV$$

$$\therefore \Delta V_{out} = \Delta V_n + \Delta V_p = -93 mV + 196 mV \\ = \underline{+103 mV}$$

$\therefore$ the final output voltage is $2.5V + 103 mV \approx 2.6V$

Chapter 8

8.10 PROBLEMS

Unless otherwise stated, assume the following:

- n-channel MOS transistors:

$$\mu_n C_{ox} = 92 \mu\text{A}/\text{V}^2$$

$$V_{tn} = 0.8 \text{ V}$$

$$\gamma = 0.5 \text{ V}^{1/2}$$

$$r_{ds} (\Omega) = 8,000 L (\mu\text{m}) / I_D (\text{mA}) \text{ in active region,}$$

$$C_j = 2.4 \times 10^{-4} \text{ pF}/(\mu\text{m})^2$$

$$C_{j\text{-sw}} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$$

$$C_{ox} = 1.9 \times 10^{-3} \text{ pF}/(\mu\text{m})^2$$

$$C_{gs(\text{overlap})} = C_{gd(\text{overlap})} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$$

- p-channel MOS transistors:

$$\mu_p C_{ox} = 30 \mu\text{A}/\text{V}^2$$

$$V_{tp} = -0.9 \text{ V}$$

$$\gamma = 0.8 \text{ V}^{1/2}$$

$$r_{ds} (\Omega) = 12,000 L (\mu\text{m}) / I_D (\text{mA}) \text{ in active region,}$$

$$C_j = 4.5 \times 10^{-4} \text{ pF}/(\mu\text{m})^2$$

$$C_{j\text{-sw}} = 2.5 \times 10^{-4} \text{ pF}/\mu\text{m}$$

$$C_{ox} = 1.9 \times 10^{-3} \text{ pF}/(\mu\text{m})^2$$

$$C_{gs(\text{overlap})} = C_{gd(\text{overlap})} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$$

- 8.1 Using the transient analysis of SPICE with a sinusoidal input signal of 2 V peak-to-peak, find the offset, gain, and linearity error of the simple S/H of Fig. 8.3.
- 8.2 For the T/H of Fig. 8.4, assume V_{in} is a 20-MHz sinusoid with a 2 V_{p-p} amplitude. Also assume that ϕ_{clk} is a 100-MHz square wave having a peak amplitude of $\pm 2.5 \text{ V}$ with rise and fall times of 1.5 ns. What is the maximum time difference between the turn-off times of the n-channel and p-channel transistors? Ignore the body effect.
- 8.3 The T/H of Fig. 8.5 has transistors that are $10 \mu\text{m}/0.8 \mu\text{m}$ and a 1-pF hold capacitor. Assume the clock waveforms are fast enough that the channel charge of the transistors is evenly distributed between the two junctions. Compare the final hold pedestal between the case when the dummy switch turns on substantially before the sampling switch turns off, and the case when it turns on substantially after the sampling switch.

- 8.4** Assume the second stage of the S/H of Fig. 8.10 has a gain of -30 and a unity-gain frequency of 100 MHz. Also assume the input signal is a 5 -MHz sinusoid with a $2\text{ V}_{\text{p-p}}$ amplitude and the clock signals are at 20 MHz and go between $+2.5\text{ V}$ and -2.5 V , with rise and fall times of 1.5 ns . What is the sampling time uncertainty?
- 8.5** Assume the S/H of Fig. 8.12 is being realized in a BiCMOS process. This means that diodes and BJT transistors are available. Design a clock-driver circuit so that the clock voltages go from two diode drops above the input voltage (in track mode) to two diode drops below the output voltage (in hold mode).
- 8.6** Assume the opamp in the S/H of Fig. 8.15 has a finite gain of A_0 . Derive the output voltage in terms of V_{in} and A_0 during hold mode (i.e., when ϕ_2 is high).
- 8.7** Derive the output voltage of the S/H of Fig. 8.19 at the end of ϕ_2 in terms of the input voltage at the end of ϕ_1 , the output voltage at the end of ϕ_1 from the previous period, and the capacitor ratio C_1/C_2 . Take the z -transform of this difference equation, and substitute $z = e^{j\omega T}$ to find the frequency-domain transfer function of the S/H. Making the assumption that $e^{j\omega T} \equiv 1 + j\omega T$ for $\omega \ll (1/T) = f_{\text{clk}}$, where f_{clk} is the sampling frequency, show that
- $$f_{-3\text{ dB}} \equiv \frac{1}{2\pi} \frac{C_1}{C_2} f_{\text{clk}}$$
- 8.8** Assume the S/H of Fig. 8.21 has each of D_5 and D_6 replaced by two series diodes. Show the voltages at all nodes for the cases of sampling a 1 V input and a -1 V input for before as well as after track mode.
- 8.9** Assume two transistors in a PTAT are biased at a current-density ratio of $8:1$ at $T = 320^\circ\text{K}$. What is the difference in their base-emitter voltages and what is its temperature dependence?
- 8.10** Assuming $V_{\text{BE}0.2} = 0.65\text{ V}$, what is the value required for K in order to get zero temperature dependence at $T = 320^\circ\text{K}$ for a current-density ratio of $8:1$ in the two transistors?
- 8.11** Prove equations (8.22) and (8.23).
- 8.12** Find values for the bipolar reference of Fig. 8.25, where $V_{\text{ref}} = 2.5\text{ V}$.
- 8.13** Prove that equation (8.41) holds for the voltage reference of Fig. 8.27(b).
- 8.14** Consider the gain cell shown in Fig. 8.29, where I_2 is fixed to a bias current of $200\text{ }\mu\text{A}$ while I_1 varies such that the gain changes from 0.5 to 4 . What are the extreme values of I_1 needed? Also, what is the maximum peak current that should be used for i_{o1} , assuming all transistors should remain in the active region with at least $20\text{ }\mu\text{A}$ of emitter current?
- 8.15** Consider the translinear multiplier shown in Fig. 8.30, where $I_1 = 100\text{ }\mu\text{A}$, $I_2 = 200\text{ }\mu\text{A}$, and i_1 is set such that the gain $i_o/i_2 = 0.5$. If a common-mode noise signal of $5\text{ }\mu\text{A}$ appears on I_1 , what is the size of the output signal due to that noise signal? Repeat for a common-mode noise signal on I_2 .
- 8.16** Repeat Problem 8.15 when the β 's of the transistors' worst-case mismatch is 5 percent.

- 8.17** Design a four-quadrant translinear multiplier having differential voltage-current conversion stages based on emitter degeneration at both inputs and a differential voltage output. All transistors should be biased at 0.2 mA. Design the multiplier so that when both differential inputs are 1 V the differential output should be 1 V. Assume a 5 V power supply is available.

Chapter 8 - Problems

8.2) Assuming V_{in} is constant during the 1.5 ns of transition, the p-channel transistor turns off when:

$$\bar{\Phi} = V_{in} - 0.9V \Rightarrow \Phi = 0.9 - V_{in}$$

The n-channel transistor turns off when:

$$\Phi = V_{in} + 0.8V$$

The difference between the Φ voltages for the two cases is:

$$\Delta\Phi = 0.1 - 2V_{in}$$

$|\Delta\Phi|$ is maximized at $V_{in} = -1$ for which $|\Delta\Phi|_{\max} = 2.1V$

$$\Rightarrow \Delta t_{\max} = \frac{2.1V}{5V} \times 1.5 \text{ ns} = \underline{\underline{0.63 \text{ ns}}}$$

8.3) Case I) the hold pedestal is only due to the sampling switch turning off. Using (8.3), the hold pedestal is:

$$\text{is : } \underline{\underline{\Delta V' = -13 \text{ mV}}}$$

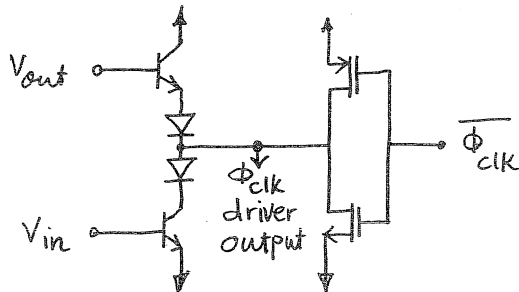
Case II) the hold pedestal due to Q_2 turning on is twice as much as the hold pedestal due to Q_1 turning off and in the opposite direction. The final hold pedestal is: $\Delta V' = 26 \text{ mV} - 13 \text{ mV} = \underline{\underline{13 \text{ mV}}}$
The result is the same hold pedestal with the opposite sign.

8.4) The gain of the second opamp at 5 MHz is $\frac{100 \text{ MHz}}{5 \text{ MHz}} = 20$.

Therefore, with a 2V p-p signal at the output, the drain of Q_1 can change from $-\frac{1V}{20}$ to $+\frac{1V}{20}$ (a total of 0.1V).

The sampling-time uncertainty is: $\frac{0.1}{5} \times 1.5 \text{ ns} = \underline{0.03 \text{ ns}}$

8.5)



8.6) During the sample mode:

$$\frac{V_o}{A} = V_{os} - V_o \Rightarrow V_o = \frac{A}{A+1} V_{os}$$

where V_{os} is the input offset voltage of the opamp.

During the sample mode, the voltage across C_H is $V_{in} - \frac{A}{A+1} V_{os}$

During the hold mode:

$$V_o = V_{in} - \frac{A}{A+1} V_{os} - \frac{V_o}{A} + V_{os}$$

$$\Rightarrow \underline{V_o = \frac{A}{A+1} V_{in} + \frac{A}{(A+1)^2} V_{os}}$$

8.7) Let's denote the end of Φ_1 by nT . The output voltage at the end of Φ_2 is equal to the output voltage (cont)

8.7) (cont.) at the end of ϕ_1 of the next period.

$$\text{Therefore: } V_o[nT] = \frac{C_2 V_o[nT-T] + C_1 V_{in}[nT-T]}{C_1 + C_2}$$

$$\text{Or, equivalently: } (C_1 + C_2) V_o[z] = C_2 z^{-1} V_o[z] + C_1 z^{-1} V_{in}[z]$$

$$\Rightarrow \frac{V_o(z)}{V_{in}(z)} = \frac{z^{-1}}{1 + \frac{C_2}{C_1} (1 - z^{-1})}$$

$$\text{Substituting } z = e^{j\omega T} \approx 1 + j\omega T :$$

$$\frac{V_o(e^{j\omega T})}{V_{in}(e^{j\omega T})} = \frac{1 - j\omega T}{1 + j\omega \frac{C_2}{C_1} T} \Rightarrow \underline{f_{3dB} = \frac{1}{2\pi} \frac{C_1}{C_2} f_{clk}}$$

8.8)

	V_{in}	V_1	V_2	V_3	V_{out}
track	1*	1.7	0.3	1	1
hold	$\begin{pmatrix} -1.1, \\ 3.1 \end{pmatrix}$	-0.4	2.4	1	1
track	-1	-0.3	-1.7	-1	-1
hold	$\begin{pmatrix} -3.1, \\ 1.1 \end{pmatrix}$	-2.4	0.4	-1	-1

* All numbers are in Volts.

8.9) Using (8.12): $\Delta V_{BE} = 0.0275 \times \ln 8 = \underline{57.2 \text{ mV}}$

$$\frac{\Delta V_{BE}}{\Delta V_T} = \frac{57.2 \text{ mV}}{320 \text{ } ^\circ\text{K}} = \underline{0.178 \frac{\text{mV}}{^\circ\text{K}}}$$

8.10) Using (8.21): $K = \frac{(1.206) + (1.3)(0.0275) - 0.65}{(0.0275) \ln(8)} = \underline{10.35}$

8.11) From (8.18): $K \frac{kT}{q} \ln\left(\frac{J_2}{J_1}\right) = \frac{T}{T_0} V_{CT0} + (m-1) \frac{kT}{q} - \frac{T}{T_0} V_{BE0-2}$

Substituting this into the last term of (8.16), we have:

$$V_{ref} = V_{CT0} + \frac{T}{T_0} (V_{BE0-2} - V_{CT0}) + (m-1) \frac{kT}{q} \ln\left(\frac{T_0}{T}\right) + \frac{T}{T_0} V_{CT0} + (m-1) \frac{kT}{q} - \frac{T}{T_0} V_{BE0-2}$$

$$\therefore V_{ref} = \underline{V_{CT0} + (m-1) \frac{kT}{q} \left(1 + \ln \frac{T_0}{T}\right)}$$

which proves (8.22).

$$\begin{aligned} \text{Also: } \frac{\partial V_{ref}}{\partial T} &= (m-1) \frac{k}{q} \left(1 + \ln \frac{T_0}{T}\right) + (m-1) \frac{kT}{q} \left(-\frac{1}{T}\right) \\ &= \underline{(m-1) \frac{k}{q} \ln \frac{T_0}{T}} \end{aligned}$$

which proves (8.23).

8.12) Using (8.34), we have:

$$V_{ref-0} = \left(1 + \frac{R_4}{R_5}\right) 1.24 \text{ V} = 2.5 \text{ V} \Rightarrow \underline{\frac{R_4}{R_5} = 1}$$

$$\text{Using (8.32): } \underline{\frac{R_1}{R_2} = 5.5}$$

Assuming $\underline{R_2 = 1 \text{ k}\Omega}$ & $\underline{R_5 = 10 \text{ k}\Omega}$ results in:

$$\underline{R_1 = 5.5 \text{ k}\Omega} \quad \& \quad \underline{R_4 = 10 \text{ k}\Omega}$$

$$\text{Also, } R_3 = \frac{R_2}{R_1} R_4 \parallel R_5 = \frac{1}{5.5} (15 \text{ k}) = \underline{900 \Omega}$$

$$8.13) \quad V_{BE1} - V_{BE2} = R_2 I_2 \quad \& \quad R_3 I_2 = R_1 I_1$$

$$V_{ref} = V_{BE1} + R_1 I_1 = V_{BE1} + R_3 I_2 = V_{BE1} + R_3 \frac{1}{R_2} (V_{BE1} - V_{BE2})$$

$$\begin{aligned} \Rightarrow V_{ref} &= V_{BE1} + \frac{R_3}{R_2} \frac{kT}{q} \ln \frac{I_1}{I_2} \\ &= \underbrace{V_{BE1} + \frac{R_3}{R_2} \frac{kT}{q} \ln \left(\frac{R_3}{R_1} \right)} \end{aligned}$$

which proves (8.41) for the circuit of Fig. 8.27(b).

8.14) Using (8.61):

$$\text{For a gain of } 0.5, \frac{I_2}{I_1} = 0.5 \Rightarrow I_1 = 2I_2 = \underline{400\mu A}$$

$$\text{" " " " } 4, \frac{I_2}{I_1} = 4 \Rightarrow I_1 = \frac{I_2}{4} = \underline{50\mu A}$$

Therefore, the extreme values of I_1 are $50\mu A$ & $400\mu A$.

The peak value of i_{o1} is limited to $I_1 - 20\mu A$.

since $I_1(\min) = 50\mu A$, i_{o1} is limited to $30\mu A$.

$$8.15) \quad \text{From (8.68): } i_o = \frac{i_1 i_2}{I_1}$$

$$\Rightarrow \ln i_o = \ln i_1 + \ln i_2 - \ln I_1$$

$$\Rightarrow \frac{\Delta i_o}{i_o} = \frac{\Delta i_1}{i_1} + \frac{\Delta i_2}{i_2} - \frac{\Delta I_1}{I_1}$$

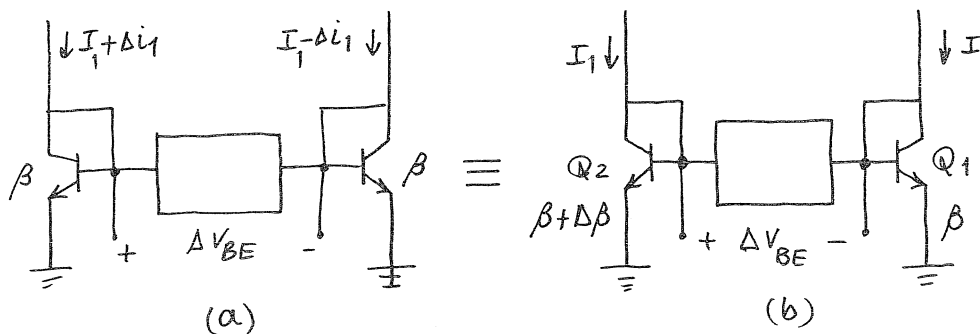
Assuming constant i_1 & i_2 to find Δi_o due to ΔI_1 :

$$\frac{\Delta i_o}{i_o} = \frac{-\Delta I_1}{I_1} \Rightarrow \Delta i_o = -i_o \frac{\Delta I_1}{I_1} = -i_o \frac{5\mu}{100\mu} = \underline{-0.05 i_o}$$

(cont.)

8.15) (cont.) Δi_o due to a change in I_2 is 0 since i_o is not a function of I_2 .

8.16)



$$\text{For (a): } \Delta V_{BE} = V_T \ln \frac{I_1 + \Delta i_1}{I_1 - \Delta i_1} = V_T \ln \left(1 + \frac{2\Delta i_1}{I_1 - \Delta i_1} \right)$$

$$\begin{aligned} \text{For (b): } \Delta V_{BE} &= V_T \ln \frac{I_{C2}}{I_{C1}} = V_T \ln \frac{(\beta + \Delta\beta)(\beta + 1)}{(\beta + \Delta\beta + 1)\beta} \\ &= V_T \ln \left(1 + \frac{\Delta\beta}{\beta(\beta + \Delta\beta + 1)} \right) \approx V_T \ln \left(1 + \frac{\Delta\beta}{\beta^2} \right) \end{aligned}$$

For the same ΔV_{BE} , we have:

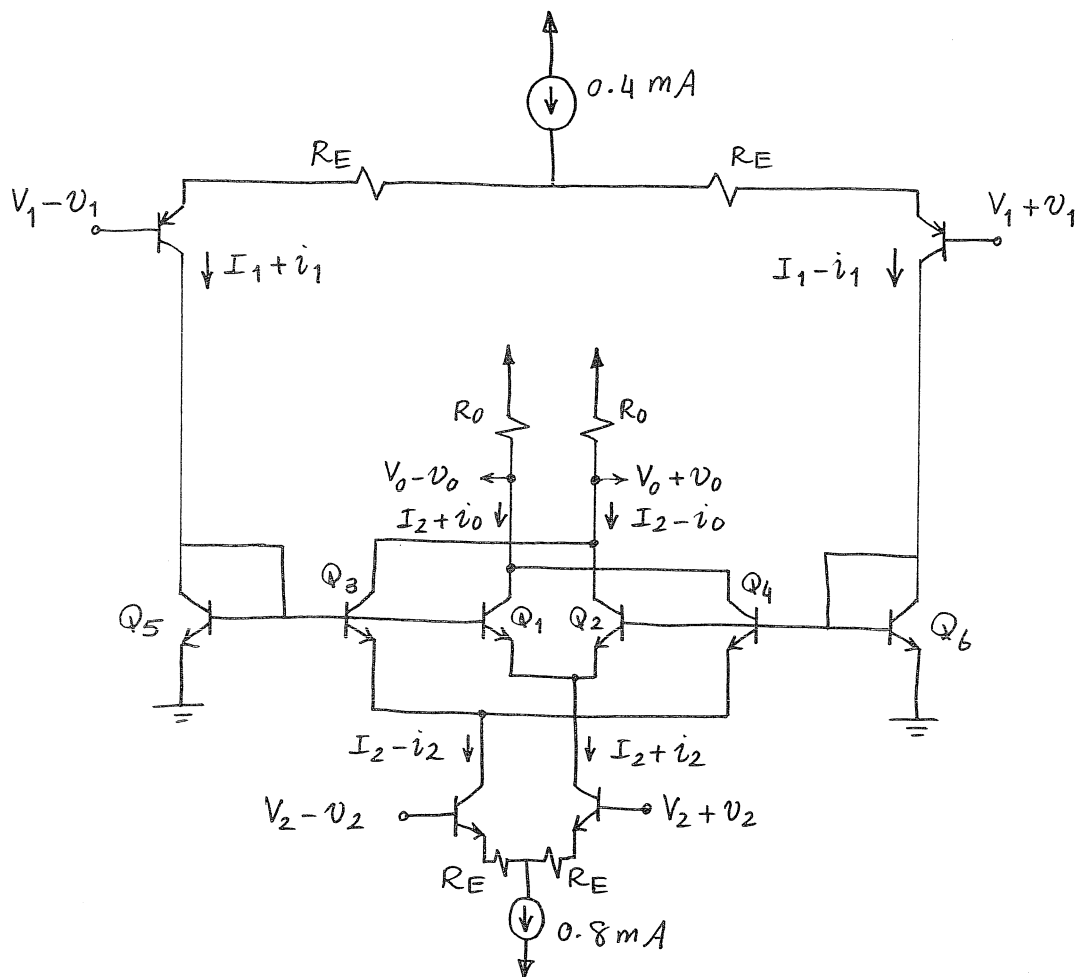
$$\frac{\Delta\beta}{\beta^2} \approx \frac{2\Delta i_1}{I_1}$$

In other words, a β -mismatch is equivalent to a

$$\Delta i_1 = \frac{I_1}{2} \frac{\Delta\beta}{\beta^2} \quad \text{Assuming } \frac{\Delta\beta}{\beta} = 0.05 \text{ results in}$$

$$\Delta i_o = \frac{0.05}{2\beta} i_2$$

8.17)



With the bias circuit shown, the bias currents of Q_1 to Q_6 are all 0.2 mA !

$$i_1 = \frac{v_1}{R_E + r_e} \quad \& \quad i_2 = \frac{v_2}{R_E + r_e}$$

Assuming $R_E \gg r_e$, we have: $i_1 = \frac{v_1}{R_E}$, $i_2 = \frac{v_2}{R_E}$

$$\text{Using (8.68): } i_0 = \frac{i_1 i_2}{I_1} = \frac{v_1 v_2}{R_E^2 I_1} \Rightarrow v_0 = R_0 \frac{v_1 v_2}{R_E^2 I_1}$$

$$v_1 = v_2 = 0.5 \text{ V} \Rightarrow v_0 = 0.5 \text{ V} \quad \therefore \quad 0.5 = R_0 \frac{0.25}{R_E^2 (0.2 \text{ mA})}$$

$$\text{Assuming } \underline{R_E = 2 \text{ k}\Omega} \Rightarrow \underline{R_0 = 1.6 \text{ k}\Omega}$$

Chapter 9

9.8 PROBLEMS

- 9.1 Show that if $x(n) \leftrightarrow X(z)$, then $x(n - k) \leftrightarrow z^{-k}X(z)$.
- 9.2 What is the maximum amount of downsampling that can be applied to $x_{s2}(n)$ in Example 9.1 before aliasing occurs?

- 9.3** Is downsampling a time-invariant operation? If so, explain. If not, give a simple counter example. Is downsampling a linear operation?
- 9.4** Repeat Example 9.3, but upsample by 8 (i.e., insert 7 zeros between samples).
- 9.5** Upsampling and downsampling are often used for converting between two sample rates. Give a block diagram showing how one might convert 0–20 kHz band-limited audio signal from a sampling rate of 50 kHz to 40 kHz.
- 9.6** Find the location of the poles and zeros of the transfer function given by $H(z)$ in (9.19).
- 9.7** Find the location of the pole required in a first-order filter that has its zero at ∞ such that its 3-dB frequency is at $f_s/10$.
- 9.8** Sketch the magnitude response (in dB) of the following transfer function, $H(z)$, by finding the gains at dc and at $\omega = \pi$, as well as the location of the 3-dB frequency.

$$H(z) = \frac{0.05z}{z - 0.95}$$

- 9.9** Consider the following difference equation, which describes a filter with input $u(n)$ and output $y(n)$.

$$y(n) = u(n) - 0.3y(n-1)$$

Determine $y(n)$ when $u(n)$ is an impulse (i.e., when the impulse is 1 at $n = 1$ and 0 otherwise) assuming $y(0) = 0$.

- 9.10** Find a first-order $H(z)$ that has a 3-dB frequency at $f_s/30$, a zero at -1 , and a dc gain of 1. Sketch the magnitude response from $\omega = 0$ to $\omega = 2\pi$.
- 9.11** Using a bilinear transform, find an $H(z)$ to realize a second-order transfer function with a dc gain of one, a maximally-flat passband (i.e., $Q = 1/\sqrt{2}$ in the continuous-time domain), and a -3 -dB frequency at 1 kHz when the sampling rate is 100 kHz.
- 9.12** Find the output sequence of a discrete-time integrator, $H(z) = 1/(z-1)$, when its input is a sinusoidal waveform of peak height 1 at $f_s/4$ and its initial state is zero. In other words, the input waveform is the series of numbers $\{1, 0, -1, 0, 1, 0, -1, \dots\}$. Using a frequency analysis, what is the magnitude and phase response for $H(z)$ at $f_s/4$? Check the frequency domain result with the time domain result.
- 9.13** A 100-Hz sinusoidal signal, $x(n)$, is digitally created using a 400-Hz sampling rate producing a sample-and-hold signal, $x_{sh}(t)$. Before any smoothing filter is applied and assuming the magnitude of $x_{sh}(t)$ at 100 Hz is $1 V_{rms}$, what is the magnitude of $x_{sh}(t)$ at 300 Hz? Also, find the magnitude of $x_{sh}(t)$ at 500 Hz and at 700 Hz.
- 9.14** Repeat Problem 9.13, but assume the 100-Hz signal is digitally created using a 10-kHz signal, and find the magnitude of $x_{sh}(t)$ at 9.9 kHz and at 10.1 kHz.
- 9.15** Repeat Example 9.8, but assume an input signal frequency of 100 Hz and find the magnitude of the output signal, $y_{sh}(t)$, at 100 Hz, 99.9 kHz, and 100.1 kHz.

- 9.16** A 0–5-kHz voiceband microphone is modelled as having a low-pass, first-order frequency response with a pole at 10 kHz. If the microphone output is sampled at 100 kHz, how much attenuation is required in the anti-aliasing filter for 80-dB noise rejection around 100 kHz?

Chapter 9 - Problems

9.1) Show that $x(n-k) \xleftrightarrow{z} z^{-k} X(z)$

The z -transform for $x(n]$ is

$$X(z) \triangleq \sum_{n=-\infty}^{\infty} x(n) z^{-n} \quad \text{for } z \triangleq e^{sT}$$
$$= \dots + x(-1) z^{-1} + x(0) z^0 + x(1) z^1 + \dots$$

Given that $x_2(n) = x(n-k)$,

$$X_2(z) = \sum_{n=-\infty}^{\infty} x_2(n) z^{-n} = \sum_{n=-\infty}^{\infty} x(n-k) z^{-n} \quad \square$$

Let $i = n-k$

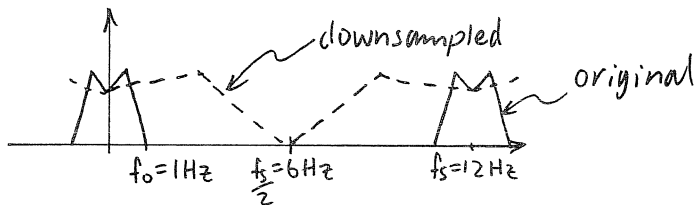
$\therefore n = i+k$ and we can rewrite $\square$ as

$$X_2(z) = \sum_{i=-\infty}^{\infty} x(i) z^{-(i+k)} = z^{-k} \underbrace{\sum_{i=-\infty}^{\infty} x(i) z^{-i}}_{= X(z)}$$

$$\therefore \underline{X_2(z) = z^{-k} X(z)}$$

Q.E.D.

9.2)



$\therefore$ bandwidth is 1 Hz , a min. sampling rate of 2 Hz is required.

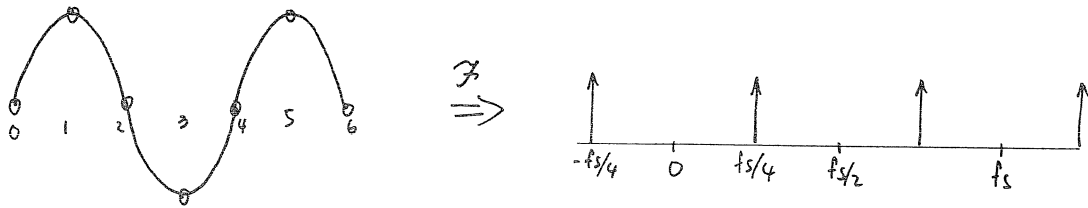
$\therefore$ The signal can be downsampled by a factor of

$$\frac{f_s}{f_{\min}} = \frac{12\text{ Hz}}{2\text{ Hz}} = \underline{6}$$

9.3) Downsampling is not time-invariant.

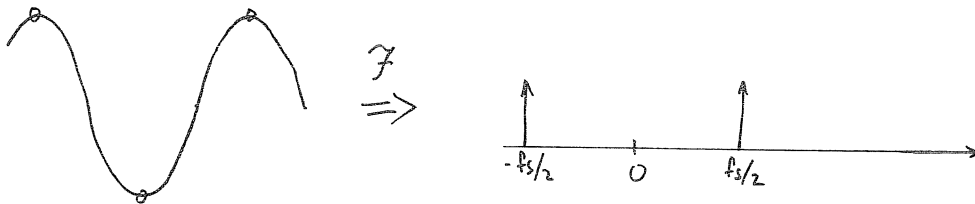
Counterexample:

Given the sampled sinusoid and its spectrum:



Consider the case where we downsample by two, done simply by discarding every other sample.

If we discard all the even samples we get, as expected,



However, by discarding all the odd samples, we are left with a constant vector of zeros, and effectively no signal.

As for linearity, downsampling is linear $\Leftrightarrow$

$$\text{Given } w(n) = ax(n) + by(n),$$

its downsampled equivalent, $w'(n)$, is given by

$$w'(n) = ax'(n) + by'(n)$$

where $x'(n)$ and $y'(n)$ are equivalently downsampled equivalents of $x(n)$ and $y(n)$.

Proof: For downsampling factor of L

$$w'(n) = w(nL+k) \quad n, L \in \mathbb{I}, k \in [0, L-1]$$

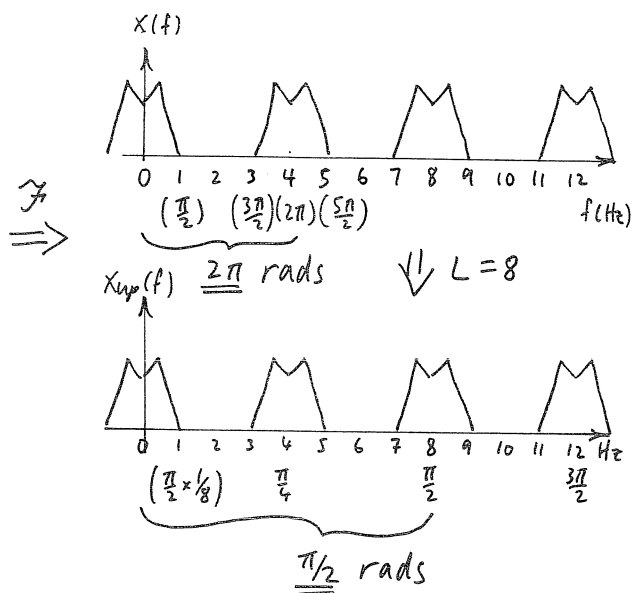
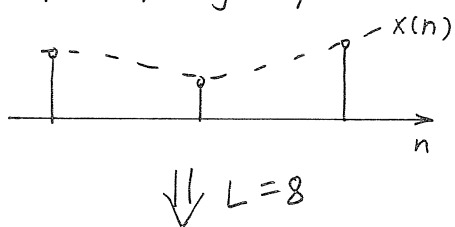
$$= a \underbrace{x(nL+k)}_{x'(n)} + b \underbrace{y(nL+k)}_{y'(n)}$$

But these are simply downsampled versions of $x(n)$ and $y(n)$, $x'(n)$ and $y'(n)$ respectively.

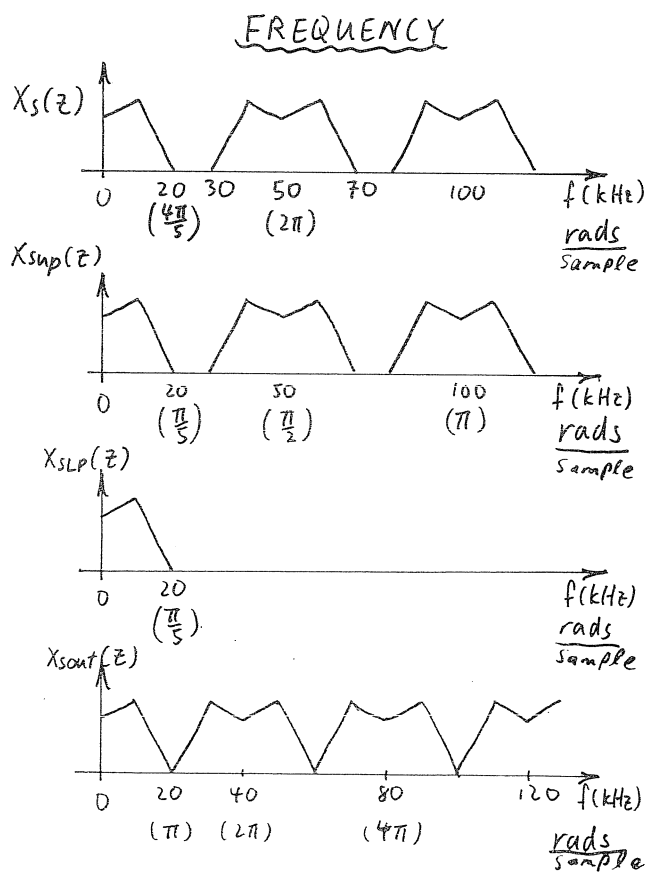
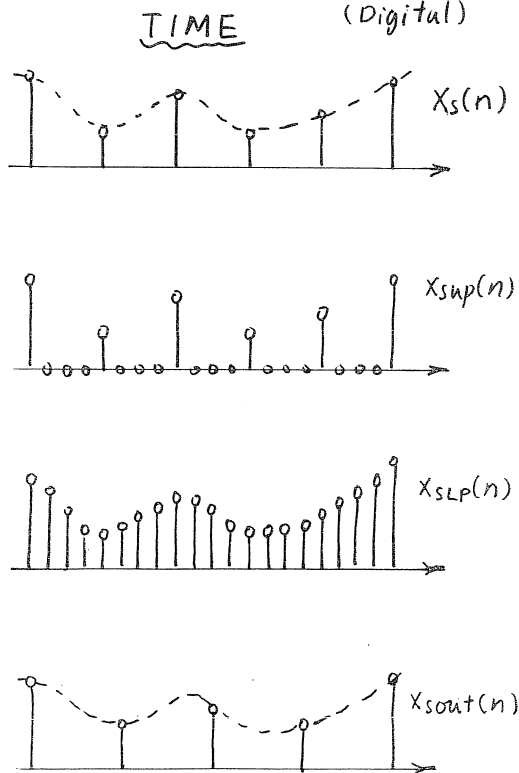
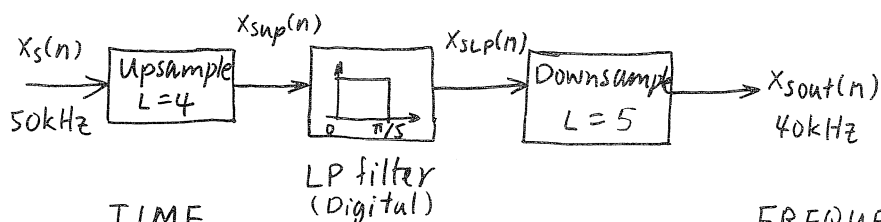
$$\therefore w'(n) = ax'(n) + by'(n)$$

Q.E.D.

9.4) Upsampling by 8 :



9.5) Block diagram of sample rate converter :



9.6) Find poles and zeros of

$$H(z) = \frac{0.05}{z^2 - 1.6z + 0.65}$$

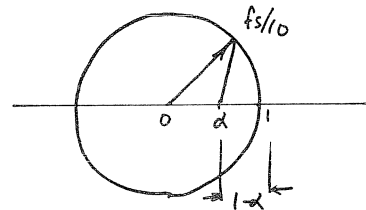
For poles : $z_p^2 - 1.6z_p + 0.65 = 0$

$$\therefore z_p = 0.8 \pm i0.1 = 0.806 \angle 7^\circ$$

By inspection, 2 zeros are at $z_z = \infty$.

9.7) Given $H(z) = \frac{K}{z - \alpha}$, $K, \alpha \in \mathbb{R}$

Find α such that $f_{-3dB} = f_s/10$.



At DC or $z = 1$

$$|H(z)|_{z=1} = \frac{K}{1-\alpha} \equiv |H(\omega)|_{\omega=0} \quad \square$$

Now $f_{-3dB} = f_s/10 \Rightarrow \omega_{-3dB} = \frac{2\pi}{10} = \pi/5$ rads/sec

$$\begin{aligned} \therefore |H(\omega_{-3dB})| &= |H(z)|_{z=e^{j\pi/5}} \\ &= \frac{K}{|e^{j\pi/5} - \alpha|} \quad \square \end{aligned}$$

Now set $|H(\omega_{-3dB})| = \frac{1}{\sqrt{2}} |H(z)|_{z=1}$

$$\begin{aligned} \therefore \frac{K}{|e^{j\pi/5} - \alpha|} &= \frac{1}{\sqrt{2}} \frac{K}{1-\alpha} \\ |e^{j\pi/5} - \alpha| &= \sqrt{2} (1-\alpha) \end{aligned}$$

$$\sqrt{(\cos(\pi/5) - \alpha)^2 + \sin^2(\pi/5)} = \sqrt{2} (1-\alpha)$$

$$\cos^2(\pi/5) - 2\alpha \cos(\pi/5) + \alpha^2 + \sin^2(\pi/5) = 2\alpha^2 - 4\alpha + 2$$

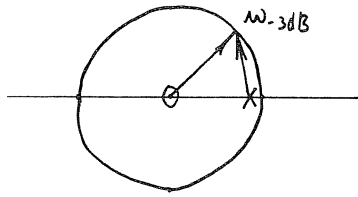
$$0 = \alpha^2 + \alpha(-4 + 2\cos(\pi/5)) + 2 - 1$$

$$0 = \alpha^2 - 2.382\alpha + 1$$

$$\therefore \alpha = \frac{1.837}{\text{unstable root}} \quad \text{or} \quad \boxed{0.544}$$

$\therefore$ the pole is located at $z = 0.544$

9.8) sketch $|H(z)|$ given $H(z) = \frac{0.05z}{z-0.95}$



∵ the zero is located at $z=0$,
it does not affect the
magnitude response

$$|H(z)|_{z=e^{j0}} = \frac{0.05}{0.05} = \underline{0 \text{ dB}}$$

$$|H(z)|_{z=e^{j\pi}} = \frac{-0.05}{-1.05} = \underline{-26.4 \text{ dB}}$$

For $\omega-3\text{dB}$,

$$|H(\omega-3\text{dB})| = \frac{1}{\sqrt{2}} |H(z)|_{z=e^{j0}}$$

$$\frac{0.05}{|e^{j\omega-3\text{dB}} - 0.95|} = \frac{1}{\sqrt{2}}$$

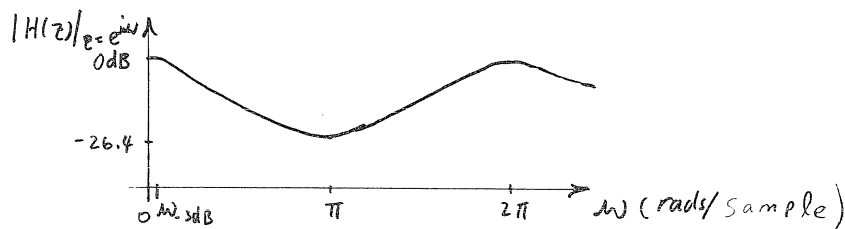
$$(\sqrt{2} \times 0.05)^2 = (\cos(\omega-3\text{dB}) - 0.95)^2 + \sin^2(\omega-3\text{dB})$$

$$0.005 = 1 - 1.9 \cos(\omega-3\text{dB}) + 0.95^2$$

$$\cos \omega-3\text{dB} = 0.99868$$

$$\therefore \underline{\omega-3\text{dB} = 0.051 \text{ rads/sample}}$$

And the frequency response is



9.9) Given $y(n] = u[n] - 0.3y[n-1]$, determine the impulse response.

$$\text{Now } u[n] = \begin{cases} 1 & n=1 \\ 0 & \text{otherwise} \end{cases}$$

$$y[0] = 0$$

$$y[1] = u[1] = 1$$

$$y[2] = -0.3y[1] = -0.3$$

$$y[3] = -0.3y[2] = (-0.3)^2$$

$\vdots$

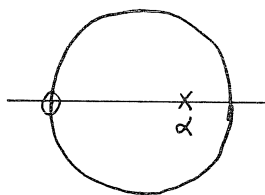
$$\therefore y[n] = \begin{cases} 0 & n=0 \\ (-0.3)^{n-1} & n>0 \end{cases} \quad \text{Impulse response}$$

9.10) Find $H(z)$.

DC gain = 1

From the basic information, we know that

$$H(z) = k \times \frac{z+1}{z-\alpha}$$



For k , $|H(z)|_{|z=1} = k \frac{2}{1-\alpha} \equiv 1$

$$\therefore k = \frac{1-\alpha}{2}$$

For α , $|H(z)|_{|z=e^{j\omega_{\text{dB}}}} = \frac{1}{\sqrt{2}} |H(z)|_{|z=e^{j0}}$ where $\omega_{\text{dB}} = \frac{2\pi}{30} = \frac{\pi}{15}$

$$\frac{1-\alpha}{2} \times \frac{|e^{j\pi/15} + 1|}{|e^{j\pi/15} - \alpha|} = \frac{1}{\sqrt{2}}$$

$$\frac{2}{(1-\alpha)^2} = \frac{(\cos \pi/15 + 1)^2 + \sin^2 \pi/15}{(\cos \pi/15 - \alpha)^2 + \sin^2 \pi/15} = \frac{3.9563}{\alpha^2 - 1.9563\alpha + 1}$$

$$0 = (3.9563 - 2)\alpha^2 + 2 \times (1.9563 - 3.9563)\alpha + (3.9563 - 2)$$

$$0 = \alpha^2 - 2.0447\alpha + 1$$

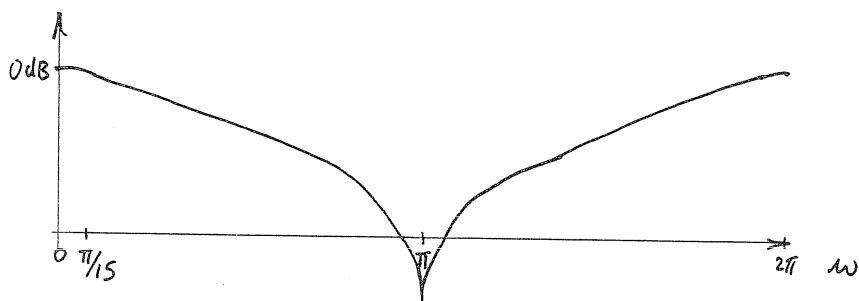
$$\therefore \alpha = 0.8098$$

and $k = \frac{1-0.8098}{2} = 0.095$

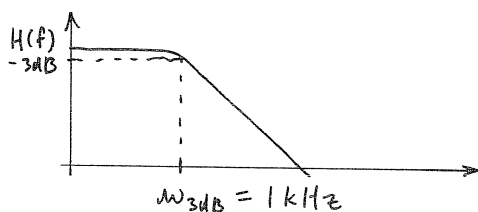
(cont.)

9.10) (cont.)

$$\therefore H(z) = \frac{0.095(z+1)}{z-0.8098}$$



9.11)



$$f_s = 100 \text{ kHz}$$

$$Q = 1/\sqrt{2}$$

$\omega_{3dB} = 1 \text{ kHz}$ corresponds to $\frac{2\pi}{100} = \pi/50$ rads/sample in the z -plane

The design is done in the p -plane where ω_{3dB} is mapped to

$$\Omega_{3dB} = \tan(\pi/50 \times 1/2)$$

$$\Omega_{3dB} = 0.03143 \text{ rads/sec}$$

$$\begin{aligned} \therefore H_p(p) &= \frac{\Omega_{3dB}^2}{p^2 + \frac{\Omega_{3dB}}{Q} p + \Omega_{3dB}^2} = \frac{0.03143^2}{p^2 + \frac{0.03143}{1/\sqrt{2}} p + 0.03143^2} \\ &= \frac{9.88 \times 10^{-4}}{p^2 + 4.44 \times 10^{-2} p + 9.88 \times 10^{-4}} \end{aligned}$$

Convert this function to the z -plane using $p = \frac{z-1}{z+1}$

$$\begin{aligned} \therefore H(z) &= 9.88 \times 10^{-4} \left[\left(\frac{z-1}{z+1} \right)^2 + 4.44 \times 10^{-2} \left(\frac{z-1}{z+1} \right) + 9.88 \times 10^{-4} \right]^{-1} \\ &= \frac{9.88 \times 10^{-4} (z+1)^2}{(1 + 9.88 \times 10^{-4} + 4.44 \times 10^{-2}) z^2 + (2 \times 9.88 \times 10^{-4} - 2) z + (1 - 4.44 \times 10^{-2} + 9.88 \times 10^{-4})} \end{aligned}$$

$$\therefore H(z) = \frac{9.45 \times 10^{-4} (z+1)^2}{z^2 - 1.91z + 0.915}$$

9.12) Given the integrator transfer function,

$$\frac{Y(z)}{X(z)} = \frac{1}{z-1}$$

determine the output to a sinusoidal input.

$$\frac{Y(z)}{X(z)} = \frac{1}{z-1}$$

$$\therefore zY(z) - Y(z) = X(z)$$

$$\Rightarrow \underline{y(n+1) = x(n) + y(n)}$$

Assume $y(0) = 0$

$$\therefore y(1) = x(0) + y(0) = 1 + 0 = 1$$

$$y(2) = 0 + 1 = 1$$

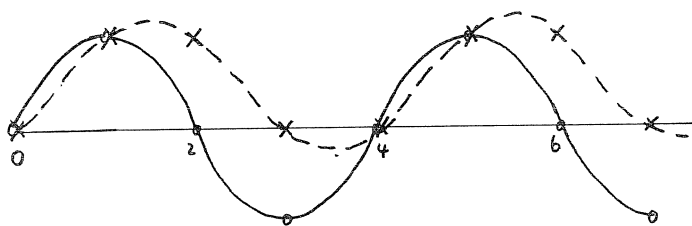
$$y(3) = -1 + 1 = 0$$

$$y(4) = 0 + 0 = 0$$

$$y(5) = 1 + 0 = 1$$

$$y(6) = 0 + 1 = 1$$

$$y(7) = -1 + 1 = 0 \dots$$



Time Domain

$$H(z)|_{z=e^{j\pi/2}} = \frac{1}{j+1} = \frac{1}{\sqrt{2}} \angle -45^\circ$$

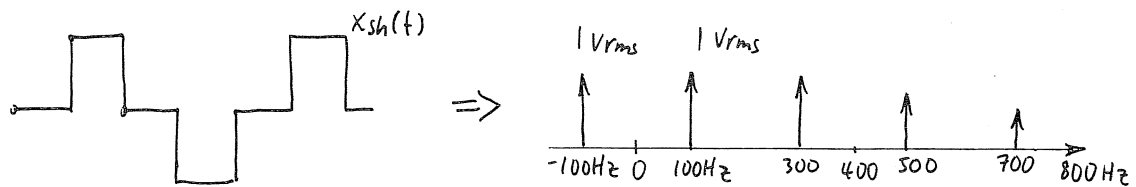
frequency domain

Interpolating the output samples, we see a phase shift of 45° and the amplitude looks about 3dB lower.

Thus our time and frequency domain results are consistent.

The dc shift is due to the initial state being 0.

9.13)



$$H_{SH}(f) \Big|_{f=100\text{Hz}} = T \frac{|\sin(\pi f/f_s)|}{\pi f/f_s} \Big|_{f=100\text{Hz}} = T \cdot \frac{|\sin(\frac{100\pi}{400})|}{100\pi/400}$$

$$= \underline{0.9003 T \equiv 1 V_{rms}}$$

$$H_{SH}(f) \Big|_{f=300\text{Hz}} = T \frac{|\sin(3\pi/4)|}{3\pi/4} = 0.3001 T = \underline{0.33 V_{rms}}$$

$$H_{SH}(f) \Big|_{f=500\text{Hz}} = T \frac{|\sin(5\pi/4)|}{5\pi/4} = 0.1801 T = \underline{0.2 V_{rms}}$$

$$H_{SH}(f) \Big|_{f=700\text{Hz}} = T \frac{|\sin(7\pi/4)|}{7\pi/4} = 0.1286 T = \underline{0.14 V_{rms}}$$

9.14) 100 Hz signal is now created with a sample rate of 10 kHz



∴ We have 100 points/period

$$H_{SH}(f) = T \frac{|\sin(\pi f/f_s)|}{\pi f/f_s}$$

$$H_{SH}(f) \Big|_{f=100\text{Hz}} = T \times \frac{|\sin(100\pi/10000)|}{100\pi/10000} = \underline{0.9998 T = 1 V_{rms}}$$

For the images,

$$H_{SH}(f) \Big|_{f=9.9\text{kHz}} = T \frac{|\sin(9900\pi/10000)|}{9900\pi/10000} = 1.01 \times 10^{-2} T = \underline{10.1 mV_{rms}}$$

$$H_{SH}(f) \Big|_{f=10.1\text{kHz}} = T \frac{|\sin(10100\pi/10000)|}{10100\pi/10000} = 0.99 \times 10^{-2} T = \underline{9.9 mV_{rms}}$$

9.15) Repeat Example 9.8 with $f_{in} = 100 \text{ Hz}$

For $|H(z)|$,

$$z_{in} = e^{j(2\pi \times 100/50000)} = e^{j\pi/250} = 0.99992 \pm j0.01257$$

$$\therefore |H(z_{in})| = \frac{0.2}{|(0.99992 - 0.8) + j0.01257|} = \frac{0.2}{0.20032} = \underline{0.9984}$$

This is the same for the images.

For the sample-and-hold circuitry,

$$H_{SH}(f)|_{f=100\text{Hz}} = T \frac{|\sin(100\pi/50000)|}{100\pi/50000} = 0.99999T$$

$$H_{SH}(f)|_{f=99.9\text{kHz}} = T \frac{|\sin(99900\pi/50000)|}{99900\pi/50000} = 1.001 \times 10^{-3}T$$

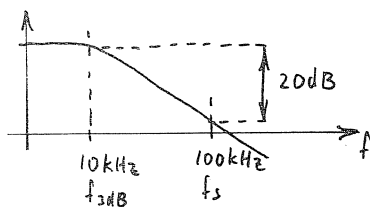
$$H_{SH}(f)|_{f=100.1\text{kHz}} = T \frac{|\sin(100100\pi/50000)|}{100100\pi/50000} = 9.99 \times 10^{-4}T$$

$$\therefore |y_{SH}(t)|_{f=100\text{Hz}} = 0.9984 \times 0.99999 \times \frac{1}{K} \times 1.0 \text{ Vrms} = \underline{998 \text{ mVrms}}$$

$$|y_{SH}(t)|_{f=99.9\text{kHz}} = 0.9984 \times 1.001 \times 10^{-3} \times 1 \text{ Vrms} = \underline{0.999 \text{ mVrms}}$$

$$|y_{SH}(t)|_{f=100.1\text{kHz}} = 0.9984 \times 9.99 \times 10^{-4} \times 1 \text{ Vrms} = \underline{0.997 \text{ mVrms}}$$

9.16)



$$|H(\omega)|_{\omega=100\text{kHz}} = \frac{1}{|10j + 1|} = \frac{1}{10.05} = \underline{-20.0 \text{ dB}}$$

- $\therefore$ 20 dB of attenuation is already provided by the microphone at 100 kHz.
- $\therefore$ Another 60 dB of attenuation is required from the anti-aliasing filter to obtain 80 dB noise rejection.

Chapter 10

10.10 PROBLEMS

- 10.1** Assuming logic signals have infinitely fast fall and rise times but are delayed by 1 ns through a logic gate, draw the output waveforms of the two phase clock generators shown in Fig. 10.3(b). Assume the delay blocks add delays of 10 ns each and a clock frequency of 10 MHz is applied to the input.
- 10.2** Ignoring the effect of parasitic capacitances, find the discrete-time transfer function of the switched-capacitor circuit shown in Fig. P10.2.

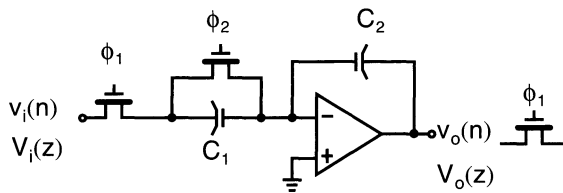


Fig. P10.2

- 10.3** Repeat Problem 10.2 but include the effect of parasitic capacitances.
- 10.4** Show that when an opamp has a finite gain of A , the transfer function for the discrete-time integrator of Fig. 10.5 is given by

$$\frac{V_o(z)}{V_i(z)} = -\left(\frac{C_1}{C_2}\right) \left(\frac{1}{z \left(1 + \frac{C_1}{C_2 A} \right) - 1} \right)$$

Also show that this transfer function has a dc gain of $-A$ and a pole that is located slightly to the left of 1 by an amount approximately equal to $(C_1/C_2)(1/A)$.

- 10.5 Repeat Example 10.3 but assume the discrete-time zero is at $z = 0$ and the pole remains at $z = 0.53327$. Note that with this change, C_1 is no longer required at the sacrifice of a nonzero gain being obtained at 50 kHz. What is the new gain at 50 kHz?
- 10.6 Find the capacitances needed for the first-order filter shown in Fig. 10.15(a) to realize a discrete-time transfer function having a dc gain of one, a zero at 0, and a 3-dB frequency of 1 kHz when a clock frequency of 50 kHz is used. Assume that $C_A = 50$ pF. What is the gain (in dB) at 25 kHz?
- 10.7 Find the transfer function of the circuit shown in Fig. 10.16 when $C_1 = 0$ pF, $C_2 = 2$ pF, $C_3 = 2$ pF, and $C_A = 20$ pF. What is the magnitude and phase of the gain at dc, $f_s/4$ and $f_s/2$?
- 10.8 Find the discrete-time transfer function of the switched capacitor circuit shown in Fig. P10.8.

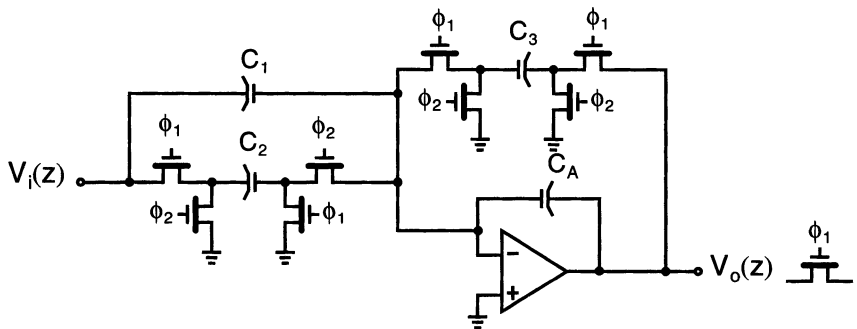
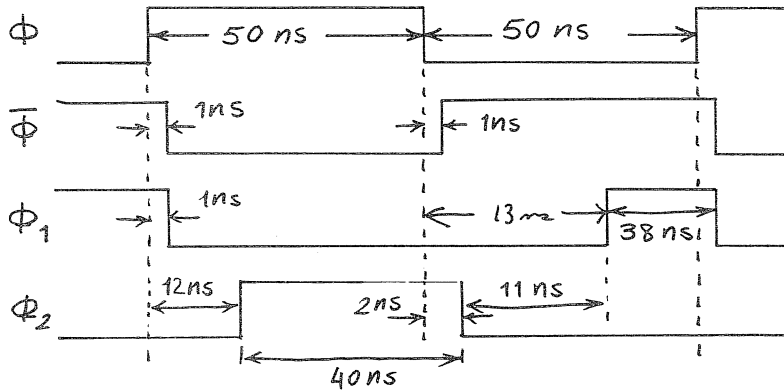


Fig. P10.8

- 10.9 Show that when $K_6 = 0$ in the biquad circuit of Fig. 10.21, if the poles are complex, they lie precisely on the unit circle (i.e., the circuit is a resonator).
- 10.10 Using the bilinear transform, design a bandpass filter with a $Q = 20$ (in the continuous-time domain) and a peak gain of one near $f_s/100$. Find the largest to smallest capacitor ratio if this transfer function is realized using the high- Q biquad circuit, and compare it to that which would be obtained if the low- Q circuit were used instead. Let $C_1 = C_2 = 1$ in both cases.
- 10.11 Sketch a typical output waveform for the capacitive-reset gain circuit of Fig. 10.31 if the opamp has an input offset of 100 mV and a gain of 10 is desired.
- 10.12 Verify that Eq. (10.104) gives the frequency of oscillation for the VCO of Fig. 10.40.

Chapter 10 - Problems

- 10.1) The clock period is $\frac{1}{10 \text{ MHz}} = 100 \text{ ns}$. Assuming 50 % duty cycle, we have the following timing diagram:



(not to scale)

10.2) $C_2 V_o(n) = C_2 V_o(n-1) - C_1 V_i(n)$

$$\Rightarrow C_2 V_o(z) = C_2 z^{-1} V_o(z) - C_1 V_i(z) \Rightarrow \frac{V_o(z)}{V_i(z)} = \frac{-C_1/C_2}{1 - z^{-1}}$$

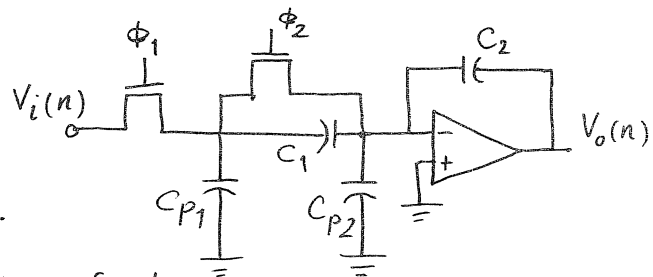
- 10.3) C_{P2} is always discharged since its voltage is virtually ground.

During Φ_1 , C_{P1} is charged to $V_i(n) C_{P1}$.

This charge will be transferred

to C_2 during Φ_2 . Therefore: $C_2 V_o(n) = C_2 V_o(n-1) - C_{P1} V_i(n-1) - C_1 V_i(n)$

$$\Rightarrow \frac{V_o(z)}{V_i(z)} = - \frac{\frac{C_1}{C_2} + \frac{C_{P1}}{C_2} z^{-1}}{1 - z^{-1}}$$



10.4) A finite gain of A implies the voltage of the inverting terminal is $-\frac{1}{A} V_o(n)$ if the output voltage is $V_o(n)$.

$$C_2 V_o(n) \left[1 + \frac{1}{A}\right] = C_2 V_o(n-1) \left[1 + \frac{1}{A}\right] - C_1 \left[V_i(n-1) + \frac{V_o(n)}{A}\right]$$

Note that at the end of Φ_2 , the voltage across C_1 , which is $V_i(n-1)$, reduces to $-\frac{V_o(n)}{A}$, not ground.

Also, assuming $\frac{1}{A} \ll 1$, we have

$$C_2 V_o(n) = C_2 V_o(n-1) - C_1 \left(V_i(n-1) + \frac{V_o(n)}{A}\right)$$

$$\Rightarrow \frac{V_o(z)}{V_i(z)} = \frac{-C_1/C_2}{z(1 + \frac{C_1}{C_2 A}) - 1}$$

At dc ($z=1$), this gain is equal to $\frac{-C_1/C_2}{1 + \frac{C_1}{C_2 A} - 1} = -A$

$$Z_P = \frac{1}{1 + \frac{C_1}{C_2 A}} = 1 - \frac{C_1}{C_2 A} \quad \text{assuming } C_1 \ll C_2 A.$$

$$10.5) \quad H(z) = \frac{Kz}{z - 0.53327}$$

$$\text{Setting } H(1) = 1 \Rightarrow H(z) = \frac{0.46673z}{z - 0.53327}$$

Equating $H(z)$ with that of Eq. (10.33) and assuming

$$C_A = 10 \text{ pF} \Rightarrow \underbrace{C_1 = 0} \quad \underbrace{C_2 = -8.752 \text{ pF}} \quad , \quad \underbrace{C_3 = 8.752 \text{ pF}}$$

$$\text{The new gain at } 50 \text{ kHz} = \underline{\underline{H(-1) = -0.304 = -10.3 \text{ dB}}}$$

$$10.6) \quad \omega_{3dB} = \frac{1 \text{ kHz}}{50 \text{ kHz}} \times 2\pi = 0.04\pi \text{ RAD/SAMPLE}$$

Since the zero is at 0 rather than -1, we cannot use a bilinear transform. From example 9.5 in chapter 9, we have

$$\omega_{3dB} = \cos^{-1} \left(2 - \frac{a}{2} - \frac{1}{2a} \right)$$

This example assumed a zero at ∞ which has same magnitude response as a zero at 0.

$$\cos(0.4\pi) = 2 - \frac{a}{2} - \frac{1}{2a} \Rightarrow a = 0.8821$$

$$H(z) = \frac{kz}{z - 0.8821} \quad \text{Forcing } H(1) = 1 \Rightarrow k = 0.1179$$

Equating coeff with (10.33) results in

$$\underline{C_1 = 0}, \quad \underline{-C_2 = C_3 = 6.683 \text{ pF}}$$

$$25 \text{ kHz corresponds to } z = -1, \quad H(-1) = 0.0626 = \underline{-24.1 \text{ dB}}$$

10.7) The transfer function is given by (10.33). Substituting the given capacitances, we have:

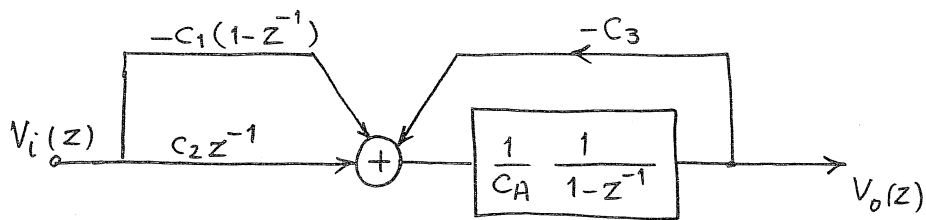
$$\underline{H(z) = \frac{0.1z}{1.1z - 1}}$$

$$\text{At dc: } z=1, \quad \underline{H(1) = 1}, \quad \cancel{H(1) = 0}$$

$$\text{At } f_{s/4}: z=j, \quad \underline{H(j) = 0.673 \angle -42.27^\circ}$$

$$\text{At } f_{s/2}: z=-1, \quad \underline{H(-1) = 0.0476 \angle 0^\circ}$$

10.8) The equivalent signal flow graph is:



$$\left\{ V_i(z) [C_2 z^{-1} - C_1 + C_1 z^{-1}] - C_3 V_o(z) \right\} \frac{1}{C_A} \frac{1}{1-z^{-1}} = V_o(z)$$

$$\Rightarrow H(z) = \frac{V_o(z)}{V_i(z)} = - \frac{C_1 - (C_1 + C_2) z^{-1}}{(C_A + C_3) - C_A z^{-1}}$$

10.9) The transfer function poles are the zeros of the denominator.

Using (10.49) with $K_6 = 0$, we have

$$z^2 + (K_4 K_5 - 2)z + 1 = 0$$

Since all the coefficients are real numbers, this equation has, in general, two complex conjugate roots, z_1 & z_1^* , with their product equal to the constant term of the equation (i.e. 1).

$$\therefore z_1 z_1^* = 1 \Rightarrow |z_1|^2 = 1 \Rightarrow |z_1| = 1$$

$\therefore$ both z_1 & z_1^* lie on the unit circle.

$$10.10) \quad \omega_0 = \frac{2\pi}{100} = 0.062832 \text{ RAD/SAMPLE} \quad \& \quad Q = 20$$

$$\text{Equivalent } \Omega_0 = \tan\left(\frac{\omega_0}{2}\right) = 0.0314263 \text{ RAD/S}$$

$$\therefore H_a(s) = \frac{\frac{\omega_0}{Q} s}{s^2 + \frac{\omega_0}{Q} s + \omega_0^2} = \frac{0.0015713 s}{s^2 + 0.0015713 s + 0.00098761}$$

Transform using the bilinear transform

$$s = \frac{z-1}{z+1}$$

$$H(z) = \frac{0.001572 z^2 - 0.001572}{1.003145 z^2 - 1.9991875 z + 1}$$

FOR Low-Q biquad from (10.51) - (10.56)

$$k_3 = -0.001572 \quad k_2 = 0.003144$$

$$k_6 = 0.003145 \quad k_4 = k_5 = 0.06291$$

$$k_1 = 0$$

FOR HIGH-Q biquad from (10.69) - (10.74)

$$H(z) = \frac{0.001567 z^2 - 0.001567}{z^2 - 1.99292 z + 0.99687}$$

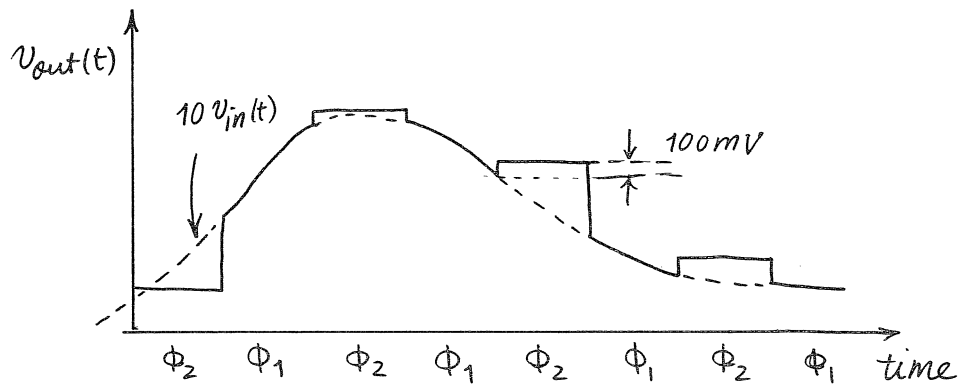
$$k_3 = 0.001567 \quad k_1 = 0 \quad k_2 = 0.0499$$

$$k_4 = k_5 = 0.06285 \quad k_6 = 0.0498$$

Ignoring k_3 (since it only sets zero at -1)

CAR ratio for low-Q is 318 while it is 20 for high-Q biquad.

10.11)



10.12) Assuming $V_{out} = -V_{ss}$,

$$\Delta V_x = -K_2 V_{ss} = -K_{in} V_{in}$$

the positive jump of V_x is still $K_1(V_{ss} + V_{DD})$ as before.

$$\Rightarrow T_{osc} = 2 \frac{K_1 (V_{ss} + V_{DD})}{K_2 V_{ss} + K_{in} V_{in}} T$$

Or, equivalently (assuming $V_{ss} = V_{DD}$)

$$f_{osc} = \frac{1}{4} \left(\frac{K_2}{K_1} + \frac{K_{in} V_{in}}{K_1 V_{DD}} \right) f$$

Chapter 11

11.6 REFERENCES

- Analog Devices. *Analog-Digital Conversion Handbook*. Prentice Hall, Englewood Cliffs, New Jersey, 1986.
- A. B. Grebene. *Bipolar and MOS Analog Integrated Circuit Design*. John Wiley & Sons, New York, 1984.
- R. van de Plassche. *Integrated Analog-to-Digital and Digital-to-Analog Converters*. Kluwer Academic Publishers, Dordrecht, the Netherlands, 1994.

11.7 PROBLEMS

- 11.1** For an ideal 10-bit, unipolar D/A converter with $V_{\text{LSB}} = 1 \text{ mV}$, what is the largest output voltage?
- 11.2** What is the SNR for an ideal 12-bit unipolar A/D converter with $V_{\text{ref}} = 3 \text{ V}$, when a sinusoidal input of $1 V_{\text{pp}}$ is applied? What size input would result in an SNR of 0 dB?
- 11.3** Find the equivalent of (11.19) for 2's-complement coding.
- 11.4** Show that, by using 4-bit 2's-complement words, the correct final sum is obtained when the numbers +5, +5, and -7 are added together while any overflow effects are ignored.
- 11.5** Starting with two 4-bit 2's-complement words, we want to add +5 and +7 to obtain the correct answer of +12 with a 5-bit word. Show how an extra bit can be added at the left of each of the 4-bit words such that numbers up to ± 15 can be represented. This approach is called *sign extension* and can be used to increase the word size of any number.
- 11.6** What is the representation of +8 and -8 in 5-bit 2's complement? Assuming a circuit added only one of these two numbers to an arbitrary 5-bit word, show a simple logic circuit (i.e., nand, nor, xnor, etc.) that would accomplish such an addition. (*Hint*: Note that the 4 LSBs in +8 and -8 are all 0 and thus do not need to be added.)
- 11.7** The following measurements are found from a 3-bit unipolar D/A converter with $V_{\text{ref}} = 8 \text{ V}$: (−0.01, 1.03, 2.02, 2.96, 3.95, 5.02, 6.00, 7.08). In units of LSBs, find the offset error, gain error, maximum DNL, and maximum INL.
- 11.8** How many bits of absolute accuracy does the converter in Problem 11.7 have? How many bits of relative accuracy does it have?
- 11.9** A 10-bit A/D converter has a reference voltage, V_{ref} , tuned to 10.24 V at 25 °C. Find the maximum allowable temperature coefficient in terms of $(\mu\text{V})/^{\circ}\text{C}$ for the reference voltage if the reference voltage is allowed to cause a maximum error of $(\pm 1/2)$ LSB over the temperature range of 0 to 50 °C.
- 11.10** Consider the following measured voltage values for a 2-bit D/A with a reference voltage of 4 V:
- $\{00 \leftrightarrow 0.01 \text{ V}\} \quad \{01 \leftrightarrow 1.02 \text{ V}\} \quad \{10 \leftrightarrow 1.97 \text{ V}\} \quad \{11 \leftrightarrow 3.02 \text{ V}\}$

In units of LSB, find the offset error, gain error, worst absolute and relative accuracies, and worst differential nonlinearity. Restate the relative accuracy in terms of an N -bit accuracy.

- 11.11** Find the maximum magnitude of quantization error for a 12-bit A/D converter having V_{ref} equal to 5 V and 0.5-LSB absolute accuracy.
- 11.12** What sampling-time uncertainty can be tolerated for a 16-bit A/D converter operating on an input signal from 0–20 kHz ?

Chapter 11 - Problems

11.1) Largest voltage output = $(2^N - 1) V_{LSB}$
= $(2^{10} - 1) 1 \text{ mV}$
= 1.023 V

11.2) Calculate SNR.

Maximum possible SNR = $6.02N + 1.76 \text{ dB}$
= $6.02 \times 12 + 1.76 \text{ dB}$
= 74 dB

This is provided the input $V_{pp} = V_{ref}$. In our case,

$$V_{ppin} = \frac{1}{3} V_{ref}$$

∴ we lose $20 \log(1/3) = 9.5 \text{ dB}$ in SNR

∴ SNR = 74 - 9.5 dB = 64.5 dB

For a SNR of 0 dB, the input signal needs to be 74 dB below its full scale level.

∴ $V_{ppin} = 3V / 10^{(74/20)} = 3/5012 = \underline{0.6 \text{ mV}_{pp}}$

For comparison, $V_{LSB} = 3V / 2^{12} = \underline{0.73 \text{ mV}}$

∴ The first quantization threshold is at $\frac{1}{2} V_{LSB} = 0.36 \text{ mV}$
and the signal will only span two quantization levels.

11.3) For 2's complement coding, we recognize that it is obtained from offset-binary coding by complementing the MSB.

∴ the output is represented by

$$V_{out} = V_{ref} [(1-b_1) 2^{-1} + b_2 2^{-2} + \dots + b_n 2^{-N}] - 0.5 V_{ref}$$

$$\underline{V_{out} = V_{ref} [-b_1 2^{-1} + b_2 2^{-2} + \dots + b_n 2^{-N}]}$$

11.4)	<u>Decimal</u>		<u>Two's Complement</u>	
	+5	→	0101	
	<u>+5</u>	→	<u>0101</u>	
	10	=	1010	← represents -6
	<u>-7</u>	→	<u>1001</u>	(temporarily incorrect)
	<u>+3</u>	=	0011	✓

∴ summation of 2's complement words results in the correct answer.

11.5)	<u>Decimal</u>	<u>Two's Complement</u>
<u>Example 1</u>	+5	0 0101
	<u>+7</u>	0 0111
	+12	0 1100
<u>Example 2</u>	-5	1 1011
	<u>-7</u>	1 1001
	-12	1 0100
<u>Example 3</u>	-5	1 1011
	<u>+7</u>	0 0111
	+2	0 0010

From the three examples, we see that we simply need to copy the MSB to increase the word size of Two's complement numbers. Math is performed as before.

11.6)

Two's Complement

+ 8

01000

- 8

11000

∴ An adder that only added one of these to another number would do the following:

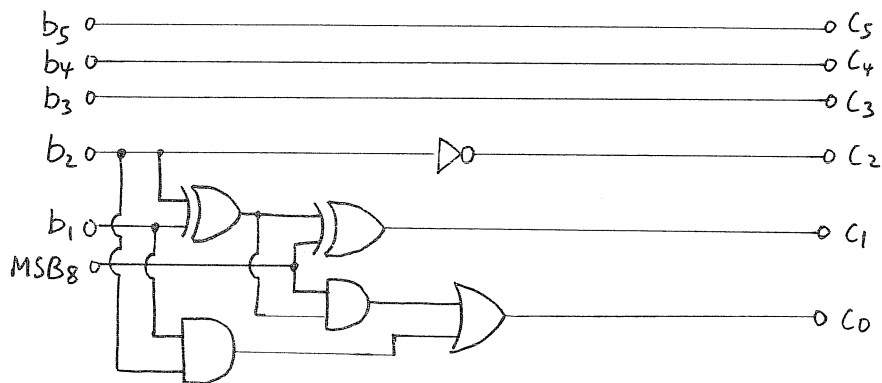
X	b ₁	b ₂	b ₃	b ₄	b ₅
+8	0	1	0	0	0
SUM	c ₀	c ₁	c ₂	c ₃	c ₄

It is clear that $C_5 = b_5$

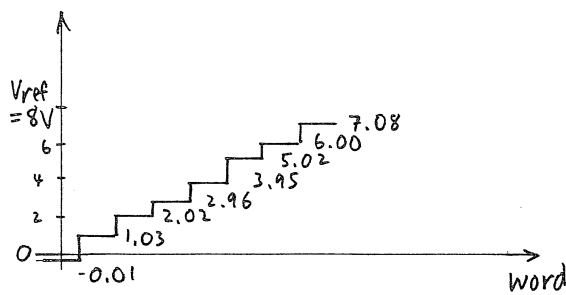
$$C_4 = b_4$$
$$C_3 = b_3$$
$$C_2 = \overline{b_2}$$
$$C_0 C_1 = b_1 + b_2 + \text{MSB}_8 \Rightarrow$$

this operation requires a full-adder circuit.

Thus the final circuit is:



11.7)



$$V_{LSB} = \frac{8V}{2^3} = 1V$$

$$\text{Err.off} = -0.01 \text{ LSB}$$

$$\begin{aligned} \text{Err.gain} &= \text{real full scale (in LSB)} - \text{ideal full scale value (in LSB)} \\ &= \frac{7.08V - (-0.01)}{1V} - 7 \text{ LSB} \end{aligned}$$

$$\text{Err.gain} = 0.09 \text{ LSB}$$

Compensating for offset and gain errors, the new steps in LSB's are given by

$$L_i = \frac{V_i}{V_{LSB}} - \text{Err.off} - \text{Err.gain} \times \frac{i}{2^N - 1}, \quad i = 0, \dots, 2^N - 1$$

$= 0, \dots, 7$

$$\therefore \begin{cases} L_0 = 0 \text{ LSB} & L_7 = 7 \text{ LSB} \\ L_1 = 1.027 \text{ LSB} & L_6 = 5.933 \text{ LSB} \\ L_2 = 2.004 \text{ LSB} & L_5 = 4.966 \text{ LSB} \\ L_3 = 2.931 \text{ LSB} & L_4 = 3.909 \text{ LSB} \end{cases} \begin{array}{l} \text{guaranteed} \\ \text{data-dependent} \end{array}$$

From these results, the integral non-linearity errors (INL) are

$$\{ 0 : +0.027 : +0.004 : -0.069 : \underline{-0.091} : -0.034 : -0.067 : 0 \} \text{ LSB}$$

maximum INL is -0.091 LSB

and the differential Non-linearity errors (DNL) are

$$\{ +0.027 : -0.023 : \underline{-0.073} : -0.022 : +0.057 : -0.033 : 0.067 \} \text{ LSB}$$

maximum DNL is -0.073 LSB

11.8) Find absolute and relative accuracies.

The absolute errors are :

$$\{-0.01, +0.03, +0.02, -0.04, -0.05, +0.02, +0.00, \underline{\underline{0.08}}\} \text{ V}$$

Thus, the largest deviation is 80mV maximum and this corresponds to 1 LSB when $V_{ref} = 8 \text{ V}$

i.e.,
$$\frac{8 \text{ V}}{2^{N_{effabs}}} = 80 \text{ mV}$$

$$2^{N_{effabs}} = 100$$

$$\underline{N_{effabs} = 6.6 \text{ bits}}$$

For relative accuracy, the maximum INL error is -0.091 LSB or -91 mV

$$\therefore \frac{8 \text{ V}}{2^{N_{effrel}}} = 91 \text{ mV}$$

$$2^{N_{effrel}} = 87.9$$

$$\underline{N_{effrel} = 6.5 \text{ bits}}$$

Thus, the converter has an absolute and relative accuracy of 6.6 bits and 6.5 bits respectively.

$$11.9) \quad V_{LSB} = \frac{V_{ref}}{2^N} = \frac{10.24V}{2^{10}} = 10mV$$

∴ We need to keep errors within $\pm \frac{1}{2} V_{LSB} = \pm 5mV$
 Now a change in the reference voltage causes the greatest error at the full range level

$$V_{full\ range} = (1 - 2^{-10}) V_{ref}$$

$$\Rightarrow \text{Err}_{full\ range} = (1 - 2^{-10}) \text{Err}_{ref}$$

$$\therefore \underline{\text{Err}_{ref}} = \frac{\text{Err}_{full\ range}}{1 - 2^{-10}} = \pm \frac{5mV}{0.999} = \underline{\underline{\pm 5.005mV}}$$

Maximum temperature coefficient

$$= \frac{\text{Err}_{max} - \text{Err}_{min}}{\text{Temp}_{max} - \text{Temp}_{min}} = \frac{2 \times 5.005mV}{50^\circ C} = \underline{\underline{200 \mu V/^\circ C}}$$

$$11.10) \quad V_{LSB} = \frac{V_{ref}}{2^N} = \frac{4V}{2^2} = 1V$$

$$\underline{\text{Err}_{offset}} = 0.01 \text{ LSB}$$

$$\begin{aligned} \text{Err}_{gain} &= \frac{V_{11} - V_{10}}{V_{LSB}} - V_{11\ ideal} = \frac{3.02 - 0.01}{1} - 3 \text{ LSB} \\ &= \underline{\underline{0.01 \text{ LSB}}} \end{aligned}$$

Absolute errors : $\{+0.01, +0.02, \underline{\underline{-0.03}}, +0.02\}$

∴ Maximum absolute error is $\underline{\underline{-0.03V = -0.03 \text{ LSB}}}$

For worse relative accuracy error, find INL errors using

$$L_i = \frac{V_{out}}{V_{LSB}} - \text{Err}_{offset} - \text{Err}_{gain} \times i/3$$

$$L_{00} = 0 \text{ LSB}$$

$$L_{11} = 3 \text{ LSB}$$

$$L_{01} = 1.007 \text{ LSB}$$

$$L_{10} = 1.953 \text{ LSB}$$

∴ INL errors = $\{0 : +0.007 : \underline{\underline{-0.047}} : 0\}$

∴ Max. relative error is $\underline{\underline{-0.047 \text{ LSB}}}$ which corresponds to 6.4 bits accuracy.

11.11) With an ideal converter, the maximum quantization error is

$$\frac{1}{2} V_{\text{LSB}} = \frac{1}{2} \frac{5V}{2^{12}} = 0.61 \text{ mV}.$$

However, given an absolute accuracy of 0.5 LSB , we add $0.5 V_{\text{LSB}}$ to the quantization error

$\therefore$ the maximum error is now $1 V_{\text{LSB}} = \underline{1.22 \text{ mV}}$

11.12) From Eq. (11.28),

$$\Delta t < \frac{1}{2^N \pi f_{\text{in}}} = \frac{1}{2^{16} \pi 20 \text{ kHz}}$$

$$\Delta t < 0.24 \text{ nsec}$$

$\therefore$ the sampling time uncertainty should be less than 0.24 nsec .

Chapter 12

12.6 PROBLEMS

- 12.1 Derive an expression for the number of switches in a general N-bit resistor-string D/A converter similar to that shown in Fig. 12.1.
- 12.2 Assume that an 8-bit resistor-string D/A converter with digital decoding (see Fig. 12.3) has a total resistor-string resistance of $400\ \Omega$, that its pass transistors have an on resistance of $400\ \Omega$, and that the drain-source capacitances to

ground of its pass transistors are 0.1 pF. Ignoring all other effects and using the open-circuit time-constant approach, estimate the worst-case settling time to 0.1 percent.

- 12.3** Derive an expression for the number of switches in an N-bit folded resistor-string D/A converter similar to that shown in Fig. 12.4 (Assume N is even).
- 12.4** Assume that the first resistor string of a 10-bit multiple-R-string D/A converter must match to 0.1 percent whereas the second string must match to 1.6 percent since the converters realize the top 4 bits and lower 6 bits, respectively. For $V_{\text{ref}} = 5 \text{ V}$, how much offsets in the opamps can be tolerated?
- 12.5** For a binary-weighted 10-bit resistor D/A converter (see Fig. 12.7), assume that R_F is chosen such that the output goes from 0 to $V_{\text{ref}} - V_{\text{LSB}}$. What is the resistor ratio between the largest and smallest resistors? What is the ratio between the currents through the switches for b_1 and b_{10} ?
- 12.6** It is desired to realize a binary-weighted 4-bit resistor D/A converter, as shown in Fig. 12.7, that must be linear to 10 bits. Ignoring all other nonidealities except for resistance mismatch error, what percentage matching accuracy is required for each of the $b_2 - b_4$ resistors relative to the b_1 resistor?
- 12.7** Consider an 8-bit D/A converter built using binary-weighted capacitors, where capacitor tolerances are ± 0.5 percent. What would be the worst-case differential nonlinearity in units of LSBs, and at what transition does it occur?
- 12.8** Consider the reduced-resistance-ratio approach shown in Fig. 12.8, where a single series resistor is applied to an N-bit converter, and where N is an even number. What is the improvement in the resistance ratio compared to using all binary-scaled resistances?
- 12.9** Draw the circuit for a reduced-resistance-ratio 8-bit D/A converter (see Fig. 12.8), where a resistor is inserted between b_4 and b_5 . Ignoring R_F , what is the resistance spread? Now draw a similar circuit, but add resistors between b_2 and b_3 as well as between b_6 and b_7 . What is this circuit's resistance spread?
- 12.10** Assuming all the switches in Fig. P12.10 are MOSFETS and are scaled so that each has 100 mV across the drain source when on (including s_4 , which always remains on), show that no accuracy is lost.
- 12.11** For the 4-bit R-2R-ladder D/A converter shown in Fig. P12.10, what is the output error (in LSBs) when $R_A = 2.01 R_B$? What is the output error when $R_C = 2.01 R$?
- 12.12** Show that the D/A converter circuit shown in Fig. 12.11 operates correctly. Estimate the speed of this circuit if the opamp has infinite bandwidth, $R = 10 \text{ k}\Omega$, and all nodes have a capacitance of 0.5 pF to ground.
- 12.13** Consider the D/A converter shown in Fig. 12.11, where $I = 1 \text{ mA}$, $R_f = 2 \text{ k}\Omega$ and $R = 10 \text{ k}\Omega$. If this converter is perfectly linear but is found to have an offset error of 0.15LSB and a gain error of 0.2LSB, find the output levels for the inputs $b_1 b_2 b_3 b_4$ equal to 0000, 1000, and 1111.

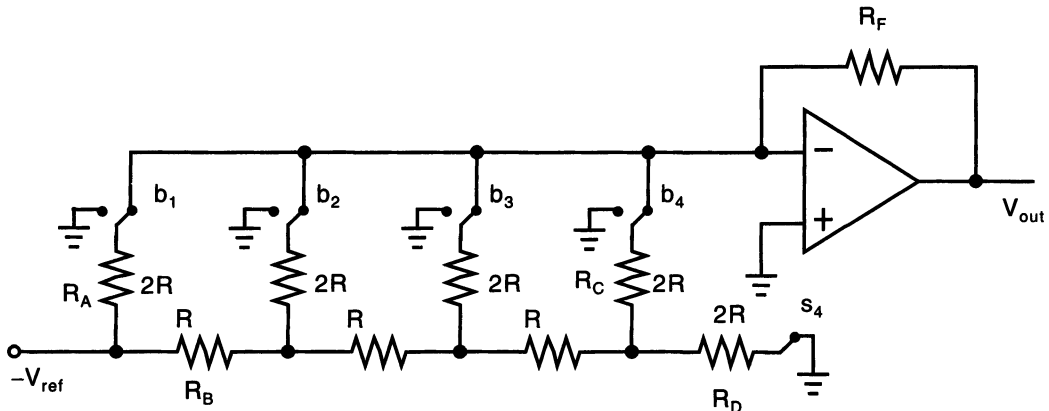
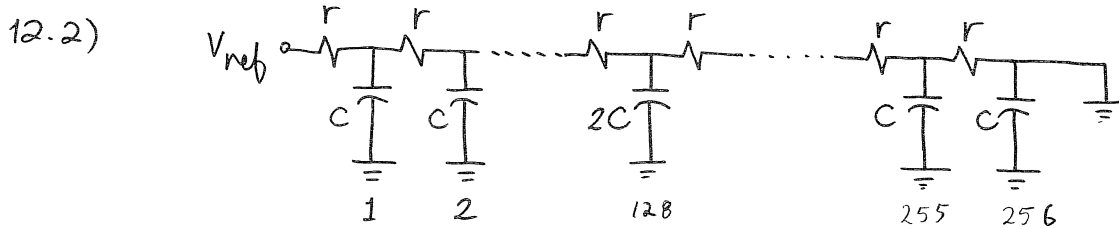


Fig. P12.10

- 12.14** A 12-bit binary-weighted D/A converter produces a glitch voltage of 0.5 V when the MSB is changed. What is the glitch voltage reduced to if a thermometer-code approach is used for the top 4 bits, whereas binary-weighting is used for the remaining 8 bits?
- 12.15** For the circuit in Fig. 12.17, sketch a typical waveform that would occur on the node connecting Q_1 , Q_2 , and Q_3 when the code signals, d_i and $\bar{d}_i$, are nonoverlapping. Repeat this sketch if the code signals are overlapping. Assume the code signals have logic swings from 0 to -5 V and that the threshold voltage for the transistors equals 1 V.
- 12.16** For the circuit in Fig. 12.18, sketch a typical waveform that would occur on the node connecting Q_1 , Q_2 , and Q_3 when the code signal, d_i , goes low-to-high and high-to-low. Assume here that the code signal has a logic swing of 0 to 5 V, that the threshold voltage for the transistors equals -1 V, and that V_{bias} is 1 V.
- 12.17** A D/A converter is realized using dynamically matched current sources, as shown in Fig. 12.20. Assuming all the transistors are ideal, find W/L for the Q_1 needed to set $V_{\text{GS}} = 3$ V when $I_{\text{ref}} = 50$ μA , $V_t = 1$ V, and $\mu_n C_{\text{ox}} = 92$ $\mu\text{A}/\text{V}^2$. If switch S_1 causes a random charge injection voltage of 1 mV, what is the expected percentage of random variation of the current being held on Q_1 ?
- 12.18** Repeat Problem 12.17 if the design does not incorporate the $0.9I_{\text{ref}}$ extra current source (i.e., if Q_1 must be the source for all of I_{ref}).

Chapter 12 - Problems

12.1) The total number of switches is $\sum_{i=1}^N 2^i = \underline{2(2^N - 1)}$



In the equivalent circuit shown, $r = \frac{400}{256} \Omega$, $C = 0.1 \text{ pF}$.

$2C$ represents the capacitance of the only switch that is ON.

This switch is considered to be in the middle of the string for the worst case time constant.

$$\begin{aligned}
 \tau &= \sum_{i=1}^{255} [(ir) \parallel (256-i)r]C + C(128r \parallel 128r) \\
 &= rc \sum_{i=1}^{255} \frac{i(256-i)}{256} + 64rc \\
 &= rc \sum_{i=1}^{255} i - \frac{rc}{256} \sum_{i=1}^{255} i^2 + 64rc \\
 &= rc \frac{(255)(256)}{2} - \frac{rc}{256} \frac{(255)(256)(510+1)}{6} + 64rc = 10987rc \\
 &= 10987 \times \frac{400}{256} \times 0.1 \text{ p} = \underline{1.7 \text{ ns}}
 \end{aligned}$$

The settling time to 0.1% is $\underline{7\tau = 12. \text{ ns}}$

12.3) The total number of switches is $2^{N/2} \cdot 2^{N/2} + 2^{N/2} = \underline{2^N + 2^{N/2}}$

12.4) For the output opamp, the offset that can be tolerated is:

$$\frac{1}{2} \frac{0.1}{100} \times V_{ref} = \frac{1}{2} \frac{1}{1000} \times 5V = \underline{2.5 \text{ mV}}$$

For the two opamp in the middle, the offset must be less than

$$\frac{1}{2} \times 2.5 \text{ mV} \times 64 = \underline{80 \text{ mV}}$$

Note that the offset introduced from the middle opamps will be divided by 64 ($=2^6$). The " $\frac{1}{2}$ " factor accounts for two opamps in the middle.

12.5) The ratio between the largest and the smallest resistor is

$$\frac{2^{10} R}{2R} = 2^9 = \underline{512}$$

the current ratio is the same as above.

12.6) The matching accuracy required for the b_2 resistor, b_3 resistor, and b_4 resistor is 2 times, 4 times, and 8 times the matching accuracy of b_1 resistor, respectively.

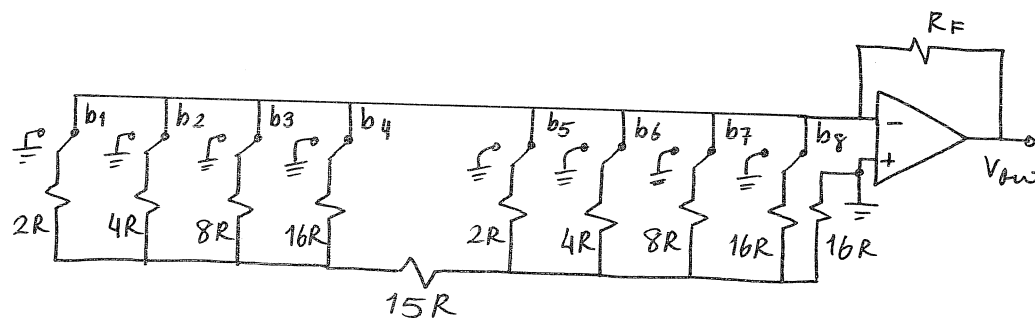
- 12.7) The worst case DNL happens at the transition from nominal 7LSB to 8LSB, assuming -0.5% error for C , $2C$, and $4C$, and $+0.5\%$ error for $8C$.

$$DNL = 8LSB(1.05) - 7LSB(0.95) - \text{LSB} = \underline{0.75 \text{ LSB}}$$

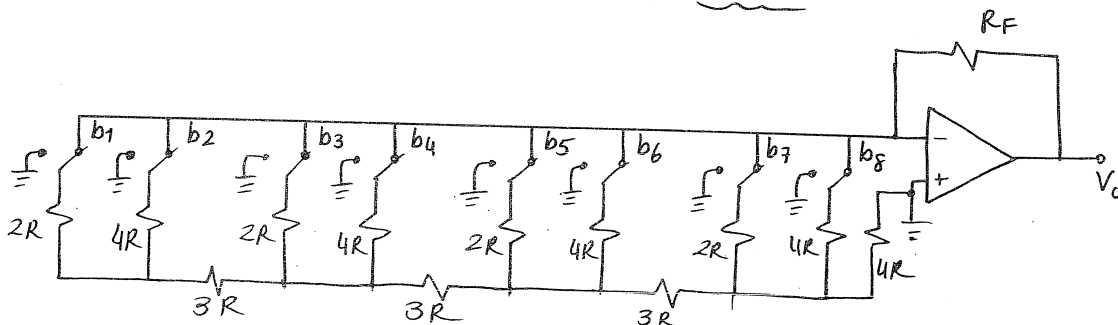
- 12.8) The largest resistance, in this case, is $2^{N/2} R$ while the smallest resistance is $2R$! Therefore, the resistance ratio is $2^{(N/2-1)}$. Noting the resistance ratio for a binary-scaled A/D is 2^{N-1} , we have:

$$\text{Resistance ratio improvement} = \underline{2^{N/2} \text{ times}}$$

12.9)

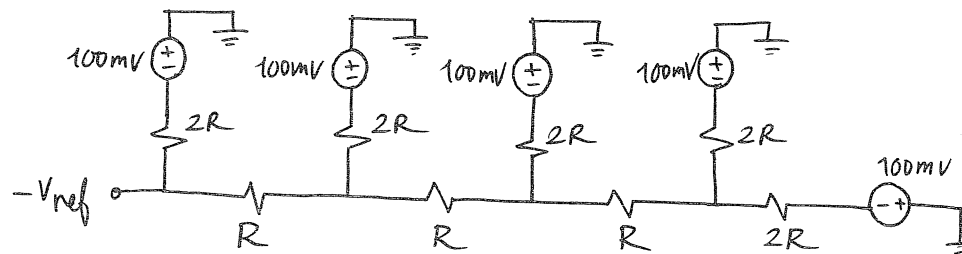


$$\text{Resistance Ratio} = 16R/(2R) = \underline{8}$$

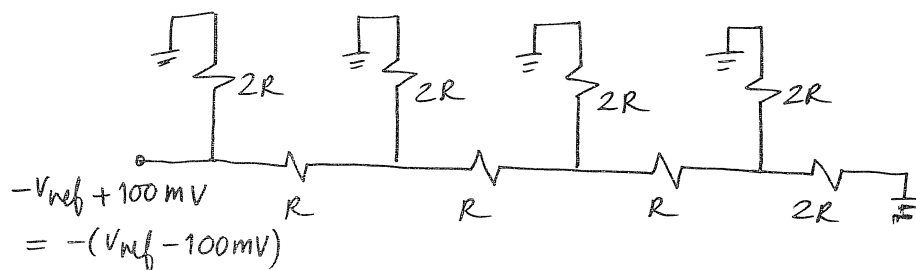


$$\text{Resistance Ratio} = 4R/(2R) = \underline{2}$$

12.10) The equivalent circuit for current calculation in $2R$ - branches is shown below:



The 100-mV voltage sources model the voltage drop of the switches. One end of $2R$ -resistor is connected to -100 mV . Therefore, they can all be connected together and the 100 mV voltage-source can be moved and added to the V_{ref} source. This will not change the current calculations of $2R$ -resistors.



∴ In effect, V_{ref} has decreased by 100 mV and the circuit still operates as if there is no voltage drop across the switches.

12.11) If $R_A = 2.01 R_B$, the output error is:

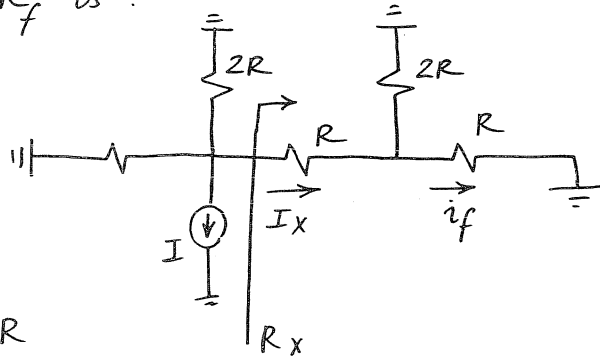
$$\left(\frac{2}{2.01} - 1 \right) \times 16 \text{ LSB} \approx \underline{0.08 \text{ LSB}}$$

If $R_C = 2.01 R$, the output error is:

$$\left(\frac{2}{2.01} - 1 \right) \times 1 \text{ LSB} = \underline{0.005 \text{ LSB}}$$

It's obvious that precision for R_A is more important than the precision for R_C .

12.12) Assuming only $b_1 = 1$, the current drawn through R_f is I .
 " " $b_2 = 1$, " " " " " " $I/2$.
 " " $b_3 = 1$, the equivalent circuit to find the current through R_f is:



$$R_x = R + \frac{2R}{3} = \frac{5}{3} R$$

$$I_x = I \frac{R}{R + \frac{5}{3} R} = \frac{3}{8} I$$

$$i_f = \frac{2R}{2R + R} I_x = \underline{\frac{1}{4} I}$$

With similar analysis, it can be shown that the current through b_4 & R_f is $\frac{I}{8}$. Therefore,

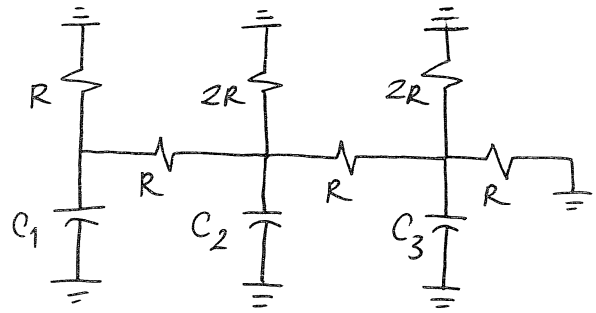
$$\underline{V_o = 2R_f I (2^{-1} b_1 + 2^{-2} b_2 + 2^{-3} b_3 + 2^{-4} b_4)}$$

(Cont.)

12.12) (cont.) The equivalent circuit for the open circuit time-constant analysis is shown:

where

$$R = 10 \text{ K}\Omega, C = 0.5 \text{ pF}$$



$$\tau = C_3 \frac{R}{2} + C_2 \frac{5R}{8} + C_1 \frac{21R}{32} = 1.78125 RC = 8.9 \text{ nsec}$$

$$\Rightarrow \underline{\omega_{3dB} \approx \frac{1}{\tau} = 2\pi \times 17.9 \text{ MHz.}}$$

12.13) Assuming $R_f = 2 \text{ K}\Omega \Rightarrow 8 \text{ LSB} = 2 \text{ V} \Rightarrow 1 \text{ LSB} = 0.25 \text{ V}$

For "0000" input, $V_o = (0 + 0.15) \text{ LSB} = \underline{0.0375 \text{ V}}$

For "1000" input, $V_o = (8 + 0.15 + \frac{0.2}{2}) \text{ LSB} = \underline{2.0625 \text{ V}}$

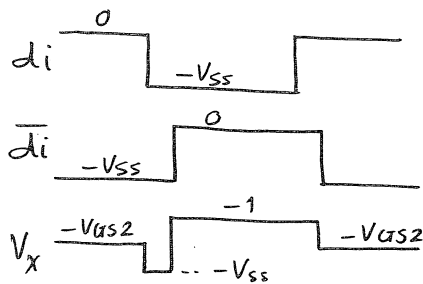
For "1111" input, $V_o = (15 + 0.15 + 0.2) \text{ LSB} = \underline{3.8375 \text{ V}}$

12.14) $\frac{0.5 \text{ V}}{16} = 0.031 \text{ V} = \underline{31 \text{ mV}}$

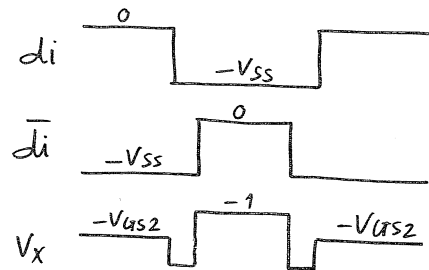
12.15) Let's denote the voltage of the node connecting Q_1 , Q_2 , & Q_3 by V_x . The following table shows V_x as a function of d_i & $\bar{d}_i$.

d_i	$\bar{d}_i$	V_x
0	0	-1
0	$-V_{ss}$	$-V_{gs2}$
$-V_{ss}$	0	-1
$-V_{ss}$	$-V_{ss}$	$-V_{ss}$

The V_x wave forms:

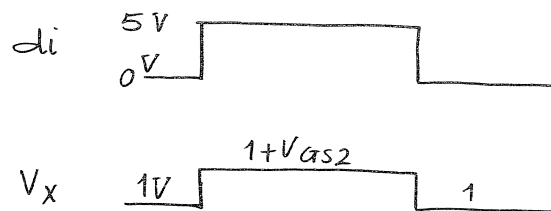


overlapping d_i & $\bar{d}_i$



non-overlapping d_i & $\bar{d}_i$

12.16) Let's denote the voltage of the node connecting Q_1 , Q_2 , and Q_3 by V_x .



12.17) Using (1.67) with $I_D = 0.1 I_{ref} = 5 \mu A$, we have

$$5 \mu = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_t)^2 = (46 \mu) \frac{W}{L} (3-1)^2$$

$$\Rightarrow \underline{\underline{\frac{W}{L} = 0.027}}$$

$$\Delta I_D = g_m \Delta V_{GS}$$

$$\text{where } g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t) = 96 \mu \times 0.027 \times 2 = 5 \frac{\mu A}{V}$$

$$\& \Delta V_{GS} = 1 mV$$

$$\Rightarrow \Delta I_D = 5 \frac{\mu A}{V} \times 1 mV = \underline{\underline{5 nA}}$$

12.18) $50 \mu = (46 \mu) \frac{W}{L} (3-1)^2 \Rightarrow \underline{\underline{\frac{W}{L} = 0.27}}$

$$g_m = \frac{2 I_D}{V_{GS} - V_t} = \frac{100 \mu}{2 V} = 50 \frac{\mu A}{V}$$

$$\Rightarrow \Delta I_D = g_m \Delta V_{GS} = 50 \frac{\mu A}{V} \times 1 mV = \underline{\underline{50 nA}}$$

Chapter 13

13.11 PROBLEMS

- 13.1 What is the worst-case conversion time for an 18-bit dual-slope integrating A/D converter when the clock rate is 5 MHz?
- 13.2 Consider an 18-bit integrating A/D converter, as shown in Fig. 13.1, where V_{ref} equals 10 volts, $C_1 = 100$ pF, and a clock frequency of 1 MHz is used. What value of R_1 should be chosen such that the opamp output never exceeds 10 volts when $0 \text{ V} < V_{\text{in}} < 10 \text{ V}$?
- 13.3 Derive the equivalent of (13.8) for Fig. 13.1 when the opamp has an input-offset voltage of V_{off1} and the comparator has an input-offset voltage of V_{off2} .

- 13.4** What is the offset error (in LSBs) of the converter described in Problem 13.2 when the integrating opamp has an input-offset voltage of 20 mV?
- 13.5** What input-signal frequencies are completely attenuated by a dual-slope 16-bit integrating A/D having a clock frequency of 1 MHz? For this same converter, what is the attenuation of an input signal at 60 Hz?
- 13.6** Repeat Problem 13.5 with the converter's clock frequency being equal to 100 kHz.
- 13.7** Consider a 4-bit unipolar DAC-based successive-approximation A/D converter, as shown in Fig. 13.5. Find the sequence of the D/A converter's output levels for an input of 3.333 volts when $V_{\text{ref}} = 8$ V. What is the final digital output?
- 13.8** Consider a 4-bit unipolar charge-redistribution A/D converter, as shown in Fig. 13.7, where $V_{\text{ref}} = 8$ V. Find the sequence of the voltage level for V_x if the input signal is 3.333 volts.
- 13.9** Repeat Problem 13.8, assuming that a parasitic capacitance of 8 C is connected between the node at V_x and ground. Is the final digital result affected?
- 13.10** Show a method for modifying the unipolar charge-redistribution A/D converter shown in Fig. 13.7 such that the threshold offsets of $0.5V_{\text{LSB}}$ are correctly realized.
- 13.11** How is the digital code obtained from the signed charge redistribution A/D converter shown in Fig. 13.8 converted to a 2's complement code?
- 13.12** What value of error voltage V_{x1} would be measured in Fig. 13.11(a) if the total array capacitance equals 64 pF, the MSB (i.e., b_1) capacitor equals 31.5 pF, and there is a parasitic capacitance to ground of 10 pF on node V_x ? For the same circuit, show that the error voltage, V_{e1} , that occurs when using the MSB capacitor during a normal conversion equals $0.5V_{x1}$.
- 13.13** For the same circuit as described in Problem 13.12, what value of error voltage V_{x2} would be measured in Fig. 13.11(b) if the MSB-1 (i.e., b_2) capacitor equals 16.4 pF? Show that the error voltage, V_{e2} , that occurs when using the MSB-1 capacitor during a normal conversion equals $0.5(V_{x2} - V_{e1})$.
- 13.14** Assuming switch resistances are all about 1 k Ω , estimate the settling time needed for a 12-bit charge-redistribution A/D converter where the total array capacitance equals 128 pF.
- 13.15** Draw a block diagram similar to that for Fig. 13.22 for a 10-bit two-step A/D converter where the first stage determines 4 bits. Indicate the accuracy needed in all the blocks.
- 13.16** Show that the circuit shown in Fig. 13.25 results in all time constants being equal when each of the latches has the same input capacitance and the amplifiers have zero output impedance.
- 13.17** Consider the clocked comparator shown in Fig. 13.18, where

$$\mu_n C_{\text{ox}} = 2 \mu_p C_{\text{ox}} = 100 \mu\text{A}/\text{V}^2$$

$$2\left(\frac{W}{L}\right)_n = \left(\frac{W}{L}\right)_p = 2$$

$$|V_{tn}| = |V_{tp}| = 1 \text{ V}$$

$$V_{DD} = 5 \text{ V}$$

$$\text{and } V_{SS} = 0 \text{ V}$$

If the MOSFETS in the inverter are ideal except for each having a $100\text{-k}\Omega$ output impedance when active, find the minimum differential input that will cause a 1-volt change (use a linear analysis). Assuming this minimum differential input corresponds to $(1/2)$ LSB, what is the maximum number of bits that this comparator could determine in a flash A/D converter?

- 13.18** In Section 13.6, it was stated that the interpolated signals will have the correct threshold crossings if V_1 and V_2 in Fig. 13.24 are linear in the region $0.25 < V_{in} < 0.5$. In fact, this region is larger than what is actually necessary. What is the minimum region of V_{iu} that V_1 and V_2 should be linear over?
- 13.19** Many very-high-speed A/D converters do not use a sample and hold since it would limit their speed, but instead allow the input signal to be applied to the array of comparators, which are all clocked “simultaneously.” Qualitatively, explain why a clocked flash or interpolating A/D converter might operate faster than a sample-and-hold circuit.
- 13.20** In an N-bit folding A/D converter (with no interpolation), what is the product of the folding rate times the number of folding blocks?
- 13.21** Find an expression for the number of latches in an N-bit folding A/D converter where the folding rate is $FR = 2^F$.
- 13.22** Assuming the input capacitance of a differential stage is the same in either a flash or folding/interpolating A/D converter, what reduction of input capacitance over a flash converter would be achieved with an 8-bit folding/interpolating A/D converter having four folding blocks, each with a folding rate of eight? If a straight interpolating A/D converter is to have the same reduction in input capacitance, how many resistors are required between “input comparators”?
- 13.23** Draw the clock waveforms of the sample-and-hold circuits for the time-interleaved A/D converter shown in Fig. 13.34.

Chapter 13 - Problems

- 13.1) The longest duration for Phase II occurs when $V_{in} = V_{ref}$ in which the required number of clock cycles is

$$2^{18} + 1 = 2.62 \times 10^5 \text{ cycles}$$

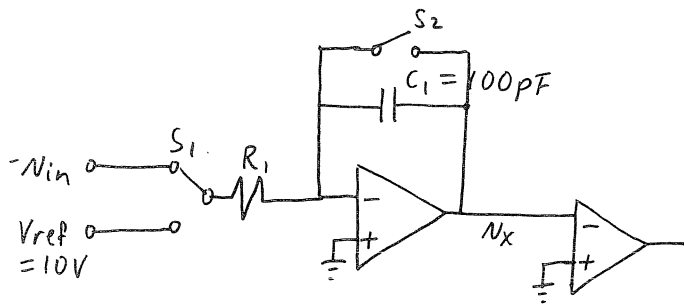
$$\therefore T_2 = 2.62 \times 10^5 T_{clk} \quad \text{where } T_{clk} = 1/5 \text{ MHz}$$

$$= 52.4 \text{ msec}$$

$$\text{and } T_1 = 2^{18} T_{clk} \approx T_2$$

$\therefore$ the total conversion time is $T_1 + T_2 = \underline{105 \text{ msec}}$ maximum

13.2)



Find R_1

Worse case situation is when $V_{in} = 10V$

$$\text{Now } T_1 = 2^{18} T_{clk} = 2^{18} \times \frac{1}{1 \text{ MHz}} = 0.262 \text{ sec}$$

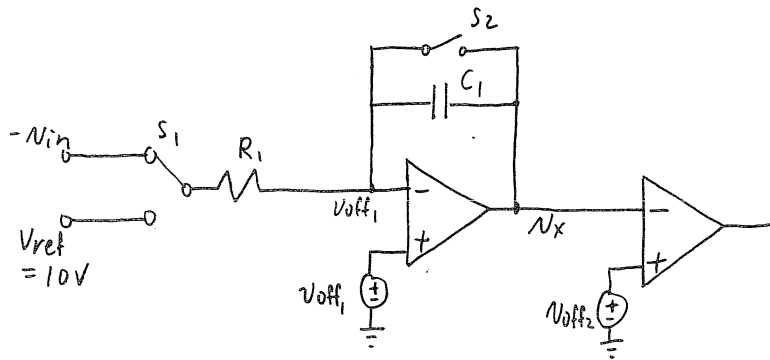
At the end of Phase I, set $v_x \equiv 10V$.

$$v_x = \frac{V_{in} T_1}{R_1 C_1} \Rightarrow R_1 = \frac{V_{in} T_1}{v_x C_1} = \frac{10 \times 0.262}{10 \times 100 \times 10^{-12}}$$

$$\underline{R_1 = 2.62 \text{ G}\Omega}$$

Note the large value required here.

13.3)



Derive Bout accounting for offset voltages.

During Phase I the input voltage is effectively $-(V_{in} + v_{off1})$

and $v_x|_{t=T_1} = \frac{(V_{in} + v_{off1}) T_1}{R_1 C_1}$ at the end of Phase I.

During Phase II, the reference voltage is effectively $V_{ref} - v_{off1}$

$$\therefore v_x(t) = -\frac{(V_{ref} - v_{off1})}{R_1 C_1} (t - T_1) + \frac{(V_{in} + v_{off1}) T_1}{R_1 C_1}$$

$$= -\frac{V_{ref}}{R_1 C_1} (t - T_1) + \frac{v_{in} T_1}{R_1 C_1} + \frac{v_{off1}}{R_1 C_1} t$$

The comparator now triggers when $v_x = v_{off2}$.

Solve for T_2 :

$$v_x(T_2) = -\frac{V_{ref}}{R_1 C_1} (T_2 - T_1) + \frac{v_{in} T_1}{R_1 C_1} + \frac{v_{off1}}{R_1 C_1} T_2 \equiv v_{off2}$$

$$\left(\frac{V_{ref}}{R_1 C_1} - \frac{v_{off1}}{R_1 C_1} \right) T_2 = \frac{v_{in} T_1}{R_1 C_1} - v_{off2}$$

$$T_2 = \frac{v_{in}}{V_{ref} - v_{off1}} T_1 - \frac{v_{off2} R_1 C_1}{V_{ref} - v_{off1}}$$

Now $T_1 = 2^N T_{clk}$

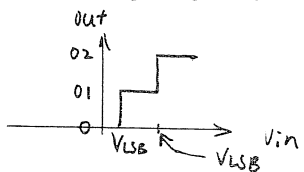
and $T_2 = 2^N \text{Bout } T_{clk} = T_1 \text{Bout}$

$$\therefore \text{Bout} = \frac{T_2}{T_1} = \frac{v_{in}}{V_{ref} - v_{off1}} - \frac{v_{off2} R_1 C_1}{V_{ref} - v_{off1}} \times \frac{1}{T_1}$$

$$\text{Bout} = \frac{v_{in} + \text{err}}{V_{ref} - v_{off1}} \quad \text{where } \text{err} = \frac{R_1 C_1}{T_1} v_{off2}$$

$\therefore v_{off1}$ is equivalent to an error in the reference voltage while v_{off2} is equivalent to an offset in the input signal.

13.4) Find offset error given $V_{off1} = 20 \text{ mV}$.



(ignore the $\frac{1}{2} V_{LSB}$ offset that is normally present)

From the above transfer characteristic we expect ideally that if $v_{in} = V_{LSB}$, the output is 00001H (Hex)

$$\text{or } B_{out} = 2^{-N} = \frac{T_2}{T_1} = \frac{T_2}{2^N T_{clk}}$$

$$\therefore \underline{T_2 = T_{clk}}$$

With the offset, however, from Problem 13.3,

$$T_2 = \frac{v_{in}}{V_{ref} - V_{off1}} T_1 - \frac{V_{off2} R_1 C_1}{V_{ref} - V_{off1}} \xrightarrow{0^\circ \text{ } V_{off2} = 0}$$

$$= \frac{V_{LSB}}{V_{ref} - V_{off1}} \times 2^N T_{clk} = \frac{V_{ref}}{(V_{ref} - V_{off1})} \times T_{clk}$$

$$= \frac{10}{(10 - 0.02)} T_{clk} = \underline{1.002 T_{clk}}$$

As each T_{clk} corresponds to 1 LSB, the offset voltage has created an offset error of $1.002 - 1 \text{ LSB} = 0.002 \text{ LSB}$

13.5) The frequencies that are completely attenuated are those with periods that are a multiple of the integration time, T_1

$$T_1 = 2^N T_{clk} = 2^{16} \times \frac{1}{10^6 \text{ Hz}} = 65.54 \text{ msec}$$

$$= \underline{1/15.3 \text{ Hz}}$$

$\therefore$ all multiples of 15.3 Hz are completely attenuated.

At 60 Hz, the attenuation is

$$\frac{|\sin(\pi f T_1)|}{\pi f T_1} = \frac{|\sin(\pi \times 60 / 15.3)|}{\pi \times 60 / 15.3} = 1.98 \times 10^{-2}$$

$$= \underline{-34 \text{ dB}}$$

13.6) Repeating Problem 13.5 with $f_{clk} = 100\text{kHz}$.

$$T_1 = 2^N T_{clk} = 2^{16} \times \frac{1}{10^5\text{Hz}} = 655.4\text{msec}$$

$$= 1/1.53\text{Hz}$$

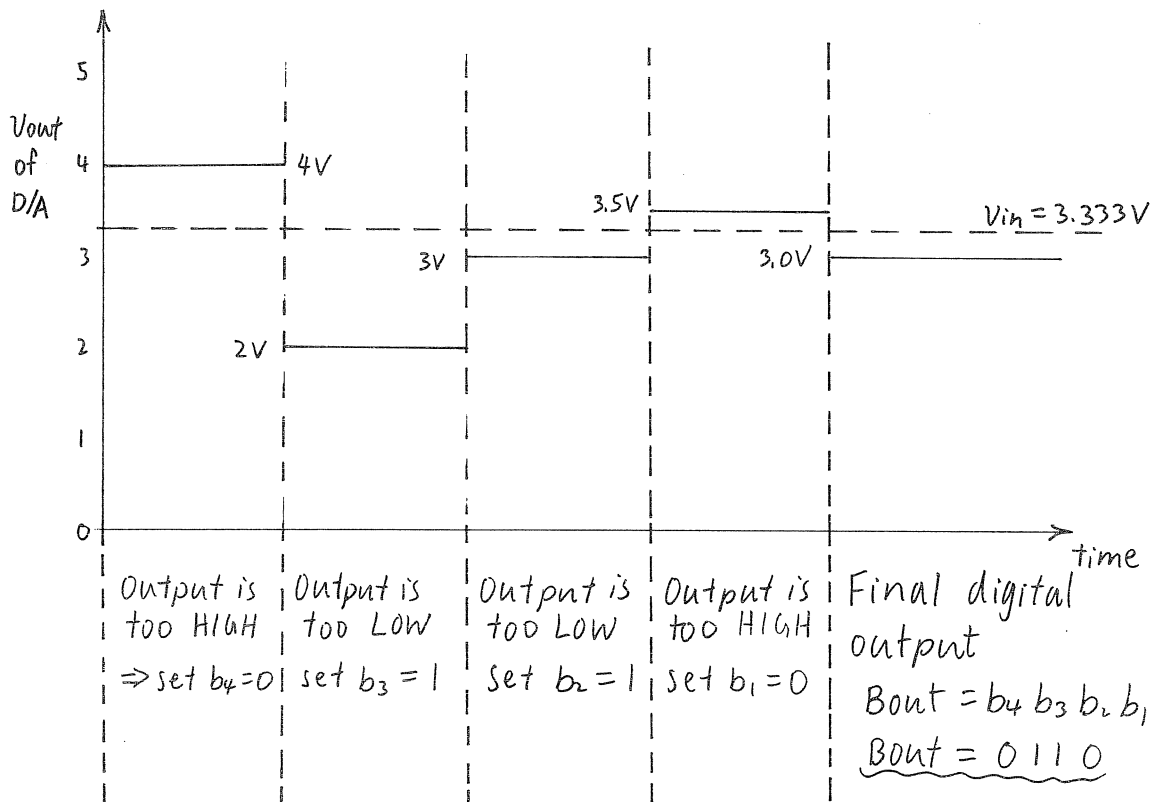
$\therefore$ all multiples of 1.53Hz are attenuated

At 60Hz , the attenuation is

$$\frac{|\sin(\pi f T_1)|}{\pi f T_1} = \frac{|\sin(\pi 60 / 1.53)|}{\pi 60 / 1.53} = 5.09 \times 10^{-3}$$

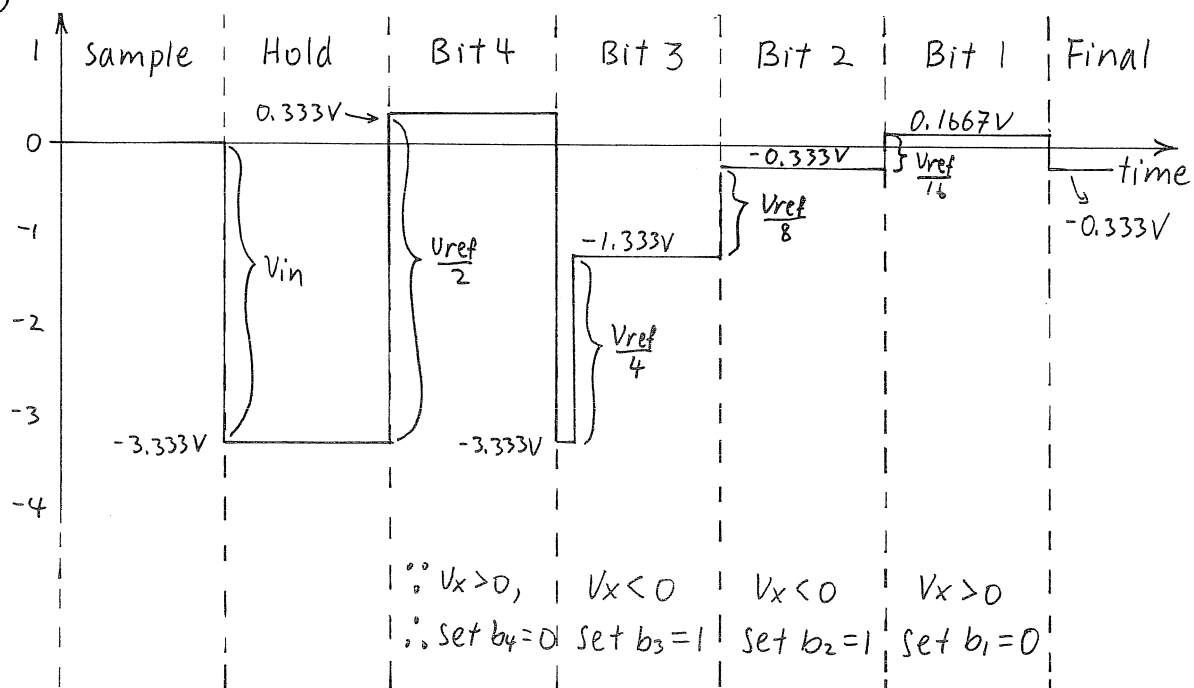
$$= \underline{\underline{-46\text{dB}}}$$

13.7)



Final digital output $Bont = 0110$

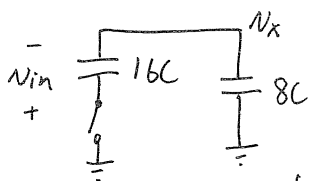
13.8)



$$\text{Final output } B_{out} = b_4 b_3 b_2 b_1 \\ = \underline{\underline{0110}}$$

13.9) Repeat Problem 13.8 assuming a parasitic capacitance of 8C.

Hold mode:

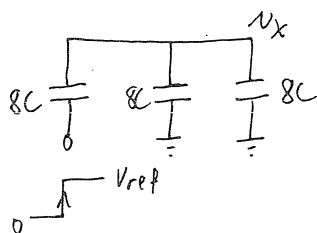


Charge conservation $Q = CV = C'V'$

$$-16C \times V_{in} = 24C \cdot V_x$$

$$\therefore V_x = -\frac{2}{3} V_{in} = -\frac{2}{3} \times 3.333V \\ = -2.222V$$

Bit 4 cycle:

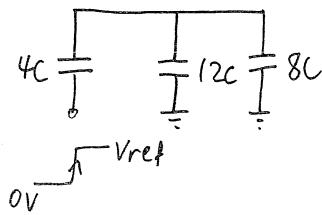


$$\Delta V_{x b_4} = \frac{8C}{24C} V_{ref} = \frac{1}{3} \times \frac{1}{2} V_{ref} = \frac{1}{3} \times 8V \\ = 2.667V$$

(Cont.)

13.9 (cont.)

Bit 3 cycle:



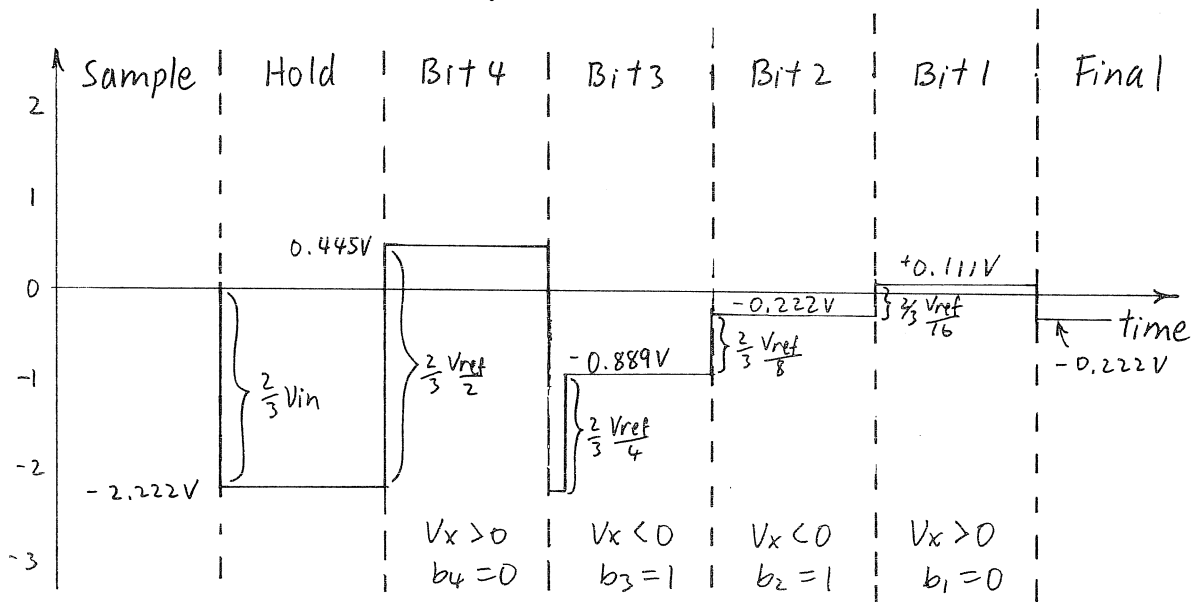
$$\Delta V_{xb3} = \frac{4C}{24C} V_{ref} = \frac{2}{3} \times \frac{1}{4} V_{ref} = \frac{1}{6} \times 8V = 1.333V$$

Similarly for

Bit 2 cycle, $\Delta V_{xb2} = \frac{2C}{24C} V_{ref} = \frac{2}{3} \times \frac{1}{8} V_{ref} = 0.667V$

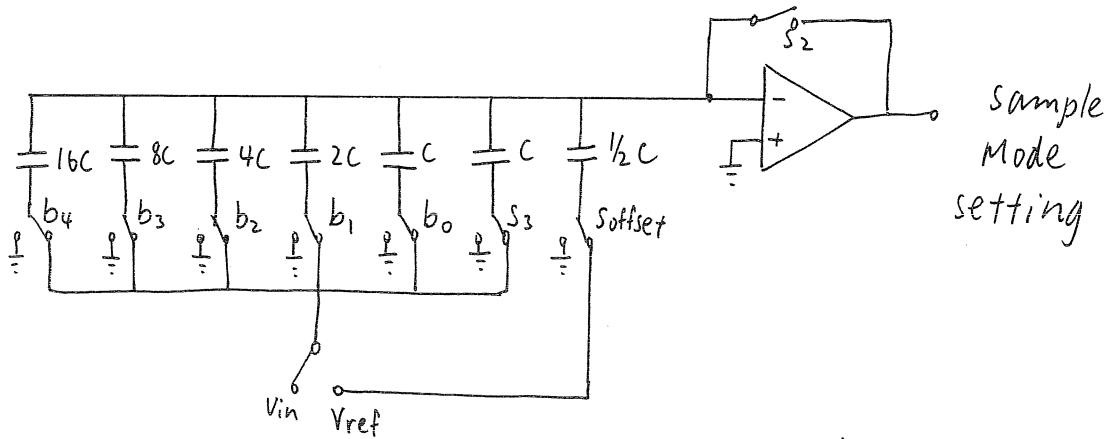
Bit 1 cycle, $\Delta V_{xb1} = \frac{C}{24C} V_{ref} = \frac{2}{3} \times \frac{1}{16} V_{ref} = 0.333V$

Note: All voltage swings are $\frac{2}{3}$ the original values but the final Bout result will remain unchanged.



Final output, Bout = $b_4 b_3 b_2 b_1 = 0110$

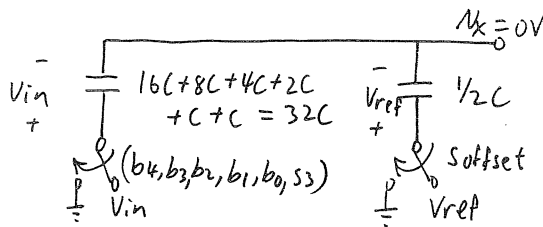
13.10)



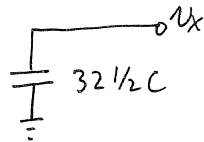
Charge redistribution A/D with offset ↗

Explanation of operation:

At the end of the sample mode, we have



During the Hold phase, switches $b_4, b_3, b_2, b_1, b_0, s_3$ and offset go to ground resulting in



By charge conservation

$$32C \times (-V_{in}) + \frac{1}{2}C \times (-V_{ref}) = 32\frac{1}{2}C \times V_x$$

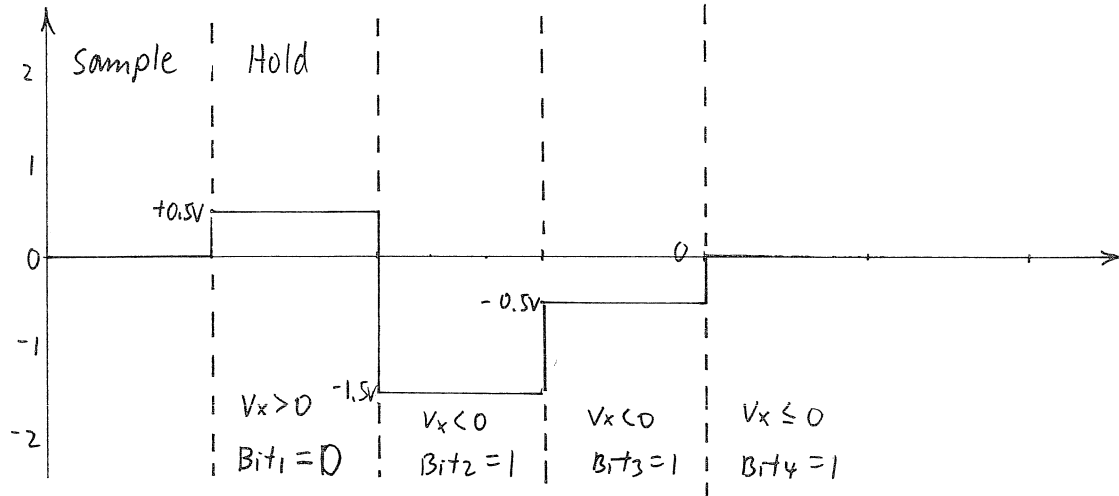
$$\therefore V_x = -\frac{32}{32\frac{1}{2}} V_{in} - \frac{\frac{1}{2}V_{ref}}{32\frac{1}{2}}$$

$$= -\frac{32}{32\frac{1}{2}} (V_{in} + \frac{1}{64}V_{ref})$$

Thus we see that we have created an offset in V_{in} of $\frac{1}{64} \times V_{ref}$ which is equivalent to $\frac{1}{2}$ LSB. Afterwards, the $\frac{1}{2}C$ cap. acts as a parasitic cap. to ground, causing ^{only} a small degradation in the signal gain ($\approx 1.5\%$).

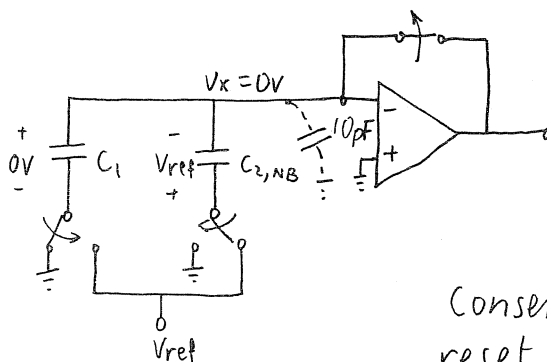
13.11) The output code from the signed charge-distribution A/D converter can be converted to a two's complement code by simply inverting the MSB.

Example: A 4-bit converter with $V_{ref} = 8V$ and $V_{in} = -1V$



$\therefore B_{out} = b_1 b_2 b_3 b_4 = \boxed{0111}$ represented by
while in 2's complement code $-1V$ is $\boxed{1111}$.

13.12) Find V_x and show $V_{e1} = \frac{1}{2} V_{x1}$



$$C_p = 10pF$$

The voltage levels and switching directions correspond to the end of the reset phase.

Conservation of charge between reset and calibration stages gives

$$C_1 \times 0V + C_{2,NB} \times (V_{ref}) + C_p \times 0V = C_1 (V_{x1} - V_{ref}) + C_{2,NB} V_{x1} + C_p V_{x1}$$

$$V_{ref} (-C_{2,NB} + C_1) = V_{x1} (C_1 + C_{2,NB} + C_p)$$

$$\text{But } C_T = C_1 + C_{2,NB} \text{ and } C_1 = \frac{C_T}{2} + \Delta C_1$$

$$\therefore -C_{2,NB} + C_1 = -\left(\frac{C_T}{2} - \Delta C_1\right) + \frac{C_T}{2} + \Delta C_1 = 2\Delta C_1$$

(cont.)

13.12 (cont.)

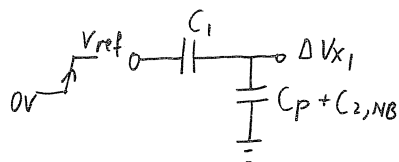
$$\therefore V_{\text{ref}} \times 2\Delta C_1 = V_{x1} (C_T + C_P)$$

$$\underline{V_{x1} = \frac{2\Delta C_1}{C_T + C_P} V_{\text{ref}}}$$

$$\text{Here } \Delta C_1 = C_1 - \frac{C_T}{2} = 31.5 - 32 \text{ pF} = -0.5 \text{ pF}$$

$$\therefore \underline{V_{x1} = \frac{2 \times (-0.5)}{64 + 10} V_{\text{ref}} = -1.35 \times 10^{-2} V_{\text{ref}}}$$

To determine V_{e1} , note that during bit 1 of the conversion process, only C_1 is switched from ground to V_{ref} . All other switches are unchanged. Hence the voltage change in V_{x1} , ΔV_{x1} , is given by a capacitive divider rule



$$\Delta V_{x1} = \frac{C_1}{C_1 + C_{2,NB} + C_P} V_{\text{ref}}$$

$$= \frac{C_T/2 + \Delta C_1}{C_T + C_P} V_{\text{ref}}$$

$$= \left[\frac{1}{2} \frac{C_T}{C_T + C_P} + \frac{\Delta C_1}{C_T + C_P} \right] V_{\text{ref}}$$

Ideally $\Delta C_1 = 0$ and

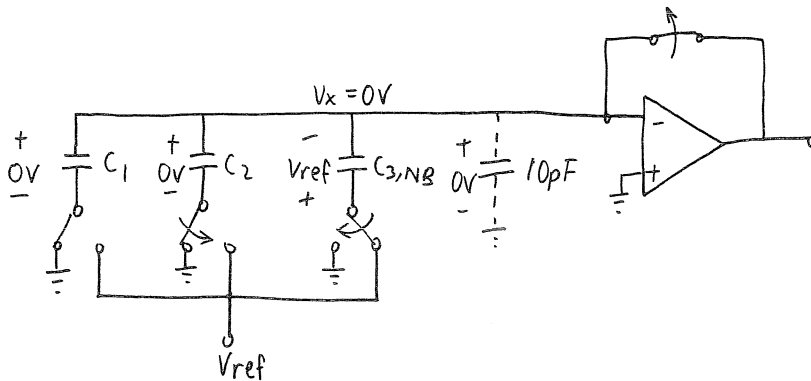
$$\Delta V_{x1} = \frac{1}{2} \frac{C_T}{C_T + C_P} V_{\text{ref}}$$

$$\therefore V_{e1} \equiv \Delta V_{x1} - \Delta V_{x1 \text{ ideal}} = \frac{\Delta C}{C_T + C_P} V_{\text{ref}}$$

$$\text{or } \underline{V_{e1} = \frac{1}{2} V_{x1}}$$

13.13) Find V_{x2} and show $V_{e2} = \frac{1}{2}(V_{x2} - V_{e1})$.

Continuing from Problem 13.12, we now analyze the calibration of C_2 .



As in Problem 13.12, the diagram shows the voltage levels and switching that occurs at the end of the reset phase. Charge conservation gives

$$C_1 \times 0V + C_2 \times 0V - C_{3,NB} V_{ref} + C_p \times 0V = C_1 V_{x2} + C_2 (V_{x2} - V_{ref}) + C_{3,NB} V_{x2} + C_p V_{x2}$$

$$(C_2 - C_{3,NB}) V_{ref} = (C_1 + C_2 + C_{3,NB} + C_p) V_{x2} \quad [1]$$

Now $C_2 = \frac{C_T}{4} + \Delta C_2$

and $C_1 + C_2 + C_{3,NB} = C_T$

$$\therefore C_{3,NB} = C_T - C_1 - C_2$$

$$= C_T - \left(\frac{C_T}{2} + \Delta C_1\right) - \left(\frac{C_T}{4} + \Delta C_2\right)$$

$$= \frac{C_T}{4} - \Delta C_1 - \Delta C_2 \quad [2]$$

[2] $\rightarrow$ [1] $\left[\left(\frac{C_T}{4} + \Delta C_2\right) - \left(\frac{C_T}{4} - \Delta C_1 - \Delta C_2\right)\right] V_{ref} = (C_T + C_p) V_{x2}$

$$V_{x2} = \frac{\Delta C_1 + 2\Delta C_2}{C_T + C_p} V_{ref}$$

$\therefore \underline{V_{x2} = \frac{1}{2} V_{x1} + \frac{2\Delta C_2}{C_T + C_p} V_{ref}} \quad [3]$

Using $C_2 = 16.4 \text{ pF}$ and $V_{x1} = -1.35 \times 10^{-2} V_{ref}$ from Problem 13.12,

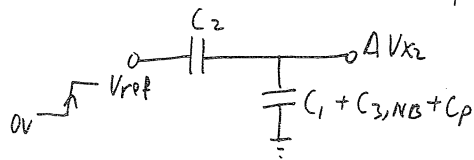
$$V_{x2} = \frac{1}{2} (-1.35 \times 10^{-2} V_{ref}) + \frac{2(16.4 - 16)}{64 + 10} V_{ref}$$

$$\underline{V_{x2} = -2.7 \times 10^{-3} V_{ref}}$$

(cont.)

13.13 (cont.)

During bit 2 of the conversion process only C_2 is switched from ground to V_{ref} . Thus the change in V_{x2} , ΔV_{x2} is given by



$$\begin{aligned}\Delta V_{x2} &= \frac{C_2}{C_1 + C_2 + C_{3,NB} + C_p} V_{ref} \\ &= \frac{C_T/4 + \Delta C_2}{C_T + C_p} V_{ref} \\ &= \frac{1}{4} \frac{C_T}{C_T + C_p} V_{ref} + \frac{\Delta C_2}{C_T + C_p} V_{ref}\end{aligned}$$

Ideally $\Delta C_2 = 0$

$$\therefore \Delta V_{x2,ideal} = \frac{1}{4} \frac{C_T}{C_T + C_p} V_{ref}$$

$$\therefore V_{e2} \triangleq \Delta V_{x2} - \Delta V_{x2,ideal} = \frac{\Delta C_2}{C_T + C_p} V_{ref} \quad [4] \quad \text{and } V_{e1} = \frac{1}{2} V_{x1} \quad [5]$$

$$[5][4] \rightarrow [3] \quad V_{x2} = V_{e1} + 2V_{e2}$$

$$\text{or } \underline{V_{e2} = \frac{1}{2} (V_{x2} - V_{e1})}$$

13.14) Find the settling time.

$$\text{Given: } R_{sw} = R_{s1} = R = R_{s2} = 1k\Omega$$

$$C_T = 128 \text{ pF}$$

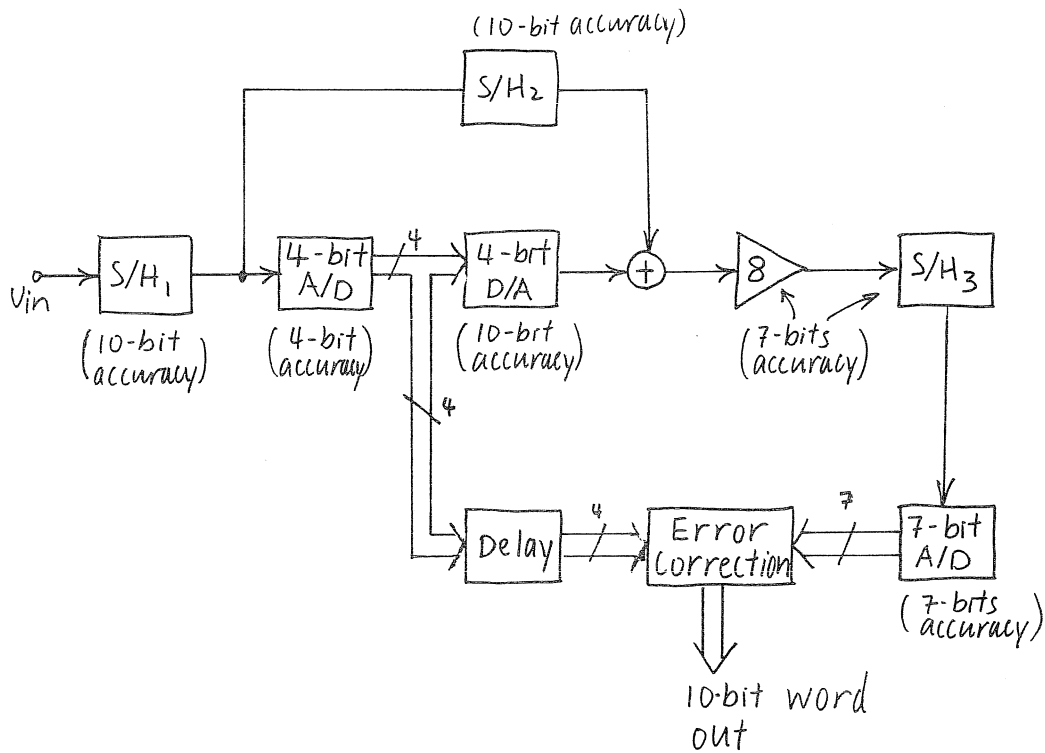
From Eq. (13.27),

$$\begin{aligned}\tau_{eq} &= (R_{s1} + R + R_{s2}) 2^N C = 3 \times R_{sw} \times C_T = 3 \times 1000 \times 128 \times 10^{-12} \\ &= 0.384 \mu\text{sec}\end{aligned}$$

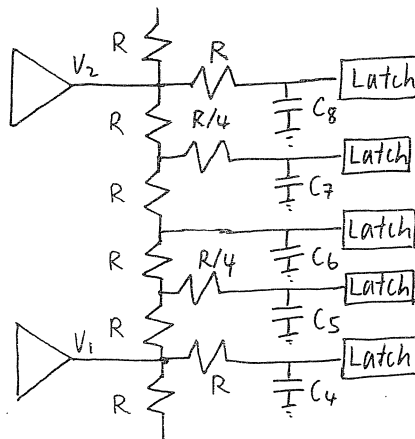
From Eq. (13.29),

$$\begin{aligned}\text{Settling time, } T &= 0.69(N+1)\tau_{eq} = 0.69 \times 13 \times 0.384 \mu\text{sec} \\ &= \underline{3.44 \mu\text{sec}}\end{aligned}$$

13.15) A 10-bit, 2-step A/D converter:



13.16) The input capacitances of the latches are shown explicitly here :



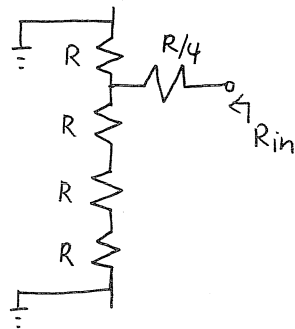
Assume nodes V_1 and V_2 are driven by low impedance op-amps (i.e., assume zero impedance outputs)

(cont.)

13.16 (cont.)

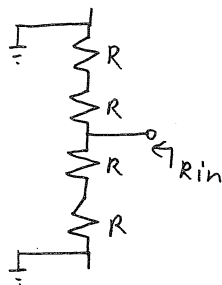
From the diagram, it is clear that C_8 sees an impedance R to node V_2 .

C_7 sees



$$\begin{aligned} R_{in} &= R/4 + (R // 3R) \\ &= R/4 + \frac{3R^2}{4R} \\ &= R \end{aligned}$$

C_6 sees



$$\begin{aligned} R_{in} &= 2R // 2R \\ &= \frac{2R \cdot 2R}{2R + 2R} \\ &= R \end{aligned}$$

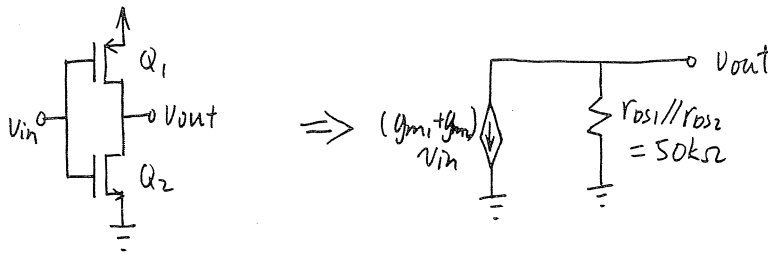
C_5 sees the identical resistor network as C_7
 $\therefore R_{in} = R$.

C_4 sees the same resistor network as C_8 and the entire structure is repeated.

$\therefore$ All latch capacitances have an impedance of R associated with them.

If we assume all capacitances are equal to C then all time constants are equal to RC .

13.17)



$$\text{gain} = - (g_{m1} + g_{m2}) \times r_{ds1} // r_{ds2}$$

$$g_{m2} = \mu_n C_{ox} (W/L)_2 V_{eff} = 100 \mu A/V^2 \times 1 \times (2.5 - 1) = 0.15 \text{ mA/V}$$

$$g_{m1} = \mu_p C_{ox} (W/L)_1 V_{eff} = 50 \mu A/V^2 \times 2 \times (2.5 - 1) = 0.15 \text{ mA/V}$$

$$\therefore \text{gain} = - 0.3 \text{ mA/V} \times 50 \text{ k}\Omega = -15$$

$\therefore$ A $\frac{1V}{15} = 67 \text{ mV}$ differential input is required to produce a 1V output change.

If $\frac{1}{2} V_{LSB} = 67 \text{ mV}$ or $V_{LSB} = 133 \text{ mV}$,

then the maximum number of bits this comparator can determine is given by N where

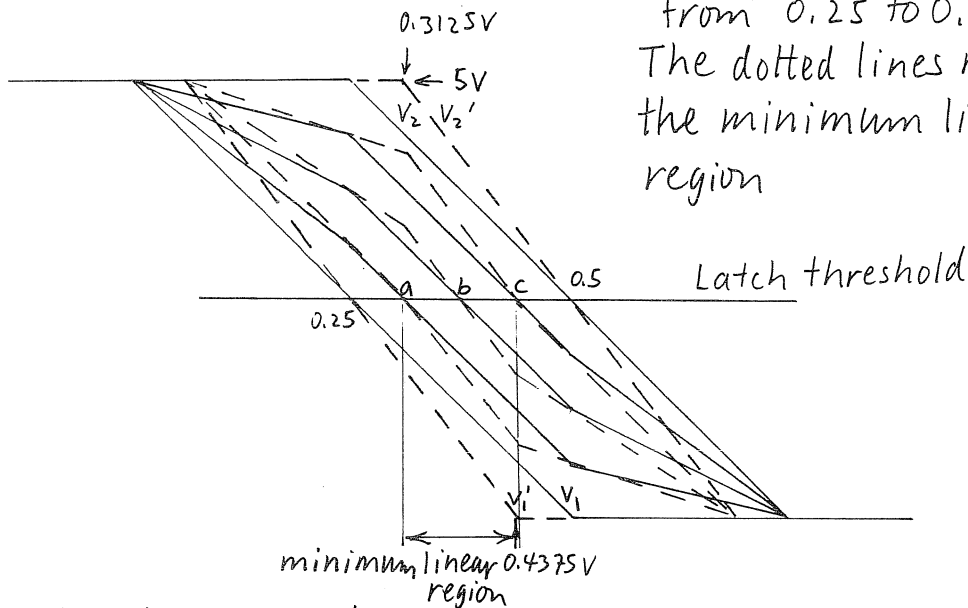
$$V_{LSB} = \frac{V_{ref}}{2^N}$$

$$\therefore 2^N = \frac{V_{ref}}{V_{LSB}} = \frac{5V}{0.133V} = 37.5$$

$$\underline{N = 5.2 \text{ bits}} \Rightarrow \underline{5 \text{ bits}}$$

13.18) For the minimum linear region, we can draw a simplified line diagram representing the linear and saturated regions of operations of the input amplifiers. The solid lines represent a linear region

from 0.25 to 0.5V. The dotted lines represent the minimum linear region



∴ The current between V_1 and V_2 is

$$i = (V_2 - V_1) / 4R$$

$$\begin{aligned} \therefore V_{2a} &= V_1 + iR = V_1 + \frac{V_2 - V_1}{4} \\ &= \frac{3}{4} V_1 + \frac{1}{4} V_2 \end{aligned}$$

Similarly

$$V_{2b} = \frac{1}{2} (V_1 + V_2)$$

$$V_{2c} = \frac{1}{4} V_1 + \frac{3}{4} V_2$$

Interpolated lines
between V_1 and V_2

These lines are shown in the above diagram and are marked a, b, and c. We can see that the lines are evenly spaced across the threshold crossing as long as the linear region covers the crossing points of V_{2a} and V_{2c}

Thus the minimum linear region is

$$0.25 + \frac{0.25}{4} < V_{in} < 0.5 - \frac{0.25}{4}$$

$$\underline{0.3125V < V_{in} < 0.4375V}$$

13.19) The key limitation to the speed of an A/D converter is its ability to sample the input signal to within $\frac{1}{2} \text{LSB}_A^{\text{of}}$ accuracy.

For a clocked flash or interpolating A/D converter, this largely relates to the ability to synchronize the sampling times of the individual components. This can be accomplished by matching impedances and clock skew within the circuitry. With sample-and-hold circuits, however, we are faced with multiple challenges such as charge injection, signal-dependent jitter and parasitic capacitances which can cause signal feedthrough at high frequencies. Combined together, these problems can often prove more difficult to resolve than those associated with flash or interpolating designs.

13.20)

$$\begin{aligned} & \text{folding rate} \times \# \text{ folding blocks} \\ &= \frac{2^N}{\# \text{ latches}} \times \frac{1 \text{ latch}}{\# \text{ folding blks}} \times \# \text{ folding blocks} \\ &= \underline{2^N} \end{aligned}$$

13.21) Given: N -bits with folding rate of 2^F

$$\text{Now } FR = \frac{2^N}{\# \text{ latches}}$$

$$\therefore \underline{\# \text{ latches}} = \frac{2^N}{FR} = \frac{2^N}{2^F} = \underline{2^{N-F}}$$

13.22)

Design (A): • 8-bit folding/interpolating A/D
• 4 folding blocks
• 5-bits MSB, 3-bits interpolation

Design (B): • straight interpolating A/D
• same input capacitive load as Design (A)

Design (C): • 8-bit flash A/D

Let C_{dp} be the input capacitance associated with a differential pair

The input capacitance of design (C), $C_{in@}$ is

$$C_{in@} = 2^8 C_{dp} = \underline{256 C_{dp}}$$

For Design (A), a folding rate of 8 gives 3-bits, and 4 folding blocks give 2-bits.

∴ An interpolation by 8 is required to obtain the remaining 3-bits needed for an 8-bit converter.

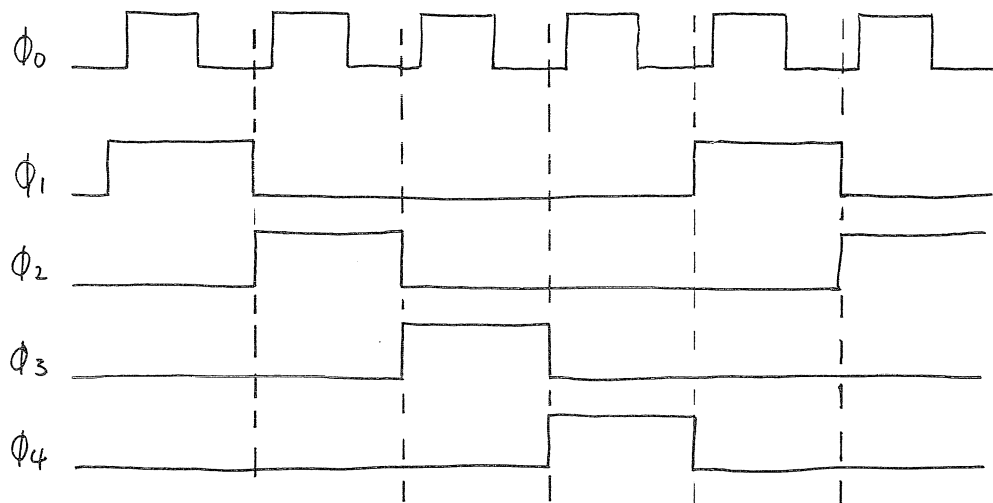
$$\begin{aligned}\therefore C_{in@} &= \# \text{folding blocks} \times C_{\text{folding block}} \\ &= 4 \times (\text{folding rate} \times C_{dp}) \\ &= \underline{32 C_{dp}}\end{aligned}$$

∴ Between the folding/interpolating design (A) and the flash design (C), there is a reduction factor of

$$\frac{C_{in@}}{C_{in@}} = \frac{256 C_{dp}}{32 C_{dp}} = \underline{8 \text{ in input capacitance.}}$$

In order for design (B) to have the same input capacitance as Design (A), an interpolation factor of 8 must be used. Thus we need 8 resistors between the input capacitors.

13.23) Assume that sampling occurs a signal is HIGH.



Note: ideally Phase ϕ_0 should slightly lag behind the other phases to ensure that the second stage S/H circuits have the most stable inputs from the first stage.

Chapter 14

14.11 PROBLEMS

- 14.1 Assuming oversampling with no noise shaping and using (14.14), find the approximate sampling rate required to obtain a maximum SNR of 80 dB on a signal with a 1-kHz bandwidth using a 1-bit quantizer.
- 14.2 Repeat Example 14.4, except use a dc input level of $1.2/3$. At what frequency would a tone appear (relative to the sampling rate, f_s)?
- 14.3 Repeat Example 14.4, except use a dc input level of 1.1 to see if the internal state of the modulator saturates.
- 14.4 Repeat Example 14.4, except use an input sequence of $\{10, -10, 10, -10, 10, -10, \dots\}$ to see if the internal state of the modulator saturates. This problem demonstrates that a large level input signal can be applied to a modulator if its signal power resides at a frequency where the gain of $H(z)$ is low.

- 14.5** Given that an 8-bit A/D converter has a SNR of 50 dB but is linear to 12 bits, what is the sampling rate required to achieve 12 bits of accuracy using straight oversampling on a signal bandwidth of 1 MHz?
- 14.6** Repeat Problem 14.5, assuming that the 8-bit A/D converter is placed inside a first-order delta-sigma modulator. What is the sampling rate if a second-order modulator is used?
- 14.7** Repeat Problem 14.1, assuming a first-order modulator is used. What is the sampling rate if a second-order modulator is used?
- 14.8** Assuming a 1-bit output signal of ± 1 and a sinusoidal signal of ± 0.5 peak-to-peak, what is the ratio of the signal power to quantization noise power before any filtering is applied?
- 14.9** What does the block $G(z) - 1$ need to be in Fig. 14.13 in order that a second-order noise transfer function is realized equal to $(1 - z^{-1})^2$?
- 14.10** Show a switched-capacitor modulator so that the second-order modulator shown in Fig. 14.10 is realized.
- 14.11** Show that the sequence of $\{1, 1, -1, 1, 1, -1, 1, 1, -1, \dots\}$ into a cascade of three integrators and differentiators shown in Fig. 14.19 gives the correct running average. Assume that the adders have infinite width and that $M = 4$.
- 14.12** At first glance, the impulse response of a running average filter of length 4 is the same as that for a hold system that holds the input signal for length 4. By producing a block diagram for each system, explain the difference between the two systems.
- 14.13** Verify (14.42).
- 14.14** Show the block diagram for a MASH architecture that realizes third-order noise shaping using a second-order modulator (as shown in Fig. 14.10) and a first-order modulator.
- 14.15** Sketch the noise transfer function from 0 to 4 MHz for the second-order band-pass modulator shown in Fig. 14.23, assuming the sampling rate, f_s , equals 4 MHz.
- 14.16** Show how the block diagram in Fig. 14.23 can be modified so that the band-pass modulator places zeros at $\pm f_s/8$ rather than $\pm f_s/4$.
- 14.17** Using the “typical” waveform pattern shown in Fig. 14.24, find the practical average outputs when 0, $1/2$, and $-1/2$ have been applied to the first-order modulator shown in Fig. 14.7.

Chapter 14 - Problems

14.1) Using (14.14) : $80 = 6.02 + 1.76 + 10 \log \text{OSR}$

$$\Rightarrow \text{OSR} = 16672472 !$$

$$f_s = (\text{OSR}) \times 2f_b = \underline{\underline{33.345 \text{ GHz} !}}$$

14.2)

n	$x(n)$	$x(n+1)$	$y(n)$	$e(n)$
0	0.1	-0.5	1	0.9
1	-0.5	0.9	-1	-0.5
2	0.9	0.3	1	0.1
3	0.3	-0.3	1	0.7
4	-0.3	1.1	-1	-0.7
5	1.1	0.5	1	-0.1
6	0.5	-0.1	1	0.5
7	-0.1	1.3	-1	-0.9
8	1.3	0.7	1	-0.3
9	0.7	0.1	1	0.3
10	0.1	-0.5	1	0.9 $\leftarrow$ repeat !

The state, $x(n)$, repeats every 10 cycle. Therefore, there

is a tone in the output stream with the frequency of $\frac{f_s}{10}$!

14.3) Since $x(n+1) = 1.1 + x(n) - y(n)$,
if $x(0) = 0.1$, $y(n)$ will
always remain 1, and $x(n)$
increments by 0.1 at each
cycle until it saturates!

n	$x(n)$	$x(n+1)$	$y(n)$
0	0.1	0.2	1
1	0.2	0.3	1
2	0.3	0.4	1
...	...	...	...

14.4)

n	u(n)	x(n)	x(n+1)	y(n)
0	10	0.1	0.1	1
1	-10	0.1	-1.9	1
2	10	-1.9	0.1	-1
3	-10	0.1	-1.9	1 $\Leftarrow$ repeat!

14.5) Doubling OSR improves the accuracy by 0.5 bit. For a 4-bit improvement, OSR will be

$$OSR = 2^{4/0.5} = 256 \Rightarrow f_s = 2f_0 \times OSR = \underline{512 \text{ MHz}}$$

14.6) For a 4-bit improvement in accuracy using a 1st-order $\Delta\Sigma$,

$$\begin{aligned} \text{we must have: } & 6.02 \times 4 + 5.17 = 30 \log(OSR) \\ (\text{i.e. } & 6.02(9) + 1.76 - 5.17 + 30 \log(OSR) = 6.02(12) + 1.76) \\ \Rightarrow OSR = & 9.44 \Rightarrow f_s = 2f_0 \times OSR = \underline{18.88 \text{ MHz}} \end{aligned}$$

Using a 2nd-order $\Delta\Sigma$, we must have:

$$6.02 \times 4 + 12.9 = 50 \log(OSR) \Rightarrow OSR = 5.49$$

$$f_s = 2f_0 \times OSR = \underline{10.98 \text{ MHz}}$$

14.7) Using a 1st-order $\Delta\Sigma \Rightarrow 80 = 6.02 + 1.76 - 5.17 + 30 \log(OSR)$

$$\Rightarrow \underline{OSR = 380} \Rightarrow \underline{f_s = 760 \text{ KHz}}$$

Using a 2nd-order $\Delta\Sigma \Rightarrow 80 = 6.02 + 1.76 - 12.9 + 50 \log(OSR)$

$$\Rightarrow \underline{OSR = 50.4} \Rightarrow \underline{f_s = 100.8 \text{ KHz}}$$

14.8) Assuming quantization noise as an independent noise signal.

output signal power = input signal power + quant. noise power

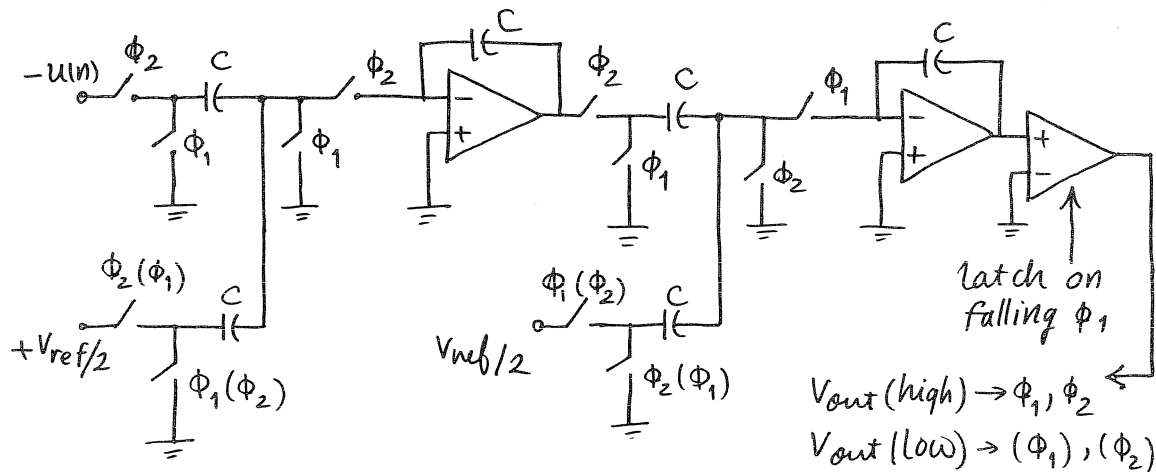
$$\Rightarrow 1W = \frac{(0.5)^2}{2} + P_e \Rightarrow P_e = 0.875W$$

$$\Rightarrow \frac{P_s}{P_e} = \frac{0.125}{0.875} = \frac{1}{7} = -8.5 \text{ dB}$$

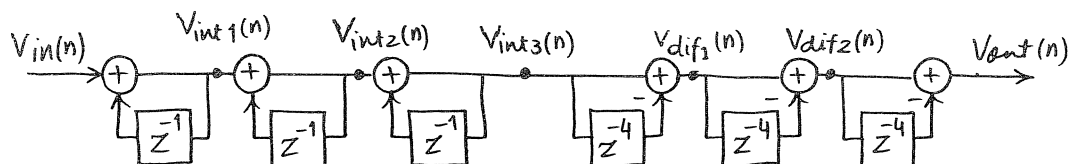
14.9) $G(z) = (1 - z^{-1})^2 \Rightarrow G(z) - 1 = (1 - z^{-1})^2 - 1$

$$\therefore G(z) - 1 = (1 - z^{-1} - 1)(1 - z^{-1} + 1) = \underline{z^{-1}(z^{-1} - 2)}$$

14.10)



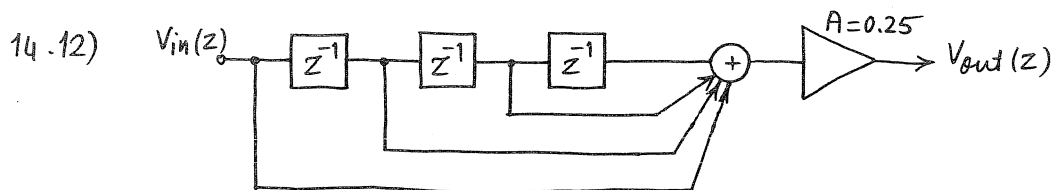
14.11)



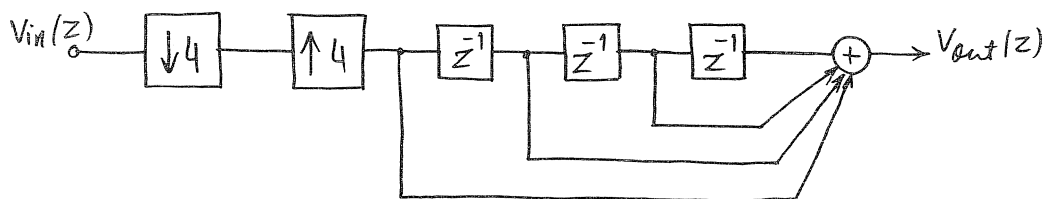
14.11) (cont.)

	0	1	2	3	4	5	6	7	8	9	10	11	12
$V_{in}(n)$	0	1	1	-1	1	1	-1	1	1	-1	1	1	-1
$V_{int1}(n)$	0	1	2	1	2	3	2	3	4	3	4	5	4
$V_{int2}(n)$	0	1	3	4	6	9	11	14	18	21	25	30	34
$V_{int3}(n)$	0	1	4	8	14	23	34	48	66	87	112	142	176
$V_{dif1}(n)$	0	1	4	8	14	22	30	40	52	64	78	94	110
$V_{dif2}(n)$	0	1	4	8	14	21	26	32	38	42	48	54	58
$V_{out}(n)$	0	1	4	8	14	20	22	24	24	21	22	22	20
$\frac{1}{64} V_{out}(n)$						0.31	0.34	0.37	0.37	0.33	0.34	0.34	0.31

The steady-state value of $\frac{1}{64} V_{out}(n)$ is close to $\frac{1}{3}$.



Block diagram of a running-average filter of length 4.



Block diagram of a hold system with length 4.

While the first system is a time-invariant system, the second system is time-varying due to the down-sampler!

$$\begin{aligned}
 14.13) \quad Y(z) &= z^{-1} [z^{-1} U(z) + (1-z^{-1}) E_1(z)] \\
 &\quad - (1-z^{-1}) [z^{-1} E_1(z) + (1-z^{-1}) E_2(z)] \\
 &= \underline{z^{-2} U(z) - (1-z^{-1})^2 E_2(z)}
 \end{aligned}$$

14.14) Using (14.27) & (14.28), for a 2nd-order $\Delta\Sigma$ mod. we have:

$$y_1(n) = z^{-1} u(n) + (1-z^{-1})^2 e_1(n)$$

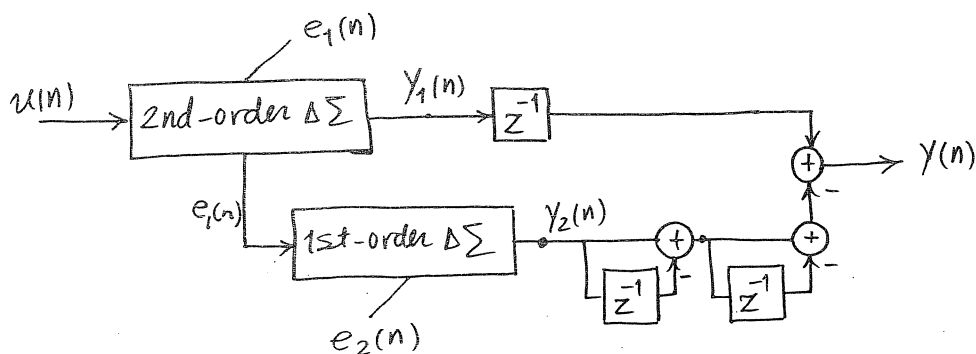
If $e_1(n)$ is fed into a 1st-order $\Delta\Sigma$ mod., the output, $y_2(n)$, will be:

$$y_2(n) = z^{-1} e_1(n) + (1-z^{-1}) e_2(n)$$

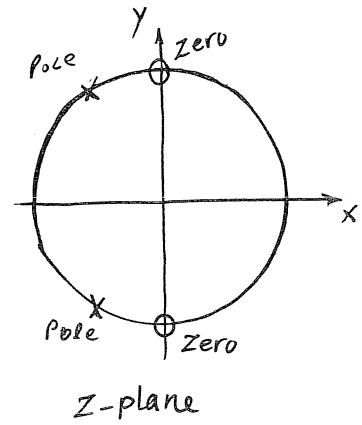
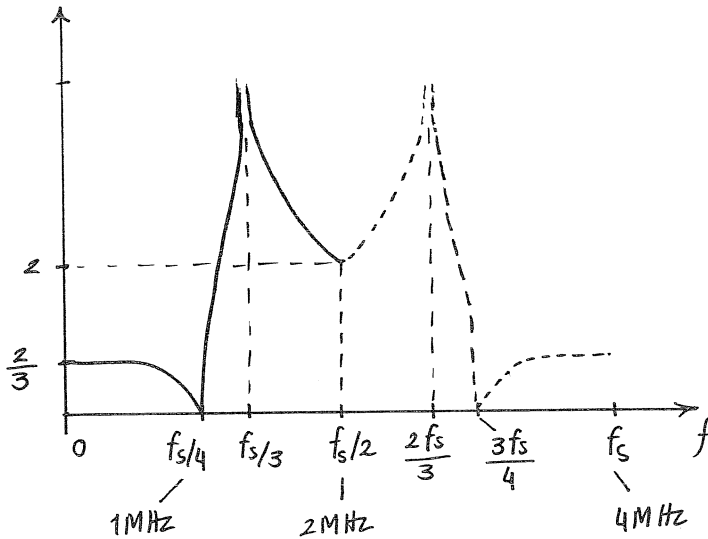
In order to eliminate $e_1(n)$ in a combination of $y_1(n)$ & $y_2(n)$, $y_1(n)$ must be multiplied by z^{-1} and $y_2(n)$ by $(1-z^{-1})^2$. The final output, $y(n)$, will be:

$$y(n) = z^{-1} y_1(n) - (1-z^{-1})^2 y_2(n) = z^{-1} u(n) - (1-z^{-1})^3 e_2(n)$$

The block diagram for this system is shown below:



$$14.15) \quad N_{TF}(z) = \frac{1}{1+H(z)} = \frac{1+z^{-2}}{1+z^{-1}+z^{-2}}$$

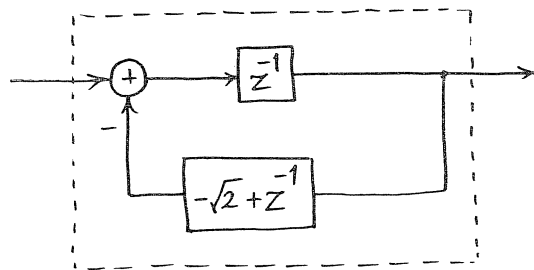


Poles should be adjusted for better stability.

14.16) $H(z)$ is required to have its pole at $e^{\pm j\pi/4}$. Therefore,

$$(z - e^{j\pi/4})(z - e^{-j\pi/4}) = z^2 - \sqrt{2}z + 1$$

$$\Rightarrow H(z) = \frac{z}{z^2 - \sqrt{2}z + 1} = \frac{z^{-1}}{1 - \sqrt{2}z^{-1} + z^{-2}}$$



$H(z)$

$$14.17) \quad 0 : \{1, -1, 1, -1, 1, -1, \dots\}$$

$$\Rightarrow V_{ave} = \frac{A_1 + A_0}{2} + \frac{\delta_1 + \delta_2}{2} = \underbrace{V_{ave-ideal}}_0 + \frac{\delta_1 + \delta_2}{2}$$

$$1/2 : \{1, 1, 1, -1, 1, 1, 1, -1, \dots\}$$

$$\Rightarrow V_{ave} = \frac{3A_1 + A_0}{4} + \frac{\delta_1 + \delta_2}{4} = \underbrace{V_{ave-ideal}}_{1/2} + \frac{\delta_1 + \delta_2}{4}$$

$$-1/2 : \{-1, -1, -1, 1, -1, -1, -1, 1, \dots\}$$

$$\Rightarrow V_{ave} = \frac{3A_0 + A_1}{4} + \frac{\delta_1 + \delta_2}{4} = \underbrace{V_{ave-ideal}}_{-1/2} + \frac{\delta_1 + \delta_2}{4}$$

The averages lie on a straight line if $\delta_1 = -\delta_2$!

Chapter 15

15.10 PROBLEMS

Unless otherwise stated, assume the following:

- npn bipolar transistors:
 $\beta = 100$
 $V_A = 80 \text{ V}$
 $\tau_b = 13 \text{ ps}$
 $\tau_s = 4 \text{ ns}$
 $r_b = 330 \Omega$
- n-channel MOS transistors:
 $\mu_n C_{ox} = 92 \mu\text{A}/\text{V}^2$
 $V_{tn} = 0.8 \text{ V}$
 $\gamma = 0.5 \text{ V}^{1/2}$
 $r_{ds}(\Omega) = 8,000L(\mu\text{m})/I_D(\text{mA})$ in active region
 $C_j = 2.4 \times 10^{-4} \text{ pF}/(\mu\text{m})^2$
 $C_{j-sw} = 2.0 \times 10^{-4} \text{ pF}/(\mu\text{m})$
 $C_{ox} = 1.9 \times 10^{-3} \text{ pF}/(\mu\text{m})^2$
 $C_{gs(\text{overlap})} = C_{gd(\text{overlap})} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$
- p-channel MOS transistors:
 $\mu_p C_{ox} = 30 \mu\text{A}/\text{V}^2$
 $V_{tp} = -0.9 \text{ V}$
 $\gamma = 0.8 \text{ V}^{1/2}$
 $r_{ds}(\Omega) = 12,000L(\mu\text{m})/I_D(\text{mA})$ in active region
 $C_j = 4.5 \times 10^{-4} \text{ pF}/\mu\text{m}^2$
 $C_{j-sw} = 2.5 \times 10^{-4} \text{ pF}/\mu\text{m}$
 $C_{ox} = 1.9 \times 10^{-3} \text{ pF}/(\mu\text{m})^2$
 $C_{gs(\text{overlap})} = C_{gd(\text{overlap})} = 2.0 \times 10^{-4} \text{ pF}/\mu\text{m}$

- 15.1** Show that the input impedance (at nodes i_o) of the Miller integrator shown in Fig. 15.6 is on the order of $2/(g_{m2}^2 r_{ds})$ at frequencies where capacitors $2C_1$ can be considered short circuits.

- 15.2** Based on the block diagram of Fig. 15.9, find transconductance values for a first-order filter with a dc gain of 10 and a pole at 15 MHz (no finite zero). Assume the integrating capacitors are sized $C_A = 5$ pF.
- 15.3** Find the transconductance and capacitance values of the second-order filter of Fig. 15.11 for a low-pass filter with a pole frequency of 10 MHz, $Q = 1$, and a dc gain of 5.
- 15.4** Derive the design equations for the biquad circuit shown in Fig. 15.24.
- 15.5** Consider the transconductor shown in Fig. 15.13. Find the transconductance of this circuit for finite opamp gains A .
- 15.6** In the transconductor of Fig. 15.12(b), find the minimum collector voltages of Q_1, Q_2 such that they remain in the active region for $v_i = \pm 1$ V around a common-mode voltage of 2 V. Repeat for the transconductor of Fig. 15.13.
- 15.7** In the transconductor of Fig. 15.14, find the new transconductance value if the base-emitter areas of Q_7, Q_8 are four times those of Q_3, Q_4 .
- 15.8** Consider the transconductor of Fig. 15.14, where we want $G_m = 1$ mA/V. What are reasonable values for I_1 and I_2 such that all transistors do not fall below 20 percent of their nominal bias current when the peak value of $v_i = v_i^+ - v_i^-$ is 500 mV?
- 15.9** Consider the linearized two-differential pairs shown in Fig. 15.20, where $I_1 = 2$ mA. Find the percentage of error between an ideal linear transconductor and the true output current when $v_i = 48$ mV.
- 15.10** Consider the circuit Fig. 15.21, with $I_1 = 100$ μ A, $I_2 = 10$ μ A, $(W/L)_9 = 2$, $V_C = 5$ V, and $v_i^- = 2.5$ V.
- Find the transconductance, G_m .
 - Assuming a perfectly linear transconductor, find i_{o1} when v_i^+ equals 2.6 V and 3 V.
 - Find the true value of i_{o1} when v_i^+ equals 2.6 V and 3 V by taking into account the triode equation for Q_9 (assume everything else is ideal). Compare your results with those in part b.
- 15.11** Choose reasonable transistor sizes for the circuit in Fig. 15.25 to realize a transconductor that has $G_m = 0.3$ mA/V, $k_1/k_3 = 6.7$, and a peak differential-input voltage of 1 V.
- 15.12** Consider the circuit in Fig. 15.30, where all transistors have $W/L = 10$ μ m/2 μ m, $V_{DD} = -V_{SS} = 2.5$ V, and $V_{C1} = -V_{C2} = 2.0$ V. Find the transconductance of the circuit.
- 15.13** Show that the transconductance of Fig. 15.32 is given by

$$G_m = \left(\frac{n}{n+1} \right) 4\sqrt{kI_B}$$

- 15.14** Consider the transconductor shown in Fig. 15.34, where all transistors have size $W/L = 10$ μ m/2 μ m, $I_B = 50$ μ A, and $V_{DD} = -V_{SS} = 3$ V. What

is the maximum differential input, $v_1 - v_2$ (centered around 0 V), that can be used while maintaining proper operation?

- 15.15** Show that the circuit of Fig. 15.36 results in

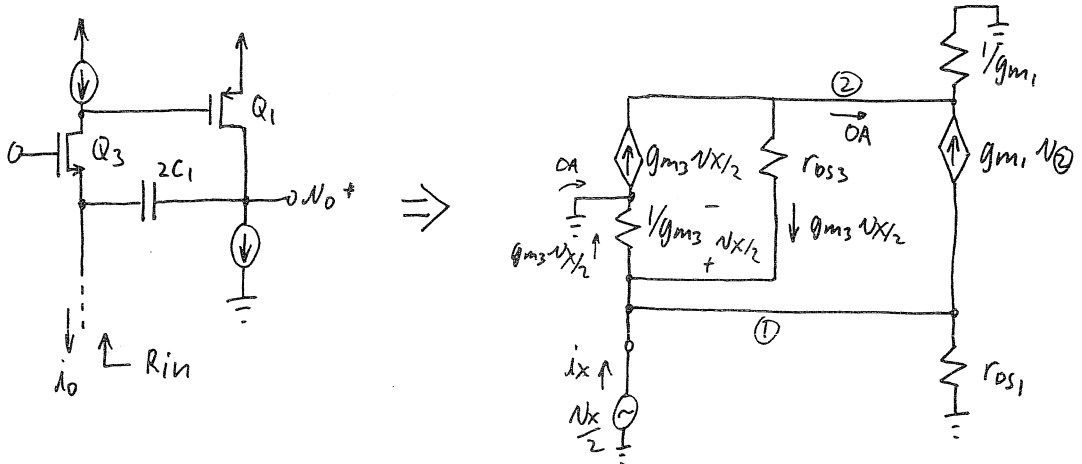
$$i_1 - i_2 = K(V_B - 2V_{tn})(2v_1 - V_B)$$

- 15.16** Using two-transistor integrators, design a MOSFET-C second-order low-pass filter with a pole frequency of 1 MHz, $Q = 1$, and a dc gain of 2. Let the integrating capacitors be 10 pF, $V_C = 3$ V, and $V_x = 0$ V.
- 15.17** Calculate the percentage THD value of a signal in which the fundamental component is at 1 MHz and has a voltage level of 1 V_{rms}, and the only significant harmonics are the first three above the fundamental. These harmonics, at 2 MHz, 3 MHz, and 4 MHz, are measured to have voltage levels of 1 mV_{rms}, 0.5 mV_{rms}, and 0.3 mV_{rms}, respectively.
- 15.18** If a circuit is measured to have $IIP_3 = 10$ dBm and has a gain of 6 dB, what output-signal level should be used such that the third-order intermodulation products are 60 dB below the fundamental?
- 15.19** At an input-signal level of -4 dBm, an intermodulation ratio of -40 dB was measured in a filter with a gain of 6 dB. Calculate the value of the input and output third-order intercept points. What input-signal level should be applied if we want an intermodulation ratio of -50 dB? If the noise power at the output is measured to be -60 dBm, what is the expected SFDR and what output-signal level does it correspond to?

Chapter 15 - Problems

15.1) Show $R_{in} \approx 2/(g_{m3}^2 r_{DS})$

$\therefore$ circuit is perfectly symmetric, we only need to look at one half.



(capacitor $2C_1$ is modelled as a short circuit)

$$\text{KCL } \textcircled{1}: i_x - g_{m3} \frac{v_x}{2} + g_{m3} \frac{v_x}{2} - g_{m1} v_{\textcircled{2}} - \frac{v_x/2}{r_{DS1}} = 0 \quad \textcircled{1}$$

$$\text{But } v_{\textcircled{2}} = \frac{v_x}{2} + g_{m3} \frac{v_x}{2} r_{DS3} \quad \textcircled{2}$$

$$\textcircled{2} \rightarrow \textcircled{1} \quad i_x - g_{m1} \left(\frac{v_x}{2} + g_{m3} r_{DS3} \frac{v_x}{2} \right) - \frac{v_x}{2 r_{DS1}} = 0$$

$$\therefore i_x = \frac{1}{2} \left(g_{m1} + g_{m1} g_{m3} r_{DS3} + \frac{1}{r_{DS1}} \right) v_x$$

$$\therefore R_{in} \triangleq \frac{v_x}{i_x} = \frac{2}{g_{m1} (1 + g_{m3} r_{DS3}) + 1/r_{DS1}} \approx \frac{2}{g_{m1} (1 + g_{m3} r_{DS3})}$$

$$\approx \frac{2}{g_{m1} g_{m3} r_{DS3}}$$

$$\therefore \underline{R_{in} \approx 2/(g_{m3}^2 r_{DS})} \quad \text{Q.E.D.}$$

(Note: g_{m3} is matched to g_{m2})

$$15.2) \quad H(s) = \frac{k_1 s + k_0}{s + \omega_0}$$

$$\therefore \text{DC gain} = \frac{k_0}{\omega_0} \equiv 10$$

$$\therefore k_0 = 10\omega_0$$

$\therefore$ no finite zeros

$$\therefore k_1 = 0$$

$$\therefore \underline{H(s) = \frac{10\omega_0}{s + \omega_0}}$$

From the design equations found in Figure 15.9
and $\omega_0 = 15 \text{ MHz} \times 2\pi = 94.25 \times 10^6 \text{ rads/sec}$

$$\underline{C_x = C_A \frac{k_1}{1 - k_1} = 0}$$

$$C_A \equiv 5 \text{ pF}$$

$$\begin{aligned} g_{m1} &= k_0 (C_A + C_x) = 10 \times 94.25 \times 10^6 (5 \times 10^{-12}) \\ &= \underline{4.71 \text{ mA/V}} \end{aligned}$$

$$\begin{aligned} g_{m2} &= \omega_0 (C_A + C_x) = 94.25 \times 10^6 (5 \times 10^{-12}) \\ &= \underline{0.47 \text{ mA/V}} \end{aligned}$$

$$15.3) \quad H(s) = \frac{k_0}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2} \quad (k_1 = k_2 = 0 \text{ for low pass})$$

$$H(0) = \frac{k_0}{\omega_0^2} \equiv 5 \quad \Rightarrow \underline{k_0 = 5\omega_0^2}$$

$$\omega_0 = 2\pi \times 10 \text{ MHz} = 62.8 \times 10^6 \text{ rads/sec}$$

$$Q = 1$$

Select reasonable values for C_A and C_B

$$\text{Let } \underline{C_A = C_B \equiv 2 \text{ pF}} \quad (C_x = 0 \because \text{LP filter})$$

$$\text{Then } g_{m1} = \omega_0 C_A = 62.8 \times 10^6 \times 2 \times 10^{-12} = \underline{0.13 \text{ mA/V}}$$

$$g_{m2} = \omega_0 C_B = \underline{0.13 \text{ mA/V}}$$

$$g_{m3} = \frac{\omega_0 C_B}{Q} = \underline{0.13 \text{ mA/V}}$$

$$g_{m4} = k_0 C_A / \omega_0 = 5\omega_0 C_A = \underline{0.63 \text{ mA/V}}$$

$$g_{m5} = \underline{0}$$

15.4) Derive design equations for Fig. 15.24.

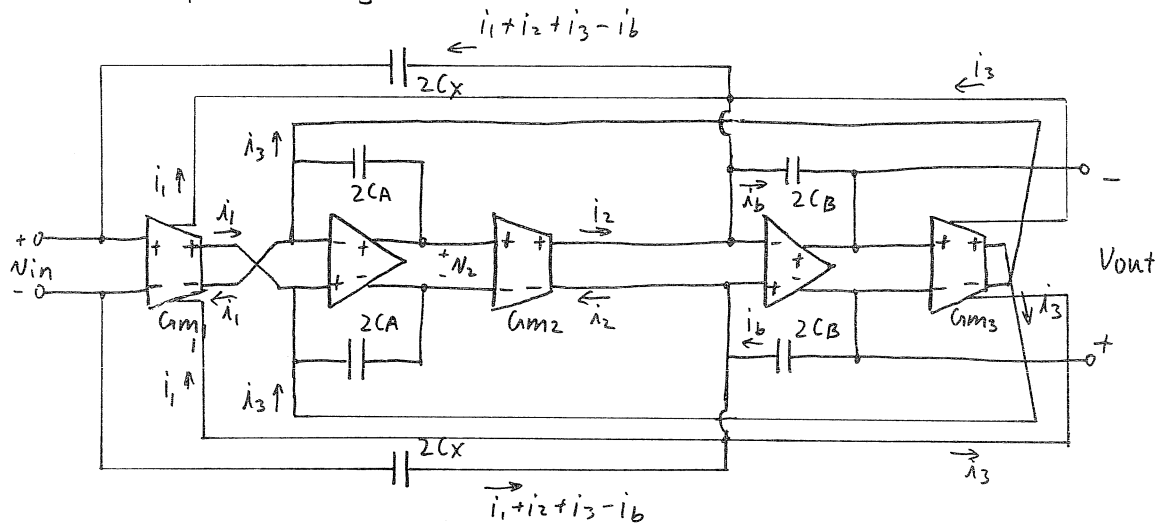
$$i_1 = g_{m1} V_{in} \quad [1]$$

$$i_3 = -g_{m3} V_{out} \quad [2]$$

$$V_2 = \frac{2(i_1 + i_3)}{2sC_A} = \frac{i_1 + i_3}{sC_A} \quad [3]$$

$$i_2 = g_{m2} V_2 \quad [4]$$

$$V_{out} = i_b / sC_B \quad [5]$$



KVL from V_{in} through $2C_x$ capacitors:

$$V_{in} + \left(\frac{i_1 + i_2 + i_3 - i_b}{2sC_x} \right) \times 2 = 0$$

$$\text{or } i_b = i_1 + i_2 + i_3 + sC_x V_{in} \quad [6]$$

$$\therefore [6] \rightarrow [5] \quad V_{out} = \frac{i_b}{sC_B} = \frac{i_1 + i_2 + i_3 + sC_x V_{in}}{sC_B} \quad [7]$$

$$[1], [2], [4] \rightarrow [7] \quad = \frac{g_{m1} V_{in} + g_{m2} V_2 - g_{m3} V_{out} + sC_x V_{in}}{sC_B}$$

$$\therefore sC_B V_{out} = g_{m1} V_{in} + \frac{g_{m2}}{sC_A} (g_{m1} V_{in} - g_{m3} V_{out}) - g_{m3} V_{out} + sC_x V_{in}$$

$$(sC_B + \frac{g_{m2} g_{m3}}{sC_A} + g_{m3}) V_{out} = (g_{m1} + \frac{g_{m1} g_{m2}}{sC_A} + sC_x) V_{in}$$

$$(s^2 C_A C_B + g_{m2} g_{m3} + s g_{m3} C_A) V_{out} = (s g_{m1} C_A + g_{m1} g_{m2} + s^2 C_x C_A) V_{in}$$

$$\therefore \frac{V_{out}}{V_{in}} = \frac{C_x C_A s^2 + C_A g_{m1} s + g_{m1} g_{m2}}{C_A C_B s^2 + C_A g_{m3} s + g_{m2} g_{m3}}$$

$$\frac{V_{out}}{V_{in}} = \frac{s^2 \frac{C_x}{C_B} + s \frac{g_{m1}}{C_B} + \frac{g_{m1} g_{m2}}{C_A C_B}}{s^2 + s \frac{g_{m3}}{C_B} + \frac{g_{m2} g_{m3}}{C_A C_B}} \quad \left[\text{which is similar to} \right]$$

Eq (15.24)

(cont.)

15.4 (cont.)

Equating our result to the generic second-order transfer function

$$H(s) = \frac{V_{out}}{V_{in}} = \frac{k_2 s^2 + k_1 s + k_0}{s^2 + \frac{\omega_0}{Q} s + \omega_0^2}$$

Gives the following design equations:

$$k_2 = \frac{C_X}{C_B} \Rightarrow \underline{C_X = k_2 C_B}$$

$$k_1 = \frac{G_{m1}}{C_B} \Rightarrow \underline{G_{m1} = k_1 C_B}$$

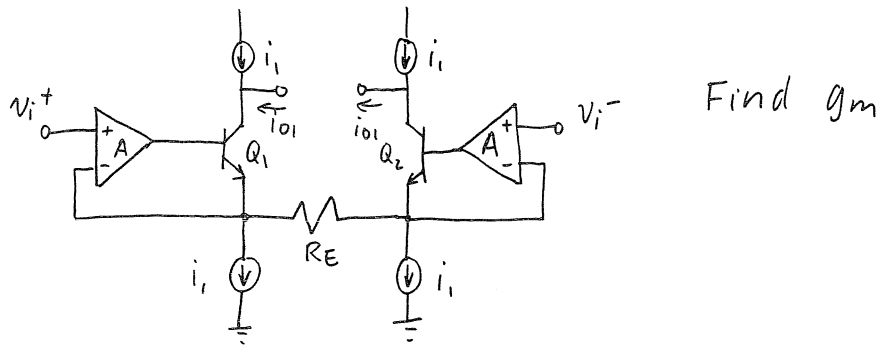
$$k_0 = \frac{G_{m1} G_{m2}}{C_A C_B} \Rightarrow \underline{G_{m2} = \frac{k_0 C_B C_A}{k_1 C_B} = \frac{k_0}{k_1} C_A} \quad (k_1 \neq 0)$$

$$\omega_0^2 = \frac{G_{m2} G_{m3}}{C_A C_B} \Rightarrow \underline{G_{m3} = \frac{C_A C_B}{G_{m2}} \omega_0^2 = \frac{\frac{k_0}{k_1} C_A C_B}{\frac{k_0}{k_1} C_A} \omega_0^2 = \frac{k_1}{k_0} C_B \omega_0^2} \quad (k_0 \neq 0)$$

$$\frac{\omega_0}{Q} = \frac{G_{m3}}{C_B} \Rightarrow \underline{\frac{\omega_0}{Q} = \frac{k_1}{k_0} \frac{C_B \omega_0^2}{C_B}}$$

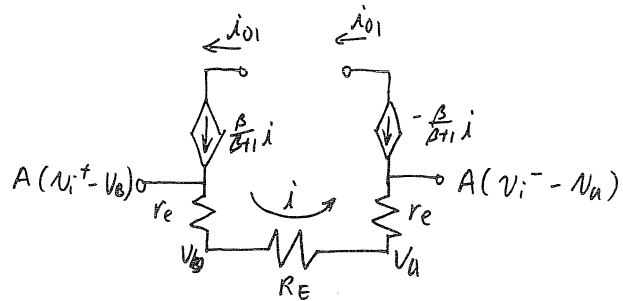
$$\underline{\omega_0 Q = \frac{k_0}{k_1}}$$

15.5)



Find g_m

Small signal model :



$$i = \frac{A(v_{i+} - V_b - v_{i-} + V_a)}{2r_e + R_E}$$

But $V_b - V_a = i R_E$

$$\therefore i = \frac{A(v_{id} - i R_E)}{2r_e + R_E} \quad \text{where } v_{id} \triangleq v_{i+} - v_{i-}$$

$$i(2r_e + R_E + A R_E) = A v_{id}$$

$$\therefore \frac{v_{id}}{i} = R_E + \frac{2r_e + R_E}{A}$$

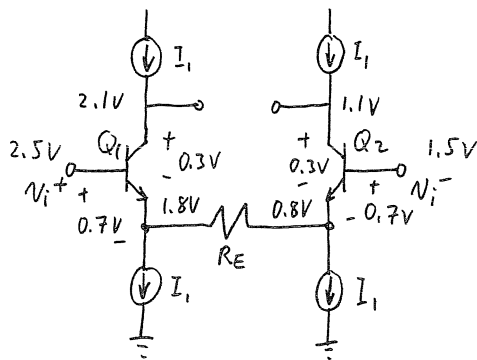
Now $i_{o1} = \frac{\beta}{\beta+1} i$

$$\therefore g_m \triangleq \frac{i_{o1}}{v_{id}} = \frac{i_{o1}}{i} \times \frac{i}{v_{id}} = \frac{\beta}{\beta+1} \times \frac{1}{R_E + \frac{2r_e + R_E}{A}}$$

Given β and $A \gg 1$

$$g_m \approx \frac{1}{R_E}$$

15.6) Find minimum collector voltages.



Given $V_i = \pm 1V$ and

$V_{cm} = 2V$

Consider the worse-case scenario where

$V_{i+} = 2.5V$ and

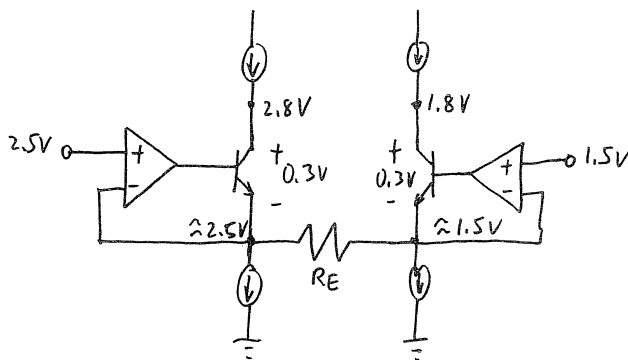
$V_{i-} = 1.5V$

(Assume $V_{cesat} = 0.3V$)

From the diagram, we need $V_{C1} \geq 2.1V$

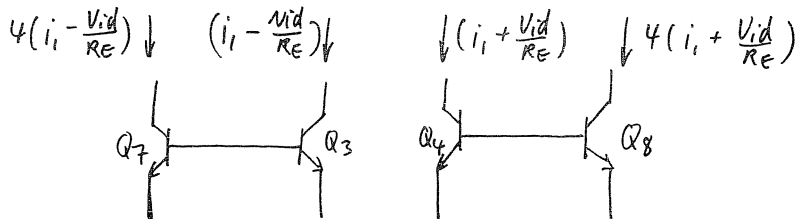
$\therefore$ the minimum collector voltage is 2.1V

For Figure 15.13, we see that $V_{C1} \geq 2.8V$



$\therefore$ the minimum collector voltage is 2.8V

15.7) Given the base-emitter areas of Q_7 & Q_8 are 4 times larger than those of Q_3 & Q_4 , we achieve a simple scaling of currents where



$$\therefore i_{o1} = \frac{4 V_{id}}{R_E} \Rightarrow \underline{G_m = 4/R_E}$$

15.8) Find I_1 and I_2

Q_1, Q_2, Q_5 and Q_6 have stable bias currents and only Q_3 and Q_4 have currents that vary significantly with the input signal.

$\therefore$ Ensure I_{C3} is within 20% of nominal which is I_1

Now when $V_i = 500\text{mV}$,

$$I_{C3} = I_1 - \frac{V_i}{R_E} \geq 0.8 I_1$$

$$\therefore I_1 \geq 5 \frac{V_i}{R_E}$$

Given

$$G_m = 1/R_E = 1\text{mA/V}$$

$$R_E = 1\text{k}\Omega$$

$$\text{and } I_1 \geq 5 \times 500\text{mV}/1\text{k}\Omega$$

$$\therefore \underline{I_1 \geq 2.5\text{mA}} \text{ which is rather large.}$$

We can reduce this by scaling Q_7 and Q_8 by 4 as in Problem 15.7. In this case,

$$G_m = 4/R_E \equiv 1\text{mA/V}$$

$$\therefore R_E = 4\text{k}\Omega$$

$$\text{and } \underline{I_1 \geq 5 \times 500\text{mV}/4\text{k}\Omega = 625\mu\text{A}}$$

The value of I_2 is not critical but should be small to prevent significant loading of bias current source I_1 .

$$\text{Let } \underline{I_2 = 1/4 I_1 = 156\mu\text{A}}$$

15.9) Find % error in output current.

From Eq (15.65),

$$G_m = \frac{8 I_1}{25 V_T} = \frac{8 \times 2 \text{ mA}}{25 \times 26 \text{ mV}} = 24.6 \text{ mA/V}$$

$$\therefore \text{Ideally, } i_o = G_m v_i = 24.6 \text{ mA/V} \cdot 0.048 \text{ V}$$

$$\underline{i_o = 1.182 \text{ mA}}$$

For the actual output current, i_o

$$v_i = V_{be1} - V_{be2} \quad \text{where}$$

$$V_{be1} = V_T \ln\left(\frac{I_{C1}}{4 I_s}\right)$$

$$V_{be2} = V_T \ln\left(\frac{I_{C2}}{I_s}\right)$$

$$\therefore v_i = V_T \left(\ln\left(\frac{I_{C1}}{4 I_s}\right) - \ln\left(\frac{I_{C2}}{I_s}\right) \right)$$

$$= V_T \ln\left(\frac{I_{C1}}{4 I_{C2}}\right)$$

$$\Rightarrow \underline{I_{C1} = 4 e^{v_i/V_T} I_{C2}}$$

$$\text{Now } I_1 = I_{E1} + I_{E2} = \frac{\beta+1}{\beta} (I_{C1} + I_{C2})$$

$$= \frac{\beta+1}{\beta} (4 e^{v_i/V_T} + 1) I_{C2}$$

Similarly

$$\therefore I_{C2} = \frac{\beta}{\beta+1} \times \frac{I_1}{1 + 4 e^{v_i/V_T}}$$

$$\text{and } I_{C1} = \frac{\beta}{\beta+1} \times \frac{I_1}{1 + 1/4 e^{-v_i/V_T}}$$

$$I_{C3} = \frac{\beta}{\beta+1} \frac{I_1}{1 + 4 e^{-v_i/V_T}}$$

$$I_{C4} = \frac{\beta}{\beta+1} \frac{I_1}{1 + 1/4 e^{v_i/V_T}}$$

$$i_o = -I_1 + I_{C1} + I_{C3} = 2 \text{ mA} \left(-1 + \frac{100}{101} \left(\frac{1}{1 + 1/4 e^{-48/26}} + \frac{1}{1 + 4 e^{-48/26}} \right) \right)$$

$$= 1.119 \text{ mA}$$

$$\therefore \underline{\% \text{ error}} = \frac{i_o - i_{o \text{ ideal}}}{i_{o \text{ ideal}}} = \frac{1.119 - 1.182}{1.182} = \underline{-5.3\%}$$

15.10)

a) Find $G_m = \mu_n C_{ox} (W/L)_q (V_{gsq} - V_{tnq})$ where

$$\begin{aligned} V_{gsq} &= V_{gq} - V_{sq} = V_c - V_{s1} = V_c - (V_{i^+} - V_{tn1}) \\ &= V_c + V_{tn1} - V_{i^+} \rightarrow \text{Assume } V_{i^+} = V_{i^-} = 2.5V \end{aligned}$$

for V_{tn1} ,

$$\begin{aligned} V_{tn1} &= V_{tn0} + \gamma (\sqrt{V_{sb1} + 2\phi_F} - \sqrt{2\phi_F}) \quad , \text{ Assume } V_{sb1} \approx V_{i^+} - 1.2V \approx 1.3V \\ &= 0.8 + 0.5 (\sqrt{1.3 + 0.7} - \sqrt{0.7}) \\ &= 1.09 V \end{aligned}$$

$$\therefore V_{sb1} = 2.5 - 1.09 V = 1.4 V \quad \leftarrow \begin{array}{l} \text{close enough} \\ \rightarrow \text{assumption accurate} \end{array}$$

$$\begin{aligned} \therefore V_{gsq} &= V_c + V_{tn1} - V_{i^+} = 5 + 1.09 - 2.5 \\ &= 3.59 V \end{aligned}$$

$$\begin{aligned} V_{tnq} &= V_{tn0} + \gamma (\sqrt{V_{sb1} + 2\phi_F} - \sqrt{2\phi_F}) \\ &= 0.8 + 0.5 (\sqrt{1.4 + 0.7} - \sqrt{0.7}) \\ &= 1.106 V \end{aligned}$$

$$\begin{aligned} \therefore G_m &= \mu_n C_{ox} (W/L)_q (V_{gsq} - V_{tnq}) \\ &= 92 \times 10^{-6} (2) (3.59 - 1.106) \end{aligned}$$

$$\therefore \underline{G_m = 0.46 \text{ mA/V}}$$

b) Find i_{o1} when $V_{i^+} = 2.6 V$ & $3V$

$$\begin{aligned} i_{o1} /_{V_{i^+} = 2.6V} &= G_m \times (V_{i^+} - V_{i^-}) \\ &= 0.46 \text{ mA/V} (2.6 - 2.5) \\ &= \underline{46 \mu A} \end{aligned}$$

$$\begin{aligned} i_{o1} /_{V_{i^+} = 3.0V} &= G_m \times (V_{i^+} - V_{i^-}) \\ &= 0.46 \times (3.0 - 2.5) \\ &= \underline{230 \mu A} \end{aligned}$$

(cont.)

15.10 (cont.)

c) Find true i_{o1} when $V_{i^+} = 2.6V$ and $3.0V$

Assume that the threshold voltages of Q_1 and Q_2 remain equal to those found in part a).

$$\text{i.e., } V_{tn1} = 1.09V$$

$$V_{tn2} = 1.106V$$

$$i_{o1} = i_{o2} = \mu_n C_{ox} (W/L)_2 \left((V_{gs2} - V_{tn2}) V_{ds2} - \frac{V_{ds2}^2}{2} \right)$$

For $V_{i^+} = 2.6V$:

$$\begin{aligned} V_{gs2} &= V_L + V_{tn1} - V_{i^+} = 5 + 1.09 - 2.6 \\ &= 3.58V \end{aligned}$$

$$\text{and } V_{ds2} = 0.1V$$

$$\begin{aligned} \therefore i_{o1} &= 92 \times 10^{-6} \times (2) \left((3.58 - 1.106) 0.1 - \frac{(0.1)^2}{2} \right) \\ &= \underline{45 \mu A} \end{aligned}$$

$$\% \text{ error} = \frac{45 - 46}{46} \approx \underline{2\% \text{ error}}$$

For $V_{i^+} = 3.0V$:

$$\begin{aligned} V_{gs2} &= 5 + 1.09 - 3.0 \\ &= 3.09V \end{aligned}$$

$$\text{and } V_{ds2} = 0.5V$$

$$\begin{aligned} \therefore i_{o1} &= 92 \times 10^{-6} (2) \left((3.09 - 1.106) 0.5 - \frac{(0.5)^2}{2} \right) \\ &= \underline{160 \mu A} \end{aligned}$$

$$\therefore \% \text{ error} = \frac{160 - 230}{230} = \underline{-30\% \text{ error}}$$

15.11) Given

$$\frac{K_1}{K_3} = 6.7 \Rightarrow \frac{\frac{\mu_n C_{ox}}{2} (W/L)_1}{\frac{\mu_n C_{ox}}{2} (W/L)_3} = 6.7$$

$$\text{If } L_1 = L_2 = L,$$

$$\therefore W_1 = 6.7 W_3$$

To find W_3 ,

$$G_m = \frac{4k_1 k_3 \sqrt{I_1}}{(k_1 + 4k_3) \sqrt{K_1}} \equiv 0.3 \text{ mA/V}$$

$$\therefore \frac{4 \times 6.7 k_3 \sqrt{I_1}}{(6.7 + 4) k_3 \sqrt{6.7 k_3}} =$$

$$\sqrt{I_1 k_3} = 0.31 \times 10^{-3}$$

$$I_1 W_3 = \frac{(0.31)^2 \times 10^{-6}}{\frac{\mu_n C_{ox}}{2} \times 1/L} \quad \text{If } L = 1 \mu\text{m}$$

$$= \frac{0.31^2 \times 2 \times 10^{-6}}{92 \times 10^{-6}} = 2.09 \times 10^{-3} \text{ mA} \cdot \mu\text{m}$$

$$\text{Let } I_1 = 200 \mu\text{A}$$

$$\therefore \underline{W_3} = \frac{2.09 \times 10^{-3}}{200 \times 10^{-6}} \approx \underline{10 \mu\text{m}} = W_4$$

$$\therefore \underline{W_1} = 6.7 W_3 = \underline{67 \mu\text{m}} = W_2$$

} and $L = 1 \mu\text{m}$

15.12) Find $G_m = 4 K_{eq} (V_{c1} - V_{teq})$

Ignore body effect.

$$\text{Now } \frac{1}{\sqrt{K_{eq}}} = \frac{1}{\sqrt{K_n}} + \frac{1}{\sqrt{K_p}} \quad \text{where } K_n = \frac{\mu_n C_{ox}}{2} \frac{W}{L} = \frac{92 \times 10^{-6}}{2} \times 10/2$$

$$= 230 \times 10^{-6}$$

$$K_p = \frac{30 \times 10^{-6}}{2} \times 10/2 = 75 \times 10^{-6}$$

$$\therefore \frac{1}{\sqrt{K_{eq}}} = 181.5$$

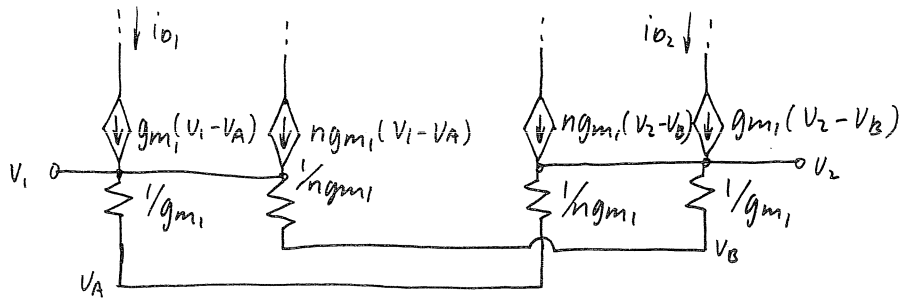
$$K_{eq} = 30 \times 10^{-6}$$

$$V_{teq} = V_{tn} - V_{tp} = 0.8 + 0.9 \text{ V} = 1.7 \text{ V}$$

$$\therefore G_m = 4 \times 30 \times 10^{-6} (2 - 1.7)$$

$$= \underline{36 \text{ mA/V}}$$

15.13) Show $G_m = \left(\frac{n}{n+1}\right) 4 \sqrt{K I_B}$



$$i_{D1} = \frac{V_1 - V_2}{\frac{1}{g_{m1}} + \frac{1}{ng_{m1}}} \quad \text{and by symmetry, } i_{D2} = -i_{D1}$$

$$\therefore i_{D1} - i_{D2} = 2 \times \frac{V_1 - V_2}{\frac{1}{g_{m1}} + \frac{1}{ng_{m1}}}$$

$\therefore$ the transconductance, G_m

$$G_m \triangleq \frac{i_{D1} - i_{D2}}{V_1 - V_2} = \frac{2}{\frac{1}{g_{m1}} + \frac{1}{ng_{m1}}} = \frac{2 g_{m1}}{1 + 1/n}$$

$$= \frac{2n}{n+1} g_{m1} \quad \text{where } g_{m1} = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_B}$$

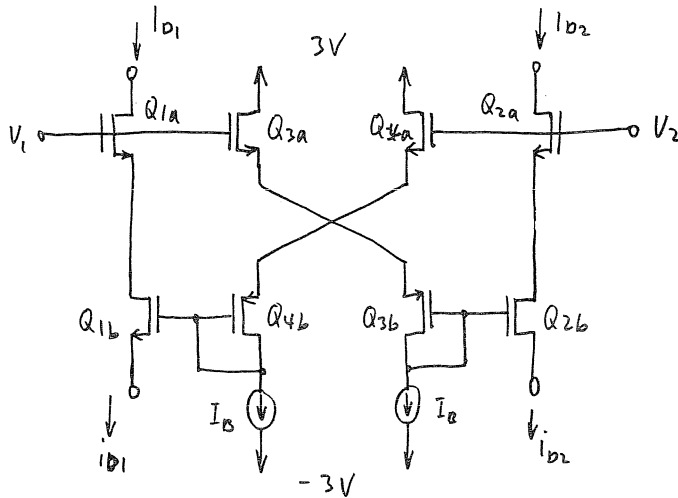
$$= \sqrt{4K I_B} \quad \text{for } K = \frac{\mu_n C_{ox}}{2} \frac{W}{L}$$

$$\therefore G_m = \frac{2n}{n+1} \sqrt{4K I_B}$$

$$\underline{G_m = \left(\frac{n}{n+1}\right) 4 \sqrt{K I_B}}$$

Q.E.D.

(5, 14)



Find the maximum differential voltage centred about 0V.

$$V_{eff4b} = V_{eff3b} = \sqrt{\frac{2 I_B}{\mu_p C_{ox} W/L}} = \sqrt{\frac{2 \times 50 \times 10^{-6}}{30 \times 10^{-6} \times 10^2}} = 0.816 V$$

$$V_{eff\ 4a} = V_{eff\ 3a} = \sqrt{\frac{2 I_a}{m \cos \theta_L}} = \sqrt{\frac{2 \times 50 \times 10^{-6}}{92 \times 10^{-6} \times 102}} = 0.466 V$$

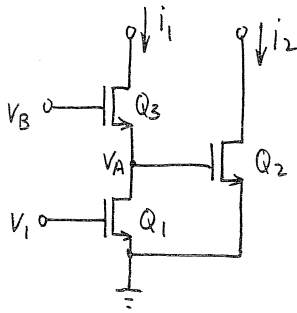
Assuming the voltage drops across current sources I_1 can be as low as 0 volts, the minimum voltage level for V_1 and V_2 is

$$\begin{aligned} V_{1,2 \min} &= V_{SS} + V_{I_{\min}} - V_{tp3,4b} + V_{eff3,4b} + V_{tn3,4a} + V_{eff3,4a} \\ &= -3 + 0 + 0.9 + 0.816 + 0.8 + 0.466 \text{ V} \\ &= -18 \text{ mV} \end{aligned}$$

$\therefore$ the minimum level for V_1 and V_2 is -18mV
 and because V_1 and V_2 are centred around 0V ,
 $\therefore$ the maximum differential voltage is

$$V_1 - V_2 = 18\text{mV} - (-18\text{mV}) = \underline{36\text{mV}}$$

15.15) Show $i_1 - i_2 = K(V_B - 2V_{tn})(2V_1 - V_B)$



$$i_{D1} = K(V_1 - V_{tn})^2$$

$$i_{D3} = K(V_B - V_A - V_{tn})^2$$

$$\text{But } i_{D1} = i_{D3} \equiv i_1$$

$$\therefore K(V_1 - V_{tn})^2 = K(V_B - V_A - V_{tn})^2$$

$$V_1 - V_{tn} = V_B - V_A - V_{tn}$$

$$\text{or } \underline{V_A = V_B - V_1} \quad (1)$$

$$i_{D2} = K(V_A - V_{tn})^2 \quad (2)$$

$$(1) \rightarrow (2) = K(V_B - V_1 - V_{tn})^2 \equiv i_2$$

$$\therefore i_1 - i_2 = K[(V_1 - V_{tn})^2 - (V_B - V_1 - V_{tn})^2]$$

$$= K[V_1^2 - 2V_1V_{tn} + V_{tn}^2 - (V_B^2 + V_1^2 + V_{tn}^2 - 2V_1V_B - 2V_BV_{tn} + 2V_1V_{tn})]$$

$$= K(-V_B^2 + 2V_1V_B - 4V_1V_{tn} + 2V_{tn}V_B)$$

$$\therefore \underline{i_1 - i_2 = K(V_B - 2V_{tn})(2V_1 - V_B)}$$

Q.E.D.

15.16) Refer to Figure 15.42 a) for schematic.

Given that we want a second-order low pass filter with a DC gain of 2,

$$H(s) = \frac{k_0}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2} = \frac{2\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2} \quad (1)$$

From Eq. (15.161)

$$\frac{V_O}{V_i} = \frac{\frac{C_1}{C_B} s^2 + \frac{G_2}{C_B} s + \frac{G_1 G_3}{C_A C_B}}{s^2 + \frac{G_5}{C_B} s + \frac{G_3 G_4}{C_A C_B}} \quad (2)$$

(cont.)

15.16 (cont.)

Equating the coefficients of [1] and [2],

$$C_1 = C_2 = 0 \quad (\text{i.e., remove components})$$

$$C_A = C_B = 10 \text{ pF}$$

$$\frac{G_5}{C_B} = \omega_0 \Rightarrow G_5 = C_B \omega_0 = 10 \times 10^{-12} \times 2\pi \times 1 \text{ MHz} \\ = 63 \text{ mA/V}$$

$$\text{Now } G_1 G_3 = 2 C_A C_B \omega_0^2 = 2 (10 \times 10^{-12})^2 (2\pi \times 1 \text{ MHz})^2 \\ = 7.90 \times 10^{-9}$$

$$\text{and } G_3 G_4 = C_B C_0 \omega_0^2 = 3.95 \times 10^{-9}$$

If we let $G_3 = 100 \text{ mA/V}$,

$$G_1 = \frac{7.90 \times 10^{-9}}{100 \times 10^{-6}} = 79 \text{ mA/V} \quad \text{and}$$

$$G_4 = \frac{3.95 \times 10^{-9}}{100 \times 10^{-6}} = 40 \text{ mA/V}$$

Now transistor sizes,

$$G_i = \mu_n C_{ox} (W/L)_i (V_c - V_x - V_{tn})$$

$$\therefore G_1 = 92 \times 10^{-6} (W/L)_1 (3 - 0.8) = 79 \text{ mA/V}$$

$$\therefore \underline{(W/L)_1 = 0.39}$$

Similarly,

$$G_3 = 92 \times 10^{-6} (W/L)_3 (3 - 0.8) = 100 \text{ mA/V}$$

$$\Rightarrow \underline{(W/L)_3 = 0.49}$$

$$\underline{(W/L)_4 = \frac{40 \text{ mA/V}}{92 \times 10^{-6} (3 - 0.8)} = 0.2}$$

$$\underline{(W/L)_5 = \frac{63 \text{ mA/V}}{92 \times 10^{-6} (3 - 0.8)} = 0.31}$$

and device G_2 is eliminated.

15.17)

$$\begin{aligned}
 \% \text{ THD} &= \left[\frac{P_{wr 2\text{MHz}} + P_{wr 3\text{MHz}} + P_{wr 4\text{MHz}}}{P_{wr \text{fundamental}}} \right]^{1/2} \\
 &= \left[\frac{V_{2\text{MHz}}^2 + V_{3\text{MHz}}^2 + V_{4\text{MHz}}^2}{V_{\text{fundamental}}^2} \right]^{1/2} \\
 &= \left[\frac{(1)^2 + (0.5)^2 + (0.3)^2}{1000^2} \right]^{1/2} = \frac{1.158}{1000}
 \end{aligned}$$

$$\therefore \% \text{ THD} = 0.12\%$$

15.18) Find output signal level.

$$\begin{aligned}
 OIP_3 &= IIP_3 + a_1 \\
 &= 10\text{dBm} + 6\text{dB} \\
 OIP_3 &= 16\text{dBm}
 \end{aligned}$$

$$\text{Let } ID_3 = -60\text{dB}$$

$$\begin{aligned}
 \therefore I_{D1} &= OIP_3 + \frac{ID_3}{2} = 16\text{dBm} - \frac{60\text{dB}}{2} \\
 &= \underline{-14\text{dBm}}
 \end{aligned}$$

Thus, an output level of -14 dBm should be used.

15.19) Given: Input at -4dBm
output at 2dBm ($= I_{D1}$)
Measured $ID_3 = -40\text{dB}$

$$\therefore OIP_3 = I_{D1} - \frac{ID_3}{2} = 2\text{dBm} + \frac{40\text{dB}}{2}$$

$$\underline{OIP_3 = 22\text{dBm}} \quad \text{and}$$

$$IIP_3 = OIP_3 - a_1 = 22\text{dBm} - 6\text{dB}$$

$$\underline{IIP_3 = 16\text{dBm}}$$

For $ID_3 = -50\text{dB}$

$$I_{D1} = OIP_3 + \frac{ID_3}{2} = 22\text{dBm} - \frac{50}{2}$$

$$\underline{I_{D1} = -3\text{dBm}}$$

This corresponds to an input level of $-3\text{dBm} - 6\text{dB} = \underline{-9\text{dBm}}$

If $N_0 = -60\text{dBm}$, then

$$SFDR = \frac{2}{3}(OIP_3 - N_0)$$

$$= \frac{2}{3}(22\text{dBm} + 60\text{dBm})$$

$$= \underline{55\text{dB}}$$

$$\text{and } I_{D1}^* = SFDR + N_0 = 55\text{dBm} - 60\text{dBm}$$

$$= \underline{-5\text{dBm}}$$

Thus the output signal level is -5dBm and the
SFDR is 55dB.

Chapter 16

16.7 PROBLEMS

- 16.1** A CMOS exclusive-OR gate is being used as phase-detector in a PLL. The power-supply voltages are 0 V and 5 V. The low-pass filter is composed of discrete components. The VCO has a 10 MHz free-running frequency for when $V_{\text{cntl}} = 2.5$ V, and it has $K_{\text{osc}} = 5$ MHz/V. Assume we want the time-constant of the loop to be 50 μs . Design the low-pass filter for $Q = 0.5$. Accurately sketch the voltage waveforms of all nodes in the loop for when the loop is in lock and the input frequency is 7 MHz, both with and without the deglitching capacitor, C_2 , of the low-pass filter. Label the maximum and minimum voltages of each node voltage in your sketch. What is the phase difference between the input signal and the oscillator's output signal? What is the lock range of the loop, assuming the VCO is linear for $1\text{ V} < V_{\text{cntl}} < 4\text{ V}$?
- 16.2** A PLL has an amplifier with a gain of A following its low-pass filter, as shown in Fig. P16.2. The conversion factor of the phase detector is 3 V/rad and the conversion factor of the VCO is 50 KHz/V. The phase detector is a multiplier. The design for the low-pass filter and the amplifier gain are to be chosen. The PLL is intended to be used to demodulate a 5-KHz signal that

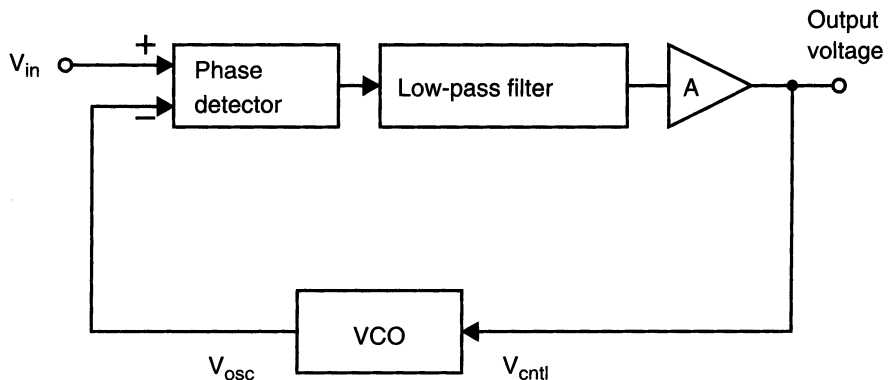


Fig. P16.2 A PLL in which an amplifier has been added after the low-pass filter.

has been frequency modulated on a 5-MHz carrier having a ± 50 -KHz deviation. Design the filter so that the magnitude of the transfer function of the small-signal model of the loop is maximally flat. The -3 -dB frequency should be the same as the band width of the signal that has been modulated. The lock range should be chosen so that the VCO's frequency never exceeds one-half the maximum possible range. Explain what happens to the lock range if the amplifier gain is increased?

- 16.3** A PLL with a multiplier phase detector has been designed so its linearized small-signal transfer function has a Q factor of $1/2$ and a resonant frequency of one-hundredth the free-running frequency of the VCO, when the VCO is at its free-running frequency. What are the resonant frequency and Q factor when the VCO's frequency is different from its free-running frequency by one-half the lock range?
- 16.4** The low-pass filter designed in Example 16.4 has V_{in} leading V_{osc} by $\pi/8$. Accurately sketch the voltage waveforms of P_u , P_d , and V_{lp} both with and without the deglitching capacitor, C_2 . Label the values of the voltages at each major transition point. (*Hint*: The general equation that describes the transient relaxation of a first-order circuit may be useful).
- 16.5** Let $K_{osc} = 2\pi \cdot 25 \text{ MHz} \cdot \text{V}^{-1}$ and $I_{ch} = 10 \text{ } \mu\text{A}$ in a charge-pump PLL. Assume the time free-running frequency of the oscillator is 100 MHz and that we want the loop to have a time constant of 50 cycles, or $0.5 \text{ } \mu\text{s}$. Design the components of the low-pass filter. The filter should include the second deglitching capacitor, C_2 .
- 16.6** Sketch the waveforms of V_{in} , V_{osc} , P_u , P_d , P_{u-dsbl} , and P_{d-dsbl} of the phase/frequency comparator for the case when V_{in} has a much smaller frequency than V_{osc} .
- 16.7** Find reasonable values for all current sources, the resistors, and C_{osc} of the emitter-coupled multivibrator in Fig. 16.12 so that the multivibrator has a free-running frequency of 10 MHz . Assuming that the on voltage of a diode changes by -2 mV for every $^\circ\text{C}$ increase in temperature, estimate the change in frequency if the temperature increases by $20 \text{ }^\circ\text{C}$.
- 16.8** Assume a ring oscillator is realized by five CMOS inverters that have a threshold voltage equal to $V_{DD}/2$. What would the change in frequency be if the power-supply voltage changes from 5 V to 5.5 V ? If this happens right at the beginning of a period, what would the corresponding time jitter of output waveform be at the end of the period?
- 16.9** Derive an equation for the frequency of operation of a ring oscillator that has delay stages of the type shown in Fig. 16.20. Assume the capacitive load of each output is constant and equal to C_L . Also, assume that, when an inverter is switching, its bias current, I_b , changes very quickly from one side to the other.
- 16.10** Shown in Fig. P16.10 is the bias circuitry and one stage of a four-stage ring oscillator. Assume the loading at each output of the differential inverter is

0.1 pF. Assume transistors are all in the active region. Given that $V_{\text{cntl}} = 1$ V, $V_{\text{ref}} = 1$ V, and $R = 100$ k Ω , derive and give a value for the frequency of oscillation and for the oscillator's gain constant (in radians/volt).

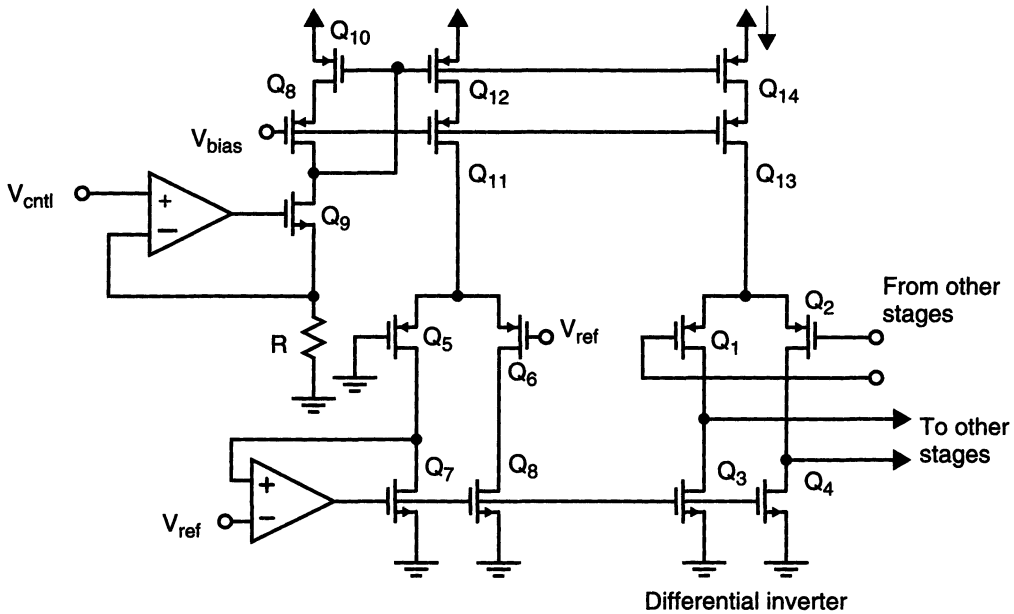


Fig. P16.10

- 16.11** Derive an equation for the frequency of operation of the CMOS relaxation oscillator shown in Fig. 16.22.

Chapter 16 - Problems

16.1) $f_{osc} = 10 \text{ MHz} + (V_{ctrl} - 2.5) 5 \text{ MHz} \Rightarrow K_{osc} = 2\pi \times 10^7 \text{ rad/V.s}$

$\tau_{PLL} = 50 \mu\text{s} \Rightarrow \omega_o = \frac{1}{\tau_{PLL}} = 20 \text{ k rad/s}$

$K_{pd} = \frac{V_{OD}}{\pi} = 1.59 \text{ V/rad} \cdot \quad K_{PLL} = \sqrt{K_{pd} K_{lp} K_{osc}} = 9995 \text{ s}^{-1/2}$

Using (16.32): $\tau_p = 0.25 \text{ sec}$

Using (16.36): $\tau_z = 0.1 \text{ msec}$

choosing arbitrarily $C_1 = 100 \text{ nF} \Rightarrow R_2 = \frac{\tau_z}{C_1} = 1 \text{ k}\Omega$

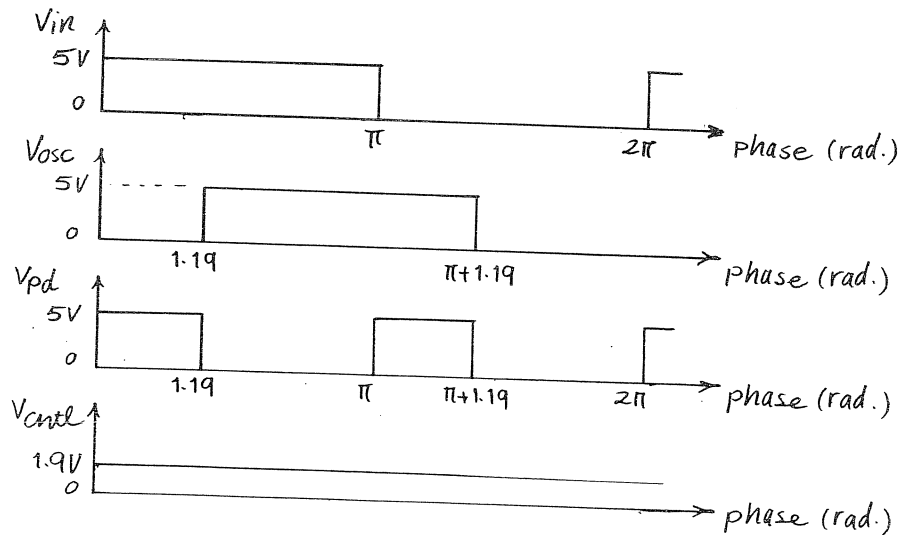
$R_1 = \frac{\tau_p}{C_1} - R_2 = 2.5 \text{ M}\Omega$

$f_{osc} = 7 \text{ MHz} \Rightarrow V_{ctrl} = 1.9 \text{ V} \Rightarrow \Delta\phi_{in} = \frac{1.9 \text{ V}}{K_{pd}} = 1.19 \text{ rad.}$

$1 < V_{ctrl} < 4 \Rightarrow 2.5 \text{ MHz} < f_{osc} < 17.5 \text{ MHz}$

Deglitching Capacitor C_2 can be chosen to be: $C_2 = \frac{C_1}{10} = 10 \text{ nF}$

C_2 has little effect on the PLL performance since the high-frequency gain of the filter is very small ($\frac{R_1}{R_2} < 0.001$)!



$$16.2) \quad K_{pd} = 3 \text{ V/rad} \quad , \quad K_{osc} = 2\pi \cdot 50 \text{ Krad/V.s}$$

$$Q = 0.707 \quad , \quad \tau_p = \frac{1}{2\pi f_p} = \underline{31.8 \text{ } \mu\text{sec}}$$

$$f_{osc} = 5 \text{ MHz} + V_{ctrl} \cdot 50 \text{ KHz}$$

In order to have a lock range of $\pm 100 \text{ KHz}$, we must have:

$$"-2 \text{ V} < V_{ctrl} < 2 \text{ V} "$$

$$K_{pd} = 3 \text{ V/rad} \Rightarrow \underline{A = 2/3}$$

$$\text{Using (16.34)} : \underline{K_{PLL} = 792.7 \text{ s}^{-1/2}}$$

$$\text{Using (16.33)} : \underline{\tau_z = 8.5 \text{ } \mu\text{sec}}$$

$$\text{Choosing arbitrarily } \underline{C_1 = 10 \text{ nF}} \Rightarrow R_2 = \frac{\tau_z}{C_1} = \underline{847 \Omega}$$

$$R_1 = \frac{\tau_p}{C_1} - R_2 = \underline{2333 \Omega}$$

The lock range will increase with an increase in gain if V_{ctrl} is not limited by the VCO design.

$$16.3) \quad \text{From (16.9)} : V_{ctrl} = K_{pd} K_{lp} \sin \phi$$

The linearized model of this equation at $\phi = 0$ is:

$$\Delta V_{ctrl} = K_{pd} K_{lp} \Delta \phi \quad (\text{at } \phi = 0)$$

At the frequency half-the-lock-range away from f_r ,

$\sin \phi = \frac{1}{2}$ or $\phi = \pi/6$. At this phase, the linearized model is:

$$\Delta V_{ctrl} = \left(\frac{\sqrt{3}}{2} K_{pd} \right) K_{lp} \Delta \phi \quad (\text{at } \phi = \pi/6)$$

Which means K_{pd} has effectively been reduced (cont.)

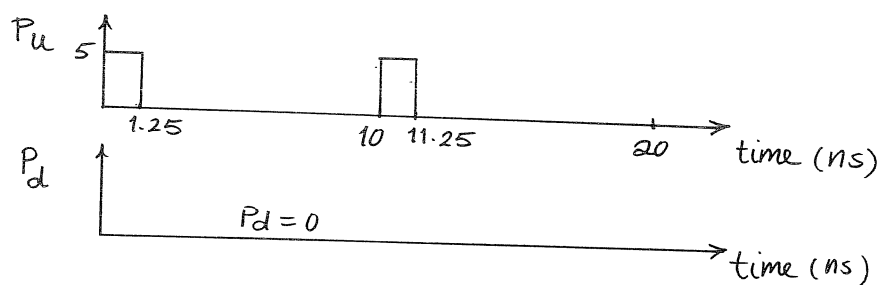
16.3) (cont.) by a factor of $\frac{\sqrt{3}}{2}$. Using (16.32) & (16.35):

$$\underline{\omega_o = 0.93 \omega_o(\text{original})} \quad \& \quad \underline{Q = 1.07 Q(\text{original})}$$

16.4) $f_{osc} = 50 \text{ MHz} \Rightarrow T_{osc} = 20 \text{ nsec.}$

Therefore, a phase difference of $\frac{\pi}{8}$ corresponds to:

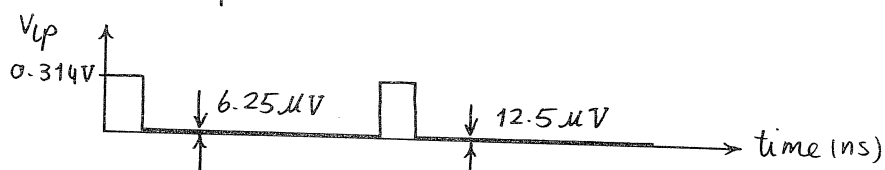
$$\frac{20 \text{ ns}}{2\pi} \times \frac{\pi}{8} = \underline{2.5 \text{ ns}}$$



Case I) without C_2 : the initial V_{lp} jump is $I_{ch}R = 0.314 \text{ V}$!

The voltage increment by V_{lp} during half a period is

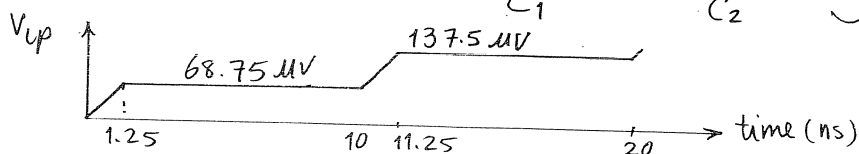
$$\frac{I_{ch} \times 1.25 \text{ ns}}{C_1} = \underline{6.25 \mu\text{V}}$$



Case II) With C_2 : the initial jump is 0 (note that

$RC_2 = 6.28 \mu\text{s}$, $\therefore$ jump during 1.25 ns is filtered)

$$\text{the voltage increment: } \frac{I_{ch} \times 1.25 \text{ ns}}{C_1} + \frac{I_{ch} \times 1.25 \text{ ns}}{C_2} = \underline{68.75 \mu\text{V}}$$



16.5) A time constant of $0.5 \mu s$ corresponds to

$$\omega_0 = \frac{1}{0.5 \mu s} = 2 \text{ M rad/s}$$

Using (16.56) and (16.60), we have:

$$C_1 = \frac{1}{\omega_0^2} \frac{I_{ch}}{2\pi} K_{osc} = \underline{62.5 \text{ pF}}$$

$$\text{Let } C_2 = C_1/10 = \underline{6.25 \text{ pF}}$$

$$\text{Let } Q = 0.4, \text{ from (16.61): } R = \frac{1}{Q} \sqrt{\frac{2\pi}{C_1 I_{ch} K_{osc}}} = \underline{20 \text{ k}\Omega}$$

16.6) Since the phase detector of Fig. 16.9 is symmetric with respect to V_{in} and V_{osc} , the voltage waveform for the case $f_{osc} \gg f_{in}$ are the same as those shown in Fig. 16.10 with V_{in} , P_u , and P_{u-dsbl} interchanged with V_{osc} , P_d , and P_{d-dsbl} , respectively.

$$16.7) f_{fr} = \frac{I_1}{4V_D C_{osc}}$$

$$\text{Assuming } V_D \approx 0.7 \text{ V} \Rightarrow 10 \text{ M} = \frac{I_1}{2.8 C_{osc}} \Rightarrow \frac{I_1}{C_{osc}} = 28 \text{ M}$$

$$\text{Choosing } \underline{C_{osc} = 0.5 \text{ pF}} \Rightarrow \underline{I_1 = 14 \mu A}$$

$$V_D (T_{nom} + 20^\circ) = 700 \text{ mV} - 20 \times 2 \text{ mV} = 0.66 \text{ V}$$

$$\Rightarrow f_{fr} (T_{nom} + 20^\circ) = \frac{14 \mu}{4 \times 0.66 \times 0.5 \text{ p}} = \underline{10.6 \text{ MHz}}$$

16.8) Since $f_{osc} \propto V_{DD}$:

$$f_{osc} \Big|_{V_{DD}=5.5V} = \frac{5.5}{5} f_{osc} \Big|_{V_{DD}=5V} = 1.1 f_{osc} \Big|_{V_{DD}=5V}$$

That is a 10 % increase in the frequency of osc.

$$\text{Time-Jitter} = T_{osc}(5V) - T_{osc}(5.5V) \approx \underline{0.1 T_{osc}(5V)}$$

16.9) Assuming a small differential voltage can switch the inverter, the delay of each stage is the time required for each output to change by $V_{ref}/2$.

$$V_o(t) = V_{ref} (1 - e^{-t/\tau_{inv}}) = V_{ref}/2 \Rightarrow \tau_{inv} = RC_L \ln 2$$

$$\text{Also, } R = V_{ref}/I_b. \text{ Therefore, } \underline{f_{osc} = \frac{I_b}{nV_{ref}RC_L \ln 2}}$$

where n is the number of stages.

16.10) Using the result of the previous problem, and noting

$$I_b = \frac{V_{cntl}}{R}, \text{ we have: } \underline{f_{osc} = \frac{V_{cntl}}{nV_{ref}RC_L \ln 2}}$$

For $n=4$, $V_{ref}=1V$, $R=100k\Omega$, $C_L=0.1pF$, and $V_{cntl}=1V$:

$$\underline{f_{osc} \approx 36 \text{ MHz}}, \quad \underline{K_{osc} = 2\pi \cdot 36 \text{ MHz/V}}$$

16.11) If $t_1 =$ "the time required to discharge C_1 from V_{DD} to V_{ref} "
 $t_2 =$ "the time " " " " C_2 " " " "

Then: $T_{osc} = t_1 + t_2$

Using $I = \frac{V_{ctrl}}{R}$ & $I = C \frac{dV}{dt} \Rightarrow t_1 = \frac{V_{ctrl}}{RC_1(V_{DD} - V_{REF})}$

& $t_2 = \frac{V_{ctrl}}{RC_2(V_{DD} - V_{REF})}$

$f_{osc} = \frac{1}{t_1 + t_2} = R \frac{C_1 C_2}{C_1 + C_2} \frac{V_{DD} - V_{REF}}{V_{ctrl}}$
